

# Story of How POSTECH Defeats KAIST

## Mathematics for Science Quiz

2026 Science War Edition  
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POSTECH–KAIST Science War  
Pohang University of Science and Technology  
vs  
Korea Advanced Institute of Science and Technology

# Preface

This book is intended to provide complete preparation for the Mathematics section of Science Quiz in the **POSTECH-KAIST** Science War. Its structure begins with past Science Quiz problems: the mathematical concepts needed to solve them are developed in Mathematical Concepts, while the names, results, terminology, history, and other information worth remembering are organized in Facts to Memorize.

Science Quiz ranges across an unusually broad mathematical landscape, from middle- and high-school mathematics to undergraduate and more advanced university subjects. A strong competitor must therefore become fluent across many different areas of mathematics and be able to move between them quickly.

Accordingly, this book goes one step beyond the past problems themselves. It draws broadly from middle- and high-school mathematics competitions and olympiads; entrance examinations for gifted schools and science high schools; university entrance examinations; university midterm and final examinations; university mathematics competitions; and national and international olympiads.

Additional practice problems and their solutions are collected separately in *Expected Problems*.

Despite this breadth, the book retains its identity as a Science Quiz mathematics handbook. It emphasizes results, memorable formulas, elegant connections, interesting examples, and decisive ideas rather than long proofs or repetitive training. Its aim is to cultivate mathematical talent, intuition, and a reliable feel for unfamiliar problems.

Studying the mathematics of Science Quiz should also produce genuine mathematical growth beyond Science Quiz itself. The knowledge and habits of thought developed here are meant to strengthen mathematical understanding, judgment, and problem-solving ability in a lasting way.

I prepared this book as the 2026 Science Quiz Mathematics representative in the hope that **POSTECH** wins this year and continues to win in the years ahead. If circumstances permit, I will continue to revise and improve the book after 2026.

I sincerely hope that this book contributes to **POSTECH**'s victory. Good luck, and let the game begin.

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# Notation and Conventions

The notation specified below is used throughout this book. When several typographic variants of the same letter exist, each variant is assigned a distinct role and the variants are not interchanged.

## General Conventions

Scalars are written in ordinary italic letters, such as  $a, b, c, \lambda, t$ . Vectors are written in bold lowercase letters, such as  $\mathbf{u}, \mathbf{v}, \mathbf{x}$ . Matrices are written in bold uppercase letters, such as  $\mathbf{A}, \mathbf{B}, \mathbf{P}$ . In particular, the identity matrix, a diagonal matrix, and a projection matrix are written as  $\mathbf{I}_n$ ,  $\mathbf{D}$ , and  $\mathbf{P}$ , respectively. Unless otherwise stated, vectors are column vectors. Number systems are written in blackboard bold, sets ordinarily use uppercase italic letters, and families and collections use calligraphic letters when useful. Named operators are set upright.

Round parentheses are preferred for ordered tuples, sequences, and matrix delimiters; square brackets are an accepted alternative for matrices. Square brackets are also used for closed intervals and for area notation such as  $[ABC]$ . Braces are used for sets. In set-builder notation, the condition is separated by  $|$ , as in  $\{x \in A \mid P(x)\}$ . Determinants may be enclosed by vertical bars because a determinant is a scalar.

In symbolic statements, logical conjunction, disjunction, and negation are written as  $\wedge$ ,  $\vee$ , and  $\neg$ . The words “and” and “or” are used in ordinary prose. The symbol  $\Delta$  follows standard mathematical and physical usage:  $\Delta q$  may denote a finite change in a quantity,  $\Delta a_n = a_{n+1} - a_n$  denotes the forward difference of a sequence, and  $\Delta f$  denotes the Laplacian when  $\Delta$  acts as a differential operator on a function. These meanings are distinguished by context and, when more than one occurs nearby, the intended meaning is stated explicitly. When several equivalent notations are listed in the same table entry, the first notation is the convention used in this book; the later forms are included only to identify common alternatives that a reader may encounter elsewhere.

| Notation                                                       | Meaning                                                               |
|----------------------------------------------------------------|-----------------------------------------------------------------------|
| $\mathbb{N}$                                                   | the set of positive integers, $\{1, 2, 3, \dots\}$                    |
| $\mathbb{N}_0$                                                 | the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$              |
| $\mathbb{Z}$                                                   | the set of integers                                                   |
| $\mathbb{Q}$                                                   | the set of rational numbers                                           |
| $\mathbb{R}$                                                   | the set of real numbers                                               |
| $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ | the set of extended real numbers                                      |
| $\mathbb{C}$                                                   | the set of complex numbers                                            |
| $\mathbb{H}$                                                   | the quaternion division ring                                          |
| $a := b$                                                       | $a$ is defined to be $b$                                              |
| s.t.                                                           | abbreviation used to introduce a condition or constraint              |
| $\varepsilon$                                                  | a positive quantity in analysis, usually arbitrary                    |
| $\epsilon_{ijk}$                                               | the Levi-Civita symbol; the glyph $\epsilon$ is reserved for this use |

| Notation        | Meaning                                                          |
|-----------------|------------------------------------------------------------------|
| $O(g(x))$       | Big-O notation                                                   |
| $o(g(x))$       | little-o notation                                                |
| $\text{sgn}(x)$ | the sign of a real number or a permutation, according to context |
| $\square$       | the end of a proof or solution                                   |

## Theorem-like Environments

| Label          | Usage                                                                               |
|----------------|-------------------------------------------------------------------------------------|
| Definition     | the precise meaning of a term, object, property, or construction                    |
| Theorem        | a principal mathematical result                                                     |
| Proposition    | a useful result of more limited scope                                               |
| Lemma          | a supporting result, usually for a later proof                                      |
| Corollary      | a result following readily from a preceding result                                  |
| Conjecture     | a statement proposed to be true but not proved in full generality                   |
| Hypothesis     | a named assumption or proposed statement                                            |
| Formula        | a distinguished computational identity or closed form                               |
| Algorithm      | an explicit step-by-step procedure                                                  |
| Fact           | a verified statement not requiring theorem status                                   |
| Example        | an illustration of a definition, construction, phenomenon, or result                |
| Counterexample | an example disproving a general assertion or showing that a hypothesis is necessary |
| Remark         | supplementary context, a warning, or a qualification                                |
| Problem        | a mathematical task intended to be solved                                           |
| Open Problem   | a problem not known to be solved in the stated form                                 |
| Idea           | a decisive conceptual insight                                                       |
| Strategy       | a broad problem-solving plan                                                        |
| Tactic         | a local step within a larger strategy                                               |

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period. The ordinary end mark is  $\square$ ;  $\blacksquare$  is reserved for substantial or especially important proofs, and Q.E.D. for exceptional paper-level proofs.

Enumeration labels are set upright. Decimal labels such as 1.1, 1.2,  $\dots$  identify independent units under a common parent; (1), (2),  $\dots$  are used for ordinary enumerations; (a), (b),  $\dots$  are used for connected subparts of one problem; and (i), (ii),  $\dots$  are reserved for formal mathematical clauses or hypotheses. Displayed equations are punctuated as parts of their surrounding sentences and are numbered only when they are referenced or need a stable label.

## Set Theory and Logic

Uppercase letters such as  $A, B, X, Y$  usually denote sets, while lowercase letters such as  $a, b, x, y$  usually denote their elements unless the context indicates another meaning. The arrow  $f: X \rightarrow Y$  specifies the domain and codomain of a function, while  $x \mapsto f(x)$  specifies its action on an element.

| Notation                    | Meaning                                                             |
|-----------------------------|---------------------------------------------------------------------|
| $x \in A$                   | $x$ is an element of $A$                                            |
| $x \notin A$                | $x$ is not an element of $A$                                        |
| $A \subseteq B$             | $A$ is a subset of $B$                                              |
| $A \subsetneq B$            | $A$ is a proper subset of $B$                                       |
| $A \supseteq B$             | $A$ is a superset of $B$                                            |
| $A \supsetneq B$            | $A$ is a proper superset of $B$                                     |
| $A \cup B$                  | the union of $A$ and $B$                                            |
| $A \cap B$                  | the intersection of $A$ and $B$                                     |
| $\bigcup_{i \in I} A_i$     | the union of an indexed family of sets                              |
| $\bigcap_{i \in I} A_i$     | the intersection of an indexed family of sets                       |
| $A \setminus B$             | the set difference of $A$ and $B$                                   |
| $A \Delta B$                | the symmetric difference of $A$ and $B$                             |
| $A^c$                       | the complement of $A$ in the understood ambient set                 |
| $\emptyset$                 | the empty set                                                       |
| $\mathcal{P}(A)$            | the power set of $A$                                                |
| $A \times B$                | the Cartesian product of $A$ and $B$                                |
| $A \sqcup B$                | the disjoint union of $A$ and $B$                                   |
| $ A $                       | the cardinality of a finite set $A$                                 |
| $\{x \in A \mid P(x)\}$     | the set of all $x \in A$ for which $P(x)$ holds                     |
| $x \sim y$                  | $x$ and $y$ are equivalent under an understood equivalence relation |
| $A / \sim$                  | the set of equivalence classes of $A$ under $\sim$                  |
| $\forall x \in A$           | for every $x \in A$                                                 |
| $\exists x \in A$           | there exists $x \in A$                                              |
| $\exists! x \in A$          | there exists a unique $x \in A$                                     |
| $\nexists x \in A$          | there does not exist $x \in A$                                      |
| $P \Rightarrow Q$           | $P$ implies $Q$                                                     |
| $P \nRightarrow Q$          | $P$ does not imply $Q$                                              |
| $P \Leftarrow Q$            | $Q$ implies $P$                                                     |
| $P \Leftrightarrow Q$       | $P$ if and only if $Q$                                              |
| $P \wedge Q$                | logical conjunction                                                 |
| $P \vee Q$                  | logical disjunction                                                 |
| $\neg P$                    | logical negation                                                    |
| $f: X \rightarrow Y$        | a function from $X$ to $Y$                                          |
| $x \mapsto f(x)$            | the elementwise rule of a function                                  |
| $\text{dom } f$             | the domain of $f$                                                   |
| $\text{codom } f$           | the codomain of $f$                                                 |
| $\text{id}_X$               | the identity map on $X$                                             |
| $g \circ f$                 | composition of functions                                            |
| $f _A$                      | restriction of $f$ to $A$                                           |
| $f: X \hookrightarrow Y$    | an injective map                                                    |
| $f: X \twoheadrightarrow Y$ | a surjective map                                                    |
| $f: X \xrightarrow{\sim} Y$ | a bijective map                                                     |
| $f: X \xrightarrow{\sim} Y$ | an isomorphism                                                      |
| $X \cong Y$                 | $X$ and $Y$ are isomorphic; in topology, this means homeomorphic    |
| $f(A)$                      | the image of $A$ under $f$                                          |
| $f^{-1}(B)$                 | the inverse image, or preimage, of $B$                              |

| Notation          | Meaning                                     |
|-------------------|---------------------------------------------|
| $\mathbf{1}_A(x)$ | the indicator function of $A$               |
| $\aleph_0$        | the cardinality of a countably infinite set |
| $\mathfrak{c}$    | the cardinality of the continuum            |

The symbol  $\mathbf{1}_A$  denotes an indicator function, not a vector. The all-ones vector is  $\mathbf{1}$ . Boldface is otherwise reserved for vectors and matrices.

## Calculus and Analysis

In polar, cylindrical, and spherical coordinates,  $\theta$  is the azimuthal angle measured counterclockwise from the positive  $x$ -axis in the  $xy$ -plane. In spherical coordinates,  $\phi$  is the polar angle measured from the positive  $z$ -axis. The glyph  $\phi$  is reserved for this spherical polar angle. The glyph  $\varphi$  is used for functions, homomorphisms, Euler's totient function, and the golden ratio whenever one of these objects is introduced.

| Notation                        | Meaning                                                      |
|---------------------------------|--------------------------------------------------------------|
| $\lim_{x \rightarrow a} f(x)$   | the limit of $f(x)$ as $x$ approaches $a$                    |
| $\lim_{x \rightarrow a^-} f(x)$ | the left-hand limit of $f(x)$ as $x$ approaches $a$          |
| $\lim_{x \rightarrow a^+} f(x)$ | the right-hand limit of $f(x)$ as $x$ approaches $a$         |
| $\min A, \max A$                | the minimum and maximum of $A$                               |
| $\inf A, \sup A$                | the infimum and supremum of $A$                              |
| $f'(a)$                         | the first derivative of $f$ at $a$                           |
| $f^{(n)}(a)$                    | the $n$ th derivative of $f$ at $a$                          |
| $\frac{dy}{dx}$                 | the derivative of $y$ with respect to $x$ , with upright d   |
| $\int_a^b f(x) dx$              | the definite integral of $f$ over $[a, b]$                   |
| $\iint_D f dA$                  | a double integral over $D$                                   |
| $\iiint_{\Omega} f dV$          | a triple integral over $\Omega$                              |
| $\sum_{k=1}^{\infty} a_k$       | a finite sum                                                 |
| $\sum_{k=1}^{\infty} a_k$       | an infinite series                                           |
| $\prod_{k=1}^n a_k$             | a finite product                                             |
| $e^x, \exp x$                   | the exponential function                                     |
| $\Gamma(z)$                     | the Gamma function                                           |
| $B(x, y)$                       | the Beta function                                            |
| $\zeta(s)$                      | the Riemann zeta function                                    |
| $\ln x$                         | the logarithm to base $e$                                    |
| $\lg x$                         | the logarithm to base 10                                     |
| $\text{lb } x$                  | the logarithm to base 2                                      |
| $\log_a x$                      | the logarithm to the explicitly specified base $a$           |
| $\arcsin x$                     | the principal inverse sine, with values in $[-\pi/2, \pi/2]$ |
| $\arccos x$                     | the principal inverse cosine, with values in $[0, \pi]$      |

| Notation                                             | Meaning                                                                          |
|------------------------------------------------------|----------------------------------------------------------------------------------|
| $\arctan x$                                          | the principal inverse tangent, with values in $(-\pi/2, \pi/2)$                  |
| $\operatorname{arcsec} x$                            | the principal inverse secant, with values in $[0, \pi] \setminus \{\pi/2\}$      |
| $\operatorname{arccsc} x$                            | the principal inverse cosecant, with values in $[-\pi/2, \pi/2] \setminus \{0\}$ |
| $\operatorname{arccot} x$                            | the principal inverse cotangent, with values in $(0, \pi)$                       |
| $\operatorname{sech} x$                              | the hyperbolic secant $1/\cosh x$                                                |
| $\operatorname{csch} x$                              | the hyperbolic cosecant $1/\sinh x$                                              |
| $\operatorname{arcsinh} x$                           | the inverse hyperbolic sine                                                      |
| $\operatorname{arccosh} x$                           | the principal inverse hyperbolic cosine, with values in $[0, \infty)$            |
| $\operatorname{arctanh} x$                           | the inverse hyperbolic tangent on $(-1, 1)$                                      |
| $\operatorname{arcsech} x$                           | the principal inverse hyperbolic secant on $(0, 1]$                              |
| $\operatorname{arccsch} x$                           | the principal inverse hyperbolic cosecant on $\mathbb{R} \setminus \{0\}$        |
| $\operatorname{arccoth} x$                           | the principal inverse hyperbolic cotangent on $(-\infty, -1) \cup (1, \infty)$   |
| $C^k(I)$                                             | the space of $k$ times continuously differentiable functions on $I$              |
| $C^\infty(I)$                                        | the space of smooth functions on $I$                                             |
| $\frac{\partial f}{\partial x_i}$                    | the partial derivative of $f$ with respect to $x_i$                              |
| $(x, y), (x, y, z)$                                  | Cartesian coordinates                                                            |
| $(r, \theta)$                                        | polar coordinates                                                                |
| $(r, \theta, z)$                                     | cylindrical coordinates                                                          |
| $(\rho, \theta, \phi)$                               | spherical coordinates                                                            |
| $\nabla f$                                           | the gradient of $f$                                                              |
| $\nabla \cdot \mathbf{F}$                            | the divergence of $\mathbf{F}$                                                   |
| $\nabla \times \mathbf{F}$                           | the curl of $\mathbf{F}$                                                         |
| $\Delta q = q_{\text{final}} - q_{\text{initial}}$   | the finite change of a quantity $q$ , when the context is discrete or physical   |
| $\Delta f$                                           | the Laplacian of $f$ , when $\Delta$ acts as a differential operator             |
| $D\mathbf{F}(\mathbf{a})$                            | the total derivative, or Jacobian matrix, of $\mathbf{F}$ at $\mathbf{a}$        |
| $\operatorname{Jac}(\mathbf{F})$                     | the Jacobian determinant of $\mathbf{F}$                                         |
| $\operatorname{Hess} f$                              | the Hessian matrix of $f$                                                        |
| $\mathbf{r}(t)$                                      | a parametrized curve                                                             |
| $ds$                                                 | an element of arc length                                                         |
| $\int_C \mathbf{F} \cdot d\mathbf{r}$                | a line integral along $C$                                                        |
| $\iint_S \mathbf{F} \cdot \mathbf{n} dS$             | a flux integral across $S$                                                       |
| $B_r(x)$                                             | the open ball of radius $r$ centered at $x$                                      |
| $\overline{B}_r(x)$                                  | the closed ball of radius $r$ centered at $x$                                    |
| $x_n \rightarrow x$                                  | convergence of a sequence                                                        |
| $a_n \nearrow L$                                     | monotone increasing convergence to $L$                                           |
| $a_n \searrow L$                                     | monotone decreasing convergence to $L$                                           |
| $\limsup a_n$                                        | the limit superior of $(a_n)$                                                    |
| $\liminf a_n$                                        | the limit inferior of $(a_n)$                                                    |
| $f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$  | pointwise convergence                                                            |
| $f_n \Rightarrow f, f_n \xrightarrow{\text{unif}} f$ | uniform convergence                                                              |
| $f_n \xrightarrow{\text{a.e.}} f$                    | convergence almost everywhere                                                    |
| $\ f\ _\infty$                                       | the supremum norm of $f$                                                         |
| $z = x + iy$                                         | a complex number with real part $x$ and imaginary part $y$                       |
| $\operatorname{Re} z, \Re z$                         | the real part of $z$                                                             |
| $\operatorname{Im} z, \Im z$                         | the imaginary part of $z$                                                        |

| Notation                                                  | Meaning                                                             |
|-----------------------------------------------------------|---------------------------------------------------------------------|
| $\bar{z}$                                                 | the complex conjugate of $z$                                        |
| $ z $                                                     | the modulus of $z$                                                  |
| $\arg z$                                                  | the multivalued argument of $z \neq 0$                              |
| $\operatorname{Arg} z$                                    | the principal argument, with $-\pi < \operatorname{Arg} z \leq \pi$ |
| $\log z$                                                  | the multivalued complex logarithm                                   |
| $\operatorname{Log} z = \ln  z  + i \operatorname{Arg} z$ | the principal complex logarithm                                     |
| $\oint_C f(z) \, dz$                                      | a contour integral around a closed curve $C$                        |
| $\operatorname{Res}(f; a)$                                | the residue of $f$ at $a$                                           |

## Linear Algebra

| Notation                                                 | Meaning                                                         |
|----------------------------------------------------------|-----------------------------------------------------------------|
| $\mathbf{A} = (a_{ij})$                                  | the matrix whose $(i, j)$ -entry is $a_{ij}$                    |
| $\mathbf{A}^\top$                                        | the transpose of $\mathbf{A}$                                   |
| $\mathbf{A}^H$                                           | the conjugate transpose of $\mathbf{A}$                         |
| $\mathbf{A}^{-1}$                                        | the inverse of $\mathbf{A}$                                     |
| $\mathbf{I}_n$                                           | the $n \times n$ identity matrix                                |
| $\mathbf{0}$                                             | a zero vector or zero matrix, with dimension clear from context |
| $\mathbf{D}$                                             | a diagonal matrix                                               |
| $\operatorname{diag}(d_1, \dots, d_n)$                   | the diagonal matrix with diagonal entries $d_1, \dots, d_n$     |
| $\mathbf{1}$                                             | the all-ones vector                                             |
| $\mathbf{J}_{m,n}, \mathbf{J}_n$                         | the $m \times n$ and $n \times n$ all-ones matrices             |
| $\mathbf{P}_\sigma$                                      | the permutation matrix associated with $\sigma$                 |
| $\det(\mathbf{A})$                                       | the determinant of $\mathbf{A}$                                 |
| $\operatorname{rk}(\mathbf{A})$                          | the rank of $\mathbf{A}$                                        |
| $\operatorname{tr}(\mathbf{A})$                          | the trace of $\mathbf{A}$                                       |
| $p_{\mathbf{A}}(\lambda)$                                | the characteristic polynomial of $\mathbf{A}$                   |
| $m_{\mathbf{A}}(\lambda)$                                | the minimal polynomial of $\mathbf{A}$                          |
| $\rho(\mathbf{A})$                                       | the spectral radius of $\mathbf{A}$                             |
| $\sigma_i(\mathbf{A})$                                   | the $i$ th singular value of $\mathbf{A}$                       |
| $M_{ij}$                                                 | the minor of the entry $a_{ij}$                                 |
| $C_{ij} = (-1)^{i+j} M_{ij}$                             | the cofactor of the entry $a_{ij}$                              |
| $\operatorname{adj}(\mathbf{A})$                         | the adjugate of $\mathbf{A}$                                    |
| $\operatorname{Col}(\mathbf{A})$                         | the column space of $\mathbf{A}$                                |
| $\operatorname{Null}(\mathbf{A})$                        | the null space of $\mathbf{A}$                                  |
| $\operatorname{Row}(\mathbf{A})$                         | the row space of $\mathbf{A}$                                   |
| $\ker T$                                                 | the kernel of a linear map $T$                                  |
| $\operatorname{im} T$                                    | the image of a linear map $T$                                   |
| $\mathbf{e}_i$                                           | the $i$ th standard basis vector                                |
| $\dim V$                                                 | the dimension of the vector space $V$                           |
| $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ | the span of the listed vectors                                  |
| $U \oplus V$                                             | the direct sum of subspaces $U$ and $V$                         |
| $\langle \mathbf{u}, \mathbf{v} \rangle$                 | the inner product of $\mathbf{u}$ and $\mathbf{v}$              |
| $\mathbf{u} \cdot \mathbf{v}$                            | the standard Euclidean inner product on $\mathbb{R}^n$          |
| $\mathbf{u}^H \mathbf{v}$                                | the standard Hermitian inner product on $\mathbb{C}^n$          |

| Notation                              | Meaning                                                                  |
|---------------------------------------|--------------------------------------------------------------------------|
| $\ \mathbf{v}\ $                      | the norm of $\mathbf{v}$                                                 |
| $U^\perp$                             | the orthogonal complement of the subspace $U$                            |
| $\text{proj}_U \mathbf{x}$            | the orthogonal projection of $\mathbf{x}$ onto $U$                       |
| $\text{proj}_{\mathbf{v}} \mathbf{x}$ | the orthogonal projection of $\mathbf{x}$ onto $\text{span}(\mathbf{v})$ |
| $\mathbf{u} \perp \mathbf{v}$         | $\mathbf{u}$ and $\mathbf{v}$ are orthogonal                             |
| $\mathbf{u} \times \mathbf{v}$        | the cross product in $\mathbb{R}^3$                                      |
| $E_\lambda(\mathbf{A})$               | the eigenspace of $\mathbf{A}$ associated with $\lambda$                 |
| $\sigma(\mathbf{A})$                  | the spectrum of $\mathbf{A}$                                             |

On a complex inner-product space, the inner product is conjugate-linear in its first argument and linear in its second argument.

## Differential Equations

The Fourier transform uses the normalization

$$\hat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\mathbf{x}, \quad f(\mathbf{x}) = \int_{\mathbb{R}^n} \hat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\boldsymbol{\xi},$$

whenever the Fourier inversion formula applies.

| Notation                    | Meaning                                                              |
|-----------------------------|----------------------------------------------------------------------|
| $y'$                        | the first derivative of $y$ with respect to the independent variable |
| $y''$                       | the second derivative of $y$                                         |
| $\dot{x}$                   | the first derivative of $x$ with respect to time $t$                 |
| $\ddot{x}$                  | the second derivative of $x$ with respect to time $t$                |
| $y_h$                       | the homogeneous part of a solution                                   |
| $y_p$                       | a particular solution                                                |
| $c_1, c_2, \dots$           | arbitrary constants in a general solution                            |
| $W(y_1, \dots, y_n)(t)$     | the Wronskian of $y_1, \dots, y_n$                                   |
| $J_\nu(x), Y_\nu(x)$        | Bessel functions of the first and second kinds of order $\nu$        |
| $\mathcal{L}\{f(t)\}(s)$    | the Laplace transform of $f(t)$                                      |
| $H(t-a), u(t-a)$            | the Heaviside unit-step function shifted by $a$                      |
| $\hat{f}(\boldsymbol{\xi})$ | the Fourier transform under the normalization above                  |

## Combinatorics and Discrete Mathematics

| Notation                                                        | Meaning                                                           |
|-----------------------------------------------------------------|-------------------------------------------------------------------|
| $[n]$                                                           | the set $\{1, 2, \dots, n\}$                                      |
| $n!$                                                            | factorial                                                         |
| $\binom{n}{k}, {}_n C_k$                                        | the number of $k$ -element subsets of an $n$ -element set         |
| $\left(\!\!\binom{n}{k}\!\!\right), {}_n H_k$                   | the number of size- $k$ multisets chosen from an $n$ -element set |
| $P(n, k)$                                                       | the number of $k$ -permutations of an $n$ -element set            |
| $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ | the Stirling number of the second kind                            |
| $a_n$                                                           | the $n$ th term of a sequence                                     |
| $\Delta a_n$                                                    | the forward difference $a_{n+1} - a_n$                            |

| Notation                              | Meaning                                      |
|---------------------------------------|----------------------------------------------|
| $S_n = \sum_{k=1}^n a_k$              | the $n$ th partial sum                       |
| $A(x) = \sum_{n \geq 0} a_n x^n$      | an ordinary generating function              |
| $[x^n]A(x)$                           | the coefficient of $x^n$ in $A(x)$           |
| $\omega_n = e^{2\pi i/n}$             | a primitive $n$ th root of unity             |
| $\sigma = (a_1 \ a_2 \ \cdots \ a_k)$ | cycle notation for a permutation             |
| $\text{inv}(\sigma)$                  | the inversion number of $\sigma$             |
| $\text{Fix}(g)$                       | the set of objects fixed by $g$              |
| $G = (V, E)$                          | a graph with vertex set $V$ and edge set $E$ |
| $\deg(v)$                             | the degree of the vertex $v$                 |
| $K_n$                                 | the complete graph on $n$ vertices           |
| $C_n$                                 | the cycle graph on $n$ vertices              |
| $P_n$                                 | the path graph on $n$ vertices               |

## Probability and Statistics

Random variables are usually denoted by uppercase letters such as  $X, Y, Z$ ; events are denoted by uppercase letters such as  $A, B, E$ .

| Notation                            | Meaning                                                                  |
|-------------------------------------|--------------------------------------------------------------------------|
| $\mathbb{P}(A)$                     | the probability of the event $A$                                         |
| $\mathbb{P}(A \mid B)$              | the conditional probability of $A$ given $B$                             |
| $\mathbb{E}[X]$                     | the expectation of $X$                                                   |
| $\text{Var}(X)$                     | the variance of $X$                                                      |
| $\text{Cov}(X, Y)$                  | the covariance of $X$ and $Y$                                            |
| $X \sim \text{Bern}(p)$             | a Bernoulli distribution                                                 |
| $X \sim \text{Bin}(n, p)$           | a binomial distribution                                                  |
| $X \sim \text{Geom}(p)$             | a geometric distribution on $\mathbb{N}$ , counting the successful trial |
| $X \sim \text{Pois}(\lambda)$       | a Poisson distribution                                                   |
| $X \sim \text{Unif}(a, b)$          | a continuous uniform distribution on $[a, b]$                            |
| $X \sim \text{Exp}(\lambda)$        | an exponential distribution with rate $\lambda$                          |
| $X \sim \mathcal{N}(\mu, \sigma^2)$ | a normal distribution with mean $\mu$ and variance $\sigma^2$            |
| $X_n \xrightarrow{\text{a.s.}} X$   | almost-sure convergence                                                  |
| $X_n \xrightarrow{\mathbb{P}} X$    | convergence in probability                                               |
| $X_n \xrightarrow{L^p} X$           | convergence in $L^p$                                                     |
| $X_n \xrightarrow{d} X$             | convergence in distribution                                              |

## Geometry

Uppercase letters such as  $A, B, C, D$  denote points. When the usual triangle notation is adopted, the corresponding lowercase letters  $a, b, c$  denote the side lengths opposite  $A, B, C$ , respectively.

| Notation            | Meaning                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| $\triangle ABC$     | the triangle with vertices $A, B, C$                                                                                                        |
| $\angle ABC$        | the angle with vertex $B$                                                                                                                   |
| $\overline{AB}$     | the segment joining $A$ and $B$                                                                                                             |
| $AB$                | the length of $\overline{AB}$                                                                                                               |
| $[ABC]$             | the area of $\triangle ABC$                                                                                                                 |
| $\text{length}(S)$  | the length of a curve or one-dimensional geometric object $S$                                                                               |
| $\text{area}(S)$    | the area of a planar region or surface $S$                                                                                                  |
| $\text{vol}(S)$     | the volume of a three-dimensional region or solid $S$                                                                                       |
| $\text{dist}(P, S)$ | the distance from the point $P$ to the set or geometric object $S$                                                                          |
| $\text{conv}(S)$    | the convex hull of $S$                                                                                                                      |
| $\mathbb{R}^n$      | $n$ -dimensional Euclidean space, in geometric contexts                                                                                     |
| $\mathbb{H}^n$      | $n$ -dimensional hyperbolic space, in geometric contexts;<br>$\mathbb{H}$ without an exponent denotes the quaternions in algebraic contexts |
| $R$                 | the circumradius of a triangle                                                                                                              |
| $r$                 | the inradius of a triangle                                                                                                                  |
| $s$                 | the semiperimeter of a triangle                                                                                                             |
| $I$                 | the number of interior lattice points in Pick's Theorem                                                                                     |
| $B$                 | the number of boundary lattice points in Pick's Theorem                                                                                     |
| $\parallel$         | parallel                                                                                                                                    |
| $\perp$             | perpendicular                                                                                                                               |

## Number Theory

Unless otherwise stated, divisibility and congruence statements are made in  $\mathbb{Z}$ .

| Notation                          | Meaning                                               |
|-----------------------------------|-------------------------------------------------------|
| $a \mid b$                        | $a$ divides $b$                                       |
| $a \nmid b$                       | $a$ does not divide $b$                               |
| $\gcd(a, b)$                      | the greatest common divisor of $a$ and $b$            |
| $\text{lcm}(a, b)$                | the least common multiple of $a$ and $b$              |
| $a \equiv b \pmod{n}$             | $a$ is congruent to $b$ modulo $n$                    |
| $\pi(x)$                          | the number of primes not exceeding $x$                |
| $[a]_n$                           | the residue class of $a$ modulo $n$                   |
| $\mathbb{Z}/n\mathbb{Z}$          | the ring of residue classes modulo $n$                |
| $(\mathbb{Z}/n\mathbb{Z})^\times$ | the multiplicative group of units modulo $n$          |
| $a^{-1} \pmod{n}$                 | the multiplicative inverse of $a$ modulo $n$          |
| $\left(\frac{a}{p}\right)$        | the Legendre symbol                                   |
| $\varphi(n)$                      | Euler's totient function                              |
| $\text{ord}_n(a)$                 | the multiplicative order of $a$ modulo $n$            |
| $v_p(n)$                          | the exponent of $p$ in the prime factorization of $n$ |
| $\tau(n)$                         | the number of positive divisors of $n$                |
| $\sigma(n)$                       | the sum of the positive divisors of $n$               |
| $\mu(n)$                          | the Möbius function                                   |
| $\text{rad}(n)$                   | the product of the distinct prime divisors of $n$     |
| $p$                               | usually a prime number in number-theoretic contexts   |
| $\mathbb{F}_q$                    | the finite field with $q$ elements                    |

## Algebra

Capital letters such as  $G, H$  denote groups,  $R, S$  denote rings, and  $F, K, L$  denote fields; lower-case letters denote their elements unless the context requires otherwise.

| Notation                | Meaning                                                                     |
|-------------------------|-----------------------------------------------------------------------------|
| $G$                     | usually a group                                                             |
| $S_n$                   | the symmetric group on $n$ symbols                                          |
| $A_n$                   | the alternating group on $n$ symbols                                        |
| $D_{2n}$                | the dihedral group of order $2n$                                            |
| $e$                     | the identity element of a group                                             |
| $ G $                   | the order of a finite group $G$                                             |
| $\langle S \rangle$     | the subgroup generated by $S$                                               |
| $H \leq G$              | $H$ is a subgroup of $G$                                                    |
| $H \trianglelefteq G$   | $H$ is a normal subgroup of $G$                                             |
| $G/H$                   | the quotient group of $G$ by $H$                                            |
| $N \rtimes_{\varphi} H$ | the semidirect product determined by $\varphi: H \rightarrow \text{Aut}(N)$ |
| $\ker \varphi$          | the kernel of a homomorphism $\varphi$                                      |
| $\text{im } \varphi$    | the image of a homomorphism $\varphi$                                       |
| $\text{Aut}(G)$         | the automorphism group of $G$                                               |
| $G \cdot x$             | the orbit of $x$ under a group action                                       |
| $G_x$                   | the stabilizer of $x$ under a group action                                  |
| $Z(G)$                  | the center of $G$                                                           |
| $C_G(x)$                | the centralizer of $x$ in $G$                                               |
| $\text{GL}_n(F)$        | the general linear group over $F$                                           |
| $\text{SL}_n(F)$        | the special linear group over $F$                                           |
| $R$                     | usually a ring                                                              |
| UFD                     | unique factorization domain                                                 |
| PID                     | principal ideal domain                                                      |
| ED                      | Euclidean domain                                                            |
| $I \trianglelefteq R$   | $I$ is an ideal of $R$                                                      |
| $R/I$                   | the quotient ring of $R$ by $I$                                             |
| $F$                     | a base field in an algebraic context                                        |
| $K$                     | a field extension of $F$ , or a second field when required                  |
| $F[x]$                  | the polynomial ring over $F$                                                |
| $\deg f$                | the degree of a polynomial $f$                                              |
| $\text{Disc}(f)$        | the discriminant of a polynomial $f$                                        |
| $\text{trdeg}_K L$      | the transcendence degree of $L$ over $K$                                    |
| $\text{char } F$        | the characteristic of $F$                                                   |
| $\text{Gal}(L/K)$       | the Galois group of the extension $L/K$                                     |

## Topology

| Notation           | Meaning             |
|--------------------|---------------------|
| $(X, \mathcal{T})$ | a topological space |
| $A^\circ$          | the interior of $A$ |
| $\overline{A}$     | the closure of $A$  |

| Notation        | Meaning                                     |
|-----------------|---------------------------------------------|
| $\partial A$    | the boundary of $A$                         |
| $\partial M$    | the boundary of a manifold $M$              |
| $X \vee Y$      | the wedge sum of pointed spaces $X$ and $Y$ |
| $\pi_1(X, x_0)$ | the fundamental group of $X$ based at $x_0$ |
| $\chi(X)$       | the Euler characteristic of $X$             |
| $X \cong Y$     | $X$ and $Y$ are homeomorphic                |
| $X \simeq Y$    | $X$ and $Y$ are homotopy equivalent         |

## Miscellaneous Topics

### Mathematical Constants

The table contains constants that occur frequently across several areas of mathematics. A defining expression is used whenever a single-letter symbol is not needed elsewhere in the book.

| Symbol     | Name                      | Meaning or definition                                                                    | Exact value or approximation                         |
|------------|---------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------|
| $i$        | Imaginary unit            | The imaginary unit, defined by the relation $i^2 = -1$ .                                 | $i^2 = -1$                                           |
| $\sqrt{2}$ | Pythagoras constant       | The positive length of the diagonal of a unit square.                                    | 1.41421356237...                                     |
| $\sqrt{3}$ | Square root of 3          | The positive square root of 3.                                                           | 1.73205080757...                                     |
| $\sqrt{5}$ | Square root of 5          | The positive square root of 5.                                                           | 2.23606797750...                                     |
| $\sqrt{7}$ | Square root of 7          | The positive square root of 7.                                                           | 2.64575131106...                                     |
| $e$        | Euler's number            | The base of the natural logarithm:<br>$e = \sum_{n=0}^{\infty} 1/n!$                     | 2.71828182846...                                     |
| $\pi$      | Pi                        | The ratio of the circumference of a Euclidean circle to its diameter.                    | 3.14159265359...                                     |
| $\varphi$  | Golden ratio              | The positive solution of $x^2 = x + 1$ .                                                 | $\frac{1 + \sqrt{5}}{2} \approx$<br>1.61803398875... |
| $\gamma$   | Euler–Mascheroni constant | $\gamma = \lim_{n \rightarrow \infty} \left( \sum_{k=1}^n \frac{1}{k} - \ln n \right)$ . | 0.577215664902...                                    |
| $\zeta(3)$ | Apéry's constant          | The value of the Riemann zeta function at 3: $\sum_{n=1}^{\infty} 1/n^3$ .               | 1.20205690316...                                     |
| $G$        | Catalan's constant        | $G = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$ .                                      | 0.915965594177...                                    |

**Other Notation**

| Notation            | Meaning                                     |
|---------------------|---------------------------------------------|
| $\lfloor x \rfloor$ | the floor of $x$                            |
| $\lceil x \rceil$   | the ceiling of $x$                          |
| $\{x\}$             | the fractional part $x - \lfloor x \rfloor$ |

**Part I**

**Mathematical Concepts**

# Chapter 1

## Set Theory and Logic

Related **POSTECH** Courses: (MATH202) Set Theory

Related Books: Introduction to Set Theory (Karel Hrbacek, Thomas Jech)

### Definition 1.1. Set and Element

A set is a collection of objects, called its elements. We write  $x \in A$  if  $x$  is an element of  $A$ , and  $x \notin A$  otherwise. The relation  $A \subseteq B$  means

$$\forall x, \quad x \in A \Rightarrow x \in B.$$

### Formula 1.2. Basic Set Operations

For sets  $A$  and  $B$  inside an ambient set  $U$ , we use

$$A \cup B = \{x \in U \mid (x \in A) \vee (x \in B)\},$$

$$A \cap B = \{x \in U \mid (x \in A) \wedge (x \in B)\},$$

and

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

### Formula 1.3. Set Algebra

For subsets  $A, B, C \subseteq U$ , the following identities are often useful:

$$A \cup \emptyset = A, \quad A \cap U = A, \quad A \cup A = A, \quad A \cap A = A.$$

The distributive laws are

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The absorption laws are

$$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A.$$

### Formula 1.4. Basic Logical Equivalences

The most useful logical equivalences are

$$(P \Rightarrow Q) \Longleftrightarrow (\neg Q \Rightarrow \neg P),$$

$$\neg(P \Rightarrow Q) \Longleftrightarrow P \wedge \neg Q,$$

$$\neg(P \wedge Q) \Longleftrightarrow (\neg P) \vee (\neg Q), \quad \neg(P \vee Q) \Longleftrightarrow (\neg P) \wedge (\neg Q),$$

and

$$\neg(\forall x \in A, P(x)) \Longleftrightarrow \exists x \in A \text{ s.t. } \neg P(x),$$

$$\neg(\exists x \in A \text{ s.t. } P(x)) \Longleftrightarrow \forall x \in A, \neg P(x).$$

*Strategy.* To prove  $P \Rightarrow Q$ , it is often enough to prove the contrapositive  $\neg Q \Rightarrow \neg P$ . To disprove a universal statement, find a counterexample.

**Formula 1.5. De Morgan's Laws**

For subsets  $A, B \subseteq U$ , we have

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

More generally,

$$\left( \bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left( \bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

**Definition 1.6. Power Set**

The power set of a set  $A$  is

$$\mathcal{P}(A) = \{B \mid B \subseteq A\}.$$

If  $|A| = n < \infty$ , then

$$|\mathcal{P}(A)| = 2^n.$$

**Definition 1.7. Function**

A function  $f: X \rightarrow Y$  assigns to each element  $x \in X$  a unique element  $f(x) \in Y$ . The set  $X$  is the domain, and  $Y$  is the codomain. The notation  $x \mapsto f(x)$  describes the rule on elements.

**Definition 1.8. Injection, Surjection, and Bijection**

A function  $f: X \rightarrow Y$  is injective if distinct elements of  $X$  have distinct images. Equivalently,

$$\forall x_1, x_2 \in X, \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Once injectivity has been established, one may write  $f: X \hookrightarrow Y$ .

The function is surjective if every element of  $Y$  has a preimage:

$$\forall y \in Y, \quad \exists x \in X \text{ s.t. } f(x) = y.$$

Once surjectivity has been established, one may write  $f: X \twoheadrightarrow Y$ . The function is bijective if it is both injective and surjective.

**Proposition 1.9. Finite Injective-Surjective Criterion**

If  $X$  and  $Y$  are finite sets with  $|X| = |Y|$ , then for a function  $f: X \rightarrow Y$ , the following are equivalent:

$$f \text{ is injective} \iff f \text{ is surjective} \iff f \text{ is bijective}.$$

**Proposition 1.10. Inverse Image Preserves Set Operations**

If  $f: X \rightarrow Y$  and  $A, B \subseteq Y$ , then

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B),$$

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B),$$

and

$$f^{-1}(Y \setminus A) = X \setminus f^{-1}(A).$$

**Definition 1.11. Relation and Equivalence Relation**

A binary relation on a set  $A$  is a subset  $R \subseteq A \times A$ . An equivalence relation  $\sim$  is reflexive, symmetric, and transitive; explicitly,

$$\forall x \in A, \quad x \sim x,$$

$$\forall x, y \in A, \quad x \sim y \Rightarrow y \sim x,$$

and

$$\forall x, y, z \in A, \quad (x \sim y \wedge y \sim z) \Rightarrow x \sim z.$$

The equivalence class of  $a \in A$  is

$$[a] = \{x \in A \mid x \sim a\}.$$

**Proposition 1.12. Equivalence Relations and Partitions**

*Equivalence relations on a set  $A$  are the same structure as partitions of  $A$ : each equivalence relation partitions  $A$  into equivalence classes, and each partition of  $A$  determines an equivalence relation.*

**Definition 1.13. Cardinality and Equipnumerosity**

Two sets  $A$  and  $B$  have the same cardinality if there exists a bijection between them. We write

$$|A| = |B| \quad \text{or} \quad A \hookleftrightarrow B.$$

Thus cardinality compares the sizes of finite and infinite sets through one-to-one correspondences rather than through geometric containment.

**Definition 1.14. Dedekind-Infinite Set**

A set  $A$  is Dedekind-infinite if there exists a proper subset  $B \subsetneq A$  such that

$$|A| = |B|.$$

In ZFC, a set is Dedekind-infinite if and only if it is infinite. Without the Axiom of Choice, this equivalence cannot be proved in full generality.

**Definition 1.15. Countability**

A set  $A$  is countable if it is finite or if there exists a bijection  $f: A \rightarrow \mathbb{N}$ . In the countably infinite case,

$$\exists f: A \hookleftrightarrow \mathbb{N}, \quad |A| = |\mathbb{N}| = \aleph_0.$$

A set is uncountable if it is not countable.

**Theorem 1.16. Basic Cardinalities**

*The standard countably infinite sets satisfy*

$$|\mathbb{N}| = |2\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| = \aleph_0,$$

whereas

$$|(0, 1)| = |\mathbb{R}| = \mathfrak{c} = 2^{\aleph_0} > \aleph_0.$$

*In particular, a proper subset of an infinite set may have the same cardinality as the whole set, while the real numbers cannot be listed by the natural numbers.*

**Theorem 1.17. Cantor's Theorem**

For every set  $A$ , there is no surjection from  $A$  onto  $\mathcal{P}(A)$ . Equivalently,

$$|A| < |\mathcal{P}(A)|.$$

**Definition 1.18. Aleph Numbers and the Continuum**

The symbol  $\aleph_0$  denotes the cardinality of a countably infinite set, and  $\aleph_1$  denotes the least uncountable cardinal. The cardinality of the continuum is

$$\mathfrak{c} = |\mathbb{R}| = 2^{\aleph_0}.$$

The Continuum Hypothesis is the assertion

$$\mathfrak{c} = \aleph_1.$$

**Theorem 1.19. Schröder–Bernstein Theorem; Cantor–Bernstein Theorem**

If there exists an injection  $f: A \hookrightarrow B$  and an injection  $g: B \hookrightarrow A$ , then there exists a bijection

$$h: A \xrightarrow{\sim} B.$$

Thus two sets have the same cardinality once each can be embedded into the other.

**Algorithm 1.20. Mathematical Induction**

To prove a statement  $P(n)$  for all integers  $n \geq n_0$ :

- (1) Prove the base case  $P(n_0)$ .
- (2) Prove the induction step: for every  $k \geq n_0$ , show that  $P(k)$  implies  $P(k+1)$ .

Then  $P(n)$  holds for all integers  $n \geq n_0$ .

**Theorem 1.21. Well-Ordering Principle**

Every nonempty subset of  $\mathbb{N}$  has a least element.

**Fact 1.22. Russell's Paradox**

The expression “the set of all sets that do not contain themselves” leads to a contradiction in naive set theory. This is one reason modern set theory is formulated axiomatically.

**Fact 1.23. ZF and ZFC**

ZF denotes Zermelo–Fraenkel set theory, and ZFC denotes ZF together with the Axiom of Choice. The axioms restrict set formation sufficiently to avoid Russell's paradox while supporting the standard foundations of modern mathematics.

**Fact 1.24. Axiom of Choice and Zorn's Lemma**

The Axiom of Choice, Zorn's Lemma, and the Well-Ordering Theorem are equivalent over the usual Zermelo–Fraenkel axioms. They are standard foundational tools in advanced algebra, analysis, and topology.

## Chapter 2

# Calculus and Analysis

Related **POSTECH** Courses: (MATH101) Calculus I, (MATH102) Calculus II, (MATH210) Applied Complex Variables, (MATH311) Analysis I, (MATH312) Analysis II

Related Books: Calculus With Applications (Peter D. Lax, Maria Shea Terrell), Multivariable Calculus with Applications (Peter D. Lax, Maria Shea Terrell), Principles of Mathematical Analysis (Walter Rudin), Complex Variables and Applications (James Ward Brown, Ruel V. Churchill)

## One-Variable Calculus

### Definition 2.1. Sequence, Series, and Partial Sum

A sequence is a function  $a: \mathbb{N} \rightarrow \mathbb{R}$  or  $a: \mathbb{N} \rightarrow \mathbb{C}$ , usually written  $(a_n)$ . A series is the formal sum

$$\sum_{n=1}^{\infty} a_n,$$

whose  $N$ th partial sum is

$$S_N = \sum_{n=1}^N a_n.$$

The series converges to  $S$  exactly when  $S_N \rightarrow S$  as  $N \rightarrow \infty$ ; otherwise it diverges. It converges absolutely if

$$\sum_{n=1}^{\infty} |a_n|$$

converges, and it converges conditionally if it converges but does not converge absolutely. Thus every convergence test for a series is a criterion for convergence of its sequence of partial sums.

### Formula 2.2. Arithmetic and Geometric Progressions

For an arithmetic progression with first term  $a$  and common difference  $d$ , we have

$$a_n = a + (n-1)d, \quad \sum_{k=1}^n a_k = \frac{n}{2}(a_1 + a_n).$$

For a geometric progression with first term  $a$  and common ratio  $r$ , we have

$$a_n = ar^{n-1}, \quad \sum_{k=0}^{n-1} ar^k = a \frac{1-r^n}{1-r} \quad (r \neq 1).$$

If  $|r| < 1$ , then

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}.$$

**Formula 2.3. Sums of Powers; Faulhaber's Formulas**

The first power sums are

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left( \frac{n(n+1)}{2} \right)^2,$$

and

$$\sum_{k=1}^n k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

More generally, for each fixed nonnegative integer  $m$ ,

$$\sum_{k=1}^n k^m$$

is a polynomial in  $n$  of degree  $m+1$  with leading coefficient  $1/(m+1)$ .

**Formula 2.4. Finite Differences; Newton's Forward Difference Formula**

For a sequence  $(a_n)$ , define the forward difference by

$$\Delta a_n = a_{n+1} - a_n.$$

If  $a_n = P(n)$  for a polynomial  $P$  of degree  $d$  with leading coefficient  $c$ , then

$$\Delta^d a_n = d!c, \quad \Delta^{d+1} a_n = 0.$$

Conversely, a sequence with constant  $d$ th differences is polynomial in  $n$  of degree at most  $d$ . For every polynomial  $P$  of degree at most  $d$ ,

$$P(n) = \sum_{k=0}^d \binom{n}{k} \Delta^k P(0).$$

*Strategy.* A table of successive differences is one of the most efficient methods to detect a hidden polynomial sequence. Constant first, second, and third differences indicate linear, quadratic, and cubic behavior, respectively.

**Formula 2.5. Linear Recurrences with Constant Coefficients**

Consider

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_m a_{n-m}.$$

Its characteristic polynomial is

$$r^m - c_1 r^{m-1} - c_2 r^{m-2} - \cdots - c_m.$$

If the roots  $r_1, \dots, r_m$  are distinct, then

$$a_n = A_1 r_1^n + \cdots + A_m r_m^n.$$

A root  $r$  of multiplicity  $q$  contributes terms

$$r^n, n r^n, \dots, n^{q-1} r^n.$$

The constants are determined from the initial values.

### Formula 2.6. First-Order Linear Recurrence

If

$$a_n = ra_{n-1} + b_n \quad (n \geq 1),$$

then iteration gives

$$a_n = r^n a_0 + \sum_{j=1}^n r^{n-j} b_j.$$

In particular, if  $b_n = b$  and  $r \neq 1$ , then

$$a_n = r^n a_0 + b \frac{r^n - 1}{r - 1}.$$

### Formula 2.7. Nonhomogeneous Linear Recurrences

For a linear recurrence with constant coefficients, the general solution is

$$a_n = a_n^{(h)} + a_n^{(p)},$$

where  $a_n^{(h)}$  solves the associated homogeneous recurrence and  $a_n^{(p)}$  is one particular solution. If the forcing term has the form

$$P(n)\alpha^n,$$

try a particular solution of the form

$$n^s Q(n)\alpha^n,$$

where  $\deg Q = \deg P$  and  $s$  is the multiplicity of  $\alpha$  as a root of the characteristic polynomial.

*Strategy.* For recurrences, first identify whether the quickest language is characteristic roots, generating functions, a transfer matrix, or telescoping after a suitable normalization.

*Tactic. Fractional-Linear Recurrences.* Suppose that a first-order recurrence has the form

$$a_{n+1} = T(a_n), \quad T(x) = \frac{\alpha x + \beta}{\gamma x + \delta}.$$

First find the fixed points of  $T$ . If  $r$  is a repeated fixed point, the substitution

$$b_n = \frac{1}{a_n - r}$$

often produces an arithmetic or first-order linear recurrence. If  $r$  and  $s$  are distinct fixed points, try

$$b_n = \frac{a_n - r}{a_n - s}.$$

This converts iteration of many fractional-linear maps into a geometric recurrence. Check separately any value that makes a denominator zero.

### Theorem 2.8. Master Theorem for Divide-and-Conquer Recurrences

Let  $T(n) = aT(n/b) + f(n)$ , where  $a \geq 1$ ,  $b > 1$ ,  $T(1) = \Theta(1)$ , and  $f$  is eventually nonnegative. Take  $n$  through powers of  $b$  and put  $d = \log_b a$ . Then:

1. If  $f(n) = O(n^{d-\varepsilon})$  for some  $\varepsilon > 0$ , then  $T(n) = \Theta(n^d)$ .
2. If  $f(n) = \Theta(n^d (\log n)^k)$  for  $k \geq 0$ , then  $T(n) = \Theta(n^d (\log n)^{k+1})$ .
3. If  $f(n) = \Omega(n^{d+\varepsilon})$  for some  $\varepsilon > 0$  and  $af(n/b) \leq cf(n)$  eventually for a constant  $c < 1$ , then  $T(n) = \Theta(f(n))$ .

Here  $\Theta$  means bounds above and below by positive constant multiples. Check the hypotheses before choosing a case; recurrences with unequal subproblem sizes need a different method.

**Definition 2.9. Limit of a Function; Epsilon–Delta Definition**

Let  $D \subseteq \mathbb{R}$ , let  $a$  be an accumulation point of  $D$ , and let  $f: D \rightarrow \mathbb{R}$ . The statement

$$\lim_{x \rightarrow a} f(x) = L$$

means that  $f(x)$  can be made arbitrarily close to  $L$  by taking  $x$  sufficiently close to, but different from,  $a$ . Formally,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \quad 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

**Definition 2.10. One-Sided Limits and Limits at Infinity**

The right-hand limit  $\lim_{x \rightarrow a^+} f(x) = L$  is defined by

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \quad 0 < x - a < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The left-hand definition is analogous, with  $0 < a - x < \delta$ . Moreover,

$$\lim_{x \rightarrow \infty} f(x) = L$$

means

$$\forall \varepsilon > 0, \quad \exists M > 0, \quad \forall x \in D, \quad x > M \Rightarrow |f(x) - L| < \varepsilon.$$

Finally,  $\lim_{x \rightarrow a} f(x) = \infty$  means

$$\forall M > 0, \quad \exists \delta > 0, \quad \forall x \in D, \quad 0 < |x - a| < \delta \Rightarrow f(x) > M.$$

**Theorem 2.11. Squeeze Theorem; Sandwich Theorem**

Suppose that  $g(x) \leq f(x) \leq h(x)$  for all  $x$  sufficiently close to  $a$ , except possibly at  $a$ , and that

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L.$$

Then

$$\lim_{x \rightarrow a} f(x) = L.$$

**Definition 2.12. Continuity at a Point**

Let  $f: D \rightarrow \mathbb{R}$  and  $a \in D$ . The function  $f$  is continuous at  $a$  if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Equivalently,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \quad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

The function is continuous on  $D$  if it is continuous at every point of  $D$ .

**Definition 2.13. Derivative**

Let  $f$  be defined near  $a$ . The derivative of  $f$  at  $a$  is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists as a finite real number. Thus  $f'(a) = L$  if and only if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall h \in \mathbb{R}, \quad 0 < |h| < \delta \Rightarrow \left| \frac{f(a+h) - f(a)}{h} - L \right| < \varepsilon.$$

**Formula 2.14. Standard Trigonometric and Exponential Limits**

The following limits are fundamental:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2},$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = 1,$$

and

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

*Strategy.* These limits are often faster than L'Hôpital's Rule. The expressions  $\sin x$ ,  $1 - \cos x$ ,  $e^x - 1$ ,  $\ln(1 + x)$ , and  $(1 + x)^{1/x}$  are strong signals for these standard patterns.

**Formula 2.15. Elementary Function Reference Table**

The following domains, ranges, symmetries, and periods are standard.

| Function                   | Domain                                                                | Range                                             | Symmetry | Period |
|----------------------------|-----------------------------------------------------------------------|---------------------------------------------------|----------|--------|
| $e^x$                      | $\mathbb{R}$                                                          | $(0, \infty)$                                     | neither  | none   |
| $\ln x$                    | $(0, \infty)$                                                         | $\mathbb{R}$                                      | neither  | none   |
| $\sin x$                   | $\mathbb{R}$                                                          | $[-1, 1]$                                         | odd      | $2\pi$ |
| $\cos x$                   | $\mathbb{R}$                                                          | $[-1, 1]$                                         | even     | $2\pi$ |
| $\tan x$                   | $\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$ | $\mathbb{R}$                                      | odd      | $\pi$  |
| $\sec x$                   | $\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$ | $(-\infty, -1] \cup [1, \infty)$                  | even     | $2\pi$ |
| $\csc x$                   | $\mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$                 | $(-\infty, -1] \cup [1, \infty)$                  | odd      | $2\pi$ |
| $\cot x$                   | $\mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$                 | $\mathbb{R}$                                      | odd      | $\pi$  |
| $\arcsin x$                | $[-1, 1]$                                                             | $[-\frac{\pi}{2}, \frac{\pi}{2}]$                 | odd      | none   |
| $\arccos x$                | $[-1, 1]$                                                             | $[0, \pi]$                                        | neither  | none   |
| $\arctan x$                | $\mathbb{R}$                                                          | $(-\frac{\pi}{2}, \frac{\pi}{2})$                 | odd      | none   |
| $\operatorname{arcsec} x$  | $(-\infty, -1] \cup [1, \infty)$                                      | $[0, \pi] \setminus \{\frac{\pi}{2}\}$            | neither  | none   |
| $\operatorname{arccsc} x$  | $(-\infty, -1] \cup [1, \infty)$                                      | $[-\frac{\pi}{2}, \frac{\pi}{2}] \setminus \{0\}$ | odd      | none   |
| $\operatorname{arccot} x$  | $\mathbb{R}$                                                          | $(0, \pi)$                                        | neither  | none   |
| $\sinh x$                  | $\mathbb{R}$                                                          | $\mathbb{R}$                                      | odd      | none   |
| $\cosh x$                  | $\mathbb{R}$                                                          | $[1, \infty)$                                     | even     | none   |
| $\tanh x$                  | $\mathbb{R}$                                                          | $(-1, 1)$                                         | odd      | none   |
| $\operatorname{sech} x$    | $\mathbb{R}$                                                          | $(0, 1]$                                          | even     | none   |
| $\operatorname{csch} x$    | $\mathbb{R} \setminus \{0\}$                                          | $\mathbb{R} \setminus \{0\}$                      | odd      | none   |
| $\operatorname{coth} x$    | $\mathbb{R} \setminus \{0\}$                                          | $(-\infty, -1) \cup (1, \infty)$                  | odd      | none   |
| $\operatorname{arsinh} x$  | $\mathbb{R}$                                                          | $\mathbb{R}$                                      | odd      | none   |
| $\operatorname{arcosh} x$  | $[1, \infty)$                                                         | $[0, \infty)$                                     | neither  | none   |
| $\operatorname{artanh} x$  | $(-1, 1)$                                                             | $\mathbb{R}$                                      | odd      | none   |
| $\operatorname{arcsech} x$ | $(0, 1]$                                                              | $[0, \infty)$                                     | neither  | none   |
| $\operatorname{arcsch} x$  | $\mathbb{R} \setminus \{0\}$                                          | $\mathbb{R} \setminus \{0\}$                      | odd      | none   |
| $\operatorname{arcoth} x$  | $(-\infty, -1) \cup (1, \infty)$                                      | $\mathbb{R} \setminus \{0\}$                      | odd      | none   |

**Formula 2.16. Exact Trigonometric Values Table**

For the standard first-quadrant angles,

| $x$      | 0         | $\frac{\pi}{6}$       | $\frac{\pi}{4}$      | $\frac{\pi}{3}$       |
|----------|-----------|-----------------------|----------------------|-----------------------|
| $\sin x$ | 0         | $\frac{1}{2}$         | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$  |
| $\cos x$ | 1         | $\frac{\sqrt{3}}{2}$  | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$         |
| $\tan x$ | 0         | $\frac{\sqrt{3}}{3}$  | 1                    | $\sqrt{3}$            |
| $\sec x$ | 1         | $\frac{2\sqrt{3}}{3}$ | $\sqrt{2}$           | 2                     |
| $\csc x$ | undefined | 2                     | $\sqrt{2}$           | $\frac{2\sqrt{3}}{3}$ |
| $\cot x$ | undefined | $\sqrt{3}$            | 1                    | $\frac{\sqrt{3}}{3}$  |

Moreover,

$$\sin \frac{\pi}{2} = 1, \quad \cos \frac{\pi}{2} = 0,$$

and the remaining exact values follow from parity, periodicity, and reference-angle identities.

### Formula 2.17. Trigonometric Identity and Synthesis Table

The principal trigonometric identities are

| Type               | Identities                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------|
| Reciprocal         | $\sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}, \quad \cot x = \frac{1}{\tan x}$       |
| Quotient           | $\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$                              |
| Pythagorean        | $\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x$             |
| Angle addition     | $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$                                                   |
| Angle addition     | $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$                                                   |
| Angle addition     | $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$                                     |
| Double angle       | $\sin 2x = 2 \sin x \cos x$                                                                         |
| Double angle       | $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$                                   |
| Double angle       | $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$                                                           |
| Half angle         | $\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}, \quad \cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}$        |
| Half angle         | $\tan^2 \frac{x}{2} = \frac{1 - \cos x}{1 + \cos x} \quad (1 + \cos x \neq 0)$                      |
| Product-to-sum     | $2 \sin x \sin y = \cos(x - y) - \cos(x + y)$                                                       |
| Product-to-sum     | $2 \cos x \cos y = \cos(x - y) + \cos(x + y)$                                                       |
| Product-to-sum     | $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$                                                       |
| Sum-to-product     | $\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$                                         |
| Sum-to-product     | $\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$                                         |
| Sum-to-product     | $\sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2}$                                         |
| Sum-to-product     | $\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}$                                        |
| Hyperbolic         | $\cosh^2 x - \sinh^2 x = 1, \quad 1 - \tanh^2 x = \operatorname{sech}^2 x$                          |
| Hyperbolic         | $\tanh x = \frac{\sinh x}{\cosh x}, \quad \coth x = \frac{\cosh x}{\sinh x}$                        |
| Inverse hyperbolic | $\operatorname{arcsinh} x = \ln \left( x + \sqrt{x^2 + 1} \right)$                                  |
| Inverse hyperbolic | $\operatorname{arccosh} x = \ln \left( x + \sqrt{x^2 - 1} \right) \quad (x \geq 1)$                 |
| Inverse hyperbolic | $\operatorname{arctanh} x = \frac{1}{2} \ln \left( \frac{1+x}{1-x} \right) \quad ( x  < 1)$         |
| Inverse hyperbolic | $\operatorname{arcsech} x = \operatorname{arccosh} \left( \frac{1}{x} \right) \quad (0 < x \leq 1)$ |

| Type               | Identities                                                                                                                                             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inverse hyperbolic | $\operatorname{arccsch} x = \operatorname{arcsinh}\left(\frac{1}{x}\right) \quad (x \neq 0)$                                                           |
| Inverse hyperbolic | $\operatorname{arccoth} x = \operatorname{arctanh}\left(\frac{1}{x}\right) \quad ( x  > 1)$                                                            |
| Synthesis          | $a \cos x + b \sin x = R \cos(x - \alpha), \quad R = \sqrt{a^2 + b^2}, \quad \cos \alpha = \frac{a}{R}, \quad \sin \alpha = \frac{b}{R} \quad (R > 0)$ |

**Theorem 2.18. L'Hôpital's Rule**

Let  $f$  and  $g$  be differentiable on some punctured interval around  $a$ , and suppose that  $g'(x) \neq 0$  there. Assume either

$$f(x) \rightarrow 0, \quad g(x) \rightarrow 0,$$

or

$$|f(x)| \rightarrow \infty, \quad |g(x)| \rightarrow \infty,$$

as  $x \rightarrow a$ . If

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L$$

exists in the extended real numbers, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

*Strategy.* Apply L'Hôpital's Rule only after identifying a  $0/0$  or  $\infty/\infty$  form. Rewrite  $0 \cdot \infty$  and  $\infty - \infty$  as a quotient first. For  $0^0$ ,  $1^\infty$ , or  $\infty^0$ , take logarithms before differentiating.

**Formula 2.19. Derivative and Antiderivative Table**

The following table records the basic one-variable forms. Affine changes of variable are handled by the chain rule and substitution. The constant of integration is omitted from the right column.

| $f(x)$            | $f'(x)$                 | $\int f(x) \, dx$                                   |
|-------------------|-------------------------|-----------------------------------------------------|
| $c$               | $0$                     | $cx$                                                |
| $x$               | $1$                     | $\frac{x^2}{2}$                                     |
| $x^a$             | $ax^{a-1}$              | $\frac{x^{a+1}}{a+1} \quad (a \neq -1)$             |
| $\frac{1}{x}$     | $-\frac{1}{x^2}$        | $\ln  x $                                           |
| $e^x$             | $e^x$                   | $e^x$                                               |
| $a^x$             | $a^x \ln a$             | $\frac{a^x}{\ln a} \quad (a > 0, a \neq 1)$         |
| $\ln x$           | $\frac{1}{x}$           | $x \ln x - x$                                       |
| $\lg x$           | $\frac{1}{x \ln 10}$    | $\frac{x \ln x - x}{\ln 10}$                        |
| $\lg x$           | $\frac{1}{x \ln 2}$     | $\frac{x \ln x - x}{\ln 2}$                         |
| $\log_a x$        | $\frac{1}{x \ln a}$     | $\frac{x \ln x - x}{\ln a} \quad (a > 0, a \neq 1)$ |
| $x \ln x$         | $\ln x + 1$             | $\frac{x^2}{2} \ln x - \frac{x^2}{4}$               |
| $\frac{\ln x}{x}$ | $\frac{1 - \ln x}{x^2}$ | $\frac{1}{2}(\ln x)^2$                              |

| $f(x)$                     | $f'(x)$                           | $\int f(x) dx$                                                  |
|----------------------------|-----------------------------------|-----------------------------------------------------------------|
| $\frac{1}{x \ln x}$        | $-\frac{\ln x + 1}{x^2(\ln x)^2}$ | $\ln  \ln x $                                                   |
| $\sin x$                   | $\cos x$                          | $-\cos x$                                                       |
| $\cos x$                   | $-\sin x$                         | $\sin x$                                                        |
| $\tan x$                   | $\sec^2 x$                        | $-\ln  \cos x $                                                 |
| $\cot x$                   | $-\csc^2 x$                       | $\ln  \sin x $                                                  |
| $\sec x$                   | $\sec x \tan x$                   | $\ln  \sec x + \tan x $                                         |
| $\csc x$                   | $-\csc x \cot x$                  | $\ln  \csc x - \cot x $                                         |
| $\sec^2 x$                 | $2 \sec^2 x \tan x$               | $\tan x$                                                        |
| $\csc^2 x$                 | $-2 \csc^2 x \cot x$              | $-\cot x$                                                       |
| $\sec x \tan x$            | $\sec x \tan^2 x + \sec^3 x$      | $\sec x$                                                        |
| $\csc x \cot x$            | $-\csc x \cot^2 x - \csc^3 x$     | $-\csc x$                                                       |
| $\arcsin x$                | $\frac{1}{\sqrt{1-x^2}}$          | $x \arcsin x + \sqrt{1-x^2}$                                    |
| $\arccos x$                | $-\frac{1}{\sqrt{1-x^2}}$         | $x \arccos x - \sqrt{1-x^2}$                                    |
| $\arctan x$                | $\frac{1}{1+x^2}$                 | $x \arctan x - \frac{1}{2} \ln(1+x^2)$                          |
| $\operatorname{arcsec} x$  | $\frac{1}{ x \sqrt{x^2-1}}$       | $x \operatorname{arcsec} x - \ln( x  + \sqrt{x^2-1})$           |
| $\operatorname{arccsc} x$  | $-\frac{1}{ x \sqrt{x^2-1}}$      | $x \operatorname{arccsc} x + \ln( x  + \sqrt{x^2-1})$           |
| $\operatorname{arccot} x$  | $-\frac{1}{1+x^2}$                | $x \operatorname{arccot} x + \frac{1}{2} \ln(1+x^2)$            |
| $\sinh x$                  | $\cosh x$                         | $\cosh x$                                                       |
| $\cosh x$                  | $\sinh x$                         | $\sinh x$                                                       |
| $\tanh x$                  | $\operatorname{sech}^2 x$         | $\ln(\cosh x)$                                                  |
| $\coth x$                  | $-\operatorname{csch}^2 x$        | $\ln  \sinh x $                                                 |
| $\operatorname{sech} x$    | $-\operatorname{sech} x \tanh x$  | $\arctan(\sinh x)$                                              |
| $\operatorname{csch} x$    | $-\operatorname{csch} x \coth x$  | $\ln \left  \tanh \frac{x}{2} \right $                          |
| $\operatorname{arcsinh} x$ | $\frac{1}{\sqrt{x^2+1}}$          | $x \operatorname{arcsinh} x - \sqrt{x^2+1}$                     |
| $\operatorname{arccosh} x$ | $\frac{1}{\sqrt{x-1}\sqrt{x+1}}$  | $x \operatorname{arccosh} x - \sqrt{x^2-1}$                     |
| $\operatorname{arctanh} x$ | $\frac{1}{1-x^2}$                 | $x \operatorname{arctanh} x + \frac{1}{2} \ln(1-x^2)$           |
| $\operatorname{arcsech} x$ | $-\frac{1}{x\sqrt{1-x^2}}$        | $x \operatorname{arcsech} x + \arcsin x$                        |
| $\operatorname{arccsch} x$ | $-\frac{1}{ x \sqrt{1+x^2}}$      | $x \operatorname{arccsch} x + \operatorname{arcsinh}  x $       |
| $\operatorname{arccoth} x$ | $\frac{1}{1-x^2}$                 | $x \operatorname{arccoth} x + \frac{1}{2} \ln(x^2-1)$           |
| $\frac{1}{x^2+a^2}$        | $-\frac{2x}{(x^2+a^2)^2}$         | $\frac{1}{a} \arctan \frac{x}{a} \quad (a > 0)$                 |
| $\frac{1}{x^2-a^2}$        | $-\frac{2x}{(x^2-a^2)^2}$         | $\frac{1}{2a} \ln \left  \frac{x-a}{x+a} \right  \quad (a > 0)$ |
| $\frac{1}{a^2-x^2}$        | $\frac{2x}{(a^2-x^2)^2}$          | $\frac{1}{2a} \ln \left  \frac{a+x}{a-x} \right  \quad (a > 0)$ |
| $\frac{1}{\sqrt{a^2-x^2}}$ | $\frac{x}{(a^2-x^2)^{3/2}}$       | $\arcsin \frac{x}{a} \quad (a > 0)$                             |
| $\frac{1}{\sqrt{x^2+a^2}}$ | $-\frac{x}{(x^2+a^2)^{3/2}}$      | $\ln  x + \sqrt{x^2+a^2} $                                      |
| $\frac{1}{\sqrt{x^2-a^2}}$ | $-\frac{x}{(x^2-a^2)^{3/2}}$      | $\ln  x + \sqrt{x^2-a^2}  \quad (a > 0)$                        |

| $f(x)$                       | $f'(x)$                         | $\int f(x) dx$                                                                       |
|------------------------------|---------------------------------|--------------------------------------------------------------------------------------|
| $\frac{x}{\sqrt{x^2 + a^2}}$ | $\frac{a^2}{(x^2 + a^2)^{3/2}}$ | $\sqrt{x^2 + a^2}$                                                                   |
| $\frac{x}{\sqrt{a^2 - x^2}}$ | $\frac{a^2}{(a^2 - x^2)^{3/2}}$ | $-\sqrt{a^2 - x^2}$                                                                  |
| $\frac{x}{\sqrt{a^2 - x^2}}$ | $-\frac{x}{\sqrt{a^2 - x^2}}$   | $\frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\arcsin \frac{x}{a}$                     |
| $\frac{x}{\sqrt{x^2 + a^2}}$ | $\frac{x}{\sqrt{x^2 + a^2}}$    | $\frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\ln \left  x + \sqrt{x^2 + a^2} \right $ |
| $\frac{x}{\sqrt{x^2 - a^2}}$ | $\frac{x}{\sqrt{x^2 - a^2}}$    | $\frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\ln \left  x + \sqrt{x^2 - a^2} \right $ |

*Tactic.* An antiderivative is often exposed by a simple substitution, symmetry, completing the square, or integration by parts. Try those before expanding into a long calculation.

### Theorem 2.20. Intermediate Value Theorem

Let  $f$  be continuous on  $[a, b]$ . If  $N$  is any number between  $f(a)$  and  $f(b)$ , then there exists  $c \in [a, b]$  such that

$$f(c) = N.$$

*Strategy.* The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values cross the desired level.

### Theorem 2.21. Extreme Value Theorem; Weierstrass Extreme Value Theorem

Let  $f$  be continuous on  $[a, b]$ . Then  $f$  attains both a maximum and a minimum on  $[a, b]$ .

*Remark.* The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

### Theorem 2.22. Rolle's Theorem

Let  $f$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . If  $f(a) = f(b)$ , then there exists  $c \in (a, b)$  such that

$$f'(c) = 0.$$

### Theorem 2.23. Mean Value Theorem; Lagrange Mean Value Theorem

Let  $f$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Idea.* The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

### Theorem 2.24. Cauchy's Mean Value Theorem; Extended Mean Value Theorem

Let  $f$  and  $g$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

If  $g'(x) \neq 0$  for every  $x \in (a, b)$ , then  $g(b) \neq g(a)$  and the result may be written as

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

**Theorem 2.25. Derivative of an Inverse Function**

Let  $I \subseteq \mathbb{R}$  be an interval, let  $f: I \rightarrow \mathbb{R}$  be one-to-one, and suppose that  $f$  is differentiable at an interior point  $x_0 \in I$  with  $f'(x_0) \neq 0$ . If the inverse  $g = f^{-1}$  is continuous at  $y_0 = f(x_0)$ , then  $g$  is differentiable at  $y_0$  and

$$g'(y_0) = \frac{1}{f'(x_0)}.$$

In particular, the continuity hypothesis is automatic when  $f$  is continuous and strictly monotone on an interval.

**Definition 2.26. Riemann Integral**

Let  $f: [a, b] \rightarrow \mathbb{R}$  be bounded. For a partition

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b,$$

choose  $\xi_i \in [x_{i-1}, x_i]$  and form the Riemann sum

$$\sum_{i=1}^n f(\xi_i)(x_i - x_{i-1}).$$

If these sums approach the same limit  $I$  as the mesh  $\max_i(x_i - x_{i-1}) \rightarrow 0$ , independently of the sample points, then  $f$  is Riemann integrable and

$$I = \int_a^b f(x) \, dx.$$

Every continuous function on a closed bounded interval is Riemann integrable.

**Theorem 2.27. Fundamental Theorem of Calculus**

If  $f$  is continuous on  $[a, b]$  and

$$F(x) = \int_a^x f(t) \, dt,$$

then  $F'(x) = f(x)$ . Conversely, if  $F' = f$  on  $[a, b]$ , then

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

**Theorem 2.28. Taylor's Theorem**

Let  $f$  be  $(n+1)$  times differentiable on an open interval containing the closed interval with endpoints  $a$  and  $x$ . Then there exists a point  $\xi$  strictly between  $a$  and  $x$  such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}.$$

The final term is the Lagrange remainder.

**Corollary 2.29. Taylor Error Bound**

If

$$|f^{(n+1)}(t)| \leq M$$

for every  $t$  between  $a$  and  $x$ , then the Taylor remainder satisfies

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}.$$

*Strategy.* Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving  $e^x$ ,  $\sin x$ ,  $\cos x$ ,  $\ln(1+x)$ , or  $\arctan x$  often invite Taylor expansion.

**Definition 2.30. Power Series and Radius of Convergence**

A power series centered at  $a$  is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n.$$

There is a radius of convergence  $R \in [0, \infty]$  such that the series converges absolutely for  $|x-a| < R$  and diverges for  $|x-a| > R$ ; behavior at the boundary  $|x-a| = R$  must be checked separately. A Taylor series is the power series with coefficients

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

When  $a = 0$ , it is called a Maclaurin series.

**Formula 2.31. Maclaurin Series**

The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1),$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \quad (-1 < x \leq 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad (|x| \leq 1).$$

Also,

$$-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} \quad (-1 \leq x < 1),$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!},$$

and

$$\operatorname{arctanh} x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \quad (|x| < 1).$$

Finally,

$$\arcsin x = \sum_{n=0}^{\infty} \frac{\binom{2n}{n}}{4^n (2n+1)} x^{2n+1} \quad (|x| \leq 1).$$

**Formula 2.32. Binomial Series**

For any real or complex number  $\beta$ ,

$$(1+x)^\beta = \sum_{n=0}^{\infty} \binom{\beta}{n} x^n \quad (|x| < 1),$$

where

$$\binom{\beta}{n} = \frac{\beta(\beta-1)\cdots(\beta-n+1)}{n!}.$$

**Formula 2.33. Chebyshev Polynomials**

The Chebyshev polynomials of the first kind are characterized by

$$T_n(\cos \theta) = \cos(n\theta).$$

They satisfy

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Hence multiple-angle trigonometric expressions can be converted into polynomial calculations almost immediately; for example,

$$\cos(3\theta) = 4\cos^3 \theta - 3\cos \theta.$$

**Formula 2.34. Classical Series Values**

The following standard series values are useful in computation:

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n^2} &= \frac{\pi^2}{6}, & \sum_{n=1}^{\infty} \frac{1}{n^4} &= \frac{\pi^4}{90}, \\ \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} &= \ln 2, & \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} &= \frac{\pi}{4}. \end{aligned}$$

Two immediate Basel-sum consequences are

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}, \quad \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

**Formula 2.35. Geometric-Series Differentiation**

Starting from

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1),$$

term-by-term differentiation gives

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} \quad (|x| < 1),$$

and hence

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2} \quad (|x| < 1).$$

**Formula 2.36. Telescoping Series**

If

$$a_n = b_n - b_{n+1},$$

then the partial sums satisfy

$$\sum_{n=1}^N a_n = b_1 - b_{N+1}.$$

Hence, if  $b_n \rightarrow L$ , then

$$\sum_{n=1}^{\infty} a_n = b_1 - L.$$

**Formula 2.37. Abel's Summation Formula; Summation by Parts**

Let

$$A_k = \sum_{j=m}^k a_j.$$

Then

$$\sum_{k=m}^n a_k b_k = A_n b_n + \sum_{k=m}^{n-1} A_k (b_k - b_{k+1}).$$

This is the discrete analogue of integration by parts and is useful when one factor has simple partial sums while the other changes slowly.

**Formula 2.38. Euler–Maclaurin Summation Formula**

Let  $p \geq 1$  and let  $f \in C^{2p}([m, n])$ , where  $m < n$  are integers. Then

$$\sum_{k=m}^n f(k) = \int_m^n f(x) dx + \frac{f(m) + f(n)}{2} + \sum_{r=1}^{p-1} \frac{B_{2r}}{(2r)!} \left( f^{(2r-1)}(n) - f^{(2r-1)}(m) \right) + R_p,$$

where  $B_{2r}$  are the Bernoulli numbers and

$$|R_p| \leq \frac{2\zeta(2p)}{(2\pi)^{2p}} \int_m^n |f^{(2p)}(x)| dx.$$

The case  $p = 1$  gives the trapezoidal approximation together with a rigorous error bound.

**Formula 2.39. Asymptotic Growth Hierarchy**

For fixed  $p, q > 0$  and  $c > 1$ , as  $n \rightarrow \infty$ ,

$$(\ln n)^p = o(n^q), \quad n^q = o(c^n), \quad c^n = o(n!), \quad n! = o(n^n).$$

*Strategy.* For fast series calculations, first try to reduce the expression to a geometric series, a differentiated geometric series, a Maclaurin series, or a known special value such as  $\pi^2/6$ ,  $\pi^4/90$ ,  $\ln 2$ , or  $\pi/4$ .

**Formula 2.40. Integration by Parts**

If  $u$  and  $v$  are differentiable functions, then

$$\int u dv = uv - \int v du.$$

Equivalently,

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

**Theorem 2.41. Leibniz Integral Rule; Differentiation under the Integral Sign**

Let  $I$  be an open interval, let  $a, b: I \rightarrow \mathbb{R}$  be continuously differentiable, and suppose that  $f$  and  $\partial f / \partial t$  are continuous on

$$\{(x, t) \mid t \in I, a(t) \leq x \leq b(t)\}.$$

Then

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = f(b(t), t)b'(t) - f(a(t), t)a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t) dx.$$

In particular, if the endpoints are constant, only the integral of the partial derivative remains.

*Tactic.* Introducing a parameter and differentiating under the integral sign is often called Feynman's trick. It is especially effective when direct integration is difficult but the parameter derivative is elementary.

**Formula 2.42. Frullani's Integral**

Let  $f: [0, \infty) \rightarrow \mathbb{R}$  be continuous, suppose that the finite limits  $f(0)$  and  $f(\infty) := \lim_{x \rightarrow \infty} f(x)$  exist, and assume that the improper integral below converges. For  $a, b > 0$ ,

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(\infty)) \ln \frac{b}{a}.$$

A standard sufficient hypothesis is that  $f$  is continuously differentiable and  $f' \in L^1(0, \infty)$ .

**Formula 2.43. Arc Length Formula**

The length of the graph  $y = f(x)$  on  $[a, b]$  is

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

More generally, if a curve is parametrized by  $\mathbf{r}(t) = (x(t), y(t), z(t))$ , then

$$L = \int_a^b \|\mathbf{r}'(t)\| dt.$$

**Formula 2.44. Viète's Infinite Product**

For every real  $x$ ,

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right),$$

where the value at  $x = 0$  is interpreted by continuity. Setting  $x = \pi/2$  gives Viète's classical product for  $2/\pi$ .

*Idea.* Viète's formula is a product formula for  $\pi$ . The key identity is the double-angle formula

$$\sin(2x) = 2 \sin x \cos x.$$

**Formula 2.45. Euler's Formula**

For every real number  $x$ ,

$$e^{ix} = \cos x + i \sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

*Remark.* The identity  $e^{i\pi} + 1 = 0$  is called Euler's identity. It connects the constants  $e$ ,  $i$ ,  $\pi$ ,  $1$ , and  $0$  in a single formula.

**Formula 2.46. Gaussian Integral**

The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Equivalently,

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

*Remark.* The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

**Formula 2.47. Fresnel Integrals**

The Fresnel integrals satisfy

$$\int_0^{\infty} \cos(x^2) dx = \int_0^{\infty} \sin(x^2) dx = \sqrt{\frac{\pi}{8}}.$$

*Remark.* Fresnel integrals appear in wave optics, diffraction, and the Cornu spiral. They are memorable examples of non-elementary definite integrals with clean values.

**Formula 2.48. Sine Integral; Dirichlet Integral**

The sine integral is

$$\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt.$$

The classical Dirichlet integral is

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

**Formula 2.49. Arctangent Integral**

For  $a > 0$ ,

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + a^2} dx = \frac{\pi}{a}, \quad \int_0^{\infty} \frac{1}{x^2 + a^2} dx = \frac{\pi}{2a}.$$

More generally,

$$\int_{-\infty}^{\infty} \frac{c}{(x-b)^2 + a^2} dx = \frac{c\pi}{a}.$$

**Definition 2.50. Gamma Function**

For  $\text{Re}(z) > 0$ , the Gamma function is defined by

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt.$$

It extends meromorphically to  $\mathbb{C}$ , with simple poles at the nonpositive integers.

**Proposition 2.51. Gamma Function and Factorials**

*The Gamma function satisfies*

$$\Gamma(z+1) = z\Gamma(z)$$

*wherever both sides are defined. In particular, for every positive integer  $n$ ,*

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

**Definition 2.52. Beta Function; Euler Beta Integral**

For  $x, y > 0$ , the Beta function is defined by

$$B(x, y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt.$$

**Formula 2.53. Beta–Gamma Identity**

The Beta and Gamma functions are related by

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

In particular, for nonnegative integers  $m, n$ ,

$$\int_0^1 x^m(1-x)^n dx = B(m+1, n+1) = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

**Formula 2.54. Gamma-Type Exponential Integrals**

For  $a > 0$  and every nonnegative integer  $n$ ,

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}.$$

More generally, for  $\operatorname{Re}(s) > 0$ ,

$$\int_0^\infty x^{s-1} e^{-ax} dx = \frac{\Gamma(s)}{a^s}.$$

**Formula 2.55. Damped Sine and Cosine Integrals**

For  $a > 0$ ,

$$\int_0^\infty e^{-ax} \cos bx dx = \frac{a}{a^2 + b^2}, \quad \int_0^\infty e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}.$$

*Strategy.* Many fast integral problems reduce to this formula after completing a square or translating the variable. A shift such as  $u = x - b$  does not change the infinite limits.

**Formula 2.56. Symmetry Trick for Definite Integrals**

If  $f$  is integrable on  $[a, b]$ , then

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$$

Hence, if

$$f(x) + f(a+b-x) = C$$

is constant, then

$$\int_a^b f(x) dx = \frac{C(b-a)}{2}.$$

*Strategy.* This is one of the most useful Integration-Bee-style tricks. Typical signals are intervals such as  $[0, \pi/2]$ , expressions involving  $\tan x$  and  $\cot x$ , or a denominator that becomes complementary after the substitution  $x \mapsto a + b - x$ .

**Formula 2.57. Parity and Periodicity Shortcuts for Integrals**

If  $f$  is integrable on  $[-a, a]$ , then

$$f(-x) = -f(x) \implies \int_{-a}^a f(x) \, dx = 0,$$

and

$$f(-x) = f(x) \implies \int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx.$$

If  $f$  has period  $T$ , then for every  $c$ ,

$$\int_c^{c+T} f(x) \, dx = \int_0^T f(x) \, dx.$$

**Formula 2.58. Reciprocal-Substitution Identity**

If the improper integrals converge, the substitution  $x = 1/t$  gives

$$\int_0^\infty f(x) \, dx = \int_0^\infty \frac{f(1/x)}{x^2} \, dx.$$

Therefore,

$$\int_0^\infty f(x) \, dx = \frac{1}{2} \int_0^\infty \left( f(x) + \frac{f(1/x)}{x^2} \right) \, dx.$$

This often reveals a cancellation or symmetry hidden in an integral on  $(0, \infty)$ .

**Formula 2.59. Weierstrass Substitution; Tangent Half-Angle Substitution**

The substitution

$$t = \tan \frac{x}{2}$$

converts rational expressions in  $\sin x$  and  $\cos x$  into rational functions of  $t$ :

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2 \, dt}{1+t^2}.$$

**Algorithm 2.60. Fast Integral Recognition Checklist**

Before expanding or integrating directly, check in this order:

- (1) parity, periodicity, and interval symmetry;
- (2) the substitutions  $x \mapsto a - x$ ,  $x \mapsto 1/x$ , or  $x \mapsto a/x$ ;
- (3) a hidden derivative for substitution or integration by parts;
- (4) a parameter that permits differentiation under the integral sign;
- (5) Beta–Gamma, Gaussian, Wallis, or a standard trigonometric integral;
- (6) the tangent half-angle substitution for rational trigonometric expressions.

**Formula 2.61. Wallis Product**

The Wallis product is

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots$$

**Formula 2.62. Euler's Reflection Formula**

For  $z \in \mathbb{C} \setminus \mathbb{Z}$ ,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

**Formula 2.63. Euler Beta Integral on the Positive Half-Line**

If  $p > 1$ , then

$$\int_0^\infty \frac{dx}{1+x^p} = \frac{1}{p} B\left(\frac{1}{p}, 1 - \frac{1}{p}\right) = \frac{\pi}{p} \csc\left(\frac{\pi}{p}\right).$$

The substitution

$$t = \frac{x^p}{1+x^p}$$

converts the improper integral into an Euler beta integral.

**Formula 2.64. Stirling's Formula**

As  $n \rightarrow \infty$ ,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

*Remark.* Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

**Formula 2.65. Trapezoidal Rule**

The trapezoidal approximation is

$$\int_a^b f(x) dx \approx \frac{b-a}{2} (f(a) + f(b)).$$

It has degree of precision 1.

**Formula 2.66. Simpson's Rule**

Simpson's rule is

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left( f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right).$$

It has degree of precision 3.

*Remark.* Simpson's rule uses quadratic interpolation, but the resulting formula is exact for every polynomial of degree at most 3.

**Theorem 2.67.  $p$ -Series Test**

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if  $p > 1$ .

**Theorem 2.68. Direct Comparison Test**

Let  $0 \leq a_n \leq b_n$  for all sufficiently large  $n$ . If  $\sum b_n$  converges, then  $\sum a_n$  converges. If  $\sum a_n$  diverges, then  $\sum b_n$  diverges.

**Proposition 2.69. Absolute Convergence Implies Convergence**

If

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then  $\sum_{n=1}^{\infty} a_n$  converges.

**Theorem 2.70. Integral Test**

Suppose  $f$  is positive, continuous, and decreasing on  $[1, \infty)$ , and let  $a_n = f(n)$ . Then

$$\sum_{n=1}^{\infty} a_n$$

and

$$\int_1^{\infty} f(x) \, dx$$

either both converge or both diverge.

**Theorem 2.71. Limit Comparison Test**

Let  $a_n, b_n > 0$ . If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$$

for some finite number  $c > 0$ , then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} b_n$$

converge or diverge together.

**Theorem 2.72. Ratio Test; d'Alembert's Ratio Test**

For a series  $\sum a_n$  with  $a_n \neq 0$  eventually, set

$$L = \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

If  $L < 1$ , then  $\sum a_n$  converges absolutely. If

$$\liminf_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1,$$

then  $a_n \not\rightarrow 0$  and the series diverges. In the common case in which the ordinary limit

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

exists, this reduces to the familiar rule:  $L < 1$  gives absolute convergence,  $L > 1$  gives divergence, and  $L = 1$  is inconclusive.

**Theorem 2.73. Root Test; Cauchy's Root Test**

For a series  $\sum a_n$ , set

$$L = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

If  $L < 1$ , then  $\sum a_n$  converges absolutely. If  $L > 1$ , then the series diverges. If  $L = 1$ , the test is inconclusive. When the ordinary limit  $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$  exists, it equals the displayed lim sup, giving the usual limit form of the test.

**Theorem 2.74. Cauchy–Hadamard Theorem**

For a power series

$$\sum_{n=0}^{\infty} a_n (x - a)^n,$$

the radius of convergence  $R$  is given by

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

Equivalently, the series converges absolutely for  $|x - a| < R$  and diverges for  $|x - a| > R$ .

*Remark.* This is the standard theorem behind the phrase “radius of convergence” for a power series. It is also the source of the name Cauchy–Hadamard.

**Theorem 2.75. Termwise Differentiation and Integration of Power Series**

Suppose

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$$

has radius of convergence  $R > 0$ . For  $|x-a| < R$ ,

$$f'(x) = \sum_{n=1}^{\infty} n a_n(x-a)^{n-1},$$

and

$$\int_a^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (x-a)^{n+1}.$$

The differentiated and integrated series have the same radius of convergence  $R$ . Convergence at the endpoints must be checked separately.

**Theorem 2.76. Alternating Series Test; Leibniz Test**

If

$$a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$$

and  $a_n \rightarrow 0$ , then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges.

**Formula 2.77. Alternating Series Error Estimate**

Under the hypotheses of the Alternating Series Test, if

$$S = \sum_{n=1}^{\infty} (-1)^{n+1} a_n, \quad S_N = \sum_{n=1}^N (-1)^{n+1} a_n,$$

then

$$|S - S_N| \leq a_{N+1}.$$

*Strategy.* Factorials and powers often suggest the Ratio Test,  $n$ th powers suggest the Root Test, and alternating signs with decreasing terms suggest the Alternating Series Test.

**Theorem 2.78. Divergence Test; Term Test**

If the series

$$\sum_{n=1}^{\infty} a_n$$

converges, then

$$\lim_{n \rightarrow \infty} a_n = 0.$$

Equivalently, if  $\lim_{n \rightarrow \infty} a_n$  does not exist or is not 0, then  $\sum_{n=1}^{\infty} a_n$  diverges.

*Strategy.* Before trying a sophisticated convergence test, first check whether the terms go to 0. If the terms do not tend to 0, the series diverges immediately.

**Algorithm 2.79. Fast Series Checklist**

For a convergence or summation problem:

- (1) Check the term limit first.
- (2) Look for telescoping, a geometric series, or a known power series.
- (3) For positive terms, compare only the dominant asymptotic factors.
- (4) Use the Ratio Test for factorials and exponentials, and the Root Test for expressions raised to the  $n$ th power.
- (5) For alternating terms, check monotonicity and use the first omitted term to control the error.

## Multivariable Calculus

**Definition 2.80. Open Sets in Euclidean Space**

For  $\mathbf{a} \in \mathbb{R}^n$  and  $r > 0$ , the open ball of radius  $r$  centered at  $\mathbf{a}$  is

$$B_r(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x} - \mathbf{a}\| < r\},$$

where the Euclidean norm is

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

A set  $U \subseteq \mathbb{R}^n$  is open if every  $\mathbf{a} \in U$  has some  $r > 0$  such that

$$B_r(\mathbf{a}) \subseteq U.$$

The Real Analysis section later gives the corresponding definition in an arbitrary metric space.

**Definition 2.81. Scalar Field, Vector Field, and Partial Derivative**

A scalar field on  $U \subseteq \mathbb{R}^n$  is a function  $f: U \rightarrow \mathbb{R}$ , while a vector field is a function  $\mathbf{F}: U \rightarrow \mathbb{R}^m$ . For a scalar-valued function  $f(x_1, \dots, x_n)$ , the partial derivative with respect to  $x_i$  is the one-variable derivative obtained by varying  $x_i$  while holding the other variables fixed:

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a})}{h},$$

whenever the limit exists.

**Definition 2.82. Gradient, Divergence, Curl, and Laplacian**

For a scalar field  $f: U \rightarrow \mathbb{R}$  and a vector field  $\mathbf{F}: U \rightarrow \mathbb{R}^3$  with

$$\mathbf{F}(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z))^T,$$

we write

$$\begin{aligned} \nabla f &= \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \\ \partial f / \partial z \end{pmatrix}, \\ \nabla \cdot \mathbf{F} &= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}, \\ \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial / \partial x & \partial / \partial y & \partial / \partial z \\ P & Q & R \end{vmatrix} = \begin{pmatrix} \partial R / \partial y - \partial Q / \partial z \\ \partial P / \partial z - \partial R / \partial x \\ \partial Q / \partial x - \partial P / \partial y \end{pmatrix}, \end{aligned}$$

and

$$\Delta f = \nabla^2 f = \nabla \cdot (\nabla f) = f_{xx} + f_{yy} + f_{zz}.$$

*Idea.* The gradient points in the direction of steepest increase. Divergence measures source or sink behavior. Curl measures local rotation. The Laplacian measures how a value differs from its surrounding average.

**Theorem 2.83. Clairaut's Theorem; Mixed Partial Derivative Theorem**

If  $f$  has continuous second partial derivatives near a point, then the mixed partial derivatives agree at that point:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

**Definition 2.84. Jacobian Matrix and Jacobian Determinant**

Let  $\mathbf{F}: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$  have component functions  $F_1, \dots, F_m$ . Its Jacobian matrix at  $\mathbf{a}$  is

$$\mathbf{J}_{\mathbf{F}}(\mathbf{a}) = \left( \frac{\partial F_i}{\partial x_j}(\mathbf{a}) \right)_{1 \leq i \leq m, 1 \leq j \leq n}.$$

When  $m = n$ , its determinant

$$\text{Jac}(\mathbf{F})(\mathbf{a}) = \det(\mathbf{J}_{\mathbf{F}}(\mathbf{a}))$$

is the Jacobian determinant.

**Definition 2.85. Differentiability; Total Derivative; Fréchet Derivative**

Let  $\mathbf{F}: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ , where  $U$  is open, and let  $\mathbf{a} \in U$ . The map  $\mathbf{F}$  is differentiable at  $\mathbf{a}$  if there exists a linear map  $D\mathbf{F}(\mathbf{a}): \mathbb{R}^n \rightarrow \mathbb{R}^m$  such that

$$\mathbf{F}(\mathbf{a} + \mathbf{h}) = \mathbf{F}(\mathbf{a}) + D\mathbf{F}(\mathbf{a})\mathbf{h} + \mathbf{r}(\mathbf{h}),$$

where

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|\mathbf{r}(\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

Equivalently, in compact symbolic form,

$$\forall \varepsilon > 0 \exists \delta > 0 \forall \mathbf{h} \in \mathbb{R}^n, \quad (\mathbf{a} + \mathbf{h} \in U \wedge 0 < \|\mathbf{h}\| < \delta) \implies \|\mathbf{F}(\mathbf{a} + \mathbf{h}) - \mathbf{F}(\mathbf{a}) - D\mathbf{F}(\mathbf{a})\mathbf{h}\| < \varepsilon \|\mathbf{h}\|.$$

The linear map  $D\mathbf{F}(\mathbf{a})$  is the total derivative, or Fréchet derivative. Here linear means that it preserves linear combinations. In standard coordinates, its matrix is precisely the Jacobian matrix  $\mathbf{J}_{\mathbf{F}}(\mathbf{a})$  defined above.

**Definition 2.86. Directional Derivative**

Let  $f: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$  be differentiable at  $\mathbf{a} \in U$ , and let  $\mathbf{u}$  be a unit vector. The directional derivative of  $f$  at  $\mathbf{a}$  in the direction  $\mathbf{u}$  is

$$D_{\mathbf{u}}f(\mathbf{a}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{a} + t\mathbf{u}) - f(\mathbf{a})}{t} = \nabla f(\mathbf{a}) \cdot \mathbf{u}.$$

Consequently,

$$|D_{\mathbf{u}}f(\mathbf{a})| \leq \|\nabla f(\mathbf{a})\|,$$

with maximum increase in the direction of  $\nabla f(\mathbf{a})$  when the gradient is nonzero.

**Theorem 2.87. Multivariable Chain Rule**

Let  $\mathbf{F}: \mathbb{R}^n \rightarrow \mathbb{R}^m$  and  $\mathbf{G}: \mathbb{R}^m \rightarrow \mathbb{R}^k$  be differentiable. Then

$$D(\mathbf{G} \circ \mathbf{F})(\mathbf{x}) = D\mathbf{G}(\mathbf{F}(\mathbf{x}))D\mathbf{F}(\mathbf{x}).$$

**Theorem 2.88. Iterated-Integral Form of Fubini's Theorem**

Let  $R \subseteq \mathbb{R}^2$  be a bounded region of the form

$$R = \{(x, y) \mid a \leq x \leq b, \ g_1(x) \leq y \leq g_2(x)\},$$

where  $g_1$  and  $g_2$  are continuous and  $g_1 \leq g_2$ . If  $f$  is continuous on  $R$ , then

$$\iint_R f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

If the same region is also described as

$$R = \{(x, y) \mid c \leq y \leq d, \ h_1(y) \leq x \leq h_2(y)\},$$

then the integral may equally be evaluated in the reverse order. Measure-theoretic extensions allow weaker hypotheses; Tonelli's Theorem is introduced later in this chapter after the relevant measure concepts.

**Theorem 2.89. Change of Variables Formula**

Let  $U, V \subseteq \mathbb{R}^n$  be open, and let  $\mathbf{F}: U \rightarrow V$  be a  $C^1$  diffeomorphism, meaning a bijection for which both  $\mathbf{F}$  and  $\mathbf{F}^{-1}$  are continuously differentiable. If  $D \subseteq U$  is a region for which the integrals below exist and  $f$  is integrable on  $\mathbf{F}(D)$ , then

$$\int_{\mathbf{F}(D)} f(\mathbf{x}) \, d\mathbf{x} = \int_D f(\mathbf{F}(\mathbf{u})) |\det D\mathbf{F}(\mathbf{u})| \, d\mathbf{u}.$$

The absolute value is essential because the integral measures unsigned volume, independently of orientation.

*Remark.* The Jacobian determinant is the local area or volume scaling factor of a change of variables.

**Formula 2.90. Polar Coordinates**

For polar coordinates  $(r, \theta)$ ,

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r \geq 0, \quad 0 \leq \theta < 2\pi.$$

The area element is

$$dA = r \, dr \, d\theta.$$

The area enclosed by a polar curve  $r = r(\theta)$  is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r(\theta)^2 \, d\theta.$$

Its arc length is

$$L = \int_{\alpha}^{\beta} \sqrt{r(\theta)^2 + (r'(\theta))^2} \, d\theta.$$

**Formula 2.91. Cylindrical and Spherical Coordinates**

Cylindrical coordinates  $(r, \theta, z)$  satisfy

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

with volume element

$$dV = r \, dr \, d\theta \, dz.$$

Spherical coordinates  $(\rho, \theta, \phi)$  satisfy

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

where

$$\rho \geq 0, \quad 0 \leq \theta < 2\pi, \quad 0 \leq \phi \leq \pi.$$

The volume element is

$$dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi.$$

Here  $\theta$  is the azimuthal angle measured from the positive  $x$ -axis in the  $xy$ -plane, and  $\phi$  is the polar angle measured from the positive  $z$ -axis.

### Formula 2.92. Differential Operators in Standard Coordinates

Let

$$\mathbf{F} = F_r \mathbf{e}_r + F_\theta \mathbf{e}_\theta + F_z \mathbf{e}_z$$

in cylindrical coordinates. Then

$$\nabla f = \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{\partial f}{\partial z} \mathbf{e}_z,$$

$$\nabla \cdot \mathbf{F} = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z},$$

and

$$\Delta f = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}.$$

In spherical coordinates  $(\rho, \theta, \phi)$ , where  $\theta$  is the azimuthal angle and  $\phi$  is the polar angle, write

$$\mathbf{F} = F_\rho \mathbf{e}_\rho + F_\theta \mathbf{e}_\theta + F_\phi \mathbf{e}_\phi.$$

Then

$$\nabla f = \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho \sin \phi} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi,$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} (\rho^2 F_\rho) + \frac{1}{\rho \sin \phi} \frac{\partial F_\theta}{\partial \theta} + \frac{1}{\rho \sin \phi} \frac{\partial}{\partial \phi} (\sin \phi F_\phi),$$

and

$$\Delta f = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left( \rho^2 \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2 \sin^2 \phi} \frac{\partial^2 f}{\partial \theta^2} + \frac{1}{\rho^2 \sin \phi} \frac{\partial}{\partial \phi} \left( \sin \phi \frac{\partial f}{\partial \phi} \right).$$

### Theorem 2.93. Inverse Function Theorem

Let  $\mathbf{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n$  be continuously differentiable near  $\mathbf{a}$ . If

$$\det D\mathbf{F}(\mathbf{a}) \neq 0,$$

then  $\mathbf{F}$  has a continuously differentiable local inverse near  $\mathbf{a}$ .

### Theorem 2.94. Implicit Function Theorem

Let  $F(x, y)$  be continuously differentiable near  $(a, b)$ , with

$$F(a, b) = 0, \quad \frac{\partial F}{\partial y}(a, b) \neq 0.$$

Then, near  $(a, b)$ , the equation  $F(x, y) = 0$  determines  $y$  as a differentiable function of  $x$ , and

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

**Formula 2.95. Second Derivative Test and Hessian**

A critical point of a differentiable function  $f$  is a point where  $\nabla f = \mathbf{0}$ . For a twice differentiable function  $f(x, y)$ , the Hessian matrix is

$$\mathbf{H}_f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

At a critical point  $(a, b)$ , let

$$\Delta_H = \det \mathbf{H}_f(a, b) = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2,$$

where the last equality uses equality of the mixed partial derivatives under the usual continuity hypothesis. If  $\Delta_H > 0$  and  $f_{xx}(a, b) > 0$ , the point is a local minimum. If  $\Delta_H > 0$  and  $f_{xx}(a, b) < 0$ , it is a local maximum. If  $\Delta_H < 0$ , it is a saddle point. If  $\Delta_H = 0$ , the test is inconclusive.

**Theorem 2.96. Lagrange Multipliers**

Suppose that  $f$  has a local maximum or minimum subject to the constraint

$$g(x_1, \dots, x_n) = 0$$

at a point where  $\nabla g \neq \mathbf{0}$ . Then there exists a scalar  $\lambda$  such that

$$\nabla f = \lambda \nabla g.$$

*Idea.* At a constrained optimum, the level surface of  $f$  is tangent to the constraint surface. Thus their normal vectors are parallel.

**Theorem 2.97. Euler's Homogeneous Function Theorem**

Suppose  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  is differentiable and homogeneous of degree  $k$ , meaning

$$f(t\mathbf{x}) = t^k f(\mathbf{x})$$

for  $t > 0$ . Then

$$\mathbf{x} \cdot \nabla f(\mathbf{x}) = k f(\mathbf{x}).$$

In coordinates,

$$\sum_{i=1}^n x_i \frac{\partial f}{\partial x_i} = k f.$$

**Formula 2.98. Parametrized Curves**

Let  $\mathbf{r}(t)$  be a twice differentiable curve. Its velocity, speed, and acceleration are

$$\mathbf{v}(t) = \mathbf{r}'(t), \quad \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|, \quad \mathbf{a}(t) = \mathbf{r}''(t).$$

If  $\mathbf{r}'(t_0) \neq \mathbf{0}$ , the tangent line at  $t_0$  is

$$\mathbf{r}(t_0) + s\mathbf{r}'(t_0), \quad s \in \mathbb{R}.$$

The speed is constant if and only if

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 0.$$

**Definition 2.99. Frenet–Serret Apparatus**

Let  $\mathbf{r}(s)$  be a  $C^3$  regular space curve parametrized by arc length, and suppose its curvature is nonzero. Define the unit tangent, principal normal, and binormal vectors by

$$\mathbf{T} = \mathbf{r}'(s), \quad \kappa = \|\mathbf{T}'(s)\|, \quad \mathbf{N} = \frac{\mathbf{T}'(s)}{\kappa}, \quad \mathbf{B} = \mathbf{T} \times \mathbf{N}.$$

The torsion is

$$\tau = -\mathbf{B}'(s) \cdot \mathbf{N}(s).$$

The collection

$$\{\mathbf{T}, \mathbf{N}, \mathbf{B}, \kappa, \tau\}$$

is called the Frenet–Serret apparatus. It satisfies

$$\mathbf{T}' = \kappa \mathbf{N}, \quad \mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad \mathbf{B}' = -\tau \mathbf{N}.$$

**Definition 2.100. Scalar Line Integral**

If a curve  $C$  is parametrized by  $\mathbf{r}(t)$  for  $a \leq t \leq b$ , then

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt.$$

The value is unchanged by any regular reparametrization.

**Definition 2.101. Line Integral**

If a curve  $C$  is parametrized by  $\mathbf{r}(t)$  for  $a \leq t \leq b$ , then the line integral of a vector field  $\mathbf{F}$  along  $C$  is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt.$$

**Theorem 2.102. Fundamental Theorem for Line Integrals; Gradient Theorem**

If  $\mathbf{F} = \nabla f$  and  $C$  is a smooth curve from  $\mathbf{a}$  to  $\mathbf{b}$ , then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}).$$

Hence the integral is path independent.

**Proposition 2.103. Conservative-Field Test**

Let  $U \subseteq \mathbb{R}^n$  be open and simply connected, and let  $\mathbf{F}: U \rightarrow \mathbb{R}^n$  be continuously differentiable. Then  $\mathbf{F}$  is conservative if and only if its Jacobian matrix is symmetric:

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \quad (1 \leq i, j \leq n).$$

In  $\mathbb{R}^3$ , this is equivalent to  $\nabla \times \mathbf{F} = \mathbf{0}$ . Simple connectedness cannot in general be omitted.

**Theorem 2.104. Green's Theorem**

Let  $D \subseteq \mathbb{R}^2$  be a bounded region whose positively oriented boundary  $C = \partial D$  is piecewise smooth, and let  $P, Q$  have continuous first partial derivatives on an open set containing  $D$ . Then

$$\oint_C P \, dx + Q \, dy = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

*Idea.* Green's Theorem converts circulation along a boundary curve into curl over the region inside. It is the two-dimensional ancestor of Stokes' Theorem.

**Definition 2.105. Surface Area, Scalar Surface Integral, and Flux**

Let a smooth oriented surface be parametrized by

$$\mathbf{r}(u, v), \quad (u, v) \in D.$$

Its surface element is

$$dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv.$$

Hence

$$\iint_S f dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| du dv,$$

and the flux of a vector field  $\mathbf{F}$  through the chosen orientation is

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv.$$

If  $S$  is the graph  $z = g(x, y)$  with upward orientation, then

$$dS = \sqrt{1 + g_x^2 + g_y^2} dx dy,$$

and

$$\mathbf{n} dS = (-g_x, -g_y, 1) dx dy.$$

If instead  $S$  is a regular level surface  $G(x, y, z) = 0$ , then a unit normal is

$$\mathbf{n} = \pm \frac{\nabla G}{\|\nabla G\|}.$$

**Theorem 2.106. Stokes' Theorem**

Let  $S$  be an oriented piecewise smooth surface with positively oriented piecewise smooth boundary  $\partial S$ , and let  $\mathbf{F}$  be continuously differentiable on an open set containing  $S$ . Then

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS.$$

The orientation of  $\partial S$  is determined from that of  $S$  by the right-hand rule.

*Idea.* Stokes' Theorem says that circulation around the boundary equals total curl through the surface. It turns a boundary integral into an interior integral.

**Theorem 2.107. Divergence Theorem; Gauss's Theorem**

Let  $V \subseteq \mathbb{R}^3$  be a bounded solid whose boundary  $S = \partial V$  is piecewise smooth and oriented outward. If  $\mathbf{F}$  is continuously differentiable on an open set containing  $V$ , then

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V \nabla \cdot \mathbf{F} dV.$$

*Idea.* The Divergence Theorem says that outward flux through the boundary equals total source strength inside the region.

*Remark.* Green's Theorem, Stokes' Theorem, and the Divergence Theorem share the same philosophy: a boundary integral is equal to an interior integral. This is the geometric heart of vector calculus.

## Real Analysis

### Theorem 2.108. Least Upper Bound Property; Completeness Axiom of $\mathbb{R}$

Every nonempty subset of  $\mathbb{R}$  that is bounded above has a least upper bound in  $\mathbb{R}$ . Equivalently, every nonempty subset bounded below has a greatest lower bound. This completeness property distinguishes  $\mathbb{R}$  from  $\mathbb{Q}$  and underlies the standard convergence theorems of analysis.

### Definition 2.109. Metric Space and Open Ball

A metric space is a pair  $(X, d)$  in which  $d: X \times X \rightarrow \mathbb{R}$  satisfies, for all  $x, y, z \in X$ ,

$$d(x, y) \geq 0, \quad d(x, y) = 0 \Leftrightarrow x = y,$$

$$d(x, y) = d(y, x), \quad d(x, z) \leq d(x, y) + d(y, z).$$

The open ball of radius  $r > 0$  centered at  $x$  is

$$B_r(x) = \{y \in X \mid d(x, y) < r\}.$$

### Fact 2.110. Common Metrics

For  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ , each of

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^n |x_k - y_k|,$$

$$d_2(\mathbf{x}, \mathbf{y}) = \left( \sum_{k=1}^n |x_k - y_k|^2 \right)^{1/2}, \quad d_\infty(\mathbf{x}, \mathbf{y}) = \max_{1 \leq k \leq n} |x_k - y_k|$$

is a metric. On any set  $X$ , the discrete metric is

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

### Definition 2.111. Open and Closed Sets in a Metric Space

A subset  $U \subseteq X$  is open if

$$\forall x \in U, \quad \exists r > 0, \quad B_r(x) \subseteq U.$$

A subset  $F \subseteq X$  is closed if  $X \setminus F$  is open.

### Definition 2.112. Point and Set Taxonomy in a Metric Space

Let  $A \subseteq (X, d)$ . An open cover of  $A$  is a collection of open sets whose union contains  $A$ . The standard point types and associated sets are summarized below.

| Term                               | Criterion                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------|
| Interior point                     | $x \in A$ and $\exists r > 0$ such that $B_r(x) \subseteq A$                                   |
| Exterior point                     | $\exists r > 0$ such that $B_r(x) \subseteq X \setminus A$                                     |
| Boundary point                     | $\forall r > 0, B_r(x) \cap A \neq \emptyset$ and $B_r(x) \cap (X \setminus A) \neq \emptyset$ |
| Accumulation point;<br>limit point | $\forall r > 0, (B_r(x) \setminus \{x\}) \cap A \neq \emptyset$                                |

| Term                  | Criterion                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Isolated point        | $x \in A$ and $\exists r > 0$ such that $B_r(x) \cap A = \{x\}$                                            |
| Interior $A^\circ$    | the set of all interior points of $A$                                                                      |
| Closure $\bar{A}$     | $A$ together with all of its accumulation points; equivalently, the smallest closed set containing $A$     |
| Boundary $\partial A$ | $\bar{A} \setminus A^\circ$                                                                                |
| Dense set             | $\bar{A} = X$                                                                                              |
| Perfect set           | $A$ is closed and has no isolated points; equivalently, every point of $A$ is an accumulation point of $A$ |
| Bounded set           | $A \subseteq B_R(x_0)$ for some $x_0 \in X$ and $R > 0$                                                    |
| Compact set           | every open cover of $A$ has a finite subcover                                                              |

In a metric space,  $A$  is open if and only if every point of  $A$  is interior, and  $A$  is closed if and only if it contains all of its accumulation points. In  $\mathbb{R}^n$ , compactness is equivalent to being closed and bounded by the Heine–Borel Theorem; this equivalence does not hold in arbitrary metric spaces.

**Fact 2.113. Standard Subsets of the Real Line**

In the usual topology on  $\mathbb{R}$ :

- $\mathbb{Z}$  is closed but not open;
- $\mathbb{Q}$  is countable and is neither open nor closed;
- the Cantor set is nonempty, compact, perfect, uncountable, and contains no nondegenerate interval.

These examples are standard tests of the distinctions among countability, openness, closedness, and perfectness.

**Definition 2.114. Limit of a Map Between Metric Spaces**

Let  $(X, d_X)$  and  $(Y, d_Y)$  be metric spaces, let  $D \subseteq X$ , let  $a$  be an accumulation point of  $D$ , and let  $f: D \rightarrow Y$ . The statement

$$\lim_{x \rightarrow a} f(x) = L$$

means

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \quad 0 < d_X(x, a) < \delta \Rightarrow d_Y(f(x), L) < \varepsilon.$$

**Definition 2.115. Convergence of a Sequence**

A sequence  $(x_n)$  in a metric space  $(X, d)$  converges to  $x \in X$ , written  $x_n \rightarrow x$ , if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad d(x_n, x) < \varepsilon.$$

**Definition 2.116. Cauchy Sequence and Complete Metric Space**

A sequence  $(x_n)$  in  $(X, d)$  is Cauchy if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad d(x_m, x_n) < \varepsilon.$$

The metric space is complete if every Cauchy sequence in  $X$  converges to a point of  $X$ .

**Definition 2.117. Continuity in Metric Spaces**

Let  $(X, d_X)$  and  $(Y, d_Y)$  be metric spaces. A map  $f: X \rightarrow Y$  is continuous at  $x_0 \in X$  if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in X, \quad d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

It is continuous on  $X$  if it is continuous at every  $x_0 \in X$ .

**Theorem 2.118. Sequential Criterion for Continuity**

A map  $f: X \rightarrow Y$  between metric spaces is continuous at  $x \in X$  if and only if

$$x_n \rightarrow x \Rightarrow f(x_n) \rightarrow f(x)$$

for every sequence  $(x_n)$  in  $X$ .

**Definition 2.119. Uniform Continuity**

A map  $f: (X, d_X) \rightarrow (Y, d_Y)$  is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x, y \in X, \quad d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Unlike ordinary continuity, the same  $\delta$  must work for every pair  $x, y \in X$ .

**Definition 2.120. Pointwise and Uniform Convergence**

Let  $f_n: E \rightarrow Y$  and  $f: E \rightarrow Y$ , where  $(Y, d)$  is a metric space. The sequence  $(f_n)$  converges pointwise to  $f$ , written  $f_n \rightarrow f$ , if

$$\forall x \in E, \quad \forall \varepsilon > 0, \quad \exists N = N(x, \varepsilon) \in \mathbb{N}, \quad \forall n \geq N, \quad d(f_n(x), f(x)) < \varepsilon.$$

It converges uniformly to  $f$ , written  $f_n \rightrightarrows f$ , if

$$\forall \varepsilon > 0, \quad \exists N = N(\varepsilon) \in \mathbb{N}, \quad \forall n \geq N, \quad \forall x \in E, \quad d(f_n(x), f(x)) < \varepsilon.$$

For real- or complex-valued functions, this is equivalent to

$$\|f_n - f\|_\infty = \sup_{x \in E} |f_n(x) - f(x)| \rightarrow 0.$$

The convergence is locally uniform on  $E$  if, for every compact set  $K \subseteq E$ , the restrictions  $f_n|_K$  converge uniformly to  $f|_K$ . Equivalently, for real- or complex-valued functions,

$$\sup_{x \in K} |f_n(x) - f(x)| \rightarrow 0$$

for every compact  $K \subseteq E$ .

**Definition 2.121. Equicontinuity and Uniform Boundedness**

A family  $\mathcal{F}$  of maps from  $(X, d_X)$  to  $(Y, d_Y)$  is equicontinuous at  $x_0 \in X$  if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall f \in \mathcal{F}, \quad \forall x \in X, \quad d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

A sequence of real-valued functions  $(f_n)$  is uniformly bounded if

$$\exists M > 0, \quad \forall n \in \mathbb{N}, \quad \forall x \in X, \quad |f_n(x)| \leq M.$$

**Definition 2.122. Limit Superior and Limit Inferior**

For a real sequence  $(a_n)$ , define

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k, \quad \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k,$$

whenever these extended real limits are allowed.

**Theorem 2.123. Monotone Convergence Theorem for Sequences**

*Every bounded monotone real sequence converges. More precisely, every increasing sequence bounded above converges to its supremum, and every decreasing sequence bounded below converges to its infimum.*

**Theorem 2.124. Bolzano–Weierstrass Theorem**

*Every bounded sequence in  $\mathbb{R}^n$  has a convergent subsequence.*

**Theorem 2.125. Heine–Borel Theorem**

*A subset of  $\mathbb{R}^n$  is compact if and only if it is closed and bounded.*

**Theorem 2.126. Heine–Cantor Theorem; Uniform Continuity on Compact Sets**

*If  $K \subseteq \mathbb{R}^n$  is compact and  $f: K \rightarrow \mathbb{R}^m$  is continuous, then  $f$  is uniformly continuous on  $K$ .*

*Strategy.* In analysis questions, compactness usually means that one may extract a convergent subsequence or that a continuous function attains extrema. In  $\mathbb{R}^n$ , the fast recognition phrase is “closed and bounded.”

**Theorem 2.127. Cauchy Criterion for Convergence**

*In a complete metric space, a sequence converges if and only if it is Cauchy. In particular, for  $(a_n)$  in  $\mathbb{R}$  or  $\mathbb{C}$ ,*

$$(a_n) \text{ converges} \Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, |a_m - a_n| < \varepsilon.$$

**Theorem 2.128. Uniform Cauchy Criterion**

*Let  $(Y, d)$  be complete and let  $f_n: E \rightarrow Y$ . Then  $(f_n)$  converges uniformly on  $E$  if and only if*

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, \forall x \in E, d(f_m(x), f_n(x)) < \varepsilon.$$

**Theorem 2.129. Weierstrass M-Test**

*Let  $(f_n)$  be a sequence of functions on a set  $E$ . If*

$$|f_n(x)| \leq M_n \quad \text{for all } x \in E$$

*and*

$$\sum_{n=1}^{\infty} M_n$$

*converges, then the series*

$$\sum_{n=1}^{\infty} f_n(x)$$

*converges uniformly on  $E$ .*

**Theorem 2.130. Uniform Limit Theorem**

If  $(f_n)$  is a sequence of real- or complex-valued continuous functions on a metric space  $E$  and  $f_n \rightrightarrows f$  on  $E$ , then  $f$  is continuous on  $E$ .

**Theorem 2.131. Term-by-Term Differentiation**

Suppose  $(f_n)$  is a sequence of continuously differentiable functions on a closed bounded interval  $[a, b]$ , and suppose  $(f_n(x_0))$  converges for some  $x_0 \in [a, b]$ . If  $f'_n \rightrightarrows g$  on  $[a, b]$ , then  $f_n \rightrightarrows f$  for a differentiable function  $f$ , and

$$f' = g = \lim_{n \rightarrow \infty} f'_n.$$

**Theorem 2.132. Arzelà–Ascoli Theorem**

Every uniformly bounded and equicontinuous sequence of real-valued continuous functions on a compact metric space has a uniformly convergent subsequence.

**Theorem 2.133. Stone–Weierstrass Theorem**

A subalgebra of  $C(K, \mathbb{R})$  that contains the constants and separates points is dense in  $C(K, \mathbb{R})$  under the supremum norm, provided  $K$  is compact.

**Theorem 2.134. Banach Fixed-Point Theorem; Contraction Mapping Theorem**

Let  $(X, d)$  be a nonempty complete metric space, and suppose that  $T: X \rightarrow X$  is a contraction: there exists  $q \in [0, 1)$  such that

$$\forall x, y \in X, \quad d(T(x), T(y)) \leq qd(x, y).$$

Then  $T$  has a unique fixed point  $x^* \in X$ , and the iteration  $x_{n+1} = T(x_n)$  converges to  $x^*$  for every initial point  $x_0 \in X$ .

**Theorem 2.135. Baire Category Theorem**

A nonempty complete metric space cannot be written as a countable union of nowhere dense closed sets. Equivalently, a countable intersection of open dense subsets of a complete metric space is dense.

**Definition 2.136. Sigma-Algebra, Measure, Measurability, and Almost-Everywhere Convergence**

A  $\sigma$ -algebra  $\mathcal{F}$  on a set  $X$  is a collection of subsets containing  $X$  and closed under complements and countable unions. A measure is a function  $\mu: \mathcal{F} \rightarrow [0, \infty]$  with  $\mu(\emptyset) = 0$  that is countably additive on pairwise disjoint sets. The triple  $(X, \mathcal{F}, \mu)$  is a measure space.

A function  $f: X \rightarrow \mathbb{R}$  is measurable if the inverse image of every Borel set is in  $\mathcal{F}$ . A property holds almost everywhere, abbreviated a.e., if it fails only on a set of measure zero. Thus  $f_n \xrightarrow{\text{a.e.}} f$  means that  $f_n(x) \rightarrow f(x)$  for almost every  $x$ .

**Theorem 2.137. Tonelli's Theorem**

Let  $(X, \mu)$  and  $(Y, \nu)$  be  $\sigma$ -finite measure spaces, and let  $f: X \times Y \rightarrow [0, \infty]$  be measurable. Then the iterated integrals and the product-space integral agree, possibly with value  $+\infty$ :

$$\int_{X \times Y} f \, d(\mu \times \nu) = \int_X \left( \int_Y f(x, y) \, d\nu(y) \right) d\mu(x) = \int_Y \left( \int_X f(x, y) \, d\mu(x) \right) d\nu(y).$$

In particular, a nonnegative series may be integrated term by term.

**Theorem 2.138. Fatou's Lemma**

If  $f_n \geq 0$  are measurable, then

$$\int \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int f_n \, d\mu.$$

**Theorem 2.139. Lebesgue Monotone Convergence Theorem; Beppo Levi Theorem**

If  $0 \leq f_1 \leq f_2 \leq \dots$  and  $f_n \xrightarrow{\text{a.e.}} f$ , then

$$\lim_{n \rightarrow \infty} \int f_n \, d\mu = \int f \, d\mu.$$

**Theorem 2.140. Lebesgue Dominated Convergence Theorem**

If  $f_n \xrightarrow{\text{a.e.}} f$  and there exists an integrable function  $g$  such that  $|f_n| \leq g$  for all  $n$ , then

$$\lim_{n \rightarrow \infty} \int f_n \, d\mu = \int f \, d\mu.$$

**Theorem 2.141. Density of Irrational Rotations**

If  $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ , then the sequence of fractional parts

$$\{n\alpha\}, \quad n = 0, 1, 2, \dots,$$

is dense in  $[0, 1]$ . Equivalently, the orbit of any point under rotation of the circle through an irrational multiple of  $2\pi$  is dense in the circle.

*Idea.* Write  $\|x\|_{\mathbb{R}/\mathbb{Z}}$  for the distance from  $x$  to the nearest integer. By the pigeonhole principle, for every  $N \geq 2$  there is a positive integer  $q$  such that

$$0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{N}.$$

Put  $\delta = \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}$  and  $M = \lfloor 1/\delta \rfloor$ . Up to reversing orientation on the circle, the integer orbit contains

$$0, \delta, 2\delta, \dots, M\delta,$$

whose successive circular gaps are at most  $\delta < 1/N$ . Thus the two-sided orbit  $\{n\alpha \bmod 1 \mid n \in \mathbb{Z}\}$  is dense.

Moreover, irrationality gives an unbounded sequence of positive integers  $q_j$  for which

$$\|q_j\alpha\|_{\mathbb{R}/\mathbb{Z}} \rightarrow 0.$$

For each fixed  $m \geq 1$ , the nonnegative multiples  $(q_j - m)\alpha$  converge modulo 1 to  $-m\alpha$ , and  $q_j - m \geq 0$  for all sufficiently large  $j$ . Hence the closure of the nonnegative orbit contains every negative multiple as well as every nonnegative multiple. It therefore contains the dense two-sided orbit, proving the claim.

## Complex Analysis

**Definition 2.142. Complex Differentiability; Holomorphic and Entire Functions**

Let  $U \subseteq \mathbb{C}$  be open and let  $f: U \rightarrow \mathbb{C}$ . The complex derivative of  $f$  at  $z_0 \in U$  is

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0},$$

provided the limit exists independently of the direction of approach. The function is holomorphic on  $U$  if it is complex differentiable at every point of  $U$ , and entire if it is holomorphic on all of  $\mathbb{C}$ .

**Theorem 2.143. Cauchy–Riemann Equations**

Write

$$f(z) = u(x, y) + iv(x, y), \quad z = x + iy.$$

If  $f$  is complex differentiable at  $z_0 = (x_0, y_0)$  and the first partial derivatives exist there, then

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0), \quad \frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0).$$

Conversely, if these partial derivatives are continuous near  $z_0$  and satisfy the equations there, then  $f$  is complex differentiable at  $z_0$ .

**Formula 2.144. Cauchy–Riemann Equations in Polar Coordinates**

Write

$$f(re^{i\theta}) = u(r, \theta) + iv(r, \theta), \quad r > 0.$$

If the relevant first partial derivatives are continuous, then the Cauchy–Riemann equations become

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}.$$

*Strategy.* To test whether a given  $f(z) = u(x, y) + iv(x, y)$  can be holomorphic, compute the four first partial derivatives and check the Cauchy–Riemann equations. Failure at one point immediately rules out holomorphicity there.

**Fact 2.145. Standard Holomorphic and Nonholomorphic Examples**

Every polynomial,  $e^z$ ,  $\sin z$ , and  $\cos z$  is entire. The function  $\tan z$  is meromorphic but not entire because it has poles at

$$z = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

The function  $z \mapsto \operatorname{Re}(z)$  is not complex differentiable at any point, since its difference quotient depends on the direction of approach.

**Proposition 2.146. Holomorphic Functions Have Harmonic Components**

If  $f = u + iv$  is holomorphic and  $u, v$  have continuous second partial derivatives, then

$$\Delta u = 0, \quad \Delta v = 0.$$

Thus the real and imaginary parts of a holomorphic function are harmonic conjugates locally.

**Formula 2.147. Complex Exponential, Trigonometric Functions, and Logarithm**

For  $z = x + iy$ ,

$$e^z = e^x(\cos y + i \sin y).$$

The complex sine and cosine satisfy

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

For  $z \neq 0$ , the complex logarithm is multivalued:

$$\log z = \ln |z| + i(\arg z + 2\pi k), \quad k \in \mathbb{Z}.$$

A single-valued holomorphic logarithm requires a choice of branch on a domain that does not wind around the origin.

**Formula 2.148.  $n$ th Roots and Roots of Unity**

If  $w = \rho e^{i\theta} \neq 0$ , then the solutions of  $z^n = w$  are

$$z_k = \rho^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, \dots, n-1.$$

With  $\omega_n = e^{2\pi i/n}$ , the  $n$ th roots of unity are  $1, \omega_n, \dots, \omega_n^{n-1}$ , and

$$x^n - 1 = \prod_{k=0}^{n-1} (x - \omega_n^k).$$

Evaluating  $(x^n - 1)/(x - 1)$  at  $x = 1$  gives

$$\prod_{k=1}^{n-1} (1 - \omega_n^k) = n.$$

**Definition 2.149. Complex Contour Integral**

Let  $\gamma: [a, b] \rightarrow \mathbb{C}$  be a piecewise  $C^1$  parametrized contour and let  $f$  be continuous on the image of  $\gamma$ . The contour integral is defined by

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

If  $\gamma$  is closed, the notation  $\oint_{\gamma} f(z) dz$  is commonly used. Reversing the orientation changes the sign of the integral.

**Theorem 2.150. Cauchy–Goursat Integral Theorem**

If  $f$  is holomorphic on a simply connected domain  $U \subseteq \mathbb{C}$ , then for every closed piecewise smooth curve  $\gamma$  in  $U$ ,

$$\oint_{\gamma} f(z) dz = 0.$$

Equivalently, contour integrals of  $f$  in  $U$  are path-independent.

**Theorem 2.151. Cauchy Integral Formula**

Let  $f$  be holomorphic on an open set containing a positively oriented simple closed contour  $\gamma$  and its interior. If  $a$  lies inside  $\gamma$ , then

$$f(a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - a} dz.$$

More generally, for every integer  $n \geq 0$ ,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - a)^{n+1}} dz.$$

*Strategy.* When a contour integral contains  $(z - a)^{-m}$  and the remaining factor is holomorphic, compare it directly with Cauchy's integral formula before attempting a parametrization of the contour.

**Theorem 2.152. Maximum Modulus Principle**

A nonconstant holomorphic function on a connected open set cannot attain a maximum of  $|f|$  at an interior point. Consequently, if  $f$  is holomorphic on a bounded domain and continuous on its closure, then the maximum of  $|f|$  occurs on the boundary.

**Theorem 2.153. Liouville's Theorem**

*Every bounded entire function is constant.*

*Remark.* Liouville's Theorem gives a short proof of the Fundamental Theorem of Algebra: if a nonconstant complex polynomial had no zero, then its reciprocal would be a bounded entire function.

**Theorem 2.154. Taylor Series for Holomorphic Functions**

*If  $f$  is holomorphic on the open disk  $|z - a| < R$ , then*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n, \quad |z - a| < R.$$

*Thus a holomorphic function is analytic: complex differentiability implies local representability by a convergent power series.*

**Theorem 2.155. Laurent Series**

*If  $f$  is holomorphic on an annulus*

$$r < |z - a| < R,$$

*then it has a unique Laurent expansion*

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n$$

*converging throughout the annulus. The coefficient  $c_{-1}$  is called the residue of  $f$  at  $a$ .*

**Definition 2.156. Isolated Singularities and Residues**

A point  $a$  is an isolated singularity of  $f$  if  $f$  is holomorphic on some punctured disk  $0 < |z - a| < R$  but is not holomorphic at  $a$ . In the Laurent expansion about  $a$ , the residue is

$$\text{Res}(f; a) = c_{-1}.$$

The singularity is removable if every negative-power coefficient is zero, a pole if only finitely many negative-power coefficients are nonzero, and essential if infinitely many are nonzero.

**Formula 2.157. Residues at Poles**

If  $a$  is a simple pole of  $f$ , then

$$\text{Res}(f; a) = \lim_{z \rightarrow a} (z - a)f(z).$$

More generally, if  $a$  is a pole of order  $m$ , then

$$\text{Res}(f; a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} ((z - a)^m f(z)).$$

If  $f(z) = g(z)/h(z)$ , where  $g$  and  $h$  are holomorphic,  $h(a) = 0$ , and  $h'(a) \neq 0$ , then

$$\text{Res}(f; a) = \frac{g(a)}{h'(a)}.$$

**Theorem 2.158. Residue Theorem**

Let  $f$  be holomorphic inside and on a positively oriented simple closed contour  $\gamma$ , except for isolated singularities  $a_1, \dots, a_m$  inside  $\gamma$ . Then

$$\oint_{\gamma} f(z) \, dz = 2\pi i \sum_{k=1}^m \operatorname{Res}(f; a_k).$$

*Strategy. Real Integrals by Residues.* For a rational integral over  $\mathbb{R}$ , extend the integrand to a complex rational function, choose a semicircular contour, sum the residues in the appropriate half-plane, and verify that the contribution from the large arc vanishes. Trigonometric integrals over  $[0, 2\pi]$  often become contour integrals under the substitution  $z = e^{i\theta}$  and

$$d\theta = \frac{dz}{iz}.$$

**Theorem 2.159. Rouché's Theorem**

Let  $f$  and  $g$  be holomorphic inside and on a simple closed contour  $\gamma$ . If

$$|g(z)| < |f(z)| \quad \text{for every } z \in \gamma,$$

then  $f$  and  $f + g$  have the same number of zeros inside  $\gamma$ , counted with multiplicity.

**Definition 2.160. Conformal Map**

A holomorphic function  $f$  is conformal at  $z_0$  if  $f'(z_0) \neq 0$ . At such a point,  $f$  preserves oriented angles and infinitesimal shapes, although it may change scale.

**Definition 2.161. Möbius Transformation; Linear Fractional Transformation**

A Möbius transformation is a map of the extended complex plane of the form

$$T(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0.$$

It is conformal wherever its derivative is finite and nonzero, and it maps generalized circles—circles and lines—to generalized circles.

**Formula 2.162. Standard Conformal Maps**

The following transformations are useful to recognize quickly:

$$z \mapsto az + b \quad (\text{rotation, dilation, and translation}),$$

$$z \mapsto \frac{1}{z} \quad (\text{inversion}), \quad z \mapsto z^n \quad (\text{angle multiplication}),$$

and

$$z \mapsto \frac{z - i}{z + i},$$

which maps the upper half-plane conformally onto the unit disk.

## Chapter 3

# Linear Algebra

Related **POSTECH** Courses: (MATH203) Applied Linear Algebra

Related Books: Linear Algebra and Its Applications (Gilbert Strang)

### Definition 3.1. Vector Space

A vector space  $V$  over a field  $F$  is a set equipped with vector addition and scalar multiplication satisfying the standard vector-space axioms. The scalars belong to  $F$ ; in this book the principal cases are  $F = \mathbb{R}$  and  $F = \mathbb{C}$ .

### Definition 3.2. Linear Combination, Span, and Linear Independence

A vector of the form

$$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k$$

is a linear combination of  $\mathbf{v}_1, \dots, \mathbf{v}_k$ . Their span is

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \{c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k : c_1, \dots, c_k \in F\}.$$

The vectors are linearly independent if

$$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$$

implies  $c_1 = \cdots = c_k = 0$ .

### Definition 3.3. Basis and Dimension

A list

$$\mathbf{v}_1, \dots, \mathbf{v}_n$$

is a basis of  $V$  if it is linearly independent and spans  $V$ . If  $V$  has a finite basis, every basis has the same number of vectors; this number is the dimension of  $V$ , denoted by  $\dim V$ .

### Theorem 3.4. Basis Extension and Reduction

*Every linearly independent list in a finite-dimensional vector space can be extended to a basis, and every spanning list contains a basis. More generally, the assertion that every vector space has a basis is equivalent to a standard form of the axiom of choice.*

### Definition 3.5. Linear Map; Linear Transformation

Let  $V$  and  $W$  be vector spaces over the same field  $F$ . A map  $T: V \rightarrow W$  is linear if

$$\forall \alpha, \beta \in F, \quad \forall \mathbf{u}, \mathbf{v} \in V, \quad T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}).$$

Equivalently,  $T$  preserves vector addition and scalar multiplication. Its kernel and image are

$$\ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}, \quad \text{im } T = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}.$$

**Formula 3.6. Coordinate Vectors, Matrix Representation, and Change of Basis**

Let

$$\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$$

be an ordered basis of  $V$ . The coordinate vector of  $\mathbf{v} \in V$  relative to  $\mathcal{B}$  is the unique column vector whose entries are the unique scalars  $c_1, \dots, c_n \in F$  satisfying

$$[\mathbf{v}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}, \quad \mathbf{v} = \sum_{i=1}^n c_i \mathbf{v}_i.$$

When  $V = F^m$ , the basis vectors may be assembled as columns of a matrix, giving

$$\mathbf{v} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n) [\mathbf{v}]_{\mathcal{B}}.$$

If  $T: V \rightarrow W$  is linear and  $\mathcal{C}$  is an ordered basis of  $W$ , then the matrix  $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$  is characterized by

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}}.$$

If  $\mathcal{B}$  and  $\mathcal{B}'$  are ordered bases of the same  $n$ -dimensional space and

$$[\mathbf{v}]_{\mathcal{B}} = \mathbf{Q} [\mathbf{v}]_{\mathcal{B}'},$$

then  $\mathbf{Q}$  is invertible. For a linear operator  $T: V \rightarrow V$ ,

$$[T]_{\mathcal{B}'} = \mathbf{Q}^{-1} [T]_{\mathcal{B}} \mathbf{Q}.$$

Thus matrices representing the same linear operator in different bases are similar.

**Definition 3.7. Column Space, Row Space, Null Space, and Rank**

Let  $\mathbf{A} \in F^{m \times n}$ . Its column space and row space are the subspaces spanned by its columns and rows:

$$\text{Col}(\mathbf{A}) \subseteq F^m, \quad \text{Row}(\mathbf{A}) \subseteq F^n.$$

Its null space is

$$\text{Null}(\mathbf{A}) = \{\mathbf{x} \in F^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\}.$$

The rank is

$$\text{rk}(\mathbf{A}) = \dim \text{Col}(\mathbf{A}).$$

Equivalently, rank is the number of pivot columns in any row-echelon form of  $\mathbf{A}$ ; the Row Rank Equals Column Rank theorem below justifies the corresponding row-space interpretation.

**Definition 3.8. Elementary Row Operations, Echelon Forms, and Pivots**

The elementary row operations are: interchange two rows; multiply a row by a nonzero scalar; and add a scalar multiple of one row to another. A matrix is in *row echelon form* if every nonzero row begins to the right of the row above it and every zero row is below every nonzero row. The first nonzero entry of a nonzero row is a *pivot*. It is in *reduced row echelon form* if, in addition, every pivot is 1 and is the only nonzero entry in its column.

**Algorithm 3.9. Gaussian Elimination and Gauss–Jordan Reduction**

For a linear system  $\mathbf{A}\mathbf{x} = \mathbf{b}$ , form the augmented matrix  $[\mathbf{A} \mid \mathbf{b}]$  and apply elementary row operations.

- (1) Reduce to row echelon form to solve by back-substitution, or continue to reduced row echelon form for a direct description of all solutions.

- (2) Pivot columns identify basic variables; nonpivot columns identify free variables. The number of pivots is  $\text{rk}(\mathbf{A})$ .
- (3) A row of the form  $[\mathbf{0}^T \mid c]$  with  $c \neq 0$  means that the system is inconsistent. Otherwise, there is a unique solution exactly when every variable column is a pivot column; free variables produce infinitely many solutions over an infinite field.
- (4) To find  $\mathbf{A}^{-1}$  for a square matrix, row-reduce  $[\mathbf{A} \mid \mathbf{I}_n]$ . The matrix is invertible exactly when the left block reduces to  $\mathbf{I}_n$ , in which case the right block becomes  $\mathbf{A}^{-1}$ .

In a timed problem, exploit zeros, symmetry, triangular or block form, and obvious linear relations before carrying out a full reduction.

**Fact 3.10. Translation Is Affine, Not Linear**

For a fixed vector  $\mathbf{v} \in \mathbb{R}^n$ , the translation

$$T_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \mathbf{v}$$

is bijective, with inverse  $T_{-\mathbf{v}}$ . It is a linear map if and only if  $\mathbf{v} = \mathbf{0}$ .

**Definition 3.11. Inner Product, Norm, Orthogonality, and Orthonormality**

For real column vectors, the standard inner product is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v},$$

while over  $\mathbb{C}$  it is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^H \mathbf{v}.$$

The induced norm is

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Vectors  $\mathbf{u}$  and  $\mathbf{v}$  are orthogonal if  $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ . A list is orthonormal if its vectors are pairwise orthogonal and each has norm 1.

**Formula 3.12. Adjoint Transfer**

For a real matrix  $\mathbf{A}$  of compatible size,

$$\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^T \mathbf{x}, \mathbf{y} \rangle.$$

Hence, if  $\mathbf{A}$  is symmetric,

$$\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle.$$

Over  $\mathbb{C}$ , replace  $\mathbf{A}^T$  by  $\mathbf{A}^H$ .

**Theorem 3.13. Rank–Nullity Theorem**

Let  $T: V \rightarrow W$  be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \ker T + \dim \text{im } T.$$

For a matrix  $\mathbf{A} \in F^{m \times n}$ , this becomes

$$n = \text{rk}(\mathbf{A}) + \dim \text{Null}(\mathbf{A}).$$

*Strategy.* Rank–Nullity connects variables, pivots, and free variables. It is the linear algebra version of accounting: nothing disappears; dimensions are redistributed.

**Theorem 3.14. Row Rank Equals Column Rank**

For every matrix  $\mathbf{A}$ ,

$$\dim \text{Row}(\mathbf{A}) = \dim \text{Col}(\mathbf{A}).$$

This common dimension is called the rank of  $\mathbf{A}$  and is denoted by  $\text{rk}(\mathbf{A})$ .

**Definition 3.15. Determinant**

The determinant is a scalar associated to a square matrix  $\mathbf{A} \in F^{n \times n}$ , denoted by  $\det(\mathbf{A})$ . Geometrically, for real matrices,  $|\det(\mathbf{A})|$  is the volume scaling factor of the linear transformation  $\mathbf{x} \mapsto \mathbf{Ax}$ .

**Definition 3.16. Minor, Cofactor, and Adjugate**

Let  $\mathbf{A} = (a_{ij}) \in F^{n \times n}$ . The minor  $M_{ij}$  is the determinant of the  $(n-1) \times (n-1)$  matrix obtained by deleting row  $i$  and column  $j$ . The corresponding cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The cofactor matrix is  $(C_{ij})$ , and the adjugate is its transpose:

$$\text{adj}(\mathbf{A}) = (C_{ij})^T.$$

Cofactors are the coefficients in Laplace expansion and are therefore the bridge between determinants and the adjugate formula for an inverse.

**Theorem 3.17. Invertible Matrix Theorem**

For a square matrix  $\mathbf{A} \in F^{n \times n}$ , the following statements are equivalent:

$$\mathbf{A} \text{ is invertible}, \quad \det(\mathbf{A}) \neq 0, \quad \text{rk}(\mathbf{A}) = n,$$

$$\text{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{Ax} = \mathbf{b}$$

has a unique solution for every  $\mathbf{b} \in F^n$ .

*Remark.* This theorem is a dictionary. It translates between determinants, rank, null spaces, and solvability of linear systems.

**Proposition 3.18. One-Sided Inverse for Square Matrices**

If  $\mathbf{A}, \mathbf{B} \in F^{n \times n}$  satisfy

$$\mathbf{AB} = \mathbf{I}_n,$$

then both matrices are invertible and

$$\mathbf{BA} = \mathbf{I}_n.$$

Thus, for square matrices, a right inverse is automatically a left inverse, and conversely.

**Formula 3.19. Inverse of a  $2 \times 2$  Matrix**

If

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and  $ad - bc \neq 0$ , then

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

**Formula 3.20. Trace and Determinant of a  $2 \times 2$  Inverse**

If  $\mathbf{A} \in F^{2 \times 2}$  is invertible, then

$$\operatorname{tr}(\mathbf{A}^{-1}) = \frac{\operatorname{tr}(\mathbf{A})}{\det(\mathbf{A})}, \quad \det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

The trace identity is special to dimension 2.

**Theorem 3.21. Cramer's Rule**

Let  $\mathbf{A} \in F^{n \times n}$  be invertible. The unique solution of  $\mathbf{Ax} = \mathbf{b}$  is given by

$$x_i = \frac{\det(\mathbf{A}_i)}{\det(\mathbf{A})},$$

where  $\mathbf{A}_i$  is obtained from  $\mathbf{A}$  by replacing its  $i$ th column with  $\mathbf{b}$ .

**Formula 3.22. Adjugate Formula for the Inverse**

For every square matrix  $\mathbf{A}$ ,

$$\mathbf{A} \operatorname{adj}(\mathbf{A}) = \operatorname{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}_n.$$

Hence, if  $\det(\mathbf{A}) \neq 0$ , then

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A}).$$

**Formula 3.23. Determinant as Volume**

If  $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^n$ , then the  $n$ -dimensional volume of the parallelepiped spanned by these vectors is

$$|\det(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n)|.$$

In  $\mathbb{R}^3$ , this is the absolute value of the scalar triple product

$$|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|.$$

**Formula 3.24. Scalar and Vector Triple Products**

For  $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ ,

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \det(\mathbf{a}, \mathbf{b}, \mathbf{c}).$$

The vector triple-product identities are

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c},$$

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}.$$

**Formula 3.25. Cross-Product Determinant Identity**

For  $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ ,

$$\det(\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}) = \det(\mathbf{a}, \mathbf{b}, \mathbf{c})^2.$$

Therefore, if  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  are linearly independent, then

$$\mathbf{a} \times \mathbf{b}, \quad \mathbf{b} \times \mathbf{c}, \quad \mathbf{c} \times \mathbf{a}$$

are also linearly independent.

**Proposition 3.26. Basic Properties of Determinants**

For square matrices  $\mathbf{A}, \mathbf{B} \in F^{n \times n}$ , we have

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}),$$

$$\det(\mathbf{A}^\top) = \det(\mathbf{A}),$$

and, if  $\mathbf{A}$  is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

**Formula 3.27. Determinant of a Triangular Matrix**

If  $\mathbf{A}$  is upper triangular or lower triangular with diagonal entries  $a_{11}, a_{22}, \dots, a_{nn}$ , then

$$\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}.$$

**Formula 3.28. Determinant of a Block Triangular Matrix**

If the diagonal blocks are square, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A}) \det(\mathbf{D}),$$

and the same formula holds for a lower block triangular matrix.

**Proposition 3.29. Multilinearity and Alternation of the Determinant**

The determinant is linear in each row separately and also linear in each column separately. If two rows or two columns are interchanged, the sign of the determinant changes. In particular, if the rows or columns of  $\mathbf{A}$  are linearly dependent, then

$$\det(\mathbf{A}) = 0.$$

*Strategy.* In short determinant problems, first look for dependent rows or columns, row or column sums, repeated patterns, triangular form, and easy row operations. Direct expansion should usually be the last resort, except for  $2 \times 2$  and  $3 \times 3$  matrices.

**Proposition 3.30. Characterization of the Determinant**

Let

$$f: (F^n)^n \rightarrow F$$

be alternating and multilinear. Then

$$f(\mathbf{v}_1, \dots, \mathbf{v}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det(\mathbf{v}_1 \cdots \mathbf{v}_n).$$

In particular, the determinant is the unique alternating multilinear function satisfying

$$\det(\mathbf{I}_n) = 1.$$

**Formula 3.31. Laplace Expansion; Cofactor Expansion**

Let  $\mathbf{A} = (a_{ij}) \in F^{n \times n}$ , and let  $C_{ij}$  be the cofactor of  $a_{ij}$ . Expanding along row  $i$  gives

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Expanding along column  $j$  gives

$$\det(\mathbf{A}) = \sum_{i=1}^n a_{ij} C_{ij}.$$

**Formula 3.32. Sarrus' Rule**

For a  $3 \times 3$  matrix,

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

**Formula 3.33. Determinant and Elementary Row Operations**

The determinant changes under elementary row operations as follows.

- Swapping two rows multiplies the determinant by  $-1$ .
- Multiplying one row by  $c$  multiplies the determinant by  $c$ .
- Adding a multiple of one row to another row does not change the determinant.

The same rules hold for columns.

**Algorithm 3.34. Computing a Determinant**

For a square matrix  $\mathbf{A}$ , choose the shortest structure-aware route rather than expanding automatically.

- (1) First test for an immediate value: a zero row or column, or linearly dependent rows or columns, gives determinant 0; triangular and block-triangular matrices reduce to products of diagonal determinants.
- (2) Use row or column operations to create zeros and approach triangular form. Track row swaps and row scalings exactly; adding a multiple of one row to another leaves the determinant unchanged.
- (3) If one row or column is sparse, Laplace expansion along that row or column can be faster than elimination. For a  $3 \times 3$  matrix, Sarrus' rule is available when it is genuinely shorter.
- (4) Before doing arithmetic, recognize standard matrices and low-rank structure. Vandermonde, Cauchy, Pascal, block, permutation, and rank-one perturbation formulas often turn a long calculation into one line.
- (5) Check the result against structural facts: a singular matrix must have determinant 0, and  $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ .

**Formula 3.35. Matrix Determinant Lemma**

If  $\mathbf{A} \in F^{n \times n}$  is invertible and  $\mathbf{u}, \mathbf{v} \in F^n$ , then

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^T) = \det(\mathbf{A}) (1 + \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u}).$$

In particular,

$$\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^T) = 1 + \mathbf{v}^T \mathbf{u}.$$

**Theorem 3.36. Sylvester's Determinant Theorem**

If  $\mathbf{A} \in F^{m \times n}$  and  $\mathbf{B} \in F^{n \times m}$ , then

$$\det(\mathbf{I}_m + \mathbf{AB}) = \det(\mathbf{I}_n + \mathbf{BA}).$$

*This allows a determinant involving a large product to be replaced by a determinant of the smaller dimension.*

**Formula 3.37. Sherman–Morrison Formula**

If  $\mathbf{A}$  is invertible and  $1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} \neq 0$ , then

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1} \mathbf{u} \mathbf{v}^\top \mathbf{A}^{-1}}{1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u}}.$$

It is the inverse counterpart of the Matrix Determinant Lemma.

**Formula 3.38. Schur Complement Determinant Formula**

If  $\mathbf{A}$  is invertible, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A}) \det(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}).$$

The matrix  $\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$  is the Schur complement of  $\mathbf{A}$ .

**Formula 3.39. Symmetric Block Determinant**

If  $\mathbf{A}$  and  $\mathbf{B}$  are square matrices of the same size, then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} = \det(\mathbf{A} + \mathbf{B}) \det(\mathbf{A} - \mathbf{B}).$$

This follows by changing to the invariant subspaces consisting of vectors of the form  $(\mathbf{x}, \mathbf{x})$  and  $(\mathbf{x}, -\mathbf{x})$ .

**Definition 3.40. Eigenvalue, Eigenvector, Eigenspace, and Characteristic Polynomial**

Let  $\mathbf{A} \in F^{n \times n}$ . A scalar  $\lambda \in F$  is an eigenvalue of  $\mathbf{A}$  if there exists a nonzero vector  $\mathbf{v} \in F^n$  such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

Such a nonzero vector  $\mathbf{v}$  is an eigenvector associated with  $\lambda$ . The eigenspace is

$$E_\lambda(\mathbf{A}) = \text{Null}(\mathbf{A} - \lambda\mathbf{I}_n).$$

The characteristic polynomial of  $\mathbf{A}$  is

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}).$$

Regarding the eigenvalues over  $\mathbb{C}$ , the spectral radius is

$$\rho(\mathbf{A}) = \max\{|\lambda| \mid \lambda \text{ is an eigenvalue of } \mathbf{A}\}.$$

**Algorithm 3.41. Computing Eigenvalues and Eigenvectors**

Let  $\mathbf{A} \in F^{n \times n}$ .

- (1) Inspect the matrix first. For triangular or block-triangular matrices, the eigenvalues come from the diagonal blocks; symmetry, all-ones structure, permutation structure, and low-rank formulas can reveal eigenvectors without computing a full characteristic polynomial.
- (2) Otherwise compute

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A})$$

and solve  $p_{\mathbf{A}}(\lambda) = 0$ . Over  $\mathbb{R}$ , remember that nonreal roots may occur; over  $\mathbb{C}$ , the characteristic polynomial splits completely.

- (3) For each eigenvalue  $\lambda$ , row-reduce  $\mathbf{A} - \lambda\mathbf{I}_n$  and find a basis of  $\text{Null}(\mathbf{A} - \lambda\mathbf{I}_n)$ . Every nonzero vector in that null space is an eigenvector for  $\lambda$ .

(4) Use

$$\sum_i \lambda_i = \text{tr}(\mathbf{A}), \quad \prod_i \lambda_i = \det(\mathbf{A})$$

with algebraic multiplicity as fast checks. Compare algebraic and geometric multiplicities when deciding whether  $\mathbf{A}$  is diagonalizable.

**Definition 3.42. Algebraic and Geometric Multiplicity**

Let  $\lambda$  be an eigenvalue of  $\mathbf{A} \in F^{n \times n}$ . Its algebraic multiplicity  $m_a(\lambda)$  is its multiplicity as a root of  $p_{\mathbf{A}}(t)$ , while its geometric multiplicity is

$$m_g(\lambda) = \dim \text{Null}(\mathbf{A} - \lambda \mathbf{I}_n).$$

For every eigenvalue,

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

**Definition 3.43. Diagonalizable Matrix**

A matrix  $\mathbf{A} \in F^{n \times n}$  is diagonalizable over  $F$  if there exist an invertible matrix  $\mathbf{P} \in F^{n \times n}$  and a diagonal matrix  $\mathbf{D}$  such that

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}.$$

Equivalently, the columns of  $\mathbf{P}$  can be chosen as a basis of eigenvectors of  $\mathbf{A}$ . Diagonalization is useful because powers and polynomial functions reduce to entrywise calculations on  $\mathbf{D}$ :

$$\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}.$$

**Definition 3.44. Conjugate Transpose; Hermitian Matrix**

For a complex matrix  $\mathbf{A} = (a_{ij})$ , its conjugate transpose, or Hermitian transpose, is

$$\mathbf{A}^H = \overline{\mathbf{A}}^T = (\overline{a_{ji}}).$$

A complex square matrix is Hermitian if

$$\mathbf{A}^H = \mathbf{A}.$$

For real matrices, the Hermitian transpose is the ordinary transpose.

**Proposition 3.45. Basic Rules for the Hermitian Transpose**

For complex matrices of compatible sizes,

$$(\mathbf{A} + \mathbf{B})^H = \mathbf{A}^H + \mathbf{B}^H, \quad (c\mathbf{A})^H = \bar{c} \mathbf{A}^H,$$

$$(\mathbf{AB})^H = \mathbf{B}^H \mathbf{A}^H, \quad (\mathbf{A}^H)^H = \mathbf{A}.$$

**Definition 3.46. Orthogonal and Unitary Matrices**

A real square matrix  $\mathbf{Q}$  is orthogonal if

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{I}_n.$$

A complex square matrix  $\mathbf{U}$  is unitary if

$$\mathbf{U}^H \mathbf{U} = \mathbf{I}_n.$$

Equivalently, their columns form orthonormal bases over  $\mathbb{R}$  and  $\mathbb{C}$ , respectively.

**Formula 3.47. Special Matrix Recognition Table**

For real or complex square matrices, the following implications permit rapid determinant, inverse, and eigenvalue calculations.

| Matrix type           | Defining relation                             | Immediate consequences                                                             |
|-----------------------|-----------------------------------------------|------------------------------------------------------------------------------------|
| Diagonal              | $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$   | eigenvalues $d_i$ ; $\det(\mathbf{D}) = \prod_i d_i$                               |
| Triangular            | $a_{ij} = 0$ for $i > j$ or for $i < j$       | eigenvalues are diagonal entries; determinant is their product                     |
| Orthogonal            | $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$   | $\mathbf{Q}^{-1} = \mathbf{Q}^\top$ ; $ \det(\mathbf{Q})  = 1$                     |
| Unitary               | $\mathbf{U}^H \mathbf{U} = \mathbf{I}_n$      | $\mathbf{U}^{-1} = \mathbf{U}^H$ ; every eigenvalue has modulus 1                  |
| Idempotent            | $\mathbf{P}^2 = \mathbf{P}$                   | eigenvalues belong to $\{0, 1\}$ ; $\text{rk}(\mathbf{P}) = \text{tr}(\mathbf{P})$ |
| Involutory            | $\mathbf{A}^2 = \mathbf{I}_n$                 | $\mathbf{A}^{-1} = \mathbf{A}$ ; eigenvalues belong to $\{-1, 1\}$                 |
| Nilpotent             | $\mathbf{N}^k = \mathbf{0}$ for some $k$      | every eigenvalue is 0; $\det(\mathbf{N}) = 0$                                      |
| Permutation           | one 1 in each row and column                  | $\mathbf{P}^{-1} = \mathbf{P}^\top$ ; determinant is $\pm 1$                       |
| Orthogonal projection | $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\top$ | $\mathbb{R}^n = \text{Col}(\mathbf{P}) \oplus \text{Null}(\mathbf{P})$             |

**Formula 3.48. Vandermonde Determinant**

For scalars  $x_1, x_2, \dots, x_n$ ,

$$\det \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

**Fact 3.49. Pascal Matrices**

Define the lower triangular Pascal matrix by

$$\mathbf{P}_n = (p_{ij})_{0 \leq i, j \leq n}, \quad p_{ij} = \begin{cases} \binom{i}{j}, & j \leq i, \\ 0, & j > i. \end{cases}$$

Since all diagonal entries are equal to 1, we have

$$\det(\mathbf{P}_n) = 1.$$

Define the symmetric Pascal matrix by

$$\mathbf{S}_n = (s_{ij})_{0 \leq i, j \leq n}, \quad s_{ij} = \binom{i+j}{i}.$$

This matrix also has determinant 1.

*Remark.* Pascal matrices have combinatorial entries while their determinants admit simple closed forms.

**Formula 3.50. Cauchy Determinant**

Let

$$c_{ij} = \frac{1}{x_i + y_j},$$

where every denominator is nonzero and the  $x_i$  and  $y_j$  are pairwise distinct within their respective families. Then

$$\det(c_{ij})_{1 \leq i, j \leq n} = \frac{\prod_{1 \leq i < j \leq n} (x_j - x_i)(y_j - y_i)}{\prod_{i=1}^n \prod_{j=1}^n (x_i + y_j)}.$$

The Hilbert matrix is a famous special case of a Cauchy matrix.

**Formula 3.51. Hilbert Matrix Determinant**

The  $n \times n$  Hilbert matrix is

$$\mathbf{H}_n = \left( \frac{1}{i+j-1} \right)_{1 \leq i, j \leq n}.$$

As a special Cauchy matrix, it satisfies

$$\det(\mathbf{H}_n) = \frac{\prod_{k=1}^{n-1} (k!)^4}{\prod_{k=1}^{2n-1} k!}.$$

**Definition 3.52. Hadamard Matrix**

A Hadamard matrix of order  $n$  is an  $n \times n$  matrix  $\mathbf{H}$  whose entries are all  $+1$  or  $-1$  and whose rows are mutually orthogonal. Equivalently,

$$\mathbf{H}\mathbf{H}^\top = n\mathbf{I}_n.$$

**Formula 3.53. Hadamard Determinant**

If  $\mathbf{H}$  is a Hadamard matrix of order  $n$ , then

$$|\det(\mathbf{H})| = n^{n/2}.$$

**Formula 3.54. Permutation Matrices**

A permutation matrix  $\mathbf{P}_\sigma$  has exactly one entry equal to 1 in each row and each column. It satisfies

$$\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^\top, \quad \det(\mathbf{P}_\sigma) = \text{sgn}(\sigma).$$

Its powers are determined by the cycle structure of  $\sigma$ .

**Formula 3.55. Eigenvalues of a Circulant Matrix**

Let  $\mathbf{C}$  be the circulant matrix whose first row is

$$(c_0, c_1, \dots, c_{n-1}).$$

With  $\omega_n = e^{-2\pi i/n}$ , its eigenvalues are

$$\lambda_j = \sum_{k=0}^{n-1} c_k \omega_n^{jk}, \quad j = 0, 1, \dots, n-1.$$

Hence  $\det(\mathbf{C}) = \prod_{j=0}^{n-1} \lambda_j$ .

**Formula 3.56. Discrete Fourier Matrix**

Let  $\omega_n = e^{2\pi i/n}$ . The normalized discrete Fourier matrix is

$$\mathbf{F}_n = \frac{1}{\sqrt{n}} \left( \omega_n^{-jk} \right)_{0 \leq j, k \leq n-1}.$$

It is unitary:

$$\mathbf{F}_n^H \mathbf{F}_n = \mathbf{I}_n.$$

Every circulant matrix is diagonalized by  $\mathbf{F}_n$ .

**Formula 3.57. Symmetric Tridiagonal Toeplitz Matrix**

For

$$\mathbf{T}_n = \begin{pmatrix} a & b & 0 & \cdots & 0 \\ b & a & b & \ddots & \vdots \\ 0 & b & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b \\ 0 & \cdots & 0 & b & a \end{pmatrix},$$

the eigenvalues are

$$a + 2b \cos \frac{k\pi}{n+1}, \quad k = 1, 2, \dots, n.$$

**Formula 3.58. Continuant Recurrence for Tridiagonal Determinants**

Let

$$\mathbf{T}_n = \begin{pmatrix} a_1 & b_1 & 0 & \cdots & 0 \\ c_1 & a_2 & b_2 & \ddots & \vdots \\ 0 & c_2 & a_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b_{n-1} \\ 0 & \cdots & 0 & c_{n-1} & a_n \end{pmatrix}.$$

If  $D_n = \det(\mathbf{T}_n)$ , then

$$D_0 = 1, \quad D_1 = a_1, \quad D_n = a_n D_{n-1} - b_{n-1} c_{n-1} D_{n-2}.$$

**Formula 3.59. Householder Reflection**

For a nonzero vector  $\mathbf{v}$ ,

$$\mathbf{H} = \mathbf{I}_n - 2 \frac{\mathbf{v} \mathbf{v}^T}{\mathbf{v}^T \mathbf{v}}$$

is symmetric, orthogonal, and involutory:

$$\mathbf{H}^T = \mathbf{H}, \quad \mathbf{H}^T \mathbf{H} = \mathbf{I}_n, \quad \mathbf{H}^2 = \mathbf{I}_n.$$

It reflects vectors across the hyperplane perpendicular to  $\mathbf{v}$ .

**Formula 3.60. Rotation and Reflection Matrices**

Rotation by angle  $\theta$  in the plane is represented by

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Reflection across the line making angle  $\theta$  with the positive  $x$ -axis is represented by

$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

**Formula 3.61. Eigenvalues of a Three-Dimensional Rotation Matrix**

A rotation of  $\mathbb{R}^3$  by angle  $\theta$  around an axis through the origin has eigenvalues

$$1, \quad e^{i\theta}, \quad e^{-i\theta}.$$

The eigenvalue 1 corresponds to the rotation axis. Therefore, if  $\theta \not\equiv 0, \pi \pmod{2\pi}$ , the rotation matrix has exactly one real eigenvalue.

**Proposition 3.62. Trace, Determinant, and Eigenvalues**

Let  $\mathbf{A} \in \mathbb{C}^{n \times n}$  have eigenvalues  $\lambda_1, \dots, \lambda_n$ , counted with algebraic multiplicity. Then

$$\operatorname{tr}(\mathbf{A}) = \lambda_1 + \dots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

**Proposition 3.63. Cyclic Property of the Trace**

Whenever the products are defined and square,

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA}).$$

More generally,

$$\operatorname{tr}(\mathbf{ABC}) = \operatorname{tr}(\mathbf{BCA}) = \operatorname{tr}(\mathbf{CAB}).$$

The factors may be cyclically permuted, but they may not in general be rearranged arbitrarily.

**Proposition 3.64. Eigenvalues and the Characteristic Polynomial**

A scalar  $\lambda$  is an eigenvalue of  $\mathbf{A}$  if and only if

$$\det(\lambda \mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of  $\mathbf{A}$  are the roots of the characteristic polynomial  $p_{\mathbf{A}}(\lambda)$ .

**Formula 3.65. All-Ones Matrix**

Let  $\mathbf{J}_n$  be the  $n \times n$  matrix whose entries are all 1. Then

$$\mathbf{J}_n \mathbf{1} = n\mathbf{1}, \quad \mathbf{J}_n \mathbf{v} = \mathbf{0} \quad \text{if} \quad v_1 + \dots + v_n = 0.$$

Hence the eigenvalues of  $\mathbf{J}_n$  are  $n$  with multiplicity 1 and 0 with multiplicity  $n - 1$ . The matrix  $\mathbf{J}_n - \mathbf{I}_n$  has eigenvalues  $n - 1$  with multiplicity 1 and  $-1$  with multiplicity  $n - 1$ .

**Formula 3.66. Constant Diagonal and Off-Diagonal Matrix**

Let  $\mathbf{A} \in F^{n \times n}$  have diagonal entries  $a$  and off-diagonal entries  $b$ . Then

$$\mathbf{A} = (a - b)\mathbf{I}_n + b\mathbf{J}_n.$$

Counting algebraic multiplicity, its eigenvalues consist of

$$a + (n - 1)b$$

once and

$$a - b$$

$n - 1$  times. If these two values coincide, they together give the single eigenvalue of multiplicity  $n$ .

**Formula 3.67. Rank-One Outer Product**

Let  $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$  be nonzero and set

$$\mathbf{A} = \mathbf{v}\mathbf{w}^\top.$$

Then  $\text{rk}(\mathbf{A}) = 1$  and

$$\mathbf{A}\mathbf{v} = (\mathbf{w}^\top \mathbf{v})\mathbf{v}.$$

The characteristic polynomial is

$$\lambda^{n-1}(\lambda - \mathbf{w}^\top \mathbf{v}).$$

Thus the eigenvalues, counted with algebraic multiplicity, are  $\mathbf{w}^\top \mathbf{v}$  and 0 repeated  $n - 1$  times. If  $\mathbf{w}^\top \mathbf{v} = 0$ , every eigenvalue is 0 and  $\mathbf{A}^2 = \mathbf{0}$ .

**Theorem 3.68. Cayley–Hamilton Theorem**

*Every square matrix satisfies its own characteristic polynomial. That is, if*

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}),$$

*then*

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

*Remark.* Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

**Definition 3.69. Minimal Polynomial**

The minimal polynomial  $m_{\mathbf{A}}(t)$  of a square matrix  $\mathbf{A}$  is the unique monic polynomial of smallest positive degree such that

$$m_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

By the Cayley–Hamilton Theorem, it divides the characteristic polynomial  $p_{\mathbf{A}}(t)$ . A short minimal polynomial is useful because it reduces high powers of  $\mathbf{A}$  to lower powers.

**Formula 3.70. Companion Matrix and Linear Recurrences**

For

$$p(x) = x^m - c_1x^{m-1} - \cdots - c_m,$$

the companion matrix

$$\mathbf{C} = \begin{pmatrix} c_1 & c_2 & \cdots & c_{m-1} & c_m \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

has characteristic polynomial  $p$ . The recurrence

$$a_n = c_1a_{n-1} + \cdots + c_ma_{n-m}$$

is encoded by multiplication by  $\mathbf{C}$ , so fast matrix exponentiation computes distant terms efficiently.

**Formula 3.71. Fibonacci  $Q$ -Matrix**

For

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

we have

$$\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \quad (n \geq 1).$$

Taking determinants gives Cassini's identity

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n.$$

**Formula 3.72. Matrix Geometric Series**

Let  $\mathbf{A}$  be an  $n \times n$  matrix. If  $\mathbf{I}_n - \mathbf{A}$  is invertible, then

$$\sum_{k=0}^{N-1} \mathbf{A}^k = (\mathbf{I}_n - \mathbf{A}^N)(\mathbf{I}_n - \mathbf{A})^{-1}.$$

If the spectral radius of  $\mathbf{A}$  is less than 1, then

$$\sum_{k=0}^{\infty} \mathbf{A}^k = (\mathbf{I}_n - \mathbf{A})^{-1}.$$

*Strategy. Fast Matrix Powers.* To compute a large power  $\mathbf{A}^N$ , first look for one of the following: diagonalization, a short minimal polynomial, idempotence, involution, nilpotence, rank one, a circulant structure, or a decomposition into  $a\mathbf{I}_n + b\mathbf{J}_n$ . Cayley–Hamilton then reduces every sufficiently high power to a linear combination of lower powers.

**Theorem 3.73. Diagonalization Criteria**

Suppose the characteristic polynomial of  $\mathbf{A} \in F^{n \times n}$  splits over  $F$ . The following are equivalent:

$\mathbf{A}$  is diagonalizable over  $F$ ,

$$\sum_{\lambda} m_g(\lambda) = n,$$

and

$$m_g(\lambda) = m_a(\lambda) \quad \text{for every eigenvalue } \lambda.$$

Equivalently, the minimal polynomial of  $\mathbf{A}$  splits over  $F$  into distinct linear factors.

**Corollary 3.74. Distinct Eigenvalues Imply Diagonalizability**

If an  $n \times n$  matrix  $\mathbf{A}$  has  $n$  distinct eigenvalues in  $F$ , then  $\mathbf{A}$  is diagonalizable over  $F$ .

*Remark.* The field matters. A real matrix can fail to be diagonalizable over  $\mathbb{R}$  because its characteristic polynomial has nonreal roots, while the same matrix may be diagonalizable over  $\mathbb{C}$ . Over  $\mathbb{C}$ , every characteristic polynomial splits, so only the multiplicity conditions remain. Stronger diagonalization results for symmetric, Hermitian, and normal matrices appear below.

**Proposition 3.75. Polynomials of a Diagonalizable Matrix**

Let  $\mathbf{A} \in F^{n \times n}$  be diagonalizable over  $F$ , and suppose

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}, \quad \mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n).$$

For a polynomial  $p(x) \in F[x]$ , we have

$$p(\mathbf{A}) = \mathbf{S}p(\mathbf{D})\mathbf{S}^{-1},$$

where

$$p(\mathbf{D}) = \text{diag}(p(\lambda_1), \dots, p(\lambda_n)).$$

In particular,

$$\det(p(\mathbf{A})) = p(\lambda_1)p(\lambda_2) \cdots p(\lambda_n).$$

*Idea.* Similar matrices represent the same linear transformation in different bases. Therefore, applying a polynomial to a diagonalizable matrix is the same as applying that polynomial to each diagonal entry after diagonalization.

*Strategy.* Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

**Theorem 3.76. Gershgorin Circle Theorem**

Let  $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{n \times n}$ . Every eigenvalue of  $\mathbf{A}$  lies in at least one disk

$$\left\{ z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right\}.$$

**Theorem 3.77. Perron–Frobenius Theorem, Positive-Matrix Form**

If every entry of a real square matrix  $\mathbf{A}$  is positive, then  $\rho(\mathbf{A})$  is a positive simple eigenvalue with a positive eigenvector, and every other eigenvalue  $\lambda$  satisfies

$$|\lambda| < \rho(\mathbf{A}).$$

**Theorem 3.78. Spectral Theorem for Real Symmetric Matrices**

If  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is symmetric, then all eigenvalues of  $\mathbf{A}$  are real, and there exists an orthogonal matrix  $\mathbf{Q}$  and a diagonal matrix  $\mathbf{D}$  such that

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^T.$$

Equivalently,  $\mathbb{R}^n$  has an orthonormal basis consisting of eigenvectors of  $\mathbf{A}$ .

*Remark.* A real symmetric matrix has real eigenvalues and admits orthogonal diagonalization.

**Definition 3.79. Normal Matrix**

A complex square matrix  $\mathbf{A}$  is normal if

$$\mathbf{A}^H \mathbf{A} = \mathbf{A} \mathbf{A}^H.$$

Hermitian, skew-Hermitian, and unitary matrices are normal.

**Theorem 3.80. Spectral Theorem for Normal Matrices**

A complex square matrix is normal if and only if it is unitarily diagonalizable. Equivalently,  $\mathbf{A}$  is normal if and only if there exist a unitary matrix  $\mathbf{U}$  and a diagonal matrix  $\mathbf{D}$  such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^H.$$

**Proposition 3.81. Hermitian Matrix Quick Facts**

If  $\mathbf{A}$  is Hermitian, then

$$\mathbf{x}^H \mathbf{A} \mathbf{x} \in \mathbb{R}$$

for every  $\mathbf{x} \in \mathbb{C}^n$ . All eigenvalues of  $\mathbf{A}$  are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.

**Theorem 3.82. Spectral Theorem for Hermitian Matrices**

If  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is Hermitian, then there exists a unitary matrix  $\mathbf{U}$  and a real diagonal matrix  $\mathbf{D}$  such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^H.$$

Equivalently,  $\mathbb{C}^n$  has an orthonormal basis consisting of eigenvectors of  $\mathbf{A}$ .

*Remark.* The real analogue of a Hermitian matrix is a real symmetric matrix. The real analogue of a unitary matrix is an orthogonal matrix.

**Definition 3.83. Markov Matrix; Stochastic Matrix**

A column-stochastic matrix is a square matrix with nonnegative entries whose columns each sum to 1. Thus

$$\mathbf{1}^T \mathbf{A} = \mathbf{1}^T.$$

Consequently, 1 is an eigenvalue of  $\mathbf{A}^T$  and therefore also of  $\mathbf{A}$ . A probability column vector evolves by

$$\mathbf{p}_{k+1} = \mathbf{A}\mathbf{p}_k.$$

A stationary distribution is a probability vector satisfying  $\mathbf{A}\mathbf{p} = \mathbf{p}$ .

*Remark.* Markov matrices are linear algebra objects behind finite-state Markov chains. A stationary distribution is an eigenvector corresponding to the eigenvalue 1.

**Theorem 3.84. Cauchy–Schwarz Inequality**

For vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ , we have

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Equivalently, for real numbers  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$ ,

$$\left( \sum_{i=1}^n a_i b_i \right)^2 \leq \left( \sum_{i=1}^n a_i^2 \right) \left( \sum_{i=1}^n b_i^2 \right).$$

*Strategy.* The expression  $\sum a_i b_i$  is a strong signal for Cauchy–Schwarz. The vector form is often the most efficient method to recognize the inequality.

**Corollary 3.85. Triangle Inequality**

For vectors  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ , we have

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

**Proposition 3.86. Properties of Orthogonal Matrices**

If  $\mathbf{Q}$  is orthogonal, then

$$\|\mathbf{Q}\mathbf{v}\| = \|\mathbf{v}\|, \quad \langle \mathbf{Q}\mathbf{u}, \mathbf{Q}\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle, \quad \det(\mathbf{Q}) = \pm 1.$$

**Definition 3.87. Positive Definite Matrix**

A Hermitian matrix  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is positive definite if

$$\mathbf{x}^H \mathbf{A} \mathbf{x} > 0$$

for every nonzero  $\mathbf{x} \in \mathbb{C}^n$ . Over  $\mathbb{R}$ , this is the familiar condition that  $\mathbf{A}$  is symmetric and

$$\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$$

for every nonzero  $\mathbf{x} \in \mathbb{R}^n$ .

**Definition 3.88. Singular Values**

Let  $\mathbf{A} \in \mathbb{C}^{m \times n}$ . The singular values of  $\mathbf{A}$  are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^H \mathbf{A}.$$

For real matrices,  $\mathbf{A}^H = \mathbf{A}^\top$ .

**Formula 3.89. Matrix Decomposition Reference Table**

The most useful matrix decompositions are summarized by their hypotheses and the computation they simplify.

| Decomposition       | Form                                          | Typical condition                                                                                                | Main use                                                       |
|---------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Rank factorization  | $\mathbf{A} = \mathbf{UV}$                    | $\text{rk}(\mathbf{A}) = r$ , with $\mathbf{U} \in F^{m \times r}$ , $\mathbf{V} \in F^{r \times n}$ of rank $r$ | expose rank and column/row spaces                              |
| LU                  | $\mathbf{A} = \mathbf{LU}$                    | elimination without row exchanges; with pivoting, $\mathbf{PA} = \mathbf{LU}$                                    | repeated linear solves, determinants                           |
| QR                  | $\mathbf{A} = \mathbf{QR}$                    | full column rank, with orthonormal columns of $\mathbf{Q}$                                                       | least squares, orthogonalization                               |
| Cholesky            | $\mathbf{A} = \mathbf{LL}^H$                  | Hermitian positive definite; over $\mathbb{R}$ , this is $\mathbf{LL}^\top$                                      | fast stable linear solves                                      |
| Diagonalization     | $\mathbf{A} = \mathbf{PDP}^{-1}$              | a basis of eigenvectors                                                                                          | powers and functions of $\mathbf{A}$                           |
| Orthogonal spectral | $\mathbf{A} = \mathbf{QDQ}^\top$              | real symmetric                                                                                                   | real orthonormal eigenbasis                                    |
| Unitary spectral    | $\mathbf{A} = \mathbf{\Omega\Lambda\Omega}^H$ | complex normal                                                                                                   | orthonormal eigenbasis over $\mathbb{C}$                       |
| Schur               | $\mathbf{A} = \mathbf{QTQ}^H$                 | every complex square matrix                                                                                      | triangularize without assuming diagonalizability               |
| SVD                 | $\mathbf{A} = \mathbf{U\Sigma V}^H$           | every real or complex matrix                                                                                     | rank, least squares, conditioning, best low-rank approximation |

For real matrices,  $\mathbf{V}^H = \mathbf{V}^\top$ . The displayed letters are matrices and therefore follow the book's bold-uppercase convention.

**Theorem 3.90. LU Decomposition**

When Gaussian elimination can be performed without row exchanges, a square matrix  $\mathbf{A}$  can be written as

$$\mathbf{A} = \mathbf{LU},$$

where  $\mathbf{L}$  is lower triangular and  $\mathbf{U}$  is upper triangular.

**Theorem 3.91. Gram–Schmidt Process; QR Decomposition**

From any linearly independent list

$$\mathbf{v}_1, \dots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \dots, \mathbf{q}_k$$

with the same span. In matrix form, this leads to the QR decomposition

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where the columns of  $\mathbf{Q}$  are orthonormal and  $\mathbf{R}$  is upper triangular.

*Remark.* Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

### Formula 3.92. Orthogonal Projection Matrices

For a nonzero vector  $\mathbf{v} \in \mathbb{R}^n$ ,

$$\text{proj}_{\mathbf{v}} \mathbf{x} = \frac{\mathbf{v}^T \mathbf{x}}{\mathbf{v}^T \mathbf{v}} \mathbf{v},$$

and the projection matrix onto  $\text{span}(\mathbf{v})$  is

$$\mathbf{P}_{\mathbf{v}} = \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T \mathbf{v}}.$$

If  $\|\mathbf{v}\| = 1$ , then

$$\mathbf{P}_{\mathbf{v}} = \mathbf{v}\mathbf{v}^T, \quad \mathbf{I}_n - \mathbf{v}\mathbf{v}^T$$

projects onto the hyperplane  $\mathbf{v}^\perp$ .

If the columns of  $\mathbf{Q} \in \mathbb{R}^{n \times k}$  are orthonormal, then the orthogonal projection matrix onto  $\text{Col}(\mathbf{Q})$  is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^T.$$

More generally, if the columns of  $\mathbf{A} \in \mathbb{R}^{n \times k}$  are linearly independent, then

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T.$$

Over  $\mathbb{C}$ , replace every transpose by the Hermitian transpose.

### Theorem 3.93. Least-Squares Normal Equations

Let  $\mathbf{A} \in \mathbb{R}^{m \times n}$  and  $\mathbf{b} \in \mathbb{R}^m$ . A vector  $\hat{\mathbf{x}}$  minimizes

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

if and only if the residual is orthogonal to  $\text{Col}(\mathbf{A})$ , equivalently,

$$\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}.$$

If the columns of  $\mathbf{A}$  are linearly independent, then the minimizer is unique and

$$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}.$$

The fitted vector is the orthogonal projection

$$\hat{\mathbf{b}} = \mathbf{A}\hat{\mathbf{x}} = \mathbf{P}\mathbf{b}, \quad \mathbf{P} = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T.$$

### Proposition 3.94. Projection Matrix Criterion

A matrix  $\mathbf{P}$  is a projection matrix if

$$\mathbf{P}^2 = \mathbf{P}.$$

Over  $\mathbb{R}$ , it is an orthogonal projection matrix if, in addition,

$$\mathbf{P}^T = \mathbf{P}.$$

Over  $\mathbb{C}$ , the corresponding condition is

$$\mathbf{P}^H = \mathbf{P}.$$

**Theorem 3.95. Sylvester's Criterion**

A real symmetric matrix is positive definite if and only if all of its leading principal minors are positive.

**Theorem 3.96. Singular Value Decomposition; SVD**

Let  $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ . Every matrix  $\mathbf{A} \in \mathbb{F}^{m \times n}$  can be written as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H,$$

where  $\mathbf{U}$  and  $\mathbf{V}$  are unitary matrices and  $\mathbf{\Sigma}$  is a rectangular diagonal matrix whose diagonal entries are the singular values of  $\mathbf{A}$ . For real  $\mathbf{A}$ , the factors may be chosen real and orthogonal, so  $\mathbf{V}^H = \mathbf{V}^T$ .

*Remark.* Singular Value Decomposition applies to every real or complex matrix, not only to square or diagonalizable matrices. This is why SVD is one of the most useful decompositions in applied linear algebra.

**Definition 3.97. Generalized Eigenvectors and Jordan Chains**

Let  $\lambda$  be an eigenvalue of a square matrix  $\mathbf{A}$ . A nonzero vector  $\mathbf{v}$  is a generalized eigenvector associated with  $\lambda$  if

$$(\mathbf{A} - \lambda\mathbf{I}_n)^k \mathbf{v} = \mathbf{0}$$

for some positive integer  $k$ . A Jordan chain  $\mathbf{v}_1, \dots, \mathbf{v}_r$  satisfies

$$(\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{v}_1 = \mathbf{0}, \quad (\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{v}_{j+1} = \mathbf{v}_j.$$

Ordinary eigenvectors are precisely the nonzero vectors satisfying the condition with  $k = 1$ .

**Theorem 3.98. Jordan Canonical Form**

Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every  $\mathbf{A} \in \mathbb{C}^{n \times n}$ , there exists an invertible matrix  $\mathbf{P}$  such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

is in Jordan canonical form.

*Remark.* Jordan canonical form describes the structure that remains when a matrix cannot be diagonalized.

# Chapter 4

## Differential Equations

Related **POSTECH** Courses: (MATH200) Differential Equations

Related Books: Elementary Differential Equations and Boundary Value Problems (William E. Boyce, Richard C. DiPrima, Douglas B. Meade)

### Definition 4.1. Differential Equation, Order, and Initial Value Problem

An ordinary differential equation (ODE) relates an unknown function of one independent variable to its derivatives. Its order is the highest derivative that occurs. An  $n$ th-order linear ODE has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

with  $a_n$  not identically zero. It is homogeneous when  $g = 0$ . An initial value problem (IVP) supplements the equation with enough values such as  $y(t_0), y'(t_0), \dots$  to select a particular solution.

### Definition 4.2. Separable, Linear, Exact, Bernoulli, and Autonomous First-Order Equations

A first-order ODE is called *separable* if it can be written as  $y' = g(t)h(y)$ , and *linear* if it has the form  $y' + p(t)y = q(t)$ . An equation

$$M(t, y) dt + N(t, y) dy = 0$$

is *exact* on a region if there exists a potential  $F$  with  $F_t = M$  and  $F_y = N$ . A *Bernoulli equation* has the form  $y' + p(t)y = q(t)y^n$  with  $n \neq 0, 1$ . An *autonomous equation* has the form  $y' = f(y)$ , with no explicit dependence on the independent variable. Recognizing the type determines the standard reduction to try.

### Formula 4.3. First-Order ODE Recognition Table

Before calculating, identify the structural form.

| Type      | Equation                               | Reduction or solution                                                                                                    |
|-----------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Separable | $y' = g(t)h(y)$                        | $\int \frac{dy}{h(y)} = \int g(t) dt + C$ on intervals where $h(y) \neq 0$ ; also check equilibrium solutions $h(y) = 0$ |
| Linear    | $y' + p(t)y = q(t)$                    | integrating factor $\mu(t) = e^{\int p(t) dt}$                                                                           |
| Exact     | $M(t, y) dt + N(t, y) dy = 0$          | find $F$ with $F_t = M$ , $F_y = N$ , then $F(t, y) = C$                                                                 |
| Bernoulli | $y' + p(t)y = q(t)y^n$ , $n \neq 0, 1$ | substitute $u = y^{1-n}$ to obtain a linear equation                                                                     |

| Type       | Equation    | Reduction or solution                                                          |
|------------|-------------|--------------------------------------------------------------------------------|
| Autonomous | $y' = f(y)$ | equilibria satisfy $f(y) = 0$ ; away from equilibria the equation is separable |

**Theorem 4.4. Picard–Lindelöf Existence and Uniqueness Theorem**

Consider the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0.$$

If  $f$  and  $\partial f / \partial y$  are continuous near  $(t_0, y_0)$ , then there exists a unique solution near  $t = t_0$ .

*Remark.* The decisive conclusion is existence and uniqueness. It tells us when an initial condition determines exactly one solution.

**Theorem 4.5. Superposition Principle**

If  $y_1, \dots, y_k$  are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1 y_1 + \dots + c_k y_k$$

is also a solution.

*Remark.* The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

**Proposition 4.6. General Solution of a Nonhomogeneous Linear Equation**

For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where  $y_h$  is the general solution of the associated homogeneous equation and  $y_p$  is one particular solution of the nonhomogeneous equation.

**Formula 4.7. Separable First-Order Equation**

A first-order equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is separable. When  $h(y) \neq 0$ , rewrite it as

$$\frac{1}{h(y)} dy = g(x) dx$$

and integrate both sides.

**Formula 4.8. First-Order Linear Equation; Integrating Factor**

The equation

$$y' + p(t)y = q(t)$$

has integrating factor

$$\mu(t) = e^{\int p(t) dt}.$$

Then

$$(\mu y)' = \mu q,$$

so

$$y(t) = \frac{1}{\mu(t)} \left( \int \mu(t)q(t) dt + C \right).$$

**Formula 4.9. Exact Differential Equation**

On a domain  $D \subseteq \mathbb{R}^2$ , an equation

$$M(x, y) dx + N(x, y) dy = 0$$

is exact if there is a potential function  $F \in C^1(D)$  such that

$$F_x = M, \quad F_y = N.$$

If  $M, N \in C^1(D)$  and  $D$  is open and simply connected, then the equation is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

In that case, the solution is given implicitly by

$$F(x, y) = C.$$

**Formula 4.10. Bernoulli Equation**

The Bernoulli equation

$$y' + p(t)y = q(t)y^n \quad (n \neq 0, 1)$$

becomes a first-order linear equation after the substitution

$$u = y^{1-n}.$$

*Tactic.* For a first-order ODE, first check whether it is separable, linear, exact, or Bernoulli. These four patterns cover many quick undergraduate differential-equation questions.

**Formula 4.11. Second-Order Constant-Coefficient Equation**

Consider the homogeneous equation

$$ay'' + by' + cy = 0, \quad a \neq 0.$$

Its characteristic equation is

$$ar^2 + br + c = 0.$$

If the roots are distinct real numbers  $r_1, r_2$ , then

$$y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

If the equation has a repeated real root  $r$ , then

$$y_h = (c_1 + c_2 t) e^{rt}.$$

If the roots are  $\alpha \pm i\beta$  with  $\beta \neq 0$ , then

$$y_h = e^{\alpha t} (c_1 \cos \beta t + c_2 \sin \beta t).$$

*Remark.* The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

**Definition 4.12. Wronskian**

For differentiable functions  $y_1, \dots, y_n$ , their Wronskian is

$$W(y_1, \dots, y_n)(t) = \det \left( y_j^{(i-1)}(t) \right)_{1 \leq i, j \leq n}.$$

For  $n = 2$ , this is  $W(y_1, y_2) = y_1 y_2' - y_1' y_2$ .

**Proposition 4.13. Wronskian and Linear Independence**

If

$$W(y_1, y_2)(t_0) \neq 0$$

for some  $t_0$ , then  $y_1$  and  $y_2$  are linearly independent.

**Formula 4.14. Abel's Identity**

For

$$y'' + p(t)y' + q(t)y = 0,$$

the Wronskian of two solutions satisfies

$$W(t) = Ce^{-\int p(t) dt}.$$

In particular, the Wronskian is either always zero or never zero on an interval where  $p$  is continuous.

**Formula 4.15. Euler–Cauchy Equation**

On an interval with  $x > 0$ , the equation

$$ax^2y'' + bxy' + cy = 0, \quad a \neq 0,$$

suggests the trial solution  $y = x^r$ , giving the indicial equation

$$ar(r-1) + br + c = 0.$$

If its roots are distinct real numbers  $r_1, r_2$ , then

$$y = c_1 x^{r_1} + c_2 x^{r_2}.$$

For a repeated root  $r$ ,

$$y = x^r(c_1 + c_2 \ln x).$$

For roots  $\alpha \pm i\beta$  with  $\beta \neq 0$ ,

$$y = x^\alpha (c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)).$$

On intervals with  $x < 0$ , the analogous real formulas use  $|x|$ .

**Definition 4.16. Bessel Equation and Bessel Functions**

Bessel's equation of order  $\nu$  is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

Its standard solutions are the Bessel functions of the first and second kinds, denoted  $J_\nu(x)$  and  $Y_\nu(x)$ . For noninteger  $\nu$ ,  $J_\nu$  and  $J_{-\nu}$  also form a fundamental pair. Bessel functions arise naturally after separation of variables in problems with cylindrical or circular symmetry.

**Algorithm 4.17. Undetermined Coefficients**

For a constant-coefficient equation with forcing term made from polynomials, exponentials, sines, or cosines, guess a particular solution of the same type. If the guess overlaps with  $y_h$ , multiply the guess by the smallest power of  $t$  that removes the overlap.

**Formula 4.18. Variation of Parameters**

For

$$y'' + p(t)y' + q(t)y = g(t)$$

with fundamental solutions  $y_1, y_2$  of the homogeneous equation, a particular solution is

$$y_p = -y_1 \int \frac{y_2 g}{W} dt + y_2 \int \frac{y_1 g}{W} dt,$$

where

$$W = y_1 y_2' - y_1' y_2.$$

**Formula 4.19. Exponential Growth and Decay**

The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If  $k > 0$ , this models exponential growth. If  $k < 0$ , this models exponential decay.

**Formula 4.20. Logistic Equation**

The logistic equation is

$$y' = ky \left(1 - \frac{y}{K}\right), \quad k > 0, \quad K > 0.$$

Its equilibrium solutions are  $y = 0$  and  $y = K$ . Every non-equilibrium solution on an interval on which it is defined has the form

$$y(t) = \frac{K}{1 + Ce^{-kt}},$$

where  $C$  is determined by the initial condition. For  $0 < y(0) < K$ , one has  $C > 0$  and  $y(t) \rightarrow K$  as  $t \rightarrow \infty$ .

*Remark.* The logistic equation is the standard model for population growth with limited resources. The parameter  $K$  represents the limiting population.

**Formula 4.21. Simple Harmonic Oscillator**

The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A \cos(\omega t - \delta)$$

for suitable constants  $A$  and  $\delta$ .

*Remark.* This is the equation behind ideal mass–spring motion and many small oscillation models in physics.

**Formula 4.22. Damped Harmonic Oscillator**

For  $m > 0$ ,  $c \geq 0$ , and  $k > 0$ , the equation

$$mx'' + cx' + kx = 0$$

has characteristic equation  $mr^2 + cr + k = 0$ . Put

$$\alpha = \frac{c}{2m}.$$

The three regimes and solution forms are

$$c^2 < 4mk : \quad x(t) = e^{-\alpha t} (c_1 \cos(\omega_d t) + c_2 \sin(\omega_d t)), \quad \omega_d = \sqrt{\frac{k}{m} - \alpha^2},$$

$$c^2 = 4mk : \quad x(t) = (c_1 + c_2 t) e^{-\alpha t},$$

and, if  $c^2 > 4mk$ ,

$$x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}, \quad r_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

These are called underdamped, critically damped, and overdamped, respectively.

**Definition 4.23. Laplace Transform**

The Laplace transform of a function  $f(t)$  is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t) dt.$$

**Formula 4.24. Laplace Transform Table**

Let  $F(s) = \mathcal{L}\{f(t)\}(s)$ . The basic transforms are

| $f(t)$           | $\mathcal{L}\{f(t)\}(s)$    | Conditions                                     |
|------------------|-----------------------------|------------------------------------------------|
| 1                | $\frac{1}{s}$               | $\operatorname{Re}(s) > 0$                     |
| $t^n$            | $\frac{n!}{s^{n+1}}$        | $n \in \mathbb{N}_0$                           |
| $e^{at}$         | $\frac{1}{s-a}$             | $\operatorname{Re}(s) > a$                     |
| $t^n e^{at}$     | $\frac{n!}{(s-a)^{n+1}}$    | $n \in \mathbb{N}_0, \operatorname{Re}(s) > a$ |
| $\sin at$        | $\frac{a}{s^2 + a^2}$       | $\operatorname{Re}(s) > 0$                     |
| $\cos at$        | $\frac{s}{s^2 + a^2}$       | $\operatorname{Re}(s) > 0$                     |
| $e^{at} \sin bt$ | $\frac{b}{(s-a)^2 + b^2}$   | $\operatorname{Re}(s) > a$                     |
| $e^{at} \cos bt$ | $\frac{s-a}{(s-a)^2 + b^2}$ | $\operatorname{Re}(s) > a$                     |
| $\sinh at$       | $\frac{a}{s^2 - a^2}$       | $\operatorname{Re}(s) >  a $                   |
| $\cosh at$       | $\frac{s}{s^2 - a^2}$       | $\operatorname{Re}(s) >  a $                   |
| $f'(t)$          | $sF(s) - f(0)$              |                                                |
| $f''(t)$         | $s^2 F(s) - sf(0) - f'(0)$  |                                                |

| $f(t)$                   | $\mathcal{L}\{f(t)\}(s)$                      | Conditions                        |
|--------------------------|-----------------------------------------------|-----------------------------------|
| $tf(t)$                  | $-F'(s)$                                      |                                   |
| $\int_0^t f(\tau) d\tau$ | $\frac{F(s)}{s}$                              |                                   |
| $u(t-a)f(t-a)$           | $e^{-as}F(s)$                                 | $a > 0$                           |
| $f(t+T) = f(t)$          | $\frac{\int_0^T e^{-st}f(t) dt}{1 - e^{-sT}}$ | $T > 0, \operatorname{Re}(s) > 0$ |

**Proposition 4.25. Laplace Shift and Convolution**

If  $F(s) = \mathcal{L}\{f(t)\}(s)$ , then

$$\mathcal{L}\{e^{at}f(t)\}(s) = F(s-a).$$

If

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau,$$

then

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f\}\mathcal{L}\{g\}.$$

**Formula 4.26. Fourier Series**

For a piecewise smooth function  $f$  on  $[-L, L]$ ,

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right),$$

where

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx.$$

If  $f$  is even, then  $b_n = 0$ ; if  $f$  is odd, then  $a_n = 0$ . At a jump discontinuity, the series converges to the midpoint of the left and right limits.

**Formula 4.27. Classical Fourier Series**

For  $-\pi < x < \pi$ ,

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx),$$

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos(nx),$$

and, for  $0 < x < \pi$ ,

$$1 = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin((2k+1)x)}{2k+1}.$$

**Definition 4.28. Fourier Transform**

For an integrable function  $f: \mathbb{R} \rightarrow \mathbb{C}$ , define

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i t \xi} dt.$$

Under suitable hypotheses, the inversion formula is

$$f(t) = \int_{-\infty}^{\infty} \widehat{f}(\xi) e^{2\pi i t \xi} d\xi.$$

**Formula 4.29. Fourier Transform Table**

With the convention above, the standard identities are

| Function or operation | Fourier transform                                     |
|-----------------------|-------------------------------------------------------|
| $f'(t)$               | $2\pi i\xi \widehat{f}(\xi)$                          |
| $f(t - a)$            | $e^{-2\pi i a \xi} \widehat{f}(\xi)$                  |
| $e^{2\pi i a t} f(t)$ | $\widehat{f}(\xi - a)$                                |
| $f(at), a \neq 0$     | $\frac{1}{ a } \widehat{f}\left(\frac{\xi}{a}\right)$ |
| $tf(t)$               | $\frac{i}{2\pi} \frac{d}{d\xi} \widehat{f}(\xi)$      |
| $(f * g)(t)$          | $\widehat{f}(\xi) \widehat{g}(\xi)$                   |
| $e^{-at^2}, a > 0$    | $\sqrt{\frac{\pi}{a}} e^{-\pi^2 \xi^2 / a}$           |

**Formula 4.30. Linear Systems and Matrix Exponential**

The homogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

has solution

$$\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}(0).$$

If  $\mathbf{A}$  is diagonalizable as  $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$ , then

$$e^{t\mathbf{A}} = \mathbf{S}e^{t\mathbf{D}}\mathbf{S}^{-1}, \quad e^{t\mathbf{D}} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

*Strategy.* For a constant-coefficient linear system, the eigenvalues of  $\mathbf{A}$  control the qualitative behavior. Negative real parts indicate decay, positive real parts indicate growth, and nonzero imaginary parts indicate rotation or oscillation.

*Remark.* The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

**Definition 4.31. Heat Equation**

The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \quad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

**Definition 4.32. Wave Equation**

The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed  $c$ .

**Theorem 4.33. d'Alembert's Formula**

Let  $c > 0$ , let  $f \in C^2(\mathbb{R})$ , and let  $g \in C^1(\mathbb{R})$ . The classical solution of

$$u_{tt} = c^2 u_{xx}, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x),$$

is

$$u(x, t) = \frac{f(x - ct) + f(x + ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) \, ds.$$

*Remark.* d'Alembert's formula shows that waves move along the characteristic lines  $x - ct = \text{constant}$  and  $x + ct = \text{constant}$ .

**Definition 4.34. Laplace Equation**

Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

**Definition 4.35. Poisson Equation**

Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

*Remark.* The heat equation, wave equation, and Laplace equation are three fundamental partial differential equations with distinct physical interpretations.

## Chapter 5

# Combinatorics and Discrete Mathematics

Related **POSTECH** Courses: (MATH261) Discrete Mathematics

Related Books: Introductory Combinatorics (Richard A. Brualdi), Invitation to Discrete Mathematics (Jiří Matoušek, Jaroslav Nešetřil)

### Definition 5.1. Permutation, Combination, and Multiset Selection

An ordered selection of  $k$  distinct objects from  $n$  distinct objects is a  $k$ -permutation, counted by

$$P(n, k) = \frac{n!}{(n - k)!}.$$

An unordered  $k$ -element selection is a  $k$ -combination, counted by

$$\binom{n}{k}.$$

If repetition is allowed and order is ignored, the number of size- $k$  multisets from  $n$  types is

$$\left(\binom{n}{k}\right) = \binom{n + k - 1}{k}.$$

These three models should be distinguished before applying a formula.

### Formula 5.2. Binomial Theorem; Newton's Binomial Formula

For every nonnegative integer  $n$ ,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

In particular,

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k.$$

*Remark.* The coefficients  $\binom{n}{k}$  are the entries of Pascal's triangle. This is one of the most recognizable bridges between algebra and counting.

*Strategy. Bijection, Double Counting, and Complementary Counting.* Before expanding a complicated sum, try to count the same objects in a second way. Bijections, double counting, and counting the complement are among the fastest and most reusable contest methods.

**Formula 5.3. Binomial Inversion**

The relations

$$b_n = \sum_{k=0}^n \binom{n}{k} a_k$$

and

$$a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k$$

are equivalent.

**Formula 5.4. Pascal's Identity and Vandermonde's Identity**

Pascal's identity is

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Vandermonde's identity is

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}.$$

**Formula 5.5. Multinomial Theorem**

For nonnegative integers  $k_1, \dots, k_m$  with  $k_1 + \dots + k_m = n$ ,

$$(x_1 + \dots + x_m)^n = \sum_{k_1 + \dots + k_m = n} \frac{n!}{k_1! \dots k_m!} x_1^{k_1} \dots x_m^{k_m}.$$

The coefficient

$$\frac{n!}{k_1! \dots k_m!}$$

is called a multinomial coefficient.

**Formula 5.6. Stars and Bars**

The number of nonnegative integer solutions of

$$x_1 + x_2 + \dots + x_n = k$$

is

$$\binom{n+k-1}{k} = \binom{\binom{n}{k}}{k}.$$

Equivalently, the expansion of  $(x_1 + x_2 + \dots + x_n)^k$  has

$$\binom{n+k-1}{k}$$

distinct monomial terms.

**Formula 5.7. Lattice Path Counting**

The number of shortest lattice paths from  $(0, 0)$  to  $(m, n)$  using only unit steps to the right and upward is

$$\binom{m+n}{m} = \binom{m+n}{n}.$$

**Theorem 5.8. Reflection Principle and Ballot-Path Formula**

The reflection principle converts a path that first crosses a boundary into an unrestricted reflected path. In particular, for integers  $a \geq b \geq 0$ , the number of paths from  $(0, 0)$  to  $(a, b)$  using right and up steps and never entering the region  $y > x$  is

$$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1} \binom{a+b}{b}.$$

When  $a = b = n$ , this becomes the Catalan number

$$\frac{1}{n+1} \binom{2n}{n}.$$

*Strategy. Tail Switching for Intersecting Paths.* When two directed lattice paths meet, exchange their tails after the first intersection. This produces a bijection with a pair of paths whose endpoints have been crossed. The method is a standard way to count or cancel intersecting path pairs.

**Formula 5.9. Ordinary Generating Function Table**

For a sequence  $(a_n)_{n \geq 0}$ , its ordinary generating function is

$$A(x) = \sum_{n=0}^{\infty} a_n x^n.$$

The basic examples are

| Sequence $a_n$                        | Ordinary generating function $A(x)$ | Conditions                    |
|---------------------------------------|-------------------------------------|-------------------------------|
| 1                                     | $\frac{1}{1-x}$                     | $ x  < 1$                     |
| $r^n$                                 | $\frac{1}{1-rx}$                    | $ rx  < 1$                    |
| $n$                                   | $\frac{x}{(1-x)^2}$                 | $ x  < 1$                     |
| $n^2$                                 | $\frac{x(1+x)}{(1-x)^3}$            | $ x  < 1$                     |
| $\binom{n}{k}$                        | $\frac{x^k}{(1-x)^{k+1}}$           | $k \in \mathbb{N}_0,  x  < 1$ |
| $\binom{n+r}{r}$                      | $\frac{1}{(1-x)^{r+1}}$             | $r \in \mathbb{N}_0,  x  < 1$ |
| $F_n$ , where $F_0 = 0$ and $F_1 = 1$ | $\frac{x}{1-x-x^2}$                 | $ x  < \frac{\sqrt{5}-1}{2}$  |
| $\frac{1}{n+1} \binom{2n}{n}$         | $\frac{1 - \sqrt{1-4x}}{2x}$        | $ x  < \frac{1}{4}$           |

*Tactic.* Generating functions are useful when a recurrence or counting problem has a repeated structure; a known generating function can replace a long derivation.

**Formula 5.10. Stirling Numbers and Bell Numbers**

The Stirling number of the second kind

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

counts partitions of an  $n$ -element set into  $k$  nonempty blocks and satisfies

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}.$$

The Bell number

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

counts all set partitions of an  $n$ -element set.

**Formula 5.11. Counting Surjections**

For finite sets with  $n$  labeled elements in the domain and  $k$  labeled elements in the codomain, the number of surjections is

$$k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n.$$

The first expression orders the  $k$  nonempty fibers. The second follows from inclusion–exclusion. The value is 0 when  $k > n$ .

**Formula 5.12. Fibonacci Numbers; Binet’s Formula**

If  $F_0 = 0$ ,  $F_1 = 1$ , and  $F_{n+1} = F_n + F_{n-1}$ , then

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

In particular,

$$\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n} = \varphi.$$

**Formula 5.13. Roots-of-Unity Filter**

Let  $\omega_m = e^{2\pi i/m}$ . Then

$$\frac{1}{m} \sum_{j=0}^{m-1} \omega_m^{j(n-r)} = \begin{cases} 1, & n \equiv r \pmod{m}, \\ 0, & n \not\equiv r \pmod{m}. \end{cases}$$

Consequently, if  $A(x) = \sum_{n \geq 0} a_n x^n$ , then the terms with indices congruent to  $r$  modulo  $m$  are extracted by

$$\sum_{n \equiv r \pmod{m}} a_n x^n = \frac{1}{m} \sum_{j=0}^{m-1} \omega_m^{-rj} A(\omega_m^j x).$$

**Formula 5.14. Tower of Hanoi**

Let  $T_n$  be the minimum number of moves needed to move  $n$  disks in the Tower of Hanoi puzzle. Then

$$T_n = 2T_{n-1} + 1, \quad T_1 = 1,$$

and hence

$$T_n = 2^n - 1.$$

*Remark.* The recurrence comes from moving the top  $n - 1$  disks away, moving the largest disk once, and then moving the  $n - 1$  disks back on top of it.

*Strategy. State Compression and Transfer Matrices.* When a counting process depends only on finitely many recent states, place the transition counts in a matrix  $\mathbf{T}$ . Counts of length  $n$  then have the form

$$\mathbf{u}^T \mathbf{T}^n \mathbf{v}.$$

This is effective for forbidden substrings, tilings, walks, and finite automata.

**Theorem 5.15. Pigeonhole Principle; Dirichlet's Box Principle**

If  $n + 1$  objects are placed into  $n$  boxes, then at least one box contains at least two objects. More generally, if  $N$  objects are placed into  $k$  boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

*Strategy.* The Pigeonhole Principle is often hidden in problems asking us to prove that two objects must share a property. The hard part is usually choosing the right “boxes.”

**Theorem 5.16. Consecutive-Sum Divisibility Lemma**

For any integers  $a_1, \dots, a_n$ , there exists a nonempty consecutive block whose sum is divisible by  $n$ . Define prefix sums

$$S_0 = 0, \quad S_k = \sum_{i=1}^k a_i.$$

If some  $S_k \equiv 0 \pmod{n}$ , use the first  $k$  terms. Otherwise, two of  $S_1, \dots, S_n$  have the same residue modulo  $n$ , and their difference is the required block sum.

**Definition 5.17. Graphs, Walks, Paths, Trails, Cycles, and Trees**

Unless stated otherwise, a graph  $G = (V, E)$  here is a finite undirected simple graph with vertex set  $V$  and edge set  $E$ : it has no loops or multiple edges. Two vertices joined by an edge are adjacent, and the degree  $\deg(v)$  is the number of edges incident to  $v$ . A walk is a sequence of adjacent vertices. A trail repeats no edge, while a path repeats no vertex. A closed trail is a circuit, and a cycle is a closed path apart from its repeated initial and terminal vertex. A graph is connected if every two vertices are joined by a path. A tree is a connected graph containing no cycle.

An Eulerian trail uses every edge exactly once; an Eulerian circuit is a closed Eulerian trail. A Hamiltonian path visits every vertex exactly once; a Hamiltonian cycle returns to its starting vertex. A graph is bipartite if its vertices can be divided into two classes with every edge joining different classes. A planar graph can be drawn in the plane without edge crossings except at common endpoints. The complete graph  $K_n$  contains every edge between distinct vertices, and the complete bipartite graph  $K_{m,n}$  contains every edge between its two vertex classes. Thus a one-stroke drawing problem is an Eulerian, not a Hamiltonian, question.

*Strategy. Coloring, Parity, and Bipartite Obstructions.* Colorings convert geometric or movement constraints into parity data. On a bipartite graph, every path alternates colors. Thus a Hamiltonian path can exist only if the two color classes differ in size by at most 1, and a Hamiltonian cycle requires the two classes to have equal size. Checkerboard colorings are especially effective for tiling and movement problems.

**Theorem 5.18. Inclusion–Exclusion Principle; Sieve Formula**

For finite sets  $A_1, \dots, A_n$ ,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|.$$

*Remark.* Inclusion–Exclusion is the standard correction mechanism for overcounting. It is also historically connected to sieve methods.

**Formula 5.19. Derangement Formula and Recurrence**

Let  $D_n$  be the number of derangements of  $n$  objects, that is, permutations with no fixed points. Then

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

The characteristic recurrence is

$$D_n = (n-1)(D_{n-1} + D_{n-2}) \quad (n \geq 2),$$

with  $D_0 = 1$  and  $D_1 = 0$ . Equivalently,

$$D_n = nD_{n-1} + (-1)^n.$$

The first recurrence comes from choosing the image of a distinguished object and separating the cases in which a resulting transposition is completed or not.

*Remark.* The approximation  $D_n \approx n!/e$  is very memorable. In probability language, a random permutation has no fixed point with probability approximately  $1/e$ .

**Formula 5.20. Catalan Numbers**

The  $n$ th Catalan number is

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

It begins as

$$1, 1, 2, 5, 14, 42, 132, \dots$$

*Remark.* Catalan numbers count many different objects, including correct parenthesizations, Dyck paths, triangulations of a polygon, and rooted full binary trees. The number 42 is a standard recognition item.

**Formula 5.21. Cayley’s Formula**

The number of labeled trees on  $n$  vertices is

$$n^{n-2} \quad (n \geq 2).$$

*Remark.* Cayley’s formula gives a compact and surprising enumeration of labeled trees.

**Definition 5.22. Finite Symmetry Action, Orbit, and Fixed Point**

For counting purposes, let  $G$  be a finite collection of symmetries that contains the identity and is closed under composition and inverses. An action of  $G$  on a finite set  $X$  assigns to each  $g \in G$  a permutation of  $X$  compatibly with composition. The orbit of  $x \in X$  is the set of objects obtainable from  $x$  by these symmetries, and

$$\text{Fix}(g) = \{x \in X \mid g \cdot x = x\}$$

is the fixed-point set of  $g$ . Chapter 9 gives the full abstract group definition and the general orbit–stabilizer framework.

**Theorem 5.23. Burnside's Lemma; Cauchy–Frobenius–Burnside Lemma**

Suppose a finite group  $G$  acts on a finite set  $X$ , meaning that each  $g \in G$  acts as a permutation of  $X$  and the action is compatible with group multiplication. The number of orbits is

$$\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|,$$

where

$$\text{Fix}(g) = \{x \in X \mid g \cdot x = x\}.$$

*Idea.* Burnside's Lemma says that the number of essentially different objects is the average number of objects fixed by a symmetry.

**Theorem 5.24. Equivalent Characterizations of a Tree**

For a finite graph  $G$  with  $n$  vertices, any two of the following conditions imply the third, and in fact all three are equivalent:

$G$  is connected and acyclic,

$G$  is connected and  $|E| = n - 1$ ,

and

every two vertices are joined by a unique path.

Equivalently, a tree is minimally connected and maximally acyclic.

**Theorem 5.25. Bipartite Graph Criterion**

A graph is bipartite if and only if it contains no odd cycle. Therefore a two-coloring obstruction and an odd-cycle obstruction are the same phenomenon.

**Lemma 5.26. Handshaking Lemma**

For a finite undirected graph  $G = (V, E)$ ,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

In particular, the number of vertices of odd degree is even.

*Remark.* The name comes from the idea that every handshake contributes 2 to the total count of hands shaken.

**Theorem 5.27. Euler's Formula for Planar Graphs**

If a connected planar graph has  $V$  vertices,  $E$  edges, and  $F$  faces, then

$$V - E + F = 2.$$

*Remark.* This is the graph-theoretic version of Euler's polyhedron formula. It is one of the most famous formulas involving planar graphs.

**Theorem 5.28. Kuratowski's Theorem**

A finite graph is planar if and only if it contains no subgraph that is a subdivision of  $K_5$  or  $K_{3,3}$ . Thus the two classical complete graphs  $K_5$  and  $K_{3,3}$  are the fundamental obstructions to planarity in the sense of subdivisions.

**Formula 5.29. Regions Cut by Hyperplanes in General Position**

For  $n$  hyperplanes in general position in dimension  $d$ , the maximum number of regions is

$$R_d(n) = \sum_{k=0}^d \binom{n}{k}.$$

In particular,  $n$  lines divide the plane into at most

$$1 + n + \binom{n}{2}$$

regions, and  $n$  planes divide  $\mathbb{R}^3$  into at most

$$1 + n + \binom{n}{2} + \binom{n}{3}$$

regions.

**Formula 5.30. Chords and Diagonals in Convex Position**

If  $n$  points lie on a circle and no three interior chords are concurrent, drawing every chord divides the disk into

$$1 + \binom{n}{2} + \binom{n}{4}$$

regions. A convex  $n$ -gon has at most

$$\binom{n}{4}$$

interior intersection points of diagonals.

**Formula 5.31. Triangulation with Interior Vertices**

Suppose a polygonal disk is triangulated using  $V$  vertices in total, of which  $b$  lie on the boundary. Then the number of edges and triangular faces are

$$E = 3V - b - 3, \quad F_{\triangle} = 2V - b - 2.$$

These values follow from Euler's formula and double counting the triangle-edge incidences.

*Strategy. Double Counting with Euler's Formula.* In a planar subdivision, count edge-face incidences first and then use  $V - E + F = 2$ . Interior edges are counted twice and boundary edges once. This is often faster and safer than trying to enumerate faces directly. For a convex polyhedron with three edges at every vertex, let  $f_k$  count its  $k$ -gonal faces. Since  $3V = 2E$  and  $\sum_{k \geq 3} k f_k = 2E$ , Euler's formula gives

$$\sum_{k \geq 3} (6 - k) f_k = 12.$$

In particular, if every face is a pentagon or a hexagon, there are exactly twelve pentagonal faces.

**Theorem 5.32. Eulerian Trail Criterion**

*A connected finite graph has an Eulerian circuit if and only if every vertex has even degree. It has an Eulerian trail but not an Eulerian circuit if and only if exactly two vertices have odd degree.*

*Remark.* This theorem is historically connected to Euler's solution of the Seven Bridges of Königsberg problem, which marks the beginning of graph theory and topology. The historical city Königsberg is now Kaliningrad, Russia.

**Definition 5.33. Matching, Perfect Matching, Vertex Cover, and Neighborhood**

In a graph, a *matching* is a set of edges no two of which share a vertex. A matching *covers* a vertex if one of its edges is incident to that vertex, and it is *perfect* if it covers every vertex. A *vertex cover* is a set of vertices meeting every edge. For  $S \subseteq V$ , its neighborhood is

$$N(S) = \{v \in V \mid v \text{ is adjacent to some } s \in S\}.$$

These notions are the basic language of assignment and pairing problems.

**Theorem 5.34. Hall's Marriage Theorem**

Let  $G = (X \cup Y, E)$  be a finite bipartite graph. There is a matching that covers every vertex of  $X$  if and only if, for every subset  $S \subseteq X$ ,

$$|N(S)| \geq |S|,$$

where  $N(S)$  is the set of neighbors of vertices in  $S$ .

*Strategy.* Hall's Marriage Theorem is the standard named theorem for proving that a perfect assignment or matching exists.

**Theorem 5.35. König's Theorem**

In every finite bipartite graph, the size of a maximum matching equals the size of a minimum vertex cover.

**Definition 5.36. Capacitated Network, Flow, and Cut**

A capacitated network is a directed graph with a distinguished source  $s$ , sink  $t$ , and nonnegative capacity  $c(e)$  on each edge. An  $s$ - $t$  flow assigns values  $f(e)$  satisfying  $0 \leq f(e) \leq c(e)$  and flow conservation at every vertex other than  $s$  and  $t$ . An  $s$ - $t$  cut is a partition  $V = S \sqcup T$  with  $s \in S$  and  $t \in T$ ; its capacity is the sum of capacities of edges directed from  $S$  to  $T$ .

**Theorem 5.37. Max-Flow Min-Cut Theorem**

In a finite capacitated network, the maximum value of a source-sink flow equals the minimum capacity of a source-sink cut.

**Theorem 5.38. Menger's Theorem, Vertex Form**

In a finite graph, for nonadjacent vertices  $u$  and  $v$ , the maximum number of pairwise internally vertex-disjoint  $u$ - $v$  paths equals the minimum number of vertices whose deletion separates  $u$  from  $v$ .

**Theorem 5.39. Kirchhoff's Matrix-Tree Theorem**

Let

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

be the Laplacian matrix of a finite graph. The number of spanning trees equals any cofactor of  $\mathbf{L}$  obtained by deleting the same row and column.

**Definition 5.40. Partially Ordered Set, Chain, and Antichain**

A partially ordered set, or poset, is a set  $P$  equipped with a relation  $\preceq$  that is reflexive, anti-symmetric, and transitive. A *chain* is a subset whose elements are pairwise comparable, while an *antichain* is a subset in which no two distinct elements are comparable. The power set  $\mathcal{P}([n])$  ordered by inclusion is the standard finite example.

**Theorem 5.41. Sperner's Theorem**

The largest family of subsets of  $[n]$  in which no member contains another has size

$$\binom{n}{\lfloor n/2 \rfloor}.$$

**Theorem 5.42. Dilworth's Theorem**

In a finite partially ordered set, the maximum size of an antichain equals the minimum number of chains covering the poset.

**Theorem 5.43. Turán's Theorem**

Among all  $n$ -vertex graphs containing no  $K_{r+1}$ , the maximum number of edges is attained by the complete  $r$ -partite graph whose part sizes differ by at most 1.

**Theorem 5.44. Dirac's and Ore's Hamiltonian Criteria**

Let  $G$  be a simple graph on  $n \geq 3$  vertices. If every vertex has degree at least  $n/2$ , then  $G$  is Hamiltonian. More generally, it is enough that

$$\deg(u) + \deg(v) \geq n$$

for every pair of nonadjacent vertices  $u, v$ .

**Theorem 5.45. Ramsey's Theorem**

For any positive integers  $r$  and  $s$ , there exists an integer  $R(r, s)$  such that every red-blue coloring of the edges of a sufficiently large complete graph contains either a red  $K_r$  or a blue  $K_s$ .

**Formula 5.46. The Party Problem**

The classical Ramsey number

$$R(3, 3) = 6$$

says that among any six people, there are either three mutual acquaintances or three mutual strangers.

*Remark.* Ramsey theory is the mathematics of unavoidable structure. Its slogan is that complete disorder is impossible in a large enough system.

**Theorem 5.47. Erdős–Szekeres Theorem**

Any sequence of

$$(r - 1)(s - 1) + 1$$

distinct real numbers contains either an increasing subsequence of length  $r$  or a decreasing subsequence of length  $s$ .

*Remark.* This is a beautiful pigeonhole-style result. It is often remembered as the monotone subsequence theorem.

## Chapter 6

# Probability and Statistics

Related **POSTECH** Courses: (MATH230) Probability & Statistics

Related Books: Probability & Statistics for Engineers & Scientists (Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, Keying Ye)

### Definition 6.1. Probability Space, Event, and Conditional Probability

A probability space is a triple  $(\Omega, \mathcal{F}, \mathbb{P})$ , where  $\Omega$  is the sample space,  $\mathcal{F}$  is a  $\sigma$ -algebra of events (closed under complements and countable unions), and  $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$  satisfies

$$\mathbb{P}(\Omega) = 1, \quad \mathbb{P}\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

for pairwise disjoint events  $A_i \in \mathcal{F}$ . If  $\mathbb{P}(B) > 0$ , the conditional probability of  $A$  given  $B$  is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

### Definition 6.2. Random Variable, Distribution, and Distribution Function

A random variable is a measurable function  $X: \Omega \rightarrow \mathbb{R}$ . Its distribution is the probability law of its values. For a discrete random variable, the probability mass function is

$$p_X(x) = \mathbb{P}(X = x), \quad \sum_x p_X(x) = 1.$$

For a continuous random variable with density  $f_X$ ,

$$\mathbb{P}(a < X \leq b) = \int_a^b f_X(x) \, dx, \quad \int_{-\infty}^{\infty} f_X(x) \, dx = 1.$$

In either case the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

### Definition 6.3. Expectation and Variance

If the relevant sum or integral converges absolutely, then

$$\mathbb{E}[X] = \begin{cases} \sum x \mathbb{P}(X = x), & X \text{ discrete,} \\ \int_{-\infty}^{\infty} x f_X(x) \, dx, & X \text{ continuous,} \end{cases}$$

and, whenever the second moment is finite,

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

**Definition 6.4. Joint, Marginal, and Conditional Distributions; Conditional Expectation**

For discrete random variables  $X$  and  $Y$ , the joint probability mass function is

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y),$$

and the marginal mass functions are obtained by summing over the other variable. If  $\mathbb{P}(Y = y) > 0$ , then

$$\mathbb{P}(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{\mathbb{P}(Y = y)},$$

and, whenever the sum converges absolutely,

$$\mathbb{E}[X | Y = y] = \sum_x x \mathbb{P}(X = x | Y = y).$$

If  $(X, Y)$  has joint density  $f_{X,Y}$ , then the marginal density of  $Y$  is

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Whenever  $f_Y(y) > 0$ ,

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad \mathbb{E}[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx.$$

The random variable  $\mathbb{E}[X | Y]$  is obtained by evaluating this conditional mean at the observed value of  $Y$ .

**Theorem 6.5. Law of Total Probability**

If  $B_1, \dots, B_n$  form a partition of the sample space and  $\mathbb{P}(B_i) > 0$  for all  $i$ , then

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

**Theorem 6.6. Law of Total Expectation; Tower Property**

If  $X$  is integrable and  $Y$  is a random variable, then

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]].$$

When  $Y$  is discrete, this becomes

$$\mathbb{E}[X] = \sum_y \mathbb{E}[X | Y = y] \mathbb{P}(Y = y),$$

where the sum is taken over values  $y$  with positive probability.

**Formula 6.7. Law of Total Variance**

If  $X$  has finite second moment, then

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]).$$

**Theorem 6.8. Bayes' Theorem**

If  $B_1, \dots, B_n$  form a partition of the sample space and  $\mathbb{P}(A) > 0$ , then

$$\mathbb{P}(B_j | A) = \frac{\mathbb{P}(A | B_j)\mathbb{P}(B_j)}{\sum_{i=1}^n \mathbb{P}(A | B_i)\mathbb{P}(B_i)}.$$

In the two-event form,

$$\mathbb{P}(B | A) = \frac{\mathbb{P}(A | B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

*Remark.* Bayes' Theorem is the standard formula for reversing conditional probability. It updates a prior probability after observing evidence.

*Remark.* In general,

$$\mathbb{P}(A | B) \neq \mathbb{P}(B | A).$$

The two quantities reverse the roles of evidence and hypothesis. Bayes' Theorem relates them; it does not permit the conditioning bar to be reversed without the corresponding prior probabilities.

**Definition 6.9. Independence**

Events  $A$  and  $B$  are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, if  $\mathbb{P}(B) > 0$ , then

$$\mathbb{P}(A | B) = \mathbb{P}(A).$$

Random variables  $X$  and  $Y$  are independent if events determined by  $X$  and events determined by  $Y$  are independent. A family of events is mutually independent if every finite subfamily has intersection probability equal to the product of its individual probabilities. In general,

$$\text{pairwise independence} \not\Rightarrow \text{mutual independence}.$$

*Tactic.* “Disjoint” and “independent” are different. Disjoint events  $A, B$  are independent exactly when  $\mathbb{P}(A)\mathbb{P}(B) = 0$ . In particular, two disjoint events of positive probability are not independent; nonempty events need not have positive probability.

**Formula 6.10. Distributions of Maxima and Minima**

Let  $X_1, \dots, X_n$  be independent with common cumulative distribution function  $F$ . For  $M = \max_i X_i$  and  $m = \min_i X_i$ ,

$$\mathbb{P}(M \leq x) = F(x)^n, \quad \mathbb{P}(m > x) = (1 - F(x))^n.$$

For a discrete distribution, subtraction of adjacent cumulative probabilities gives the mass function; in general,

$$\mathbb{P}(M = x) = F(x)^n - F(x^-)^n,$$

where  $F(x^-) = \mathbb{P}(X_1 < x)$ . For independent variables with different distribution functions  $F_i$ , replace the powers by  $\prod_{i=1}^n F_i(x)$  and  $\prod_{i=1}^n (1 - F_i(x))$ , respectively.

**Formula 6.11. Complement Rule and Inclusion–Exclusion for Probability**

The complement rule is

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

For two events,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

For three events,

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ &\quad - \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(C \cap A) \\ &\quad + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

**Proposition 6.12. Linearity of Expectation**

If  $X_1, \dots, X_n$  are integrable random variables and  $a_1, \dots, a_n$  are scalars, then

$$\mathbb{E} \left[ \sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

Independence is not required.

*Strategy.* Linearity of expectation is often the most efficient method for counting an expected number of objects. Introduce indicator random variables and add their expectations.

**Formula 6.13. Tail-Sum Formula**

If  $X$  takes values in  $\mathbb{N}_0$ , then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

The identity also holds when both sides are infinite. For waiting times, count the probability that the process has not yet stopped.

**Proposition 6.14. Expectation of a Product of Independent Random Variables**

If  $X_1, \dots, X_n$  are independent integrable random variables, then their product is integrable and

$$\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n].$$

In particular, if they are also identically distributed, the right-hand side is  $(\mathbb{E}[X_1])^n$ .

**Formula 6.15. Common Probability Distributions**

The most frequently used distributions are summarized below. Unless otherwise indicated,  $0 < p < 1$ ,  $n, r \in \mathbb{N}$ ,  $\lambda > 0$ ,  $\alpha, \beta > 0$ ,  $a < b$ ,  $\sigma > 0$ , and  $\nu \in \mathbb{N}$ . For the negative binomial row,  $X$  is the trial number on which the  $r$ th success occurs; the Gamma distribution uses the rate parameter  $\lambda$ .

| Distribution                | Mass or density function                                                          | Mean                | Variance              |
|-----------------------------|-----------------------------------------------------------------------------------|---------------------|-----------------------|
| Bern( $p$ )                 | $\mathbb{P}(X = x) = p^x(1-p)^{1-x}$ , $x \in \{0, 1\}$                           | $p$                 | $p(1-p)$              |
| Bin( $n, p$ )               | $\mathbb{P}(X = k) = \binom{n}{k} p^k(1-p)^{n-k}$ ,<br>$k \in \{0, 1, \dots, n\}$ | $np$                | $np(1-p)$             |
| Geom( $p$ )                 | $\mathbb{P}(X = k) = (1-p)^{k-1}p$ , $k \in \mathbb{N}$                           | $\frac{1}{p}$       | $\frac{1-p}{p^2}$     |
| Negative binomial( $r, p$ ) | $\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r(1-p)^{k-r}$ ,<br>$k \geq r$             | $\frac{r}{p}$       | $\frac{r(1-p)}{p^2}$  |
| Pois( $\lambda$ )           | $\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$ , $k \in \mathbb{N}_0$    | $\lambda$           | $\lambda$             |
| Unif( $a, b$ )              | $f(x) = \frac{1}{b-a}$ , $a \leq x \leq b$                                        | $\frac{a+b}{2}$     | $\frac{(b-a)^2}{12}$  |
| Exp( $\lambda$ )            | $f(x) = \lambda e^{-\lambda x}$ , $x \geq 0$                                      | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |

| Distribution                 | Mass or density function                                                          | Mean                            | Variance                                                     |
|------------------------------|-----------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------|
| Gamma( $\alpha, \lambda$ )   | $f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, x > 0$ | $\frac{\alpha}{\lambda}$        | $\frac{\alpha}{\lambda^2}$                                   |
| Beta( $\alpha, \beta$ )      | $f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, 0 < x < 1$          | $\frac{\alpha}{\alpha + \beta}$ | $\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$ |
| $\chi^2(\nu)$                | $f(x) = \frac{x^{\nu/2-1} e^{-x/2}}{2^{\nu/2} \Gamma(\nu/2)}, x > 0$              | $\nu$                           | $2\nu$                                                       |
| $\mathcal{N}(\mu, \sigma^2)$ | $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$                   | $\mu$                           | $\sigma^2$                                                   |

**Formula 6.16. Classical Sampling-Distribution Constructions**

Let  $Z, Z_1, \dots, Z_\nu \sim \mathcal{N}(0, 1)$  be independent. Then

$$\sum_{i=1}^{\nu} Z_i^2 \sim \chi^2(\nu).$$

If  $U \sim \chi^2(\nu)$  is independent of  $Z$ , then

$$\frac{Z}{\sqrt{U/\nu}} \sim t(\nu).$$

If  $U_1 \sim \chi^2(d_1)$  and  $U_2 \sim \chi^2(d_2)$  are independent, then

$$\frac{U_1/d_1}{U_2/d_2} \sim F(d_1, d_2).$$

These three constructions are the standard recognition rules for the chi-square, Student  $t$ , and  $F$  distributions.

*Strategy.* Waiting-until questions usually start with a geometric distribution. If a running sum is also involved, combine the expected number of trials with the expected contribution per trial, or condition on the first step.

**Formula 6.17. Variance, Covariance, and Correlation**

For random variables with finite second moments,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

and

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

If  $\text{Var}(X) > 0$  and  $\text{Var}(Y) > 0$ , their correlation coefficient is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}, \quad -1 \leq \rho_{X,Y} \leq 1.$$

If  $X$  and  $Y$  are independent, then  $\text{Cov}(X, Y) = 0$ , so their variances add. In general,

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ are independent.}$$

**Formula 6.18. Hypergeometric Distribution**

Suppose a population has  $N$  objects, of which  $K$  are successes. If  $n$  objects are chosen without replacement and  $X$  is the number of successes chosen, then

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

Its mean and variance are

$$\begin{aligned}\mathbb{E}[X] &= n \frac{K}{N}, \\ \text{Var}(X) &= n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}.\end{aligned}$$

The final factor is the finite-population correction caused by sampling without replacement.

**Theorem 6.19. Poisson Limit Theorem; Law of Small Numbers**

Let  $p_n \in [0, 1]$  satisfy  $np_n \rightarrow \lambda > 0$ . For each fixed nonnegative integer  $k$ ,

$$\binom{n}{k} p_n^k (1 - p_n)^{n-k} \rightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

Thus  $\text{Bin}(n, p_n)$  converges in distribution to  $\text{Pois}(\lambda)$ .

*Remark.* This theorem explains why the Poisson distribution models rare events: many trials, small success probability, and finite expected count.

**Formula 6.20. Empirical Rule; 68–95–99.7 Rule**

For a normal distribution, approximately

68%

of the mass lies within one standard deviation of the mean,

95%

lies within two standard deviations, and

99.7%

lies within three standard deviations.

**Definition 6.21. Modes of Convergence for Random Variables**

Let  $X_n$  and  $X$  be random variables on the same probability space. The principal modes of convergence used in probability are:

(i)  $X_n \xrightarrow{\text{a.s.}} X$  if

$$\mathbb{P}(\{\omega \mid X_n(\omega) \rightarrow X(\omega)\}) = 1;$$

(ii)  $X_n \xrightarrow{\mathbb{P}} X$  if

$$\forall \varepsilon > 0, \quad \mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0;$$

(iii)  $X_n \xrightarrow{L^p} X$  for  $p \geq 1$  if

$$\mathbb{E}[|X_n - X|^p] \rightarrow 0;$$

(iv)  $X_n \xrightarrow{d} X$  if

$$F_{X_n}(x) \rightarrow F_X(x)$$

at every continuity point  $x$  of the distribution function  $F_X$ .

Almost-sure convergence implies convergence in probability, and convergence in probability implies convergence in distribution.

**Theorem 6.22. Borel–Cantelli Lemmas**

If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then  $\mathbb{P}(A_n \text{ infinitely often}) = 0$ . If the  $A_n$  are independent and the series diverges, then

$$\mathbb{P}(A_n \text{ infinitely often}) = 1.$$

**Theorem 6.23. Weak Law of Large Numbers**

Let  $X_1, X_2, \dots$  be independent and identically distributed random variables with finite mean  $\mu$  and finite variance  $\sigma^2$ . Then

$$\frac{X_1 + \dots + X_n}{n} \xrightarrow{\mathbb{P}} \mu.$$

This follows directly from Chebyshev's inequality because the variance of the sample mean is  $\sigma^2/n$ .

*Idea.* The Law of Large Numbers says that the sample average stabilizes near the true mean when the number of trials becomes large.

**Theorem 6.24. Central Limit Theorem**

Let  $X_1, X_2, \dots$  be independent identically distributed random variables with mean  $\mu$  and variance  $\sigma^2 > 0$ . Then

$$\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} Z, \quad Z \sim \mathcal{N}(0, 1).$$

Thus the standardized sum converges in distribution to the standard normal distribution.

*Remark.* The Central Limit Theorem explains why the normal distribution appears so often in nature and statistics.

**Theorem 6.25. Markov's Inequality**

If  $X$  is a nonnegative random variable and  $a > 0$ , then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

**Theorem 6.26. Chebyshev's Inequality**

If  $X$  has finite mean  $\mu$  and finite variance  $\sigma^2 > 0$ , then for every  $k > 0$ ,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Equivalently, for every  $\varepsilon > 0$ ,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

*Remark.* Markov's inequality uses only the mean, while Chebyshev's inequality uses the variance. Both are probability bounds that work without assuming a normal distribution.

**Theorem 6.27. Chernoff Bound for a Binomial Random Variable**

If  $X \sim \text{Bin}(n, p)$  and  $\mu = np$ , then for  $\delta > 0$ ,

$$\mathbb{P}(X \geq (1 + \delta)\mu) \leq \left( \frac{e^\delta}{(1 + \delta)^{1 + \delta}} \right)^\mu.$$

For  $0 < \delta < 1$ ,

$$\mathbb{P}(|X - \mu| \geq \delta\mu) \leq 2e^{-\delta^2\mu/3}.$$

**Definition 6.28. Convex and Concave Functions**

Let  $I \subseteq \mathbb{R}$  be an interval. A function  $\varphi: I \rightarrow \mathbb{R}$  is convex if, for all  $x, y \in I$  and  $t \in [0, 1]$ ,

$$\varphi((1-t)x + ty) \leq (1-t)\varphi(x) + t\varphi(y).$$

It is concave if the reverse inequality holds. Equivalently,  $\varphi$  is concave if  $-\varphi$  is convex.

**Theorem 6.29. Jensen's Inequality**

Let  $X$  be integrable and take values in an interval  $I$ , and let  $\varphi: I \rightarrow \mathbb{R}$  be convex with  $\varphi(X)$  integrable. Then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

If  $\varphi$  is concave under the analogous integrability assumptions, the inequality is reversed.

*Idea.* Jensen's inequality says that, for a convex function, applying the function after averaging gives a value no larger than averaging after applying the function.

**Formula 6.30. Exact Birthday Problem**

Under the standard model of 365 equally likely birthdays and no leap day, the probability that at least two people among  $n$  share a birthday is

$$1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \quad (0 \leq n \leq 365).$$

For  $n \geq 366$ , the probability is 1 by the pigeonhole principle.

**Formula 6.31. Birthday Problem Approximation**

In a group of  $n$  people, assuming 365 equally likely birthdays, the probability that at least two people share a birthday is approximately

$$1 - \exp\left(-\frac{n(n-1)}{730}\right).$$

For  $n = 23$ , this probability is already greater than  $1/2$ .

*Remark.* The birthday problem is famous because the answer is surprisingly small: only 23 people are needed for a better-than-even chance of a shared birthday.

**Formula 6.32. Expected Number of Runs in Coin Tosses**

In  $n$  tosses of a fair coin, a run is a maximal consecutive block of identical outcomes. If  $R_n$  is the number of runs, then

$$\mathbb{E}[R_n] = 1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

*Idea.* The first toss starts one run. For each of the remaining  $n-1$  positions, a new run begins exactly when the current toss differs from the previous toss, which has probability  $1/2$ .

**Formula 6.33. Monty Hall Problem**

There are three doors, exactly one of which hides a prize. The contestant first chooses a door. A host who knows the prize location always opens an unchosen door that hides no prize and then offers the unique remaining unopened door. When the host has two eligible doors, the protocol fixes how one is selected. Under this protocol, staying wins with probability  $1/3$ , whereas switching wins with probability  $2/3$ .

*Remark.* The host's action is not random information: the host always reveals a losing door. This is why the original  $2/3$  probability of having chosen incorrectly transfers to the remaining unopened door.

**Formula 6.34. Coupon Collector**

If there are  $n$  equally likely coupon types, the expected number of trials needed to see all  $n$  types is

$$nH_n = n \left( 1 + \frac{1}{2} + \cdots + \frac{1}{n} \right).$$

*Strategy.* For a fair die, the expected number of rolls needed to see all six faces is  $6H_6$ . If a question asks about the order in which the first appearances occur, ignore repeated rolls and look only at the random ordering of the six faces.

**Formula 6.35. Gambler's Ruin**

Suppose a gambler starts with  $i$  dollars and stops when reaching either 0 dollars or  $N$  dollars. If the probability of winning each round is  $p$  and  $q = 1 - p$ , then the probability of reaching  $N$  before 0 is

$$\frac{1 - (q/p)^i}{1 - (q/p)^N} \quad (p \neq q).$$

In the fair case  $p = q = 1/2$ , the probability is

$$\frac{i}{N}.$$

*Remark.* Gambler's ruin is a classic random walk result. It is a good example where the fair case has a very simple answer.

# Chapter 7

## Geometry

### Synthetic Geometry

#### Fact 7.1. Euclid's *Elements* and the Axiomatic Method

Euclid's *Elements* consists of thirteen books and organizes geometry and number theory through definitions, common notions, postulates, propositions, and proofs. Book I begins with five postulates; the fifth is the parallel postulate.

#### Fact 7.2. Euclid's Fifth Postulate and Playfair's Axiom

Euclid's fifth postulate states that if a transversal makes the two interior angles on one side sum to less than two right angles, then the two lines meet on that side when extended. In the usual axiomatic setting for Euclidean plane geometry, this is equivalent to Playfair's axiom: through a point not on a given line, there is exactly one line parallel to the given line.

#### Definition 7.3. Euclidean, Hyperbolic, Spherical, and Elliptic Geometry

The three constant-curvature geometries are distinguished by their parallel behavior and triangle angle sums.

| Geometry   | Lines through a point off a given line | Triangle angle sum |
|------------|----------------------------------------|--------------------|
| Euclidean  | Exactly one parallel                   | Equal to $\pi$     |
| Hyperbolic | Infinitely many disjoint lines         | Less than $\pi$    |
| Spherical  | No disjoint great circle               | Greater than $\pi$ |
| Elliptic   | No disjoint geodesic line              | Greater than $\pi$ |

Spherical geometry uses the sphere itself, where antipodal points are distinct. Elliptic geometry identifies antipodal points; the two models share positive curvature and the same local triangle-angle behavior but are not the same global geometry.

#### Theorem 7.4. Parallel Postulate and Triangle Angle Sum

*Within standard neutral geometry, the Euclidean parallel postulate is equivalent to the assertion that every triangle has angle sum*

$$\pi.$$

*Consequently, the familiar triangle angle-sum theorem is not a consequence of Euclid's first four postulates alone.*

**Formula 7.5. Geodesic Triangle Angle Sum on Constant Curvature**

Let a geodesic triangle lie on a surface of constant Gaussian curvature  $K$ . If its angles are  $\alpha, \beta, \gamma$ , then the Gauss–Bonnet formula gives

$$\alpha + \beta + \gamma - \pi = K \text{ area}(\triangle ABC).$$

Thus positive curvature produces angle excess, zero curvature gives the Euclidean angle sum, and negative curvature produces angle defect.

**Definition 7.6. Convex Set, Convex Combination, and Convex Hull**

A set  $C \subseteq \mathbb{R}^n$  is convex if for every  $\mathbf{x}, \mathbf{y} \in C$  and every  $t \in [0, 1]$ ,

$$(1 - t)\mathbf{x} + t\mathbf{y} \in C.$$

More generally, a convex combination of points  $\mathbf{x}_1, \dots, \mathbf{x}_m$  is a point

$$\sum_{i=1}^m \lambda_i \mathbf{x}_i, \quad \lambda_i \geq 0, \quad \sum_{i=1}^m \lambda_i = 1.$$

The convex hull  $\text{conv}(S)$  of a set  $S \subseteq \mathbb{R}^n$  is the smallest convex set containing  $S$ ; equivalently, it is the set of all finite convex combinations of points of  $S$ .

**Formula 7.7. Basic Plane and Solid Geometry Table**

The basic measurement formulas are

| Figure                  | Perimeter or surface area | Area or volume                                  |
|-------------------------|---------------------------|-------------------------------------------------|
| Triangle                | $a + b + c$               | $\frac{1}{2}bh$                                 |
| Parallelogram           | $2(a + b)$                | $bh$                                            |
| Trapezoid               | $a + b + c + d$           | $\frac{1}{2}(a + c)h$                           |
| Regular $n$ -gon        | $na$                      | $\frac{1}{2}(na)r = \frac{na^2}{4 \tan(\pi/n)}$ |
| Circle                  | $2\pi r$                  | $\pi r^2$                                       |
| Circular sector         | $r\theta$                 | $\frac{1}{2}r^2\theta$                          |
| Rectangular box         | $2(ab + bc + ca)$         | $abc$                                           |
| Right circular cylinder | $2\pi r(r + h)$           | $\pi r^2 h$                                     |
| Right circular cone     | $\pi r(r + \ell)$         | $\frac{1}{3}\pi r^2 h$                          |
| Sphere                  | $4\pi r^2$                | $\frac{4}{3}\pi r^3$                            |

In the regular-polygon row,  $r$  is the apothem; in the cone row,  $\ell = \sqrt{r^2 + h^2}$  is the slant height. Sector angles are measured in radians.

**Theorem 7.8. Triangle Angle Sum and Exterior Angle Theorem**

For every triangle  $\triangle ABC$ ,

$$\angle A + \angle B + \angle C = \pi.$$

An exterior angle equals the sum of the two nonadjacent interior angles.

**Proposition 7.9. Triangle Congruence and Similarity Criteria**

*Two triangles are congruent under any of the criteria*

$$\text{SSS}, \quad \text{SAS}, \quad \text{ASA}, \quad \text{AAS}, \quad \text{RHS}.$$

*They are similar under any of the criteria*

$$\text{AA}, \quad \text{SAS}, \quad \text{SSS},$$

*where the latter two use proportional corresponding side lengths.*

*Remark.* For right triangles, RHS (right angle–hypotenuse–side), also called HL in some conventions, is a valid congruence criterion. A condition sometimes called RHA (right angle–hypotenuse–acute angle) is also sufficient, but it is already an ASA/AAS case because the two acute angles of a right triangle are complementary. By contrast, SSA is not a congruence criterion in general; the right-triangle hypotheses are essential in the familiar hypotenuse-based special cases.

**Theorem 7.10. Midpoint Theorem and Centroid Ratio**

*The segment joining the midpoints of two sides of a triangle is parallel to the third side and has half its length. The three medians are concurrent at the centroid  $G$ , and  $G$  divides every median in the ratio*

$$2 : 1$$

*measured from the vertex.*

*Strategy. Reverse Engineering in Geometric Constructions.* First draw a completed configuration satisfying the desired conditions. Then identify which points, perpendiculars, parallels, angle bisectors, circles, or reflections in the completed picture can be constructed from the original data. Familiar configurations involving triangle centers, similar triangles, and power of a point are especially useful targets.

**Fact 7.11. Huzita–Hatori Origami Axioms**

The standard single-fold model of mathematical origami is described by seven folding axioms commonly called the Huzita–Hatori axioms. They specify folds determined by prescribed incidences among points and lines. Unlike straightedge-and-compass constructions, single-fold origami can solve certain cubic equations; in particular, it can carry out classical constructions such as general angle trisection and doubling the cube.

**Definition 7.12. Classical Triangle Centers**

For a triangle  $\triangle ABC$ , the most important classical centers are:

$$O = \text{circumcenter}, \quad I = \text{incenter}, \quad H = \text{orthocenter}, \quad G = \text{centroid}.$$

Here  $O$  is the center of the circumcircle,  $I$  is the center of the incircle,  $H$  is the intersection of the altitudes, and  $G$  is the intersection of the medians.

**Definition 7.13. Incircle, Excircles, Inradius, and Exradii**

The incircle of a triangle is the circle tangent to all three sides; its center is the incenter  $I$ , and its radius is the inradius  $r$ . An excircle is tangent to one side of the triangle and to the extensions of the other two sides. The corresponding excenters and exradii are denoted by  $I_a, I_b, I_c$  and  $r_a, r_b, r_c$ , respectively, according to the opposite vertex.

**Theorem 7.14. Gergonne Point and Nagel Point**

The cevians joining the vertices of a triangle to the contact points of its incircle with the opposite sides are concurrent at the Gergonne point. The cevians joining the vertices to the contact points of the opposite excircles are concurrent at the Nagel point.

**Definition 7.15. Isogonal Lines and Symmedians**

Two lines through a vertex of a triangle are isogonal if they are reflections of one another across the internal angle bisector at that vertex. The  $A$ -symmedian of  $\triangle ABC$  is the line isogonal to the  $A$ -median; the other symmedians are defined cyclically.

**Theorem 7.16. Symmedian Lemma and Lemoine Point**

Let the  $A$ -symmedian of  $\triangle ABC$  meet  $BC$  at  $K$ . Then

$$\frac{BK}{KC} = \frac{AB^2}{AC^2}.$$

The three symmedians are concurrent at the Lemoine point, also called the symmedian point.

**Theorem 7.17. Fermat Point; Torricelli Point**

If every angle of  $\triangle ABC$  is less than  $120^\circ$ , there is a unique point  $P$  minimizing

$$PA + PB + PC.$$

It satisfies

$$\angle APB = \angle BPC = \angle CPA = 120^\circ.$$

If one angle is at least  $120^\circ$ , the minimizing point is that vertex.

**Formula 7.18. Extended Law of Sines**

For a triangle  $\triangle ABC$  with side lengths

$$a = BC, \quad b = CA, \quad c = AB,$$

and circumradius  $R$ , we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

**Formula 7.19. Law of Cosines**

For a triangle  $\triangle ABC$  with side lengths

$$a = BC, \quad b = CA, \quad c = AB,$$

we have

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

The analogous formulas hold after cyclic permutation of  $a, b, c$ .

**Formula 7.20. Heron's Formula**

Let  $\triangle ABC$  have side lengths  $a, b, c$  and semiperimeter

$$s = \frac{a + b + c}{2}.$$

Then its area  $[ABC]$  is

$$[ABC] = \sqrt{s(s-a)(s-b)(s-c)}.$$

Also,

$$[ABC] = rs \quad \text{and} \quad [ABC] = \frac{abc}{4R},$$

where  $r$  is the inradius and  $R$  is the circumradius.

*Strategy.* The formulas

$$[ABC] = rs, \quad [ABC] = \frac{abc}{4R}$$

are often more useful than Heron's formula when incircles or circumcircles appear.

**Formula 7.21. Triangle Area Formulas**

For a triangle with side lengths  $a, b, c$ , angles  $A, B, C$ , circumradius  $R$ , and inradius  $r$ ,

$$[ABC] = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C,$$

and

$$[ABC] = rs = \frac{abc}{4R}.$$

**Theorem 7.22. Weitzenböck's Inequality**

For a triangle with side lengths  $a, b, c$  and area  $[ABC]$ ,

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}[ABC],$$

with equality if and only if the triangle is equilateral.

**Formula 7.23. Inradius and Exradii**

For a triangle  $\triangle ABC$  with semiperimeter  $s$ , the inradius is

$$r = \frac{[ABC]}{s}.$$

If  $r_a, r_b, r_c$  are the exradii opposite  $A, B, C$ , respectively, then

$$r_a = \frac{[ABC]}{s-a}, \quad r_b = \frac{[ABC]}{s-b}, \quad r_c = \frac{[ABC]}{s-c}.$$

**Formula 7.24. Median Length Formula; Apollonius' Theorem**

If  $m_a$  is the median from  $A$  to side  $a = BC$ , then

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}.$$

Equivalently, if  $M$  is the midpoint of  $BC$ , then

$$AB^2 + AC^2 = 2(AM^2 + BM^2).$$

**Theorem 7.25. Stewart's Theorem**

Let  $D$  lie on side  $BC$  of a triangle  $\triangle ABC$ . Write

$$a = BC, \quad b = CA, \quad c = AB, \quad m = BD, \quad n = DC, \quad d = AD,$$

so that  $a = m + n$ . Then

$$b^2m + c^2n = a(d^2 + mn).$$

Equivalently,

$$d^2 = \frac{b^2m + c^2n}{a} - mn.$$

*Remark.* Stewart's Theorem is the standard general formula for the length of a cevian. The median-length formula is the special case  $m = n = a/2$ .

**Formula 7.26. Internal Angle-Bisector Length**

Let the internal angle bisector from  $A$  meet  $BC$  at  $D$ , and denote its length by  $\ell_a = AD$ . For

$$a = BC, \quad b = CA, \quad c = AB,$$

we have

$$\ell_a^2 = bc \left( 1 - \frac{a^2}{(b+c)^2} \right) = \frac{4bc s(s-a)}{(b+c)^2},$$

where  $s = (a+b+c)/2$ .

**Theorem 7.27. Angle Bisector Theorem**

In a triangle  $\triangle ABC$ , suppose the internal angle bisector of  $\angle A$  meets  $BC$  at  $D$ . Then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

**Theorem 7.28. Ceva's Theorem**

Let  $D, E, F$  lie on the lines  $BC, CA, AB$ , respectively. Then the cevians  $AD, BE, CF$  are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$

using directed lengths.

*Remark.* Ceva's Theorem is the standard theorem for proving concurrence of three lines in a triangle.

**Theorem 7.29. Trigonometric Form of Ceva's Theorem**

Let  $D, E, F$  lie on the lines  $BC, CA, AB$ , respectively. The cevians  $AD, BE, CF$  are concurrent if and only if

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} = 1,$$

with the usual directed-angle convention when points lie on extended sides.

*Strategy. Mass Points.* In a cevian configuration, assign masses to vertices so that a point dividing a segment balances the endpoint masses inversely to the segment lengths. For example, if  $D \in BC$ , then

$$\frac{BD}{DC} = \frac{m_C}{m_B}.$$

Consistent masses convert several ratio computations into weighted averages and often recover Ceva-type relations immediately.

**Theorem 7.30. Menelaus' Theorem**

Let  $D, E, F$  lie on the lines  $BC, CA, AB$ , respectively. Then the points  $D, E, F$  are collinear if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1,$$

using directed lengths.

*Remark.* Ceva is for concurrence, while Menelaus is for collinearity. This pair is one of the most important toolkits in olympiad geometry.

**Theorem 7.31. Euler Line Theorem**

*In any non-equilateral triangle, the circumcenter  $O$ , centroid  $G$ , and orthocenter  $H$  are collinear. Moreover,*

$$OG : GH = 1 : 2.$$

*The line passing through these points is called the Euler line.*

**Theorem 7.32. Nine-Point Circle Theorem; Euler Circle**

*In any triangle, the following nine points lie on one circle: the three side midpoints, the three feet of the altitudes, and the three midpoints between the orthocenter and the vertices. This circle is called the nine-point circle.*

*Remark.* The center of the nine-point circle is the midpoint of the segment joining the circumcenter  $O$  and the orthocenter  $H$ .

**Theorem 7.33. Feuerbach's Theorem**

*The nine-point circle of a triangle is internally tangent to the incircle and externally tangent to each of the three excircles.*

**Theorem 7.34. Euler's Formula for the Incenter and Circumcenter**

*For a triangle with circumradius  $R$ , inradius  $r$ , circumcenter  $O$ , and incenter  $I$ , we have*

$$OI^2 = R^2 - 2Rr.$$

*In particular,*

$$R \geq 2r.$$

*Remark.* The inequality  $R \geq 2r$  is called Euler's inequality for triangles. Equality holds exactly for an equilateral triangle.

*Strategy. Spiral Similarity.* When two segments must be compared through both an angle and a ratio, look for a center of spiral similarity. It often reveals the required similar triangles immediately.

**Definition 7.35. Directed Angles Modulo  $\pi$**

When directed angles modulo  $\pi$  are used,  $\angle ABC$  denotes the directed angle from the line  $BA$  to the line  $BC$ , taken modulo  $\pi$ . Reversing the orientation of either line changes an ordinary oriented angle by  $\pi$  and therefore does not change its class modulo  $\pi$ . This convention avoids separate case distinctions for many collinearity and cyclicity arguments.

**Theorem 7.36. Inscribed Angle Theorem**

*An inscribed angle equals half the central angle subtending the same arc. Consequently, angles subtending the same chord are equal.*

**Theorem 7.37. Cyclic Quadrilateral Criteria**

*Let  $A, B, C, D$  be four distinct points with no three collinear. Using directed angles modulo  $\pi$ , the points are concyclic if and only if*

$$\angle BAC \equiv \angle BDC \pmod{\pi}.$$

*If  $ABCD$  is a convex quadrilateral in cyclic order, this is equivalent to*

$$\angle ABC + \angle ADC = \pi.$$

**Theorem 7.38. Tangent–Chord Theorem; Alternate Segment Theorem**

*The angle between a tangent to a circle and a chord through the point of tangency equals the inscribed angle subtending that chord in the opposite arc.*

**Theorem 7.39. Power of a Point Theorem**

*Let  $P$  be a point and let a line through  $P$  meet a circle at  $A$  and  $B$ . The product*

$$PA \cdot PB$$

*is independent of the choice of the line through  $P$ . It is called the power of  $P$  with respect to the circle.*

**Formula 7.40. Tangent–Secant Theorem**

*If  $PT$  is tangent to a circle at  $T$  and a secant through  $P$  meets the circle at  $A$  and  $B$ , then*

$$PT^2 = PA \cdot PB.$$

**Theorem 7.41. Radical Axis Theorem**

*For two nonconcentric circles, the set of points having equal powers with respect to the two circles is a line. This line is called the radical axis of the two circles.*

*Remark.* Radical axes are powerful because they turn circle configurations into collinearity statements.

**Theorem 7.42. Ptolemy's Theorem**

*Let  $A, B, C, D$  be the vertices of a convex cyclic quadrilateral in this cyclic order. Then*

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

*Conversely, equality in Ptolemy's inequality for a convex quadrilateral characterizes cyclicity.*

**Theorem 7.43. Ptolemy's Inequality**

*For any four points  $A, B, C, D$  in the Euclidean plane,*

$$AC \cdot BD \leq AB \cdot CD + BC \cdot DA.$$

*Equality holds in the cyclic convex case and in the corresponding degenerate collinear case.*

**Formula 7.44. Brahmagupta's Formula**

*If a cyclic quadrilateral has side lengths  $a, b, c, d$  and semiperimeter*

$$s = \frac{a + b + c + d}{2},$$

*then its area is*

$$\text{area}(ABCD) = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

*Heron's formula is the limiting triangular case.*

**Theorem 7.45. Varignon's Theorem**

*The midpoints of the four sides of any quadrilateral form a parallelogram. Its area is half the area of the original quadrilateral.*

**Theorem 7.46. Pitot's Theorem**

If a convex quadrilateral with consecutive side lengths  $a, b, c, d$  has an incircle tangent to all four sides, then

$$a + c = b + d.$$

**Theorem 7.47. Euler's Quadrilateral Theorem**

If  $M$  and  $N$  are the midpoints of the diagonals of a quadrilateral with side lengths  $a, b, c, d$  and diagonal lengths  $p, q$ , then

$$a^2 + b^2 + c^2 + d^2 = p^2 + q^2 + 4MN^2.$$

The parallelogram law is the case  $M = N$ .

**Theorem 7.48. Simson Line Theorem**

Let  $P$  be a point on the circumcircle of  $\triangle ABC$ . The feet of the perpendiculars from  $P$  to the three lines  $AB, BC, CA$  are collinear. This line is called the Simson line of  $P$  with respect to  $\triangle ABC$ .

**Theorem 7.49. Miquel's Theorem**

In a complete quadrilateral, the circumcircles of the four triangles formed by choosing three of the four lines pass through a common point. This point is called the Miquel point.

**Theorem 7.50. British Flag Theorem**

For any point  $P$  in the plane of a rectangle  $ABCD$ ,

$$PA^2 + PC^2 = PB^2 + PD^2.$$

**Theorem 7.51. Van Aubel's Theorem**

Construct squares externally on the four sides of a quadrilateral. The segments joining the centers of opposite squares are equal in length and perpendicular.

**Theorem 7.52. Viviani's Theorem**

For any point  $P$  inside an equilateral triangle of altitude  $h$ , the sum of the perpendicular distances from  $P$  to the three sides is  $h$ .

**Theorem 7.53. Desargues' Theorem**

If two triangles are perspective from a point, then the three intersections of corresponding sides are collinear. Conversely, in the projective plane, collinearity of those intersections implies concurrence of the three lines joining corresponding vertices.

**Theorem 7.54. Pappus' Hexagon Theorem**

Let  $A, B, C$  lie on one line and  $A', B', C'$  lie on another. Then the three points

$$AB' \cap A'B, \quad AC' \cap A'C, \quad BC' \cap B'C$$

are collinear.

**Theorem 7.55. Homothety Centers of Two Circles**

For two nonconcentric circles, the two external common tangents meet at the external homothety center, and the two internal common tangents meet at the internal homothety center, when these intersections exist. Each homothety center lies on the line joining the two circle centers.

**Formula 7.56. Circle Inversion**

Under inversion with center  $O$  and radius  $\rho$ , a point  $P \neq O$  is sent to the point  $P'$  on the ray  $OP$  satisfying

$$OP \cdot OP' = \rho^2.$$

Inversion preserves angles in magnitude, sends circles through  $O$  to lines not through  $O$ , and sends such lines back to circles through  $O$ .

**Formula 7.57. Regular Polygon Tiling Angle Condition**

If  $q$  regular  $p$ -gons meet edge-to-edge at each vertex of a Euclidean tiling, then

$$q \left( 1 - \frac{2}{p} \right) \pi = 2\pi,$$

equivalently,

$$(p-2)(q-2) = 4.$$

Thus the only tilings by one type of regular polygon are

$$(p, q) = (3, 6), (4, 4), (6, 3).$$

**Theorem 7.58. Classification of Platonic Solids**

There are exactly five Platonic solids:

*tetrahedron, cube, octahedron, dodecahedron, icosahedron.*

**Analytic Geometry****Formula 7.59. Coordinate Geometry Table**

For points  $A = (x_1, y_1)$  and  $B = (x_2, y_2)$ ,

| Quantity                                           | Formula                                                           |
|----------------------------------------------------|-------------------------------------------------------------------|
| Distance                                           | $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$                       |
| Midpoint                                           | $\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$         |
| Internal division<br>$AP : PB = m : n$             | $\left( \frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n} \right)$ |
| Slope                                              | $\frac{y_2 - y_1}{x_2 - x_1}$ , when $x_1 \neq x_2$               |
| Point-slope line                                   | $y - y_1 = m(x - x_1)$                                            |
| General line                                       | $ax + by + c = 0$                                                 |
| Distance from $(x_0, y_0)$ to<br>$ax + by + c = 0$ | $\frac{ ax_0 + by_0 + c }{\sqrt{a^2 + b^2}}$                      |

**Formula 7.60. Triangle Centers in Coordinates**

For  $A = (x_1, y_1)$ ,  $B = (x_2, y_2)$ , and  $C = (x_3, y_3)$ , the centroid is

$$G = \left( \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$$

If the side lengths opposite  $A, B, C$  are  $a, b, c$ , respectively, then the incenter is

$$I = \frac{aA + bB + cC}{a + b + c}.$$

**Formula 7.61. Barycentric Coordinates**

Let  $A, B, C$  be noncollinear. Every point  $P$  in their affine plane has unique coordinates  $(\alpha, \beta, \gamma)$  satisfying

$$\alpha + \beta + \gamma = 1, \quad \mathbf{p} = \alpha \mathbf{a} + \beta \mathbf{b} + \gamma \mathbf{c}.$$

For  $P$  inside  $\triangle ABC$ ,

$$\alpha = \frac{[PBC]}{[ABC]}, \quad \beta = \frac{[PCA]}{[ABC]}, \quad \gamma = \frac{[PAB]}{[ABC]}.$$

**Formula 7.62. Shoelace Formula; Gauss's Area Formula**

For a polygon with vertices

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

in cyclic order, its area is

$$\frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right|,$$

where

$$(x_{n+1}, y_{n+1}) = (x_1, y_1).$$

*Remark.* The Shoelace Formula is one of the cleanest examples of analytic geometry: a geometric area becomes a determinant-like coordinate calculation.

**Formula 7.63. Oriented Area and Collinearity Test**

For points  $A = (x_1, y_1)$ ,  $B = (x_2, y_2)$ , and  $C = (x_3, y_3)$ , the oriented double area is

$$2[ABC]_{\text{or}} = \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

Hence  $A, B, C$  are collinear if and only if this determinant is 0.

**Formula 7.64. Plane Rotation**

Counterclockwise rotation through angle  $\theta$  about the origin sends

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In complex notation, the same transformation is simply

$$z \mapsto e^{i\theta} z.$$

**Formula 7.65. Reflection across a Line through the Origin**

If a line through the origin makes angle  $\theta$  with the positive  $x$ -axis, reflection across that line is

$$z \mapsto e^{2i\theta} \bar{z}$$

in complex notation.

*Strategy. Affine Normalization.* Affine transformations preserve collinearity, parallelism, ratios along one line, concurrence, and ratios of areas. Therefore, when a problem depends only on these properties, one may send a triangle to convenient coordinates such as

$$A = (0, 0), \quad B = (1, 0), \quad C = (0, 1).$$

Angles, lengths, and circles are not generally preserved.

**Theorem 7.66. Cavalieri's Principle**

*If two plane regions lie between the same pair of parallel lines and every line parallel to them cuts the regions in segments whose lengths have a constant ratio  $c$ , then their areas have ratio  $c$ . The three-dimensional analogue replaces segment lengths by cross-sectional areas and concludes the same ratio for volumes.*

*Strategy. Unfolding Method for Reflection Problems.* A broken path reflecting from a line or face becomes a straight segment after reflecting the region across each boundary in succession. Thus a shortest reflected path is found by measuring a straight-line distance in the unfolded picture and then folding the path back.

**Formula 7.67. Lattice Segment and Cell Counts**

A segment from  $(0, 0)$  to  $(m, n)$ , where  $m, n \in \mathbb{N}$ , passes through

$$m + n - \gcd(m, n)$$

unit squares. The number of lattice points on the segment, including its endpoints, is

$$\gcd(m, n) + 1.$$

In three dimensions, a segment from  $(0, 0, 0)$  to  $(a, b, c)$  passes through

$$a + b + c - \gcd(a, b) - \gcd(a, c) - \gcd(b, c) + \gcd(a, b, c)$$

unit cubes.

**Formula 7.68. Vector Test for Parallel Segments**

Directed segments  $AB$  and  $CD$  are parallel if and only if

$$\mathbf{b} - \mathbf{a} = \lambda(\mathbf{d} - \mathbf{c})$$

for some scalar  $\lambda$ . Equality with  $\lambda = 1$  identifies equal directed segments and is often the most efficient method to detect a hidden parallelogram.

**Formula 7.69. Vector Projection**

If  $\mathbf{u} \neq \mathbf{0}$ , then the projection of  $\mathbf{v}$  onto the line spanned by  $\mathbf{u}$  is

$$\text{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}.$$

The scalar component of  $\mathbf{v}$  in the direction of  $\mathbf{u}$  is

$$\frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|}.$$

**Formula 7.70. Vector Equations of Lines and Planes**

A line through a point  $\mathbf{p}$  with direction vector  $\mathbf{v} \neq \mathbf{0}$  is

$$\mathbf{x} = \mathbf{p} + t\mathbf{v} \quad (t \in \mathbb{R}).$$

A plane through a point  $\mathbf{p}$  with normal vector  $\mathbf{n} \neq \mathbf{0}$  is

$$(\mathbf{x} - \mathbf{p}) \cdot \mathbf{n} = 0.$$

**Formula 7.71. Angles and Volumes in  $\mathbb{R}^3$** 

If nonzero vectors  $\mathbf{u}, \mathbf{v}$  are directions of two lines, their acute angle  $\theta$  satisfies

$$\cos \theta = \frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

If  $\mathbf{n}_1, \mathbf{n}_2$  are normals to two planes, the acute angle between the planes is given by the same formula with the normals. If  $\mathbf{v}$  is a line direction and  $\mathbf{n}$  is a plane normal, the acute line-plane angle  $\alpha$  satisfies

$$\sin \alpha = \frac{|\mathbf{v} \cdot \mathbf{n}|}{\|\mathbf{v}\| \|\mathbf{n}\|}.$$

The volume of the parallelepiped spanned by  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  is

$$|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|,$$

and the tetrahedron with the same three edge vectors from one vertex has one sixth of this volume.

**Formula 7.72. Distance between Skew Lines**

Let

$$L_1 : \mathbf{x} = \mathbf{p}_1 + t\mathbf{u}, \quad L_2 : \mathbf{x} = \mathbf{p}_2 + s\mathbf{v},$$

with  $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$ . Then

$$\text{dist}(L_1, L_2) = \frac{|(\mathbf{p}_2 - \mathbf{p}_1) \cdot (\mathbf{u} \times \mathbf{v})|}{\|\mathbf{u} \times \mathbf{v}\|}.$$

**Formula 7.73. Distance from a Point to a Line**

In  $\mathbb{R}^3$ , the distance from a point  $\mathbf{p}_0$  to the line

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}$$

is

$$\frac{\|(\mathbf{p}_0 - \mathbf{p}) \times \mathbf{v}\|}{\|\mathbf{v}\|}.$$

In  $\mathbb{R}^2$ , the distance from  $(x_0, y_0)$  to the line  $ax + by + c = 0$  is

$$\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

**Formula 7.74. Distance from a Point to a Plane**

The distance from a point  $\mathbf{p}_0 = (x_0, y_0, z_0)$  to the plane

$$ax + by + cz + d = 0$$

is

$$\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

**Formula 7.75. Sphere Equation**

A sphere with center  $\mathbf{c} = (a, b, c)$  and radius  $r$  has equation

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

Equivalently,

$$\|\mathbf{x} - \mathbf{c}\| = r.$$

**Formula 7.76. Standard Forms of Common Quadrics in  $\mathbb{R}^3$** 

With  $a, b, c > 0$ , the principal-axis standard forms include:

| Surface               | Standard form                                              |
|-----------------------|------------------------------------------------------------|
| Ellipsoid             | $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  |
| Elliptic paraboloid   | $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$                    |
| Hyperbolic paraboloid | $z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$                    |
| One-sheet hyperboloid | $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$  |
| Two-sheet hyperboloid | $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ |
| Elliptic cone         | $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$  |
| Elliptic cylinder     | $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$                    |

More generally, a quadric can be written

$$\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c_0 = 0,$$

with  $\mathbf{A}$  real symmetric. Translation and orthogonal diagonalization of  $\mathbf{A}$  reduce the quadratic part to principal axes, linking the geometry directly to the spectral theorem.

*Strategy. Line–Circle Intersection by a Discriminant.* Substitute a parametric line into a circle equation. The resulting quadratic equation has two, one, or no real solutions according as its discriminant is positive, zero, or negative. The zero-discriminant case is the tangency condition.

**Formula 7.77. Circle Equation**

The circle with center  $(a, b)$  and radius  $r$  has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$

The general equation

$$x^2 + y^2 + Dx + Ey + F = 0$$

can be converted to center-radius form by completing the square.

**Theorem 7.78. Three Perpendiculars Theorem**

Let  $\Pi$  be a plane, let  $P \notin \Pi$ , and let  $H$  be the orthogonal projection of  $P$  onto  $\Pi$ . Let  $Q \in \Pi$  with  $Q \neq H$ , and let  $\ell \subseteq \Pi$  be a line through  $Q$ . If  $HQ \perp \ell$ , then

$$PQ \perp \ell.$$

*Equivalently, a line in a plane that is perpendicular to the orthogonal projection of an oblique line is perpendicular to the oblique line itself.*

*Remark.* In coordinate geometry, the three perpendiculars theorem is often replaced by dot products and projections. Still, it is a standard spatial-geometry fact from the classical school curriculum.

**Definition 7.79. Conic Section**

A conic section is a curve obtained by intersecting a plane with a double cone. The nondegenerate conics are:

$$\text{ellipse,} \quad \text{parabola,} \quad \text{hyperbola.}$$

Equivalently, a conic can be described using a focus, a directrix, and an eccentricity  $e$ .

**Formula 7.80. Focus–Directrix Definition of Conics**

A conic is the set of points  $P$  satisfying

$$\frac{\text{dist}(P, \text{focus})}{\text{dist}(P, \text{directrix})} = e.$$

It is an ellipse if

$$0 < e < 1,$$

a parabola if

$$e = 1,$$

and a hyperbola if

$$e > 1.$$

**Formula 7.81. Standard Forms of Conics**

After translation and, when necessary, rotation of coordinates, every nondegenerate real conic can be put into one of the following standard forms.

| Conic     | Standard form                                                                  | Basic condition and feature                              |
|-----------|--------------------------------------------------------------------------------|----------------------------------------------------------|
| Circle    | $(x - h)^2 + (y - k)^2 = r^2$                                                  | center $(h, k)$ , radius $r > 0$                         |
| Ellipse   | $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1, a \geq b > 0$              | $c^2 = a^2 - b^2$ , foci at distance $c$ from the center |
| Parabola  | $(y - k)^2 = 4p(x - h)$ or $(x - h)^2 = 4p(y - k)$                             | focus–directrix distance parameter $p \neq 0$            |
| Hyperbola | $\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$ or with the signs reversed | $c^2 = a^2 + b^2$ , foci at distance $c$ from the center |

The general second-degree equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

is classified, in the nondegenerate case, by the quadratic part; in particular  $B^2 - 4AC < 0, = 0$ , and  $> 0$  correspond to ellipse-type, parabola-type, and hyperbola-type conics, respectively.

**Formula 7.82. Eccentricity of an Ellipse and a Hyperbola**

For an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a \geq b > 0),$$

the eccentricity is

$$e = \frac{c}{a}, \quad c^2 = a^2 - b^2.$$

For a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

the eccentricity is

$$e = \frac{c}{a}, \quad c^2 = a^2 + b^2.$$

**Formula 7.83. Tangent Lines to Standard Conics**

At a point  $(x_1, y_1)$  on the indicated conic, the tangent line is

$$xx_1 + yy_1 = r^2 \quad \text{for } x^2 + y^2 = r^2,$$

$$yy_1 = 2a(x + x_1) \quad \text{for } y^2 = 4ax,$$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \quad \text{for } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

and

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \text{for } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

**Theorem 7.84. Reflection Property of Conics**

*At a point of a nondegenerate conic, the tangent makes equal angles with the two relevant focal directions. Consequently, a ray emitted from one focus of an ellipse reflects toward the other focus; a ray parallel to the axis of a parabola reflects through its focus; and a ray directed toward one focus of a hyperbola reflects along a ray whose backward extension passes through the other focus.*

*Remark.* The reflection property gives a geometric reason why conics appear in optics, satellite dishes, whispering galleries, and orbital mechanics.

**Theorem 7.85. Quadratic Classification of Conics**

*Consider a real quadratic equation*

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

*The discriminant*

$$B^2 - 4AC$$

*determines the type of the nondegenerate conic:*

$$B^2 - 4AC < 0 \quad \Rightarrow \quad \text{ellipse-type},$$

$$B^2 - 4AC = 0 \quad \Rightarrow \quad \text{parabola-type},$$

*and*

$$B^2 - 4AC > 0 \quad \Rightarrow \quad \text{hyperbola-type}.$$

*Remark.* This is the analytic-geometry counterpart of classifying conic sections by their shapes.

**Formula 7.86. Polar Equation of a Conic**

With a focus at the pole, a conic has polar equation

$$r = \frac{\ell}{1 + e \cos \theta}$$

after choosing a suitable axis. Here  $e$  is the eccentricity and  $\ell$  is the semi-latus rectum.

*Remark.* The same equation describes ellipses, parabolas, and hyperbolas depending on whether  $e < 1$ ,  $e = 1$ , or  $e > 1$ .

**Definition 7.87. Apollonius Circle**

Given two fixed points  $A$  and  $B$  and a positive constant  $k \neq 1$ , the locus of points  $P$  satisfying

$$\frac{PA}{PB} = k$$

is a circle. This circle is called an Apollonius circle.

*Remark.* Apollonius circles turn a ratio of distances into a concrete circle, which is a common hidden structure in geometry problems.

**Theorem 7.88. Descartes' Circle Theorem; Soddy Circle Theorem**

Suppose four circles are mutually tangent, and let their curvatures be

$$k_1, k_2, k_3, k_4,$$

where curvature means the reciprocal of the radius, with sign convention for an enclosing circle. Then

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

*Remark.* Descartes' Circle Theorem is a striking formula: tangency of four circles becomes one quadratic equation in their curvatures.

*Remark. Boundary Lattice Points for Pick's Theorem.* Apply the lattice-segment formula above to each polygon edge. Summing  $\gcd(|\Delta x|, |\Delta y|)$  over all edges gives the number  $B$  of boundary lattice points without double-counting vertices.

**Theorem 7.89. Pick's Theorem**

Let  $P$  be a simple polygon whose vertices lie on lattice points in the plane. If  $I$  is the number of lattice points strictly inside  $P$  and  $B$  is the number of lattice points on the boundary of  $P$ , then

$$\text{area}(P) = I + \frac{B}{2} - 1.$$

*Remark.* Pick's Theorem is a beautiful bridge between geometry and arithmetic: area is determined by counting lattice points.

## Chapter 8

# Number Theory

Related **POSTECH** Courses: (MATH304) Introduction to Number Theory

Related Books: Introduction to Number Theory (William W. Adams, Larry Joel Goldstein)

### Definition 8.1. Divisibility, Prime, Greatest Common Divisor, and Congruence

For  $a, b \in \mathbb{Z}$ , write  $a \mid b$  if

$$\exists k \in \mathbb{Z}, \quad b = ak.$$

An integer  $p > 1$  is prime if its only positive divisors are 1 and  $p$ . For integers  $a, b$ , not both zero,  $\gcd(a, b)$  is their greatest positive common divisor. For  $n \in \mathbb{N}$ ,

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

### Theorem 8.2. Fundamental Theorem of Arithmetic

*Every integer  $n > 1$  can be written as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. Equivalently,*

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where  $p_1, \dots, p_k$  are distinct primes and  $\alpha_1, \dots, \alpha_k$  are positive integers, and this expression is unique up to reordering.

*Remark.* This theorem is often called unique prime factorization. It is the reason prime numbers are treated as the basic building blocks of integers.

### Theorem 8.3. Euclid's Theorem; Infinitude of Primes

*There are infinitely many prime numbers.*

*Idea.* If  $p_1, \dots, p_n$  are finitely many primes, consider

$$p_1 p_2 \cdots p_n + 1.$$

This number is not divisible by any prime on the list.

### Theorem 8.4. Brun's Theorem

*The reciprocal sum over twin-prime pairs converges:*

$$\sum_{\substack{p \text{ prime} \\ p+2 \text{ prime}}} \left( \frac{1}{p} + \frac{1}{p+2} \right) < \infty.$$

*Its value is called Brun's constant. This contrasts with Euler's theorem that the sum of the reciprocals of all primes diverges.*

**Theorem 8.5. Bézout's Identity; Bézout's Lemma**

Let  $a, b \in \mathbb{Z}$ , not both zero. Then there exist integers  $x, y \in \mathbb{Z}$  such that

$$ax + by = \gcd(a, b).$$

In particular, if  $\gcd(a, b) = 1$ , then there exist integers  $x, y \in \mathbb{Z}$  such that

$$ax + by = 1.$$

*Remark.* Bézout's identity is the algebraic heart of the Euclidean algorithm and modular inverses.

**Algorithm 8.6. Euclidean Algorithm**

To compute  $\gcd(a, b)$  for integers  $a, b \in \mathbb{Z}$ , not both zero, begin with  $x = \max\{|a|, |b|\}$  and  $y = \min\{|a|, |b|\}$ . While  $y > 0$ , write

$$x = qy + r, \quad 0 \leq r < y,$$

and replace  $(x, y)$  by  $(y, r)$ . When  $y = 0$ , the current value of  $x$  is  $\gcd(a, b)$ . In particular, if one input is zero, then  $\gcd(a, 0) = |a|$ .

**Theorem 8.7. GCD of Exponential Expressions**

For every integer  $a > 1$  and positive integers  $m, n$ ,

$$\gcd(a^m - 1, a^n - 1) = a^{\gcd(m, n)} - 1.$$

The proof applies the Euclidean algorithm to the exponents. Expressions involving  $a^m + 1$  require separate parity conditions.

**Definition 8.8. Gaussian Integers**

The Gaussian integers are

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}.$$

Their norm is

$$N(a + bi) = a^2 + b^2,$$

and it is multiplicative:

$$N(zw) = N(z)N(w).$$

With the usual addition and multiplication,  $\mathbb{Z}[i]$  inherits its arithmetic from  $\mathbb{C}$ . In the terminology of the Algebra chapter,  $\mathbb{Z}[i]$  is a Euclidean domain with respect to this norm, so a Euclidean algorithm, greatest common divisors, and unique factorization are available in  $\mathbb{Z}[i]$  just as they are in  $\mathbb{Z}$ .

**Proposition 8.9. Basic Rules of Congruences**

If

$$a \equiv b \pmod{n}, \quad c \equiv d \pmod{n},$$

then

$$a + c \equiv b + d \pmod{n}, \quad ac \equiv bd \pmod{n}.$$

If  $\gcd(c, n) = 1$ , then

$$ac \equiv bc \pmod{n} \implies a \equiv b \pmod{n}.$$

**Theorem 8.10. Linear Congruence Criterion**

The congruence

$$ax \equiv b \pmod{n}$$

has a solution if and only if

$$\gcd(a, n) \mid b.$$

If  $d = \gcd(a, n)$  and  $d \mid b$ , then there are exactly  $d$  solutions modulo  $n$ .

**Formula 8.11. Least Common Multiple from Prime Factorization**

If

$$a = \prod_p p^{\alpha_p}, \quad b = \prod_p p^{\beta_p},$$

then

$$\gcd(a, b) = \prod_p p^{\min(\alpha_p, \beta_p)}, \quad \text{lcm}(a, b) = \prod_p p^{\max(\alpha_p, \beta_p)}.$$

More generally, the least common multiple of several integers is found by taking the largest exponent of each prime appearing in the list.

**Formula 8.12. Basic Divisibility Tests**

For a decimal integer  $N$ , the following tests are useful.

- Divisibility by 2, 4, and 8 depends on the last 1, 2, and 3 digits, respectively.
- Divisibility by 3 and 9 depends on the digit sum.
- Divisibility by 5 depends on the last digit.
- Divisibility by 11 depends on the alternating sum of the digits.

*Strategy. Digit problems often combine an lcm condition, a digit-sum condition, and a last-digits condition. First minimize the number of digits, then arrange the final few digits to satisfy the strongest divisibility conditions.*

**Definition 8.13. Complete Residue System**

A set of  $n$  integers is a complete residue system modulo  $n$  if its elements represent every residue class modulo  $n$  exactly once. To prove that  $a_0, \dots, a_{n-1}$  form such a system, it is enough to prove

$$a_i \equiv a_j \pmod{n} \implies i = j.$$

*Strategy. Modular Casework and Base Representation.* When a problem repeats division by  $N$ , rewrite the integer in base  $N$ . The quotient deletes the final digit and the remainder reveals it. For divisibility or existence problems, first split integers into residue classes and look for a complete residue system rather than checking values individually.

*Strategy. Infinite Descent.* Assume a positive-integer solution exists and choose one minimizing a natural positive quantity. Construct a strictly smaller solution of the same type, contradicting well-ordering.

*Strategy. Vieta Jumping.* If an integer equation is quadratic in one variable, regard that variable as one root of a monic quadratic. Viète's formulas give the other integral root. Prove that it yields a smaller positive solution and apply infinite descent.

**Theorem 8.14. Frobenius Coin Theorem for Two Denominations**

If  $a$  and  $b$  are coprime positive integers, the largest integer not representable as

$$ax + by, \quad x, y \in \mathbb{N}_0,$$

is  $ab - a - b$ . Exactly

$$\frac{(a-1)(b-1)}{2}$$

positive integers are not representable.

**Theorem 8.15. Chinese Remainder Theorem; CRT**

Let  $m_1, \dots, m_k$  be pairwise relatively prime positive integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \quad x \equiv a_2 \pmod{m_2}, \quad \dots, \quad x \equiv a_k \pmod{m_k}$$

has a solution. Moreover, the solution is unique modulo

$$M = m_1 m_2 \cdots m_k.$$

*Strategy.* CRT is the standard theorem for combining several congruence conditions into one congruence.

**Definition 8.16. Arithmetic and Multiplicative Functions**

An arithmetic function is a function  $f: \mathbb{N} \rightarrow \mathbb{C}$ . It is multiplicative if

$$\gcd(m, n) = 1 \implies f(mn) = f(m)f(n).$$

Important multiplicative arithmetic functions include Euler's totient  $\varphi(n)$ , the divisor-counting function  $\tau(n)$ , the divisor-sum function  $\sigma(n)$ , and the Möbius function  $\mu(n)$ . Multiplicativity reduces their evaluation to prime powers.

**Formula 8.17. Euler's Phi Function Formula**

If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

is the prime factorization of  $n$ , then

$$\varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

In particular, for a prime power,

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1).$$

**Formula 8.18. Sum-of-Divisors Function**

If

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

then the number of positive divisors of  $n$  is

$$\tau(n) = (\alpha_1 + 1) \cdots (\alpha_k + 1),$$

and the sum of positive divisors of  $n$  is

$$\sigma(n) = \prod_{i=1}^k \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

**Definition 8.19.  $p$ -Adic Valuation,  $p$ -Adic Absolute Value, and  $p$ -Adic Numbers**

Fix a prime  $p$ . For a nonzero rational number  $x$ , write uniquely

$$x = p^{v_p(x)} \frac{a}{b}, \quad p \nmid ab,$$

where  $v_p(x) \in \mathbb{Z}$  is the  $p$ -adic valuation of  $x$ . For a nonzero integer,  $v_p(n)$  is therefore the exponent of  $p$  in its prime factorization. Define

$$|x|_p = p^{-v_p(x)}, \quad |0|_p = 0.$$

The  $p$ -adic absolute value satisfies the strong triangle inequality

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

The field  $\mathbb{Q}_p$  of  $p$ -adic numbers is the completion of  $\mathbb{Q}$  with respect to this absolute value, just as  $\mathbb{R}$  is a completion of  $\mathbb{Q}$  with respect to the usual absolute value.

**Formula 8.20.  $p$ -Adic Valuation of a Factorial; Legendre's Formula**

For a prime  $p$  and a positive integer  $n$ ,

$$v_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Only finitely many terms are nonzero. This formula gives the exponent of  $p$  in  $n!$  and is especially useful for trailing-zero and divisibility questions.

**Definition 8.21. Möbius Function**

The Möbius function is

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & p^2 \mid n \text{ for some prime } p, \\ (-1)^k, & n \text{ is a product of } k \text{ distinct primes.} \end{cases}$$

It satisfies

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

**Theorem 8.22. Möbius Inversion Formula**

If arithmetic functions  $f$  and  $g$  satisfy

$$g(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right).$$

**Theorem 8.23. Fermat's Little Theorem**

Let  $p$  be a prime number. If  $p \nmid a$ , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer  $a$ ,

$$a^p \equiv a \pmod{p}.$$

**Theorem 8.24. Euler's Theorem; Euler–Fermat Theorem**

If  $\gcd(a, n) = 1$ , then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

*Remark.* Fermat's Little Theorem is the special case of Euler's Theorem when  $n = p$  is prime, since  $\varphi(p) = p - 1$ .

**Proposition 8.25. Multiplicative Order**

If  $\gcd(a, n) = 1$ , the multiplicative order  $\text{ord}_n(a)$  is the least positive integer  $t$  such that

$$a^t \equiv 1 \pmod{n}.$$

The invertible residue classes modulo  $n$  form a set of size  $\varphi(n)$  and, in the group terminology of the Algebra chapter, Lagrange's theorem gives

$$\text{ord}_n(a) \mid \varphi(n).$$

Consequently, for every positive integer  $k$ ,

$$a^k \equiv 1 \pmod{n} \iff \text{ord}_n(a) \mid k.$$

**Definition 8.26. Primitive Root Modulo  $n$**

If  $\gcd(g, n) = 1$ , then  $g$  is a primitive root modulo  $n$  if

$$\text{ord}_n(g) = \varphi(n).$$

Equivalently, in the group terminology of the Algebra chapter, the residue class  $[g]_n$  generates  $(\mathbb{Z}/n\mathbb{Z})^\times$ .

**Theorem 8.27. Primitive Root Theorem; Gauss's Theorem on Primitive Roots**

With the convention that the trivial group  $(\mathbb{Z}/1\mathbb{Z})^\times$  is cyclic, a primitive root modulo  $n$  exists if and only if

$$n \in \{1, 2, 4, p^k, 2p^k\},$$

where  $p$  is an odd prime and  $k \geq 1$ . Equivalently, in the language of the Algebra chapter,  $(\mathbb{Z}/n\mathbb{Z})^\times$  is cyclic exactly for these values of  $n$ .

**Definition 8.28. Finite Field**

A finite field is a finite set equipped with addition and multiplication satisfying the field axioms stated in the Algebra chapter. Its number of elements is called its order.

**Theorem 8.29. Existence and Uniqueness of Finite Fields**

Every finite field has  $q = p^n$  elements for some prime  $p$  and positive integer  $n$ . Conversely, for each prime power  $q = p^n$ , there exists a field  $\mathbb{F}_q$  with  $q$  elements, unique up to isomorphism. Its nonzero elements form a cyclic multiplicative group  $\mathbb{F}_q^\times$  of order  $q - 1$ .

**Proposition 8.30. Power Maps on Finite Cyclic Groups**

Let  $G$  be a finite cyclic group of order  $N$ , and let  $r \in \mathbb{N}$ . The map

$$G \rightarrow G, \quad x \mapsto x^r,$$

is bijective if and only if  $\gcd(r, N) = 1$ . Consequently,  $x \mapsto x^r$  permutes  $\mathbb{F}_q^\times$  if and only if  $\gcd(r, q - 1) = 1$ . The same condition characterizes when it permutes the whole field  $\mathbb{F}_q$ .

**Algorithm 8.31. Modular Exponentiation by Period Reduction**

To compute  $a^n \pmod{m}$ , first reduce  $a$  modulo  $m$ . If  $\gcd(a, m) = 1$ , determine a convenient exponent  $t > 0$  such that

$$a^t \equiv 1 \pmod{m},$$

for example from the multiplicative order of  $a$ , Euler's theorem, or Fermat's little theorem when applicable. Write

$$n = qt + r, \quad 0 \leq r < t,$$

and reduce

$$a^n \equiv a^r \pmod{m}.$$

If  $a$  is not a unit modulo  $m$ , use repeated squaring directly or analyze prime-power factors separately rather than applying a unit-group period without justification.

**Algorithm 8.32. Textbook RSA**

Choose distinct primes  $p, q$  and set

$$N = pq, \quad \varphi(N) = (p-1)(q-1).$$

Choose  $e$  with  $\gcd(e, \varphi(N)) = 1$ , and find  $d$  such that

$$ed \equiv 1 \pmod{\varphi(N)}.$$

The public key is  $(N, e)$  and the private exponent is  $d$ . Textbook encryption and decryption are

$$c \equiv m^e \pmod{N}, \quad m \equiv c^d \pmod{N}.$$

Practical RSA uses randomized padding and must not use this unpadded form directly.

*Strategy.* Last-digit and last-two-digit questions are modular arithmetic problems. Work modulo 10 or 100, not with the full number.

**Theorem 8.33. Wilson's Theorem**

A positive integer  $p > 1$  is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

*Remark.* Wilson's theorem is famous because it gives a clean factorial characterization of primes, even though it is not efficient as a primality test.

**Theorem 8.34. Lucas's Theorem**

Let  $p$  be prime and write

$$n = \sum_{i=0}^r n_i p^i, \quad k = \sum_{i=0}^r k_i p^i$$

in base  $p$ . Then

$$\binom{n}{k} \equiv \prod_{i=0}^r \binom{n_i}{k_i} \pmod{p}.$$

**Theorem 8.35. Kummer's Theorem**

The valuation

$$v_p \left( \binom{n}{k} \right)$$

equals the number of carries when  $k$  and  $n - k$  are added in base  $p$ .

**Theorem 8.36. Lifting the Exponent Lemma; LTE**

Let  $p$  be an odd prime, and suppose that  $p \mid x - y$  while  $p \nmid xy$ . Then, for every positive integer  $n$ ,

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

A standard companion form is: if  $n$  is odd,  $p \mid x + y$ , and  $p \nmid xy$ , then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

**Fact 8.37. Small-Modulus Residue Filters**

Small powers have restricted residues modulo small integers:

$$x^2 \equiv 0, 1 \pmod{4}, \quad x^2 \equiv 0, 1, 4 \pmod{8}.$$

Also,

$$x^3 \equiv 0, \pm 1 \pmod{9}, \quad x^4 \equiv 0, 1 \pmod{16}.$$

In particular, if  $x$  is odd, then  $x^2 \equiv 1 \pmod{8}$ . Hence an integer congruent to 2 or 3 modulo 4 cannot be a square, and an integer congruent to 2, 3, 5, 6, 7 modulo 8 cannot be a square. These tests can often rule out a proposed square, cube, or fourth power before any large calculation is attempted.

*Tactic.* For Diophantine equations, first reduce modulo 4, 8, 3, 5, or 9. Many impossible equations are eliminated by the small list of possible square residues.

**Definition 8.38. Legendre Symbol**

Let  $p$  be an odd prime. The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a nonzero quadratic residue modulo } p, \\ -1, & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0, & \text{if } p \mid a. \end{cases}$$

**Theorem 8.39. Euler's Criterion**

Let  $p$  be an odd prime. For any integer  $a$ ,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Equivalently, if  $p \nmid a$ , then

$$a^{(p-1)/2} \equiv \begin{cases} 1 \pmod{p}, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 \pmod{p}, & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

**Theorem 8.40. Law of Quadratic Reciprocity; Gauss's Theorem of Quadratic Reciprocity**

Let  $p$  and  $q$  be distinct odd primes. Then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless both  $p$  and  $q$  are congruent to 3 modulo 4.

*Remark.* Quadratic reciprocity is one of the crown jewels of elementary number theory. Gauss called it the golden theorem.

**Theorem 8.41. Fermat's Two-Square Theorem; Fermat's Theorem on Sums of Two Squares**

*An odd prime  $p$  can be written as a sum of two squares,*

$$p = a^2 + b^2$$

*for some integers  $a, b \in \mathbb{Z}$ , if and only if*

$$p \equiv 1 \pmod{4}.$$

*Also,*

$$2 = 1^2 + 1^2.$$

**Theorem 8.42. Sum of Two Squares Criterion**

*A positive integer*

$$n = 2^a \prod_{p \equiv 1 \pmod{4}} p^{\alpha_p} \prod_{q \equiv 3 \pmod{4}} q^{\beta_q}$$

*is representable as  $n = x^2 + y^2$  with integers  $x, y$  if and only if every exponent  $\beta_q$  is even.*

**Formula 8.43. Brahmagupta–Fibonacci Identity**

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2 = (ac + bd)^2 + (ad - bc)^2.$$

**Formula 8.44. Sophie Germain's Identity**

$$a^4 + 4b^4 = (a^2 - 2ab + 2b^2)(a^2 + 2ab + 2b^2).$$

**Theorem 8.45. Lagrange's Four-Square Theorem**

*Every nonnegative integer can be written as a sum of four integer squares. That is, for every  $n \in \mathbb{N}_0$ , there exist integers  $a, b, c, d \in \mathbb{Z}$  such that*

$$n = a^2 + b^2 + c^2 + d^2.$$

*Remark.* The two-square theorem has a congruence condition. The four-square theorem has no exception at all.

**Theorem 8.46. Continued-Fraction Best Approximation Theorem**

*Let  $p_k/q_k$  be the convergents of the simple continued fraction of an irrational number  $\alpha$ . Then*

$$\left| \alpha - \frac{p_k}{q_k} \right| < \frac{1}{q_k q_{k+1}}.$$

*Moreover, the convergents are best approximations of the second kind: if  $p, q \in \mathbb{Z}$ ,  $0 < q < q_{k+1}$ , and  $p/q \neq p_k/q_k$ , then*

$$|q_k \alpha - p_k| < |q \alpha - p|.$$

*Here “second kind” refers to minimizing  $|q\alpha - p|$ , rather than the absolute error  $|\alpha - p/q|$  itself.*

*Strategy.* Problems asking for an unusually accurate fraction with the smallest possible denominator often signal a continued-fraction expansion.

**Definition 8.47. Pell Equation**

A Pell equation has the form

$$x^2 - Dy^2 = 1,$$

where  $D$  is a positive nonsquare integer. If  $(x_1, y_1)$  is the fundamental positive solution, then all positive solutions satisfy

$$x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n.$$

Continued fractions of  $\sqrt{D}$  produce the fundamental solution.

**Theorem 8.48. Dirichlet's Theorem on Arithmetic Progressions**

If  $a$  and  $d$  are relatively prime positive integers, then the arithmetic progression

$$a, a + d, a + 2d, a + 3d, \dots$$

contains infinitely many primes.

*Remark.* Dirichlet's theorem is a major result connecting primes and arithmetic progressions. The condition  $\gcd(a, d) = 1$  is necessary.

**Theorem 8.49. Bertrand's Postulate; Chebyshev's Theorem**

For every integer  $n > 1$ , there exists a prime  $p$  such that

$$n < p < 2n.$$

**Theorem 8.50. Prime Number Theorem**

Let  $\pi(x)$  be the number of primes less than or equal to  $x$ . Then

$$\pi(x) \sim \frac{x}{\ln x} \quad \text{as } x \rightarrow \infty.$$

*Idea.* The Prime Number Theorem says that the “probability” that a large integer near  $x$  is prime is roughly  $1/\ln x$ . This is a mental model, not a literal probability statement.

**Definition 8.51. Riemann Zeta Function**

For  $\operatorname{Re}(s) > 1$ , the Riemann zeta function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It also has the Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

**Fact 8.52. Analytic Continuation and Special Values of the Zeta Function**

The Riemann zeta function extends meromorphically to the complex plane, with a single simple pole at  $s = 1$ . Its functional equation may be written symmetrically as

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \pi^{-(1-s)/2} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s).$$

Important special values include

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}, \quad \zeta(-2m) = 0 \quad (m \in \mathbb{N}).$$

The value  $\zeta(-1) = -1/12$  comes from analytic continuation; it is not the ordinary sum of  $1 + 2 + 3 + \dots$ .

*Remark.* The Euler product is the bridge between the zeta function and prime numbers. The Riemann Hypothesis asserts that the nontrivial zeros of  $\zeta(s)$  all have real part  $1/2$ .

**Definition 8.53. Perfect Number and Mersenne Prime**

A positive integer  $N$  is perfect if it equals the sum of its positive proper divisors, equivalently

$$\sigma(N) = 2N.$$

A Mersenne prime is a prime of the form

$$2^p - 1.$$

If  $2^p - 1$  is prime, then  $p$  must be prime.

**Theorem 8.54. Euclid–Euler Theorem on Perfect Numbers**

*An even positive integer  $N$  is perfect if and only if*

$$N = 2^{p-1}(2^p - 1),$$

*where  $2^p - 1$  is a Mersenne prime.*

**Theorem 8.55. Fermat’s Last Theorem**

*For every integer  $n \geq 3$ , there do not exist positive integers  $a, b, c$  such that*

$$a^n + b^n = c^n.$$

*Remark.* Fermat’s Last Theorem was proved by Andrew Wiles. It is one of the most famous theorems in the history of mathematics.

## Chapter 9

# Algebra

Related **POSTECH** Courses: (MATH301) Modern Algebra I, (MATH302) Modern Algebra II  
Related Books: Abstract Algebra (David S. Dummit, Richard M. Foote)

### Definition 9.1. Group

A group is a set  $G$  with a binary operation  $*$ :  $G \times G \rightarrow G$  such that

$$\forall a, b, c \in G, \quad (a * b) * c = a * (b * c),$$

$$\exists e \in G \text{ s.t. } \forall a \in G, \quad e * a = a * e = a,$$

and

$$\forall a \in G, \quad \exists a^{-1} \in G \text{ s.t. } a * a^{-1} = a^{-1} * a = e.$$

The group is abelian if

$$\forall a, b \in G, \quad a * b = b * a.$$

### Definition 9.2. Subgroup, Homomorphism, and Kernel

A subset  $H \subseteq G$  is a subgroup, written  $H \leq G$ , if it is a group under the operation of  $G$ . Writing the group operations multiplicatively, a map  $\varphi: G \rightarrow K$  is a group homomorphism if

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \forall a, b \in G.$$

Its kernel and image are

$$\ker \varphi = \{g \in G \mid \varphi(g) = e_K\}, \quad \text{im } \varphi = \{\varphi(g) \mid g \in G\}.$$

A bijective homomorphism is an *isomorphism*. An isomorphism  $G \rightarrow G$  is an *automorphism*. Isomorphic groups have the same group-theoretic structure up to relabeling.

### Definition 9.3. Generated Subgroup, Generating Set, and Finitely Generated Group

If  $A \subseteq G$ , the subgroup generated by  $A$ , denoted  $\langle A \rangle$ , is the smallest subgroup of  $G$  containing  $A$ . Equivalently, it consists of all finite products of elements of  $A$  and their inverses. The set  $A$  is a *generating set* for  $G$  if  $G = \langle A \rangle$ . The group  $G$  is *finitely generated* if it has a finite generating set. For a single element  $g$ , the notation  $\langle g \rangle$  denotes the cyclic subgroup generated by  $g$ .

### Proposition 9.4. Countability of Finitely Generated Groups

*Every finitely generated group is at most countable. Every finite group is finitely generated. Consequently, an uncountable group cannot be finitely generated.*

**Definition 9.5. Normal Subgroup and Simple Group**

A subgroup  $N \leq G$  is normal, written  $N \trianglelefteq G$ , if

$$gNg^{-1} = N \quad \text{for every } g \in G.$$

A nontrivial group  $G$  is simple if its only normal subgroups are  $\{e\}$  and  $G$ .

**Definition 9.6. Coset and Index**

If  $H \leq G$  and  $g \in G$ , the left and right cosets of  $H$  represented by  $g$  are

$$gH = \{gh \mid h \in H\}, \quad Hg = \{hg \mid h \in H\}.$$

The number of left cosets is the index of  $H$  in  $G$ , denoted  $[G : H]$ . For finite groups,

$$[G : H] = \frac{|G|}{|H|}.$$

**Definition 9.7. Quotient Group**

If  $N \trianglelefteq G$ , the quotient group  $G/N$  is the set of left cosets of  $N$ ,

$$G/N = \{gN \mid g \in G\},$$

with multiplication

$$(gN)(hN) = ghN.$$

Normality is exactly the condition that makes this multiplication well-defined.

**Definition 9.8. Order of a Group and of an Element**

The order of a finite group  $G$  is its cardinality  $|G|$ . The order of an element  $g \in G$ , denoted  $\text{ord}(g)$ , is the least positive integer  $m$  such that  $g^m = e$ , if such an integer exists; otherwise  $g$  has infinite order. Equivalently,

$$\text{ord}(g) = |\langle g \rangle|.$$

**Definition 9.9. Permutation, Symmetric Group, and Alternating Group**

A permutation of  $\{1, \dots, n\}$  is a bijection from the set to itself. Under composition, all such permutations form the symmetric group  $S_n$ . A transposition swaps two elements and fixes the others. A permutation is *even* if it is a product of an even number of transpositions and *odd* otherwise. The even permutations form the alternating group  $A_n \leq S_n$ , with

$$|S_n| = n!, \quad |A_n| = \frac{n!}{2} \quad (n \geq 2).$$

**Fact 9.10. Smallest Nonabelian Simple Group**

The alternating group  $A_5$  is simple and has order

$$|A_5| = 60.$$

It is the smallest nonabelian simple group. This fact, together with the Sylow theorems, is a standard shortcut in questions asking which small integers can occur as the order of a simple group.

**Theorem 9.11. Disjoint-Cycle Decomposition of a Permutation**

Every permutation  $\sigma \in S_n$  can be written as a product of disjoint cycles, uniquely up to the order of the cycles and cyclic rotation within each cycle. If  $c(\sigma)$  is the number of cycles, including fixed points, then the minimum number of transpositions needed to express  $\sigma$  is

$$n - c(\sigma).$$

**Formula 9.12. Inversions and the Sign of a Permutation**

The inversion number is

$$\text{inv}(\sigma) = |\{(i, j) \mid i < j, \sigma(i) > \sigma(j)\}|.$$

The sign satisfies

$$\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)} = (-1)^{n-c(\sigma)}.$$

Thus any transposition reverses parity.

**Definition 9.13. Group Presentation**

A group presentation

$$G = \langle S \mid R \rangle$$

specifies a generating set  $S$  together with defining relations  $R$ . Presentations give compact descriptions of groups whose elements may be difficult to list explicitly.

**Fact 9.14. Dihedral Group**

The symmetries of a regular  $n$ -gon form the dihedral group

$$D_{2n} = \langle r, s \mid r^n = e, s^2 = e, srs = r^{-1} \rangle.$$

Its elements are

$$e, r, r^2, \dots, r^{n-1}, \quad s, rs, r^2s, \dots, r^{n-1}s.$$

This presentation is the standard algebraic model for rotation and reflection counting with Burnside's Lemma.

**Definition 9.15. Cyclic Group**

A group  $G$  is cyclic if there exists an element  $g \in G$  such that every element of  $G$  is a power of  $g$ . In this case, we write

$$G = \langle g \rangle.$$

**Theorem 9.16. Lagrange's Theorem**

If  $G$  is a finite group and  $H \leq G$ , then

$$|H| \mid |G|.$$

Equivalently, the order of every subgroup divides the order of the group.

**Corollary 9.17. Order of an Element**

If  $G$  is a finite group and  $g \in G$ , then the order of  $g$  divides  $|G|$ . In particular,

$$g^{|G|} = e.$$

**Theorem 9.18. Cauchy's Theorem**

If  $G$  is a finite group and a prime  $p$  divides  $|G|$ , then  $G$  contains an element of order  $p$ .

**Corollary 9.19. Groups of Prime Order**

Every group of prime order  $p$  is cyclic and isomorphic to  $\mathbb{Z}_p$ .

**Proposition 9.20. Subgroups and Generators of a Finite Cyclic Group**

If  $G = \langle g \rangle$  has order  $n$ , then for every divisor  $d \mid n$  there is exactly one subgroup of order  $d$ . Moreover,  $G$  has exactly  $\varphi(n)$  generators, and

$$\text{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

**Theorem 9.21. First Isomorphism Theorem for Groups**

If  $\varphi: G \rightarrow H$  is a group homomorphism, then

$$G / \ker \varphi \cong \text{im } \varphi.$$

**Definition 9.22. Group Action, Orbit, and Stabilizer**

A group action of  $G$  on a set  $X$  is a map

$$G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x,$$

satisfying  $e \cdot x = x$  and  $(gh) \cdot x = g \cdot (h \cdot x)$ . For  $x \in X$ , the orbit and stabilizer are

$$G \cdot x = \{g \cdot x \mid g \in G\}, \quad G_x = \{g \in G \mid g \cdot x = x\}.$$

Group actions translate algebraic symmetry into permutations of a set.

**Theorem 9.23. Cayley's Theorem**

Every group is isomorphic to a subgroup of a symmetric group. In particular, every finite group of order  $n$  is isomorphic to a subgroup of  $S_n$ .

**Theorem 9.24. Orbit–Stabilizer Theorem**

If a finite group  $G$  acts on a set  $X$  and  $x \in X$ , then

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|},$$

where  $G \cdot x$  is the orbit of  $x$  and  $G_x$  is its stabilizer.

**Definition 9.25. Conjugacy, Center, and Centralizer**

Elements  $x, y \in G$  are conjugate if  $y = gxg^{-1}$  for some  $g \in G$ . The conjugacy class of  $x$  is

$$\text{Cl}_G(x) = \{gxg^{-1} \mid g \in G\}.$$

The center and centralizer are

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}, \quad C_G(x) = \{g \in G \mid gx = xg\}.$$

The center consists exactly of the elements whose conjugacy classes have size 1.

**Theorem 9.26. Class Equation**

Let  $G$  be a finite group, and choose one representative  $x_i$  from each conjugacy class not contained in the center  $Z(G)$ . Then

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)].$$

**Theorem 9.27. Sylow Theorems**

Let  $G$  be a finite group and suppose

$$|G| = p^k m, \quad p \nmid m,$$

where  $p$  is prime. Then  $G$  has a subgroup of order  $p^k$ , called a Sylow  $p$ -subgroup. Moreover, all Sylow  $p$ -subgroups are conjugate, and the number  $n_p$  of Sylow  $p$ -subgroups satisfies

$$n_p \equiv 1 \pmod{p}, \quad n_p \mid m.$$

**Theorem 9.28. Basic Finite  $p$ -Group Facts**

If  $G$  is a finite group of order  $p^n$  for a prime  $p$ , then its center  $Z(G)$  is nontrivial. In particular, every group of order  $p^2$  is abelian. These are standard quick consequences of the class equation.

*Strategy.* For a finite group, subgroup orders and element orders must divide the group order. When a prime divides  $|G|$ , Cauchy's theorem guarantees an element of that prime order.

**Fact 9.29. Quaternion Group**

The quaternion group is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with

$$i^2 = j^2 = k^2 = ijk = -1.$$

It is nonabelian, but every subgroup of  $Q_8$  is normal.

*Remark.* The quaternion group is a standard small group with behavior that is less intuitive than cyclic or dihedral groups, so it is a natural source of quick algebra questions.

**Definition 9.30. Direct Product and Free Group**

For groups  $G$  and  $H$ , the direct product  $G \times H$  consists of pairs  $(g, h)$  with componentwise multiplication. A free group  $F(S)$  on a set  $S$  consists of reduced words in the symbols of  $S$  and their formal inverses; it has no relations beyond the group axioms. A free group on  $n$  generators is denoted  $F_n$ .

**Definition 9.31. Semidirect Product**

Let  $N$  and  $H$  be groups, and let

$$\varphi: H \rightarrow \text{Aut}(N)$$

be a homomorphism. The semidirect product  $N \rtimes_{\varphi} H$  is the set  $N \times H$  with multiplication

$$(n_1, h_1)(n_2, h_2) = (n_1\varphi(h_1)(n_2), h_1h_2).$$

The direct product is the special case in which the action is trivial.

*Strategy.* In a finite-group classification problem, first use the Sylow theorems to locate normal Sylow subgroups. If  $N \trianglelefteq G$  and a complementary subgroup  $H$  exists, possible structures are controlled by homomorphisms  $H \rightarrow \text{Aut}(N)$ . Distinct actions can still yield isomorphic groups, so counting actions alone is not a classification.

**Theorem 9.32. Fundamental Theorem of Finitely Generated Abelian Groups**

Every finitely generated abelian group is isomorphic to

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}, \quad n_1 \mid n_2 \mid \cdots \mid n_k,$$

for uniquely determined  $r \geq 0$  and invariant factors  $n_i \geq 2$ , up to isomorphism. The integer  $r$  is the free rank. In particular, a finite abelian group has  $r = 0$  and can equivalently be written uniquely, up to reordering, as a direct product of cyclic groups of prime-power order.

**Definition 9.33. Ring and Field**

A ring is a set  $R$  with operations  $+$  and  $\cdot$  such that  $(R, +)$  is an abelian group, multiplication is associative, and multiplication distributes over addition on both sides. Throughout this book, rings are assumed to have a multiplicative identity  $1_R$  unless stated otherwise. A field is a commutative ring in which  $1_R \neq 0_R$  and every nonzero element has a multiplicative inverse.

**Definition 9.34. Unit, Zero Divisor, and Characteristic**

In a ring  $R$  with identity, a *unit* is an element  $u$  having a multiplicative inverse. A nonzero element  $a$  is a *left zero divisor* if  $ab = 0$  for some nonzero  $b$ , and a *right zero divisor* if  $ba = 0$  for some nonzero  $b$ . In a commutative ring these notions coincide and  $a$  is simply called a *zero divisor*. The characteristic of  $R$  is the least positive integer  $n$  such that

$$n1_R = 0,$$

if such an integer exists; otherwise the characteristic is 0.

**Definition 9.35. Ideal, Ring Homomorphism, and Quotient Ring**

An ideal  $I$  of a commutative ring  $R$  is an additive subgroup such that

$$r \in R \wedge a \in I \implies ra \in I.$$

A ring homomorphism  $\varphi: R \rightarrow S$  preserves addition, multiplication, and the multiplicative identity under the conventions of this book. Its kernel is an ideal. If  $I$  is an ideal of  $R$ , the quotient ring  $R/I$  consists of additive cosets with the induced ring operations.

**Definition 9.36. Integral Domain**

An integral domain is a commutative ring with identity in which there are no zero divisors. In other words, if  $ab = 0$ , then  $a = 0$  or  $b = 0$ .

**Theorem 9.37. Every Finite Integral Domain Is a Field**

Every finite integral domain is a field. Indeed, for each nonzero  $a$ , multiplication by  $a$  is injective and hence, on a finite set, surjective; therefore some element maps to 1.

**Definition 9.38. Irreducible and Prime Elements**

Let  $R$  be an integral domain. A nonzero nonunit  $p \in R$  is *irreducible* if  $p = ab$  implies that  $a$  or  $b$  is a unit. It is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b.$$

Every prime element is irreducible. In a UFD, the two notions coincide.

**Definition 9.39. UFD, PID, and Euclidean Domain**

A unique factorization domain, abbreviated UFD, is an integral domain in which every nonzero nonunit factors into irreducibles and this factorization is unique up to order and multiplication by units.

A principal ideal domain, abbreviated PID, is an integral domain in which every ideal has the form

$$(a) = \{ra \mid r \in R\}$$

for some  $a \in R$ .

A Euclidean domain is an integral domain equipped with a function

$$\delta: R \setminus \{0\} \rightarrow \mathbb{N}_0$$

such that for all  $a, b \in R$  with  $b \neq 0$ , there exist  $q, r \in R$  satisfying

$$a = bq + r, \quad r = 0 \vee \delta(r) < \delta(b).$$

**Theorem 9.40. Factorization-Domain Hierarchy**

*The standard implications are*

$$\text{field} \implies \text{Euclidean domain} \implies \text{PID} \implies \text{UFD} \implies \text{integral domain}.$$

*None of the reverse implications holds in general.*

**Definition 9.41. Quaternion Algebra**

The quaternions form the real algebra

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\},$$

where

$$i^2 = j^2 = k^2 = ijk = -1.$$

Quaternion multiplication is associative but not commutative; for example,

$$ij = k, \quad ji = -k.$$

If

$$q = a + bi + cj + dk,$$

then

$$\bar{q} = a - bi - cj - dk, \quad |q|^2 = q\bar{q} = a^2 + b^2 + c^2 + d^2.$$

Every nonzero quaternion has inverse

$$q^{-1} = \frac{\bar{q}}{|q|^2}.$$

Thus  $\mathbb{H}$  is a noncommutative division ring, not a field under the convention that fields are commutative.

**Formula 9.42. Basic Factorization Identities**

The most frequently used polynomial factorizations include

$$a^2 - b^2 = (a - b)(a + b),$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2), \quad a^3 + b^3 = (a + b)(a^2 - ab + b^2).$$

More generally,

$$a^n - b^n = (a - b) \sum_{k=0}^{n-1} a^{n-1-k} b^k,$$

and, when  $n$  is odd,

$$a^n + b^n = (a + b) \sum_{k=0}^{n-1} (-1)^k a^{n-1-k} b^k.$$

**Formula 9.43. Cubic Factorization Identity**

For all  $a, b, c$ ,

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca),$$

or equivalently,

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a + b + c)((a - b)^2 + (b - c)^2 + (c - a)^2).$$

In particular, if  $a + b + c = 0$ , then

$$a^3 + b^3 + c^3 = 3abc.$$

**Algorithm 9.44. Polynomial Division Algorithm**

Let  $F$  be a field. If  $f(x), g(x) \in F[x]$  and  $g(x) \neq 0$ , then there exist unique polynomials  $q(x), r(x) \in F[x]$  such that

$$f(x) = q(x)g(x) + r(x), \quad r(x) = 0 \vee \deg r < \deg g.$$

**Algorithm 9.45. Euclidean Algorithm for Polynomials**

Over a field  $F$ , repeatedly apply polynomial division

$$f = qg + r, \quad r = 0 \vee \deg r < \deg g,$$

and replace  $(f, g)$  by  $(g, r)$ . The last nonzero remainder, normalized to be monic, is  $\gcd(f, g)$ . Back-substitution expresses the gcd as

$$af + bg = \gcd(f, g)$$

for some  $a, b \in F[x]$ .

**Theorem 9.46. Factor Theorem**

For a polynomial  $f(x) \in F[x]$  and an element  $a \in F$ ,

$$f(a) = 0 \iff x - a \text{ divides } f(x).$$

**Proposition 9.47. Root-Counting Principle for Polynomials**

A nonzero polynomial of degree  $n$  over a field has at most  $n$  roots. Hence two polynomials of degree at most  $n$  that agree at  $n + 1$  distinct points are identical.

**Formula 9.48. Lagrange Interpolation Formula**

For distinct  $x_0, \dots, x_n$ , the unique polynomial of degree at most  $n$  satisfying  $P(x_i) = y_i$  is

$$P(x) = \sum_{i=0}^n y_i \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}.$$

**Formula 9.49. Viète's Formulas**

If

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = a_n \prod_{i=1}^n (x - r_i),$$

where the roots are counted with multiplicity, then

$$\sum_{i=1}^n r_i = -\frac{a_{n-1}}{a_n}, \quad \prod_{i=1}^n r_i = (-1)^n \frac{a_0}{a_n}.$$

More generally, the elementary symmetric sums of the roots are the coefficients with alternating signs, divided by  $a_n$ .

**Formula 9.50. Newton's Identities; Newton–Girard Formulas**

Let  $e_k$  be the elementary symmetric polynomials in roots  $r_1, \dots, r_n$ , and let

$$p_k = r_1^k + \dots + r_n^k.$$

With  $e_0 = 1$ , for  $1 \leq k \leq n$ ,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \dots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0.$$

**Formula 9.51. Polynomial Discriminant**

If

$$f(x) = a_n \prod_{i=1}^n (x - r_i),$$

then

$$\text{Disc}(f) = a_n^{2n-2} \prod_{1 \leq i < j \leq n} (r_i - r_j)^2.$$

It vanishes if and only if  $f$  has a repeated root.

**Definition 9.52. Algebraic Number, Minimal Polynomial, and Algebraic Conjugates**

A complex number  $\alpha$  is algebraic over  $\mathbb{Q}$  if it is a root of a nonzero polynomial in  $\mathbb{Q}[x]$ . Its minimal polynomial  $m_{\alpha, \mathbb{Q}}(x)$  is the unique monic irreducible polynomial in  $\mathbb{Q}[x]$  having  $\alpha$  as a root. A complex number that is not algebraic over  $\mathbb{Q}$  is transcendental over  $\mathbb{Q}$ . The other roots of the minimal polynomial are the algebraic conjugates of  $\alpha$  over  $\mathbb{Q}$ .

**Theorem 9.53. Conjugate Root Theorems**

*The standard conjugate-root principles are as follows.*

- (i) If  $f \in \mathbb{R}[x]$  and  $f(\alpha) = 0$ , then  $f(\bar{\alpha}) = 0$ . Hence non-real roots of a real-coefficient polynomial occur in complex-conjugate pairs, with the same multiplicity.
- (ii) If  $f \in \mathbb{Q}[x]$ ,  $a, b \in \mathbb{Q}$ , and  $d$  is a squarefree integer, then

$$f(a + b\sqrt{d}) = 0 \implies f(a - b\sqrt{d}) = 0.$$

*Thus quadratic surd roots occur with their quadratic conjugates.*

- (iii) More generally, if  $f \in \mathbb{Q}[x]$  and  $f(\alpha) = 0$ , then  $m_{\alpha, \mathbb{Q}}$  divides  $f$  in  $\mathbb{Q}[x]$ . Therefore every algebraic conjugate of  $\alpha$  over  $\mathbb{Q}$  is also a root of  $f$ . The same conclusion applies to integer-coefficient polynomials because  $\mathbb{Z}[x] \subseteq \mathbb{Q}[x]$ .

**Theorem 9.54. Rational Root Theorem**

Let

$$f(x) = a_n x^n + \dots + a_0 \in \mathbb{Z}[x].$$

If a rational number  $p/q$  in lowest terms is a root of  $f$ , then

$$p \mid a_0, \quad q \mid a_n.$$

**Theorem 9.55. Cauchy Root Bound**

If

$$f(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0, \quad a_n \neq 0,$$

then every complex root  $z$  of  $f$  satisfies

$$|z| \leq 1 + \max_{0 \leq k < n} \left| \frac{a_k}{a_n} \right|.$$

Thus all roots lie in an explicitly computable disk before any root is solved for exactly.

**Theorem 9.56. Eisenstein's Irreducibility Criterion**

Let

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x].$$

If there exists a prime  $p$  such that

$$p \nmid a_n, \quad p \mid a_i \quad (0 \leq i < n), \quad p^2 \nmid a_0,$$

then  $f$  is irreducible over  $\mathbb{Q}$ .

**Theorem 9.57. Fundamental Theorem of Symmetric Polynomials**

Every symmetric polynomial can be expressed uniquely as a polynomial in the elementary symmetric polynomials.

**Theorem 9.58. Combinatorial Nullstellensatz, Coefficient Form**

Let  $P \in F[x_1, \dots, x_n]$  have total degree  $t_1 + \cdots + t_n$ , and suppose the coefficient of  $x_1^{t_1} \cdots x_n^{t_n}$  is nonzero. If

$$|S_i| > t_i$$

for every finite set  $S_i \subseteq F$ , then some  $(s_1, \dots, s_n) \in S_1 \times \cdots \times S_n$  satisfies

$$P(s_1, \dots, s_n) \neq 0.$$

*Strategy. Polynomial Method.* Encode a forbidden configuration as zeros of a polynomial, then use a degree bound, interpolation, factorization, or a nonzero coefficient to prove that the polynomial cannot vanish everywhere.

**Theorem 9.59. Fundamental Theorem of Algebra**

Every nonconstant complex polynomial has at least one complex root. Equivalently, every polynomial of degree  $n \geq 1$  over  $\mathbb{C}$  splits into  $n$  linear factors over  $\mathbb{C}$ , counted with multiplicity.

**Definition 9.60. Algebraically Closed Field**

A field  $F$  is algebraically closed if every nonconstant polynomial in  $F[x]$  has a root in  $F$ .

**Definition 9.61. Field Extension and Galois Group**

If  $F \subseteq K$  are fields, then  $K/F$  is a field extension and its degree is

$$[K : F] = \dim_F K.$$

The extension is finite if this dimension is finite. The group

$$\text{Gal}(K/F) = \{\sigma \in \text{Aut}(K) \mid \sigma(a) = a \ \forall a \in F\}$$

is the Galois group. For a finite Galois extension, intermediate fields and subgroups of the Galois group are related by the Galois correspondence.

**Theorem 9.62. Fundamental Theorem of Galois Theory; Galois Correspondence**

For a finite Galois extension  $L/K$ , there is an inclusion-reversing correspondence between intermediate fields  $K \subseteq E \subseteq L$  and subgroups  $H \leq \text{Gal}(L/K)$ , given by

$$E \mapsto \text{Gal}(L/E), \quad H \mapsto L^H.$$

**Theorem 9.63. Abel–Ruffini Theorem**

*There is no formula for the roots of a general polynomial of degree 5 or higher using only the field operations and radicals.*

*Remark.* The Abel–Ruffini theorem does not say that every quintic equation is unsolvable. It says that no universal radical formula exists for the general quintic.

# Chapter 10

## Topology

Related **POSTECH** Courses: (MATH321) General Topology

Related Books: Topology (James R. Munkres)

### Definition 10.1. Topology

A topology on a set  $X$  is a collection  $\mathcal{T}$  of subsets of  $X$  whose elements are called open sets, satisfying:

- (i)  $\emptyset, X \in \mathcal{T}$ ;
- (ii) arbitrary unions of sets in  $\mathcal{T}$  belong to  $\mathcal{T}$ ;
- (iii) finite intersections of sets in  $\mathcal{T}$  belong to  $\mathcal{T}$ .

### Definition 10.2. Basis for a Topology

A collection  $\mathcal{B}$  of subsets of  $X$  is a basis if every  $x \in X$  lies in some  $B \in \mathcal{B}$  and, whenever  $x \in B_1 \cap B_2$  with  $B_1, B_2 \in \mathcal{B}$ , there exists  $B_3 \in \mathcal{B}$  such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

The topology generated by  $\mathcal{B}$  consists of arbitrary unions of basis elements. Open balls form the standard basis for the metric topology of a metric space.

### Definition 10.3. Open Sets, Closed Sets, and Neighborhoods

If  $(X, \mathcal{T})$  is a topological space, a subset  $U \subseteq X$  is open if  $U \in \mathcal{T}$ . A subset  $F \subseteq X$  is closed if  $X \setminus F \in \mathcal{T}$ . A neighborhood of  $x \in X$  is a set that contains an open set containing  $x$ . A local basis at  $x$  is a family of neighborhoods with the property that every neighborhood of  $x$  contains one of them.

### Definition 10.4. First- and Second-Countability

A topological space is first-countable if every point has a countable local neighborhood basis. It is second-countable if its topology has a countable basis. Every second-countable space is first-countable, but not conversely in general.

### Definition 10.5. Compactness

An open cover of  $X$  is a family of open sets whose union contains  $X$ . A topological space  $X$  is compact if every open cover has a finite subcover. Equivalently, for every family  $\mathcal{U} \subseteq \mathcal{T}$ ,

$$X \subseteq \bigcup_{U \in \mathcal{U}} U \Rightarrow \exists U_1, \dots, U_n \in \mathcal{U} \text{ s.t. } X \subseteq U_1 \cup \dots \cup U_n.$$

**Definition 10.6. Connectedness**

A topological space  $X$  is connected if it cannot be written as the union of two disjoint nonempty open sets. Equivalently, the only subsets of  $X$  that are both open and closed are  $\emptyset$  and  $X$ .

**Definition 10.7. Continuity in Topological Spaces**

A map  $f: (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$  is continuous if the inverse image of every open set is open:

$$\forall V \in \mathcal{T}_Y, \quad f^{-1}(V) \in \mathcal{T}_X.$$

**Definition 10.8. Homeomorphism**

A homeomorphism is a bijective continuous map whose inverse is also continuous. Equivalently,  $f: X \xrightarrow{\sim} Y$  is a homeomorphism if  $f$  and  $f^{-1}$  are continuous. Two spaces are homeomorphic, written  $X \cong Y$ , if such a map exists.

**Theorem 10.9. Basic Topological Invariants under Continuous Maps**

*A continuous image of a compact space is compact, and a continuous image of a connected space is connected. Consequently, compactness and connectedness are preserved by homeomorphism.*

**Definition 10.10. Hausdorff Space**

A topological space  $X$  is Hausdorff if distinct points can be separated by disjoint open neighborhoods:

$$x \neq y \implies \exists U, V \text{ open such that } x \in U, y \in V, U \cap V = \emptyset.$$

Every metric space is Hausdorff.

**Theorem 10.11. Compact Subsets of Hausdorff Spaces**

*Every compact subset of a Hausdorff space is closed. In particular, a continuous bijection from a compact space onto a Hausdorff space is automatically a homeomorphism.*

**Definition 10.12. Path, Path-Connectedness, and Homotopy**

A path from  $x_0$  to  $x_1$  in  $X$  is a continuous map

$$\gamma: [0, 1] \rightarrow X, \quad \gamma(0) = x_0, \quad \gamma(1) = x_1.$$

The space  $X$  is path-connected if every two points can be joined by a path. Two maps  $f, g: X \rightarrow Y$  are homotopic, written  $f \simeq g$ , if there exists a continuous map

$$H: X \times [0, 1] \rightarrow Y$$

with  $H(x, 0) = f(x)$  and  $H(x, 1) = g(x)$ . A loop is a path with the same initial and terminal point.

**Theorem 10.13. Path-Connectedness Implies Connectedness**

*Every path-connected space is connected. The converse is false in general, so connectedness and path-connectedness should not be interchanged without additional hypotheses.*

**Theorem 10.14. Jordan Curve Theorem**

*Every simple closed curve in the plane separates the plane into exactly two connected components, an interior and an exterior, and is the common boundary of both.*

**Theorem 10.15. Brouwer Fixed-Point Theorem**

*Every continuous map from a closed ball in  $\mathbb{R}^n$  to itself has at least one fixed point.*

**Theorem 10.16. Borsuk–Ulam Theorem**

Every continuous map

$$f: S^n \rightarrow \mathbb{R}^n$$

takes some pair of antipodal points to the same value.

**Corollary 10.17. Ham Sandwich Theorem**

Any  $n$  bounded measurable subsets of  $\mathbb{R}^n$  can be bisected simultaneously by one hyperplane.

**Theorem 10.18. Hairy Ball Theorem**

Every continuous tangent vector field on an even-dimensional sphere  $S^{2m}$  vanishes at some point.

**Theorem 10.19. Invariance of Domain**

If  $U \subseteq \mathbb{R}^n$  is open and  $f: U \rightarrow \mathbb{R}^n$  is continuous and injective, then  $f(U)$  is open and  $f$  is a homeomorphism from  $U$  onto  $f(U)$ .

**Definition 10.20. Manifold**

An  $n$ -dimensional topological manifold is a Hausdorff, second-countable topological space in which every point has a neighborhood homeomorphic to an open subset of  $\mathbb{R}^n$ .

**Definition 10.21. Boundary, Puncture, and Wedge Sum**

An  $n$ -manifold with boundary is locally modeled on the closed half-space

$$\mathbb{H}^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}.$$

Its boundary  $\partial M$  consists of the points modeled on  $x_n = 0$ . A puncture means removal of a point. For pointed spaces  $(X, x_0)$  and  $(Y, y_0)$ , the wedge sum  $X \vee Y$  is obtained by identifying  $x_0$  with  $y_0$  in the disjoint union.

**Definition 10.22. Fundamental Group**

Fix a base point  $x_0 \in X$ . The fundamental group  $\pi_1(X, x_0)$  is the set of endpoint-preserving homotopy classes of loops based at  $x_0$ , with multiplication induced by concatenation of loops. It is a basic algebraic invariant of a topological space. A path-connected space is simply connected if

$$\pi_1(X, x_0) \cong \{e\},$$

equivalently, every loop can be contracted to a point.

**Fact 10.23. Fundamental Groups of Familiar Spaces**

The following basic examples are worth recognizing:

$$\pi_1(S^1) \cong \mathbb{Z}, \quad \pi_1(S^n) \cong \{e\} \quad (n \geq 2), \quad \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2.$$

More generally,

$$\pi_1\left(\bigvee_{j=1}^r S^1\right) \cong F_r.$$

The wedge point is generally not a manifold point.

*Strategy.* For introductory topology questions, the most common quick checks are: compactness means every open cover has a finite subcover, connectedness means the space cannot be separated into two disjoint nonempty open sets, and a homeomorphism is a continuous bijection with continuous inverse.

**Formula 10.24. Euler Characteristic of a Surface**

For a triangulated surface, the Euler characteristic is

$$\chi = V - E + F.$$

For a compact connected orientable surface  $S_{g,b}$  of genus  $g$  with  $b$  boundary components,

$$\chi(S_{g,b}) = 2 - 2g - b.$$

For a compact connected nonorientable surface  $N_{k,b}$  with  $k$  projective-plane summands and  $b$  boundary components,

$$\chi(N_{k,b}) = 2 - k - b.$$

**Theorem 10.25. Classification of Closed Surfaces**

*Every connected closed surface is homeomorphic to either an orientable surface of genus  $g \geq 0$  or a nonorientable surface obtained as a connected sum of  $k \geq 1$  projective planes.*

*Remark. Removing a Point versus Removing a Disk.* Removing one point from a closed surface produces a noncompact surface without boundary. Removing the interior of an embedded closed disk instead produces a compact surface with one boundary component. These spaces can be homotopy equivalent without being homeomorphic. Thus an informal “hole” must be identified as a point, disk, or boundary removal before applying a topological invariant.

**Theorem 10.26. Poincaré Conjecture**

*Every closed simply connected 3-manifold is homeomorphic to the 3-sphere  $S^3$ .*

*Remark.* The Poincaré conjecture was proved by Grigori Perelman using Ricci flow. The fundamental correspondence is

$$\text{Poincaré conjecture} \longleftrightarrow \text{Perelman} \longleftrightarrow \text{Ricci flow}.$$

**Theorem 10.27. Thurston Geometrization Theorem**

*Every closed orientable 3-manifold can be decomposed along embedded 2-spheres and incompressible tori into pieces whose interiors admit geometric structures modeled on one of the eight Thurston geometries:*

$$S^3, \mathbb{R}^3, \mathbb{H}^3, S^2 \times \mathbb{R}, \mathbb{H}^2 \times \mathbb{R}, \widetilde{\text{SL}}_2(\mathbb{R}), \text{Nil}, \text{Sol}.$$

*The sphere decomposition is the prime decomposition, and the torus decomposition is the JSJ decomposition of the irreducible pieces.*

*Remark.* The recognition cue is “eight Thurston geometries”; the theorem above gives the complete list once, without repeating it in a separate fact.

# Chapter 11

## Miscellaneous Topics

### Definition 11.1. Category

A category  $\mathcal{C}$  consists of objects and morphisms between objects. For objects  $A, B, C$ , morphisms can be composed,

$$A \xrightarrow{f} B \xrightarrow{g} C \quad \longmapsto \quad g \circ f: A \rightarrow C,$$

composition is associative, and every object  $A$  has an identity morphism  $\text{id}_A$ . For every morphism  $f: A \rightarrow B$ , the identity laws are

$$f \circ \text{id}_A = f, \quad \text{id}_B \circ f = f.$$

Category theory studies mathematical structures by emphasizing objects, structure-preserving maps, and their composition.

### Definition 11.2. Nash Equilibrium

In a strategic game, a strategy profile is a Nash equilibrium if no player can improve their payoff by changing only their own strategy while every other player's strategy is held fixed. Equivalently, every player's chosen strategy is a best response to the strategies chosen by the others.

### Definition 11.3. Self-Similar Dimension

Suppose a self-similar set is the union of  $N$  geometrically similar copies of itself, each scaled by a factor  $r$  with  $0 < r < 1$ , and the usual separation assumptions hold. Its similarity dimension  $d$  is determined by

$$Nr^d = 1, \quad d = \frac{\ln N}{\ln(1/r)}.$$

This gives a standard noninteger dimension for many classical fractals.

### Fact 11.4. Menger Sponge

At each stage of the Menger sponge construction, a cube is replaced by 20 copies scaled by  $1/3$ . Hence its similarity dimension is

$$\boxed{\frac{\ln 20}{\ln 3}}.$$

*Strategy. Symmetry and Without Loss of Generality.* Before splitting into cases, identify transformations that preserve the problem. If several cases differ only by a permutation, reflection, sign change, or cyclic relabeling, choose one representative and state explicitly why the remaining cases follow by symmetry. “Without loss of generality” is justified only after this invariance has been identified.

*Strategy. Parity and Modular Obstructions.* Before searching for a construction or integer solution, reduce the relevant quantities modulo a small modulus. Parity, residues of powers, and invariant congruence classes often rule out entire families of possibilities with almost no calculation.

*Strategy. Telescoping and Local Cancellation.* For a sum or product indexed consecutively, try to rewrite each term so that neighboring factors cancel. Typical forms are

$$a_k = b_k - b_{k+1} \quad \text{or} \quad a_k = \frac{b_{k+1}}{b_k}.$$

Then

$$\sum_{k=m}^n a_k = b_m - b_{n+1}, \quad \prod_{k=m}^n a_k = \frac{b_{n+1}}{b_m}.$$

*Strategy. Descent and Minimal Counterexample.* If a statement about positive integers or finite objects is suspected to hold universally, assume a counterexample exists and choose one minimal under a natural size parameter. Use the defining relations to construct a strictly smaller counterexample. This contradiction is the abstract form of infinite descent and is useful far beyond number theory.

*Strategy. Complex Numbers as Plane Geometry.* Identify the Euclidean plane with  $\mathbb{C}$  when rotations, similarities, roots of unity, or cyclic configurations are prominent. Multiplication by  $e^{i\theta}$  is rotation by  $\theta$ , multiplication by a positive scalar is dilation, and complex conjugation is reflection across the real axis. Translate back to ordinary geometry only after the algebraic relation has been simplified.

*Strategy. Polynomial Encoding.* Encode a finite set of numbers as roots of a polynomial, or encode a sequence by a generating polynomial or generating function, when sums, products, symmetric expressions, or recurrence relations are involved. Root-coefficient relations, interpolation, and coefficient extraction can then replace case-by-case manipulation.

*Tactic. Floor and Fractional-Part Identities.* For every real  $x$ ,

$$x = \lfloor x \rfloor + \{x\}, \quad 0 \leq \{x\} < 1.$$

Moreover,

$$\lfloor x \rfloor + \lfloor -x \rfloor = \begin{cases} 0, & x \in \mathbb{Z}, \\ -1, & x \notin \mathbb{Z}, \end{cases}$$

and for  $n \in \mathbb{Z}$ ,

$$\lfloor x + n \rfloor = \lfloor x \rfloor + n.$$

These identities convert many apparently analytic floor problems into integer casework.

*Strategy. Iterated Knowledge Elimination.* Let  $\Omega_0$  be a finite set of possible states. For each agent  $i$  and state  $\omega \in \Omega$ , let

$$\mathcal{I}_i^\Omega(\omega) = \{\omega' \in \Omega \mid \omega' \text{ gives agent } i \text{ the same information as } \omega\}.$$

Relative to the current state space  $\Omega$ , agent  $i$  knows the state precisely when

$$|\mathcal{I}_i^\Omega(\omega)| = 1.$$

A truthful public announcement restricts  $\Omega$  to exactly the states at which that announcement is true. Repeating this restriction models successive statements about knowledge or ignorance. In particular, the statement “I knew that  $Q$ ” means

$$\forall \omega' \in \mathcal{I}_i^\Omega(\omega), \quad Q(\omega').$$

*Strategy. Transform the Given Objects.* Replace the original configuration by an equivalent one whose structure is easier to see: partition a region, introduce prefix sums, encode a movement by vectors, pass to residues, diagonalize a matrix, or complete a partial object. The transformed problem should preserve the quantity or property being asked about.

*Strategy. Invariant, Monovariant, and Extremal Principles.* An invariant remains unchanged under every allowed move. A monovariant moves strictly in one direction and therefore prevents cycles or proves termination. The extremal principle chooses a largest, smallest, first, or last object and exploits the extra restrictions forced by extremality. These are among the most common contest ideas hidden behind elementary statements.

*Strategy. Complete the Configuration and Work Backward.* First imagine a completed solution, then determine which parts are forced and which choices remain free. This is effective in geometric constructions, permutation completions, lattice paths, tilings, and assignment problems. Fixing one object under a symmetry often removes redundant cases before the remaining count begins.

*Strategy. Rotate, Reflect, and Superimpose.* A rotation, reflection, or translated copy can convert a broken path into a straight one, expose equal vectors, pair terms, or turn a local pattern into a global invariant. When two configurations overlap, compare what cancels and what remains rather than recomputing each separately.

*Strategy. Homogenization and Normalization.* If an inequality is homogeneous, scale the variables to impose a simple condition such as  $a + b + c = 1$ ,  $abc = 1$ , or  $x^2 + y^2 = 1$ .

*Strategy. Equality Cases and Smoothing.* Identify the expected equality case before selecting an inequality. For a symmetric expression, replacing two variables by their average often reveals the extremal configuration.

### Theorem 11.5. Triangle and Reverse Triangle Inequalities

For real or complex numbers  $x, y$ ,

$$||x| - |y|| \leq |x - y| \leq |x| + |y|.$$

More generally, every norm satisfies

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

### Formula 11.6. Basic Sum-of-Squares Identities

For real  $a, b, c$ ,

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2} ((a - b)^2 + (b - c)^2 + (c - a)^2) \geq 0.$$

Consequently,

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

and

$$(a + b + c)^2 \leq 3(a^2 + b^2 + c^2).$$

### Formula 11.7. Frequently Used Two-Variable Inequalities

For real  $x, y$ ,

$$x^2 + y^2 \geq 2xy.$$

For positive  $x, y$ ,

$$\frac{x}{y} + \frac{y}{x} \geq 2, \quad x + \frac{1}{x} \geq 2.$$

Equality holds exactly when the two compared positive quantities are equal.

**Theorem 11.8. Lagrange's Identity**

For real numbers  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$ ,

$$\left( \sum_{i=1}^n a_i^2 \right) \left( \sum_{i=1}^n b_i^2 \right) - \left( \sum_{i=1}^n a_i b_i \right)^2 = \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2.$$

The nonnegativity of the right-hand side gives the Cauchy–Schwarz inequality immediately.

**Theorem 11.9. QM–AM–GM–HM Inequalities**

For positive  $x_1, \dots, x_n$ ,

$$\sqrt{\frac{x_1^2 + \dots + x_n^2}{n}} \geq \frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \dots x_n} \geq \frac{n}{\frac{1}{x_1} + \dots + \frac{1}{x_n}}.$$

Equality holds if and only if all variables are equal.

**Theorem 11.10. Weighted AM–GM Inequality**

If  $w_i \geq 0$ ,  $\sum_{i=1}^n w_i = 1$ , and  $x_i > 0$ , then

$$\sum_{i=1}^n w_i x_i \geq \prod_{i=1}^n x_i^{w_i}.$$

Equality holds when all  $x_i$  with positive weight are equal.

**Theorem 11.11. Power Mean Inequality**

For positive  $x_1, \dots, x_n$  and real  $r \neq 0$ , define

$$M_r = \left( \frac{x_1^r + \dots + x_n^r}{n} \right)^{1/r},$$

and define  $M_0 = (x_1 \dots x_n)^{1/n}$ . If  $r < s$ , then

$$M_r \leq M_s.$$

The harmonic, geometric, arithmetic, and quadratic means correspond to  $r = -1, 0, 1, 2$ , respectively.

**Theorem 11.12. Young's Inequality**

If  $x, y \geq 0$ ,  $p, q > 1$ , and

$$\frac{1}{p} + \frac{1}{q} = 1,$$

then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

**Theorem 11.13. Cauchy–Schwarz Inequality, Engel Form**

For real  $a_i$  and positive  $b_i$ ,

$$\sum_{i=1}^n \frac{a_i^2}{b_i} \geq \frac{(a_1 + \cdots + a_n)^2}{b_1 + \cdots + b_n}.$$

**Theorem 11.14. Hölder's Inequality**

If  $p, q > 1$  and  $1/p + 1/q = 1$ , then

$$\sum_{i=1}^n |a_i b_i| \leq \left( \sum_{i=1}^n |a_i|^p \right)^{1/p} \left( \sum_{i=1}^n |b_i|^q \right)^{1/q}.$$

**Theorem 11.15. Minkowski's Inequality**

For  $p \geq 1$ ,

$$\left( \sum_{i=1}^n |a_i + b_i|^p \right)^{1/p} \leq \left( \sum_{i=1}^n |a_i|^p \right)^{1/p} + \left( \sum_{i=1}^n |b_i|^p \right)^{1/p}.$$

**Theorem 11.16. Rearrangement Inequality**

If

$$a_1 \leq \cdots \leq a_n, \quad b_1 \leq \cdots \leq b_n,$$

then for every permutation  $\sigma \in S_n$ ,

$$\sum_{i=1}^n a_i b_{n+1-i} \leq \sum_{i=1}^n a_i b_{\sigma(i)} \leq \sum_{i=1}^n a_i b_i.$$

**Theorem 11.17. Chebyshev's Sum Inequality**

If  $a_1 \leq \cdots \leq a_n$  and  $b_1 \leq \cdots \leq b_n$ , then

$$\frac{1}{n} \sum_{i=1}^n a_i b_i \geq \left( \frac{1}{n} \sum_{i=1}^n a_i \right) \left( \frac{1}{n} \sum_{i=1}^n b_i \right).$$

**Theorem 11.18. Schur's Inequality**

For  $a, b, c \geq 0$  and  $r \geq 0$ ,

$$\sum_{\text{cyc}} a^r (a - b)(a - c) \geq 0.$$

**Formula 11.19. Nesbitt's Inequality**

For positive  $a, b, c$ ,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

**Theorem 11.20. Bernoulli's Inequality**

If  $x \geq -1$  and  $r \geq 1$ , then

$$(1+x)^r \geq 1+rx.$$

*Strategy. Comparing Powers by Logarithms.* To compare two positive numbers such as  $a^b$  and  $b^a$ , take logarithms:

$$a^b > b^a \iff b \ln a > a \ln b \iff \frac{\ln a}{a} > \frac{\ln b}{b}.$$

The function  $\ln x/x$  is decreasing for  $x > e$ . This quickly handles comparisons such as  $11^9$  versus  $9^{11}$  or  $e^\pi$  versus  $\pi^e$ .

**Formula 11.21. Elementary Telescoping Series**

Two common fast calculations are

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1, \quad \sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}.$$

Both follow by partial fractions and telescoping. Classical zeta and alternating-series values are collected in the Calculus chapter.

*Strategy.* These values often appear after a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2} \left( 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right) = \frac{\ln 2}{2}.$$

**Part II**

**Facts to Memorize**

# Chapter 12

## Mathematical Prizes and Congresses

Science Quiz often asks for the short association between a prize, a congress, a laureate, and a mathematical achievement. This chapter records the facts that should be recognized quickly: not full biographies, but the names, years, quiz anchors, and historical landmarks most likely to become quiz answers.

### Fact 12.1. Science Quiz Prize Triad

The three mathematics prizes that should be recognized first are

Fields Medal,      Abel Prize,      Wolf Prize in Mathematics.

Recognize each prize through the association

prize  $\longleftrightarrow$  laureate  $\longleftrightarrow$  achievement or area.

*Strategy.* History questions often ask for a short association rather than a full biography. Current prominence, a recent award, or a connection to the POSTECH–KAIST context makes such an association especially relevant.

### Fact 12.2. Why There Is No Nobel Prize in Mathematics

Nobel’s will named prizes in physics, chemistry, physiology or medicine, literature, and peace. The popular claim that mathematics was excluded because of a romantic rivalry is unsupported.

| Prize        | Origin and first award                                                  | Namesake                                                                                                                                  | Administration and cycle                                                                                 |
|--------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Fields Medal | Proposed at the 1924 ICM in Toronto; first awarded in 1936.             | John Charles Fields, Canadian mathematician and organizer of the 1924 ICM, who donated funds for the medals.                              | Awarded by the IMU at the ICM, normally every four years; subject to the age condition.                  |
| Abel Prize   | Established by the Norwegian Parliament in 2002; first awarded in 2003. | Niels Henrik Abel, Norwegian mathematician known for algebra, analysis, and the impossibility of solving the general quintic by radicals. | Awarded annually by the Norwegian Academy of Science and Letters; presented in Oslo; no age restriction. |

| Prize                     | Origin and first award                                                              | Namesake                                                                                              | Administration and cycle                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Wolf Prize in Mathematics | The Wolf Foundation was established in 1975; the prizes were first awarded in 1978. | Ricardo Wolf, a German-born Cuban inventor, diplomat, and philanthropist; he was not a mathematician. | Awarded by the Wolf Foundation in Israel; normally annual, although mathematics is not necessarily awarded every year; no age restriction. |

## Fields Medal and ICM

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. At the 1924 ICM in Toronto, a resolution proposed two gold medals for outstanding mathematical achievement. John Charles Fields later donated funds for the medals that bear his name. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress. Since 1966, up to four medals may be awarded at an ICM.

| Year | ICM location                  | Medalists                                                           | Quiz anchors                                                                                                              |
|------|-------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1936 | Oslo, Norway                  | Lars Ahlfors; Jesse Douglas                                         | Complex analysis, Riemann surfaces; minimal surfaces, Plateau problem.                                                    |
| 1950 | Cambridge, United States      | Laurent Schwartz; Atle Selberg                                      | Distribution theory; analytic number theory, sieve methods, zeta functions.                                               |
| 1954 | Amsterdam, Netherlands        | Kunihiko Kodaira; Jean-Pierre Serre                                 | Complex geometry, algebraic geometry; algebraic topology, sheaf theory.                                                   |
| 1958 | Edinburgh, United Kingdom     | Klaus Roth; René Thom                                               | Diophantine approximation; cobordism, differential topology.                                                              |
| 1962 | Stockholm, Sweden             | Lars Hörmander; John Milnor                                         | Partial differential equations; differential topology, exotic spheres.                                                    |
| 1966 | Moscow, Soviet Union (USSR)   | Michael Atiyah; Paul Cohen; Alexander Grothendieck; Stephen Smale   | K-theory and index theorem; forcing and continuum hypothesis; modern algebraic geometry; generalized Poincaré conjecture. |
| 1970 | Nice, France                  | Alan Baker; Heisuke Hironaka; Serge Novikov; John G. Thompson       | Transcendental number theory; resolution of singularities; topology and geometry; finite group theory.                    |
| 1974 | Vancouver, Canada             | Enrico Bombieri; David Mumford                                      | Number theory, complex analysis, minimal surfaces; moduli spaces and algebraic geometry.                                  |
| 1978 | Helsinki, Finland             | Pierre Deligne; Charles Fefferman; Grigory Margulis; Daniel Quillen | Weil conjectures; several complex variables; Lie groups and ergodic theory; algebraic K-theory.                           |
| 1982 | Warsaw, Poland (held in 1983) | Alain Connes; William Thurston; Shing-Tung Yau                      | Noncommutative geometry; 3-manifolds and low-dimensional topology; Calabi conjecture and geometric analysis.              |

Continued on the next page.

| Year | ICM location                                                                       | Medalists                                                                  | Quiz anchors                                                                                                                                                         |
|------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1986 | Berkeley, United States                                                            | Simon Donaldson; Gerd Faltings; Michael Freedman                           | Gauge theory and 4-manifolds; Mordell conjecture and arithmetic geometry; topological 4-dimensional Poincaré conjecture.                                             |
| 1990 | Kyoto, Japan                                                                       | Vladimir Drinfeld; Vaughan Jones; Shigefumi Mori; Edward Witten            | Quantum groups and Langlands program; Jones polynomial; minimal model program; mathematical physics and quantum field theory.                                        |
| 1994 | Zurich, Switzerland                                                                | Jean Bourgain; Pierre-Louis Lions; Jean-Christophe Yoccoz; Efim Zelmanov   | Analysis and Banach spaces; partial differential equations; dynamical systems; restricted Burnside problem.                                                          |
| 1998 | Berlin, Germany                                                                    | Richard Borcherds; W. Timothy Gowers; Maxim Kontsevich; Curtis T. McMullen | Moonshine and automorphic forms; functional analysis and combinatorics; algebraic geometry and mathematical physics; complex dynamics.                               |
| 2002 | Beijing, China                                                                     | Laurent Lafforgue; Vladimir Voevodsky                                      | Langlands correspondence over function fields; motivic cohomology and $\mathbb{A}^1$ -homotopy theory.                                                               |
| 2006 | Madrid, Spain                                                                      | Andrei Okounkov; Grigori Perelman; Terence Tao; Wendelin Werner            | Probability, representation theory, algebraic geometry; Ricci flow; PDE, harmonic analysis, additive combinatorics; SLE and Brownian motion.                         |
| 2010 | Hyderabad, India                                                                   | Elon Lindenstrauss; Ngô Bảo Châu; Stanislav Smirnov; Cédric Villani        | Ergodic theory and number theory; Fundamental Lemma; conformal invariance; Boltzmann equation and optimal transport.                                                 |
| 2014 | Seoul, Republic of Korea                                                           | Artur Avila; Manjul Bhargava; Martin Hairer; Maryam Mirzakhani             | Dynamical systems; geometry of numbers; stochastic PDE; Riemann surfaces and moduli spaces.                                                                          |
| 2018 | Rio de Janeiro, Brazil                                                             | Caucher Birkar; Alessio Figalli; Peter Scholze; Akshay Venkatesh           | Fano varieties and minimal model program; optimal transport; $p$ -adic geometry; number theory and homogeneous dynamics.                                             |
| 2022 | Originally Saint Petersburg, Russia; virtual congress; awards in Helsinki, Finland | Hugo Duminil-Copin; June Huh; James Maynard; Maryna Viazovska              | Phase transitions in statistical physics; Hodge theory in combinatorics and matroids; prime numbers; sphere packing and the $E_8$ lattice.                           |
| 2026 | Philadelphia, United States                                                        | Yu Deng; John Pardon; Jacob Tsimerman; Hong Wang                           | PDE and kinetic theory; symplectic geometry and topology; arithmetic geometry and o-minimality; harmonic analysis, geometric measure theory, and the Kakeya problem. |

**Fact 12.3. ICM and Fields Medal landmarks**

The following associations are especially useful.

- ICM stands for International Congress of Mathematicians.

- IMU stands for International Mathematical Union.
- The 1936 ICM in Oslo hosted the first Fields Medal ceremony.
- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- The 1966 Fields Medal was the first edition with four laureates.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal; as an undergraduate, he majored in history rather than mathematics.
- At the 1998 ICM, Andrew Wiles received a special IMU silver plaque for his proof of Fermat's Last Theorem.
- At the 2006 ICM, Grigori Perelman declined the Fields Medal.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- Hong Wang became the third woman Fields Medalist at the 2026 ICM.
- The 2014 ICM was held in Seoul, Republic of Korea.
- The 2022 ICM was originally scheduled for Saint Petersburg, Russia. It was held as a virtual congress, and the prize ceremony took place in Helsinki, Finland.
- 2026 ICM: Philadelphia, United States, July 23–30, 2026.
- There is no ICM scheduled for 2027: the regular cycle proceeds from 2026 to 2030.
- The 2030 ICM is scheduled for Glasgow, United Kingdom; the associated IMU General Assembly is scheduled for Dublin, Ireland.

#### **Fact 12.4. Fields Medal National and Regional Firsts**

The following short associations are useful when a quiz asks for a national or regional first. The labels refer to the laureates' country of origin or the national mathematical community with which the record is conventionally associated.

- Kunihiko Kodaira, 1954  $\leftrightarrow$  first Japanese Fields Medalist.
- Vaughan Jones, 1990  $\leftrightarrow$  first Fields Medalist from the Southern Hemisphere and the only New Zealander to date.
- Terence Tao, 2006  $\leftrightarrow$  first Australian Fields Medalist.
- Ngô Bảo Châu, 2010  $\leftrightarrow$  first Vietnamese Fields Medalist.
- Artur Avila, 2014  $\leftrightarrow$  first Brazilian and first Latin American Fields Medalist.
- Maryam Mirzakhani, 2014  $\leftrightarrow$  first Iranian and first woman Fields Medalist.
- June Huh, 2022  $\leftrightarrow$  first Fields Medalist of Korean descent and the first laureate associated with the Korean mathematical community.

Because citizenship and institutional affiliation may change during a mathematician's career, the person's name and year are safer quiz anchors than an oversimplified nationality label.

**Fact 12.5. Fields Medal design**

The obverse of the Fields Medal bears a profile of Archimedes and the Latin inscription

$$TRANSIRE SUUM PECTUS MUNDOQUE POTIRI.$$

The reverse shows a sphere inscribed in a cylinder, recalling the result that Archimedes requested for his tomb. For the corresponding sphere and circumscribed cylinder,

$$V_{\text{cylinder}} : V_{\text{sphere}} = 3 : 2,$$

equivalently

$$\frac{V_{\text{sphere}}}{V_{\text{cylinder}}} = \frac{2}{3}.$$

**Abel Prize**

The Abel Prize is awarded by the Norwegian Academy of Science and Letters. It is named after Niels Henrik Abel. The Norwegian Parliament established the modern prize in 2002, the bicentenary of Abel's birth, and the first award was made in 2003. Unlike the Fields Medal, the Abel Prize has no age restriction. For Science Quiz, the Abel Prize is often asked through the pattern

$$\text{year} \longleftrightarrow \text{laureate} \longleftrightarrow \text{famous theorem or area.}$$

| Year | Laureate(s)                    | Quiz anchors                                                                         |
|------|--------------------------------|--------------------------------------------------------------------------------------|
| 2003 | Jean-Pierre Serre              | Algebraic topology, algebraic geometry, number theory; first Abel Prize laureate.    |
| 2004 | Michael Atiyah; Isadore Singer | Atiyah–Singer index theorem.                                                         |
| 2005 | Peter D. Lax                   | Partial differential equations and applied mathematics.                              |
| 2006 | Lennart Carleson               | Harmonic analysis; convergence of Fourier series.                                    |
| 2007 | Srinivasa S. R. Varadhan       | Probability theory; large deviations.                                                |
| 2008 | John G. Thompson; Jacques Tits | Finite groups, classification of finite simple groups; buildings and group theory.   |
| 2009 | Mikhail Gromov                 | Geometry, geometric group theory, symplectic geometry.                               |
| 2010 | John Tate                      | Number theory and arithmetic geometry.                                               |
| 2011 | John Milnor                    | Topology, geometry, dynamics; exotic spheres.                                        |
| 2012 | Endre Szemerédi                | Discrete mathematics and theoretical computer science; Szemerédi's theorem.          |
| 2013 | Pierre Deligne                 | Algebraic geometry; Weil conjectures.                                                |
| 2014 | Yakov Sinai                    | Dynamical systems, ergodic theory, mathematical physics.                             |
| 2015 | John Nash; Louis Nirenberg     | Nonlinear PDE, geometry, Nash embedding theorem.                                     |
| 2016 | Andrew Wiles                   | Fermat's Last Theorem; modularity of elliptic curves.                                |
| 2017 | Yves Meyer                     | Wavelet theory; applications from signal compression to gravitational-wave analysis. |
| 2018 | Robert Langlands               | Langlands program; representation theory and number theory.                          |
| 2019 | Karen Uhlenbeck                | Geometric analysis and gauge theory; first woman Abel Prize laureate.                |

Continued on the next page.

| Year | Laureate(s)                          | Quiz anchors                                                                             |
|------|--------------------------------------|------------------------------------------------------------------------------------------|
| 2020 | Hillel Furstenberg; Gregory Margulis | Probability, dynamics, ergodic theory, Lie groups.                                       |
| 2021 | László Lovász; Avi Wigderson         | Discrete mathematics, theoretical computer science, complexity theory.                   |
| 2022 | Dennis Sullivan                      | Topology, geometric topology, dynamical systems.                                         |
| 2023 | Luis Caffarelli                      | Nonlinear partial differential equations, free-boundary problems, Monge–Ampère equation. |
| 2024 | Michel Talagrand                     | Probability theory, functional analysis, spin glasses, concentration inequalities.       |
| 2025 | Masaki Kashiwara                     | Algebraic analysis, $D$ -modules, representation theory, crystal bases.                  |
| 2026 | Gerd Faltings                        | Arithmetic geometry; Mordell conjecture, Lang’s conjectures, Diophantine geometry.       |

**Fact 12.6. Abel Prize landmarks**

The following associations summarize the prize’s main quiz anchors.

- Abel Prize  $\leftrightarrow$  Norwegian Academy of Science and Letters.
- Abel Prize  $\leftrightarrow$  Niels Henrik Abel.
- First Abel Prize laureate  $\leftrightarrow$  Jean-Pierre Serre.
- First woman Abel Prize laureate  $\leftrightarrow$  Karen Uhlenbeck.
- Since 2019, the Abel Prize has carried a monetary award of NOK 7.5 million, on the order of one million US dollars.
- John Milnor  $\leftrightarrow$  1962 Fields Medal and 2011 Abel Prize.
- Andrew Wiles  $\leftrightarrow$  2016 Abel Prize  $\leftrightarrow$  Fermat’s Last Theorem.
- Yves Meyer  $\leftrightarrow$  2017 Abel Prize  $\leftrightarrow$  wavelets.
- Robert Langlands  $\leftrightarrow$  2018 Abel Prize  $\leftrightarrow$  Langlands program.
- Gerd Faltings  $\leftrightarrow$  1986 Fields Medal and 2026 Abel Prize.

**Wolf Prize in Mathematics**

The Wolf Prize is awarded by the Wolf Foundation, established in Israel in 1975 by Ricardo Wolf. Wolf was a German-born Cuban inventor, diplomat, and philanthropist, not a mathematician. The prizes were first awarded in 1978. In the sciences, the Wolf Prize is awarded in Medicine, Agriculture, Mathematics, Chemistry, and Physics. It is especially important historically because many Wolf Prize in Mathematics laureates later received, or had already received, other major mathematical prizes. Unlike the Fields Medal, it is not restricted by age, and mathematics need not be awarded every year.

| Year    | Laureate(s)                                      | Quiz anchors                                                                                |
|---------|--------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1978    | Israel Gelfand; Carl L. Siegel                   | Representation theory, functional analysis; number theory, several complex variables.       |
| 1979    | Jean Leray; André Weil                           | PDE and algebraic topology; algebraic geometry and number theory.                           |
| 1980    | Henri Cartan; Andrey Kolmogorov                  | Several complex variables and algebraic topology; probability theory and dynamical systems. |
| 1981    | Lars Ahlfors; Oscar Zariski                      | Complex analysis and Riemann surfaces; algebraic geometry.                                  |
| 1982    | Mark Krein; Hassler Whitney                      | Functional analysis; differential topology and Whitney embedding.                           |
| 1983    | Shiing-Shen Chern                                | Global differential geometry, Chern classes.                                                |
| 1984    | Paul Erdős                                       | Combinatorics, number theory, probabilistic method.                                         |
| 1985    | Kunihiko Kodaira; Hans Lewy                      | Complex geometry; partial differential equations.                                           |
| 1986    | Samuel Eilenberg; Atle Selberg                   | Algebraic topology and category theory; analytic number theory and Selberg trace formula.   |
| 1987    | Kiyoshi Itô; Peter D. Lax                        | Stochastic calculus; partial differential equations and applied mathematics.                |
| 1988    | Lars Hörmander; Friedrich Hirzebruch             | Partial differential equations; topology, characteristic classes, Hirzebruch–Riemann–Roch.  |
| 1989    | Alberto Calderón; John Milnor                    | Harmonic analysis and singular integrals; topology and exotic spheres.                      |
| 1990    | Ennio De Giorgi; Ilya Piatetski-Shapiro          | Minimal surfaces and regularity theory; automorphic forms.                                  |
| 1992    | Lennart Carleson; John G. Thompson               | Harmonic analysis; finite group theory.                                                     |
| 1993    | Mikhail Gromov; Jacques Tits                     | Geometry and geometric group theory; buildings and group theory.                            |
| 1995/96 | Robert Langlands; Andrew Wiles                   | Langlands program; Fermat’s Last Theorem.                                                   |
| 1996/97 | Joseph Keller; Yakov Sinai                       | Applied mathematics and waves; dynamical systems and mathematical physics.                  |
| 1999    | László Lovász; Elias M. Stein                    | Discrete mathematics and algorithms; harmonic analysis.                                     |
| 2000    | Raoul Bott; Jean-Pierre Serre                    | Topology and geometry; algebraic topology, algebraic geometry, number theory.               |
| 2001    | Vladimir Arnold; Saharon Shelah                  | Dynamical systems and singularity theory; mathematical logic and set theory.                |
| 2002/03 | Mikio Sato; John Tate                            | Algebraic analysis and hyperfunctions; arithmetic geometry.                                 |
| 2005    | Gregory Margulis; Serge Novikov                  | Lie groups, ergodic theory, discrete subgroups; topology and geometry.                      |
| 2006/07 | Stephen Smale; Hillel Furstenberg                | Differential topology and dynamical systems; ergodic theory and probability.                |
| 2008/09 | Pierre Deligne; Phillip Griffiths; David Mumford | Algebraic geometry, Hodge theory, moduli spaces.                                            |
| 2010    | Shing-Tung Yau; Dennis Sullivan                  | Geometric analysis; topology and dynamics.                                                  |

Continued on the next page.

| Year | Laureate(s)                            | Quiz anchors                                                                          |
|------|----------------------------------------|---------------------------------------------------------------------------------------|
| 2012 | Michael Aschbacher; Luis Caffarelli    | Finite groups; nonlinear PDE and free-boundary problems.                              |
| 2013 | George Mostow; Michael Artin           | Rigidity, geometry, Lie groups; algebraic geometry.                                   |
| 2014 | Peter Sarnak                           | Number theory, analysis, geometry, automorphic forms.                                 |
| 2015 | James Arthur                           | Trace formula, automorphic representations.                                           |
| 2017 | Richard Schoen; Charles Fefferman      | Geometric analysis; analysis and partial differential equations.                      |
| 2018 | Alexander Beilinson; Vladimir Drinfeld | Geometric representation theory, mathematical physics, Langlands program.             |
| 2019 | Jean-François Le Gall; Gregory Lawler  | Probability theory, random walks, Brownian motion, SLE.                               |
| 2020 | Simon Donaldson; Yakov Eliashberg      | Differential geometry, topology, gauge theory; symplectic and contact topology.       |
| 2022 | George Lusztig                         | Representation theory and related areas.                                              |
| 2023 | Ingrid Daubechies                      | Wavelet theory, time-frequency analysis, signal processing.                           |
| 2024 | Adi Shamir; Noga Alon                  | Mathematical cryptography; combinatorics, graph theory, theoretical computer science. |

**Fact 12.7. Wolf Prize landmarks**

The following associations summarize the prize's main quiz anchors.

- Wolf Prize  $\leftrightarrow$  Wolf Foundation.
- The name honors Ricardo Wolf; it does not refer to a mathematician named Wolf.
- Wolf Prize in Mathematics  $\leftrightarrow$  no Fields-style age restriction.
- Paul Erdős  $\leftrightarrow$  1984 Wolf Prize  $\leftrightarrow$  combinatorics and probabilistic method.
- John Milnor  $\leftrightarrow$  1989 Wolf Prize; together with the Fields Medal and Abel Prize, this makes him a standard example connected to all three major mathematics prizes.
- Andrew Wiles and Robert Langlands shared the 1995/96 Wolf Prize in Mathematics.
- Ingrid Daubechies  $\leftrightarrow$  2023 Wolf Prize  $\leftrightarrow$  wavelets.
- Adi Shamir and Noga Alon  $\leftrightarrow$  2024 Wolf Prize  $\leftrightarrow$  cryptography and combinatorics.

**Fact 12.8. Prize recognition**

For concise review, remember the following map first.

- Fields Medal  $\leftrightarrow$  ICM  $\leftrightarrow$  IMU  $\leftrightarrow$  every four years  $\leftrightarrow$  age condition.
- Abel Prize  $\leftrightarrow$  Norwegian Academy of Science and Letters  $\leftrightarrow$  no age restriction.
- Wolf Prize  $\leftrightarrow$  Wolf Foundation  $\leftrightarrow$  no Fields-style age restriction.
- Fields Medal questions often ask for the year, ICM location, laureate, and keyword.
- Abel and Wolf questions often ask for the laureate–area association.

## Chapter 13

# Mathematicians and Mathematical Institutions

Science Quiz repeatedly asks for a mathematician from a birth year, a named theorem, a field, a book, or a short biographical clue, and it also uses universities, societies, unions, and research institutes as identifying information. The goal is rapid recognition through a few unmistakable associations.

## Mathematicians

The following table is ordered by birth. Ancient dates are approximate, and “fl.” indicates the period in which a person is known to have worked. Living mathematicians are written with an open-ended life span, such as “1980–”. The third column records origin and major career locations. Nationality or citizenship is included only when explicitly stated. An arrow records chronological movement only; it does not by itself assert a change of citizenship. When a modern national label would be anachronistic, the historical polity is given together with its present geographical location. Botong Wang is included without a birth year because no reliable public birth year was identified.

| Mathematician       | Life              | Origin and career locations           | Main fields             | Quiz anchors                                                                                                                                  |
|---------------------|-------------------|---------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Thales of Miletus   | c. 624–c. 546 BCE | Ancient Greece (Miletus; now Türkiye) | Geometry                | Early deductive geometry; Thales’ theorem; a right angle in a semicircle.                                                                     |
| Pythagoras          | c. 570–c. 495 BCE | Ancient Greece (Samos)                | Geometry, number theory | Pythagorean theorem; Pythagorean school; numerical ratios in music.                                                                           |
| Euclid              | fl. c. 300 BCE    | Hellenistic Egypt                     | Geometry, foundations   | <i>Elements</i> in thirteen books; axiomatic method; five postulates and the parallel postulate; Euclidean algorithm; infinitely many primes. |
| Archimedes          | c. 287–212 BCE    | Greek Sicily (now Italy)              | Geometry, mechanics     | Method of exhaustion; bounds for $\pi$ ; lever; buoyancy; sphere and circumscribed cylinder.                                                  |
| Apollonius of Perga | c. 240–c. 190 BCE | Ancient Greece (Perga; now Türkiye)   | Geometry                | <i>Conics</i> ; systematic study of ellipses, parabolas, and hyperbolas.                                                                      |

Continued on the next page.

| Mathematician           | Life               | Origin and career locations    | Main fields                         | Quiz anchors                                                                                                |
|-------------------------|--------------------|--------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Diophantus              | c. 200–c. 284      | Roman Egypt                    | Number theory, algebra              | <i>Arithmetica</i> ; Diophantine equations.                                                                 |
| Liu Hui                 | c. 225–c. 295      | China                          | Arithmetic, geometry                | Commentary on <i>The Nine Chapters</i> ; polygon method for $\pi$ ; elimination procedures.                 |
| Hypatia                 | c. 355–415         | Roman Egypt                    | Geometry, astronomy                 | Alexandrian teacher and commentator; associated with editions of Diophantus and Apollonius.                 |
| Zu Chongzhi             | 429–500            | China                          | Numerical mathematics, astronomy    | $3.1415926 < \pi < 3.1415927$ ; the approximation $\pi \approx 355/113$ .                                   |
| Aryabhata               | 476–550            | India                          | Arithmetic, trigonometry, astronomy | <i>Aryabhatiya</i> ; sine tables; place-value computation and astronomical algorithms.                      |
| Brahmagupta             | 598–c. 668         | India                          | Algebra, number theory              | Rules for zero and negative numbers; Brahmagupta's formula; quadratic equations.                            |
| Muhammad al-Khwarizmi   | c. 780–c. 850      | Khwarazm (now Uzbekistan)      | Algebra, arithmetic                 | Systematic algebra; <i>al-jabr</i> and <i>al-muqabala</i> ; the words <i>algebra</i> and <i>algorithm</i> . |
| Omar Khayyam            | 1048–1131          | Persia (now Iran)              | Algebra, geometry                   | Classification of cubic equations and geometric solutions by conic sections.                                |
| Bhāskara II             | 1114–1185          | India                          | Algebra, number theory, astronomy   | <i>Līlāvati</i> ; indeterminate equations; cyclic methods related to Pell equations.                        |
| Fibonacci               | c. 1170–after 1240 | Italy                          | Arithmetic, number theory           | <i>Liber Abaci</i> ; spread of Hindu–Arabic numerals in Europe; Fibonacci sequence.                         |
| Madhava of Sangamagrama | c. 1340–c. 1425    | India                          | Infinite series, trigonometry       | Kerala school; series for $\pi$ , sine, and cosine.                                                         |
| Jamshid al-Kashi        | c. 1380–1429       | Persia (now Iran)              | Numerical mathematics, astronomy    | Decimal fractions; high-precision computation of $\pi$ ; law of cosines.                                    |
| Niccolò Tartaglia       | c. 1499–1557       | Republic of Venice (now Italy) | Algebra                             | A method for depressed cubic equations; the Cardano–Tartaglia priority dispute.                             |
| Gerolamo Cardano        | 1501–1576          | Duchy of Milan (now Italy)     | Algebra, probability                | <i>Ars Magna</i> ; cubic and quartic equations; early probability.                                          |
| Robert Recorde          | c. 1512–1558       | Wales                          | Arithmetic, algebra                 | Introduced the equality sign $=$ in <i>The Whetstone of Witte</i> (1557).                                   |
| Lodovico Ferrari        | 1522–1565          | Italy                          | Algebra                             | General solution of the quartic equation, published in Cardano's <i>Ars Magna</i> .                         |
| Rafael Bombelli         | 1526–1572          | Italy                          | Algebra                             | <i>L'Algebra</i> ; systematic rules for complex numbers in solutions of cubic equations.                    |
| François Viète          | 1540–1603          | France                         | Algebra, trigonometry               | Systematic symbolic algebra; Viète's formulas.                                                              |
| Simon Stevin            | 1548–1620          | Flanders (now Belgium)         | Arithmetic, mechanics               | Promoted decimal fractions in Europe; work on statics and hydrostatics.                                     |
| John Napier             | 1550–1617          | Scotland                       | Computation, logarithms             | Introduced logarithms in print in 1614; Napier's bones.                                                     |

Continued on the next page.

| Mathematician             | Life         | Origin and career locations                                                      | Main fields                             | Quiz anchors                                                                                                                                       |
|---------------------------|--------------|----------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Thomas Harriot            | c. 1560–1621 | England                                                                          | Algebra, astronomy                      | Introduced the inequality signs $<$ and $>$ in work published posthumously in 1631.                                                                |
| Galileo Galilei           | 1564–1642    | Italy                                                                            | Mathematical physics                    | Mathematical description of motion; telescopic astronomy.                                                                                          |
| Johannes Kepler           | 1571–1630    | Holy Roman Empire (now Germany)                                                  | Geometry, astronomy                     | Laws of planetary motion; Kepler conjecture on sphere packing.                                                                                     |
| Marin Mersenne            | 1588–1648    | France                                                                           | Number theory, acoustics                | Mersenne primes $2^p - 1$ ; scientific correspondence network in seventeenth-century Europe.                                                       |
| Girard Desargues          | 1591–1661    | France                                                                           | Projective geometry                     | Desargues' theorem; foundational ideas of projective geometry.                                                                                     |
| René Descartes            | 1596–1650    | France (birthplace or historical origin) → Dutch Republic (later residence/work) | Analytic geometry, algebra              | Cartesian coordinates; <i>La Géométrie</i> ; rule of signs.                                                                                        |
| Bonaventura Cavalieri     | 1598–1647    | Italy                                                                            | Geometry                                | Method of indivisibles, an important precursor to integral calculus.                                                                               |
| Pierre de Fermat          | c. 1607–1665 | France                                                                           | Number theory, geometry, probability    | Fermat's Last Theorem; little theorem; analytic geometry; probability correspondence with Pascal.                                                  |
| John Wallis               | 1616–1703    | England                                                                          | Analysis, geometry                      | Introduced the symbol $\infty$ ; Wallis product for $\pi$ ; infinitesimal methods.                                                                 |
| Blaise Pascal             | 1623–1662    | France                                                                           | Probability, geometry                   | Pascal's triangle; foundations of probability with Fermat; projective geometry.                                                                    |
| Christiaan Huygens        | 1629–1695    | Dutch Republic (Netherlands)                                                     | Probability, mechanics                  | First printed treatise on probability; pendulum and wave theory.                                                                                   |
| Isaac Barrow              | 1630–1677    | England                                                                          | Geometry, analysis                      | Geometric form of the fundamental theorem of calculus; Newton's teacher at Cambridge.                                                              |
| James Gregory             | 1638–1675    | Scotland                                                                         | Analysis, geometry                      | Infinite series; Gregory series for $\arctan x$ ; anticipated the general Taylor-series method in correspondence of 1671.                          |
| Isaac Newton              | 1643–1727    | England                                                                          | Calculus, mechanics                     | Fluxions; fundamental theorem of calculus; generalized binomial theorem; laws of motion and gravitation.                                           |
| Gottfried Wilhelm Leibniz | 1646–1716    | Holy Roman Empire (now Germany)                                                  | Calculus, logic                         | Independent development of calculus; $dx$ and $\int$ notation; binary arithmetic.                                                                  |
| Choi Seok-jeong           | 1646–1715    | Joseon Korea                                                                     | Combinatorics, recreational mathematics | <i>Gusuryak</i> ; orthogonal Latin squares of order 9; magic squares; Jisuguimundo, or the Hexagonal Tortoise Problem.                             |
| Michel Rolle              | 1652–1719    | France                                                                           | Analysis, algebra                       | Rolle's theorem; early critic of infinitesimal calculus who later accepted it; adopted the now-standard indexed radical notation for $n$ th roots. |

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| Mathematician           | Life         | Origin and career locations                                                                        | Main fields                        | Quiz anchors                                                                                                      |
|-------------------------|--------------|----------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Jakob Bernoulli         | 1655–1705    | Switzerland                                                                                        | Probability, analysis              | Bernoulli numbers; law of large numbers; <i>Ars Conjectandi</i> .                                                 |
| Guillaume de l'Hôpital  | 1661–1704    | France                                                                                             | Analysis                           | First calculus textbook; l'Hôpital's rule, developed from lessons by Johann Bernoulli.                            |
| Abraham de Moivre       | 1667–1754    | France (birthplace or historical origin) → England (later residence/work)                          | Probability, complex numbers       | De Moivre's formula; normal approximation to the binomial distribution.                                           |
| Johann Bernoulli        | 1667–1748    | Switzerland                                                                                        | Analysis, mechanics                | Brachistochrone problem; early calculus; teacher of Euler and l'Hôpital.                                          |
| William Jones           | 1675–1749    | Wales                                                                                              | Mathematics, navigation            | Introduced the symbol $\pi$ for the circle ratio in 1706; Euler later popularized it.                             |
| Brook Taylor            | 1685–1731    | England                                                                                            | Analysis                           | Taylor's theorem and Taylor series; finite differences.                                                           |
| Christian Goldbach      | 1690–1764    | Prussia (now Russia) (birthplace or historical origin) → Russia (later residence/work)             | Number theory                      | Goldbach conjecture, proposed in correspondence with Euler.                                                       |
| James Stirling          | 1692–1770    | Scotland                                                                                           | Analysis, combinatorics            | Stirling's approximation; Stirling numbers.                                                                       |
| Colin Maclaurin         | 1698–1746    | Scotland                                                                                           | Analysis, geometry                 | Maclaurin series; <i>Treatise of Fluxions</i> .                                                                   |
| Daniel Bernoulli        | 1700–1782    | Netherlands (birthplace or historical origin) → Switzerland (later residence/work)                 | Probability, fluid mechanics       | Bernoulli principle; expected utility and the St. Petersburg paradox.                                             |
| Thomas Bayes            | c. 1701–1761 | England                                                                                            | Probability                        | Bayes' theorem; posthumous 1763 publication.                                                                      |
| Leonhard Euler          | 1707–1783    | Switzerland (birthplace or historical origin) → Russia and Prussia (later residence/work)          | Analysis, number theory, geometry  | Basel problem; Euler formula and identity; graph theory; polyhedron formula; standardized much modern notation.   |
| Jean le Rond d'Alembert | 1717–1783    | France                                                                                             | Analysis, mechanics                | Wave equation; d'Alembert's principle; work on the fundamental theorem of algebra.                                |
| Maria Gaetana Agnesi    | 1718–1799    | Italy                                                                                              | Analysis                           | <i>Instituzioni analitiche</i> ; one of the earliest comprehensive calculus textbooks; the witch of Agnesi curve. |
| Joseph-Louis Lagrange   | 1736–1813    | Sardinia (now Italy) (birthplace or historical origin) → Prussia and France (later residence/work) | Analysis, mechanics, number theory | Lagrangian mechanics; multipliers; four-square theorem.                                                           |
| Gaspard Monge           | 1746–1818    | France                                                                                             | Geometry                           | Descriptive geometry; Monge formulation of optimal transport.                                                     |
| John Playfair           | 1748–1819    | Scotland                                                                                           | Geometry, geology                  | Playfair's axiom, the standard modern reformulation of Euclid's parallel postulate.                               |
| Pierre-Simon Laplace    | 1749–1827    | France                                                                                             | Probability, celestial mechanics   | Laplace transform; Laplacian; analytic theory of probability.                                                     |

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| Mathematician                  | Life      | Origin and career locations                                                             | Main fields                         | Quiz anchors                                                                                                 |
|--------------------------------|-----------|-----------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Adrien-Marie Legendre          | 1752–1833 | France                                                                                  | Number theory, analysis             | Legendre symbol; elliptic integrals; least squares.                                                          |
| Joseph Fourier                 | 1768–1830 | France                                                                                  | Analysis, PDE                       | Fourier series and transform; heat equation.                                                                 |
| Sophie Germain                 | 1776–1831 | France                                                                                  | Number theory, elasticity           | Work on Fermat’s Last Theorem and vibrating plates; used the pseudonym M. LeBlanc.                           |
| Carl Friedrich Gauss           | 1777–1855 | Brunswick (now Germany)                                                                 | Number theory, geometry, statistics | <i>Disquisitiones Arithmeticae</i> ; quadratic reciprocity; Gaussian distribution; least squares; curvature. |
| Siméon Denis Poisson           | 1781–1840 | France                                                                                  | Analysis, probability, physics      | Poisson distribution and equation; mathematical physics and potential theory.                                |
| Bernard Bolzano                | 1781–1848 | Bohemia (now Czechia)                                                                   | Analysis, logic                     | Early rigorous analysis; intermediate value theorem; Bolzano–Weierstrass theorem.                            |
| Augustin-Louis Cauchy          | 1789–1857 | France                                                                                  | Analysis                            | Rigorous limits and convergence; Cauchy sequences; integral theorem; determinant theory.                     |
| August Ferdinand Möbius        | 1790–1868 | Saxony (now Germany)                                                                    | Geometry, number theory             | Möbius strip; Möbius transformations and inversion function.                                                 |
| Charles Babbage                | 1791–1871 | England                                                                                 | Computation, numerical mathematics  | Difference Engine and Analytical Engine.                                                                     |
| Nikolai Lobachevsky            | 1792–1856 | Russia                                                                                  | Geometry                            | Independent creation of hyperbolic, non-Euclidean geometry.                                                  |
| George Green                   | 1793–1841 | England                                                                                 | Analysis, mathematical physics      | Green’s theorem, Green’s identities, and Green functions.                                                    |
| Niels Henrik Abel              | 1802–1829 | Norway                                                                                  | Algebra, analysis                   | Impossibility of a radical formula for the general quintic; elliptic functions; Abel Prize namesake.         |
| János Bolyai                   | 1802–1860 | Transylvania (now Romania); Hungarian                                                   | Geometry                            | Independent creation of hyperbolic geometry.                                                                 |
| Carl Gustav Jacobi             | 1804–1851 | Prussia (now Germany)                                                                   | Analysis, mechanics, number theory  | Jacobian determinant; elliptic functions; Hamilton–Jacobi theory.                                            |
| Peter Gustav Lejeune Dirichlet | 1805–1859 | France (now Germany); German                                                            | Number theory, analysis             | Primes in arithmetic progressions; Dirichlet characters; modern function concept.                            |
| William Rowan Hamilton         | 1805–1865 | Ireland                                                                                 | Algebra, mechanics                  | Quaternions; Hamiltonian mechanics; Hamilton–Jacobi equation.                                                |
| Augustus De Morgan             | 1806–1871 | British India (birthplace or historical origin) → United Kingdom (later residence/work) | Logic, algebra                      | De Morgan’s laws; formal logic; mathematical induction terminology.                                          |
| Joseph Liouville               | 1809–1882 | France                                                                                  | Analysis, number theory             | First proof that transcendental numbers exist; Liouville’s theorem.                                          |
| Hermann Grassmann              | 1809–1877 | Prussia (now Poland); German                                                            | Algebra, geometry                   | <i>Ausdehnungslehre</i> ; vector spaces and exterior algebra.                                                |

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| Mathematician                            | Life      | Origin and career locations                                                       | Main fields                               | Quiz anchors                                                                                                                       |
|------------------------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Ernst Kummer                             | 1810–1893 | Prussia (now Poland); German                                                      | Number theory, algebra                    | Ideal numbers; regular primes; major progress on Fermat’s Last Theorem.                                                            |
| Évariste Galois                          | 1811–1832 | France                                                                            | Algebra                                   | Galois theory; solvability by radicals through groups.                                                                             |
| James Joseph Sylvester                   | 1814–1897 | England                                                                           | Algebra, number theory                    | Matrix and invariant theory; coined the term “matrix”; Sylvester’s law of inertia.                                                 |
| Karl Weierstrass                         | 1815–1897 | Prussia (now Germany)                                                             | Analysis                                  | $\varepsilon$ – $\delta$ rigor; uniform convergence; nowhere-differentiable function.                                              |
| George Boole                             | 1815–1864 | England (birthplace or historical origin) → Ireland (later residence/work)        | Logic, algebra                            | Boolean algebra; algebraic treatment of logic.                                                                                     |
| Ada Lovelace                             | 1815–1852 | England                                                                           | Computation                               | Notes on Babbage’s Analytical Engine; an algorithm for Bernoulli numbers often described as the first published computer program.  |
| George Gabriel Stokes                    | 1819–1903 | Ireland (birthplace or historical origin) → United Kingdom (later residence/work) | Analysis, mathematical physics            | Stokes’ theorem; fluid equations; optics.                                                                                          |
| Pafnuty Chebyshev                        | 1821–1894 | Russia                                                                            | Number theory, approximation, probability | Chebyshev polynomials and inequality; work on prime distribution.                                                                  |
| Arthur Cayley                            | 1821–1895 | England                                                                           | Algebra, geometry                         | Matrix algebra; Cayley’s theorem; Cayley–Hamilton theorem.                                                                         |
| Charles Hermite                          | 1822–1901 | France                                                                            | Analysis, number theory                   | Proved $e$ transcendental in 1873; Hermite polynomials and normal forms.                                                           |
| Leopold Kronecker                        | 1823–1891 | Prussia (now Poland); German                                                      | Algebra, number theory                    | Kronecker product and delta; arithmetic approach to algebra.                                                                       |
| Bernhard Riemann                         | 1826–1866 | Kingdom of Hanover (now Germany)                                                  | Analysis, geometry, number theory         | Riemann surfaces and integral; Riemannian geometry; zeta function and hypothesis.                                                  |
| Richard Dedekind                         | 1831–1916 | Brunswick (now Germany)                                                           | Algebra, foundations, number theory       | Dedekind cuts; ideals; rigorous construction of the real numbers.                                                                  |
| Charles Lutwidge Dodgson (Lewis Carroll) | 1832–1898 | England                                                                           | Logic, geometry, recreational mathematics | Oxford mathematics lecturer; wrote <i>Alice’s Adventures in Wonderland</i> under the pen name Lewis Carroll; Dodgson condensation. |
| Camille Jordan                           | 1838–1922 | France                                                                            | Algebra, analysis                         | Jordan normal form; Jordan curve theorem; permutation groups.                                                                      |
| Sophus Lie                               | 1842–1899 | Norway                                                                            | Geometry, differential equations          | Lie groups and Lie algebras; continuous symmetries.                                                                                |
| Georg Cantor                             | 1845–1918 | Russian Empire (birthplace or historical origin) → Germany (later residence/work) | Set theory                                | Founder of set theory; diagonal argument; cardinalities; Cantor set.                                                               |
| Wilhelm Killing                          | 1847–1923 | Prussia (now Germany)                                                             | Lie theory, geometry                      | Classification program for simple Lie algebras; Killing form.                                                                      |

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| Mathematician                     | Life                   | Origin and career locations                                                                       | Main fields                       | Quiz anchors                                                                                |
|-----------------------------------|------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------|
| Gottlob Frege                     | 1848–1925              | Germany                                                                                           | Logic, foundations                | Predicate logic; <i>Begriffsschrift</i> ; logicist foundations of arithmetic.               |
| Felix Klein                       | 1849–1925              | Prussia (now Germany)                                                                             | Geometry, group theory            | Erlangen program; Klein bottle; Klein four-group.                                           |
| Ferdinand Georg Frobenius         | 1849–1917              | Prussia (now Germany)                                                                             | Algebra, analysis                 | Frobenius theorem and endomorphism; character theory of finite groups.                      |
| Sofia Kovalevskaya                | 1850–1891              | Russia (birthplace or historical origin) → Sweden (later residence/work)                          | Analysis, PDE                     | Cauchy–Kovalevskaya theorem; rigid-body dynamics.                                           |
| Ferdinand von Lindemann           | 1852–1939              | Germany                                                                                           | Number theory, analysis           | Proved $\pi$ transcendental in 1882, establishing the impossibility of squaring the circle. |
| Henri Poincaré                    | 1854–1912              | France                                                                                            | Topology, dynamics, geometry      | Poincaré conjecture; qualitative dynamics; foundational topology.                           |
| Andrey Markov                     | 1856–1922              | Russia                                                                                            | Probability, number theory        | Markov chains; Markov inequality; stochastic dependence.                                    |
| Giuseppe Peano                    | 1858–1932              | Italy                                                                                             | Logic, foundations                | Peano axioms; symbolic logic; space-filling Peano curve.                                    |
| David Hilbert                     | 1862–1943              | Prussia (Königsberg; now Russia); German                                                          | Foundations, algebra, geometry    | Twenty-three problems; Hilbert spaces and basis theorem; axiomatic geometry.                |
| John Charles Fields               | 1863–1932              | Canada                                                                                            | Complex analysis                  | Canadian mathematician who proposed and endowed the Fields Medal.                           |
| Hermann Minkowski                 | 1864–1909              | Russian Empire (now Lithuania) (birthplace or historical origin) → Germany (later residence/work) | Geometry, number theory           | Geometry of numbers; Minkowski spacetime; convex body theorem.                              |
| Jacques Hadamard                  | 8 Dec 1865–17 Oct 1963 | France                                                                                            | Analysis, number theory           | Prime number theorem; Cauchy–Hadamard theorem; Hadamard matrices.                           |
| Charles-Jean de la Vallée Poussin | 1866–1962              | Belgium                                                                                           | Analysis, number theory           | Independent proof of the prime number theorem in 1896.                                      |
| Felix Hausdorff                   | 1868–1942              | Germany                                                                                           | Topology, set theory              | Hausdorff spaces; Hausdorff measure and dimension.                                          |
| Élie Cartan                       | 1869–1951              | France                                                                                            | Differential geometry, Lie theory | Lie groups and algebras; differential forms; moving frames.                                 |
| Ernst Zermelo                     | 1871–1953              | Germany                                                                                           | Set theory, foundations           | Axiomatization of set theory; well-ordering theorem; Zermelo–Fraenkel set theory.           |
| Bertrand Russell                  | 1872–1970              | Wales, United Kingdom                                                                             | Logic, foundations                | Russell’s paradox; <i>Principia Mathematica</i> with Whitehead.                             |
| Alfred Young                      | 1873–1940              | England                                                                                           | Algebra, representation theory    | Young diagrams and Young tableaux for symmetric groups.                                     |
| Henri Lebesgue                    | 1875–1941              | France                                                                                            | Analysis, measure theory          | Lebesgue measure and integral; dominated convergence theorem.                               |

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| Mathematician       | Life      | Origin and career locations                                                                                        | Main fields                              | Quiz anchors                                                                                         |
|---------------------|-----------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------|
| G. H. Hardy         | 1877–1947 | England                                                                                                            | Analysis, number theory                  | Hardy–Weinberg principle; Hardy–Littlewood circle method; collaboration with Ramanujan.              |
| Maurice Fréchet     | 1878–1973 | France                                                                                                             | Analysis, topology                       | Metric spaces; Fréchet derivative; abstract functional spaces.                                       |
| Frigyes Riesz       | 1880–1956 | Austria-Hungary (now Hungary)                                                                                      | Functional analysis                      | Riesz representation theorem; $L^p$ spaces and operator theory.                                      |
| L. E. J. Brouwer    | 1881–1966 | Netherlands                                                                                                        | Topology, foundations                    | Brouwer fixed-point theorem; invariance of domain; intuitionism.                                     |
| Wacław Sierpiński   | 1882–1969 | Russian Empire (now Poland); Polish                                                                                | Set theory, topology                     | Sierpiński triangle and carpet; continuum and descriptive set theory.                                |
| Emmy Noether        | 1882–1935 | Germany (birthplace or historical origin) → United States (later residence/work)                                   | Algebra, mathematical physics            | Noetherian rings and modules; symmetry–conservation theorem.                                         |
| J. E. Littlewood    | 1885–1977 | England                                                                                                            | Analysis, number theory                  | Hardy–Littlewood circle method and conjectures; analytic number theory.                              |
| Hermann Weyl        | 1885–1955 | Germany (birthplace or historical origin) → Switzerland and United States (later residence/work)                   | Geometry, representation theory, physics | Weyl groups and character formula; gauge ideas; foundations.                                         |
| Hugo Steinhaus      | 1887–1972 | Austria-Hungary (now Ukraine); Polish                                                                              | Analysis, probability                    | Lwów school; Steinhaus theorem; co-founder of <i>Studia Mathematica</i> .                            |
| George Pólya        | 1887–1985 | Austria-Hungary (now Hungary) (birthplace or historical origin) → United States (later residence/work)             | Combinatorics, probability               | Pólya enumeration theorem; random walks; <i>How to Solve It</i> .                                    |
| Srinivasa Ramanujan | 1887–1920 | British India                                                                                                      | Number theory, analysis                  | Partitions, modular forms, mock theta functions; 1729.                                               |
| Richard Courant     | 1888–1972 | Prussia (now Poland) (birthplace or historical origin) → Germany and United States (later residence/work)          | Analysis, PDE                            | Courant Institute namesake; finite elements; Courant–Friedrichs–Lewy condition.                      |
| Ludwig Wittgenstein | 1889–1951 | Austria (birthplace or historical origin) → United Kingdom (later residence/work)                                  | Logic, philosophy                        | <i>Tractatus</i> ; language games; major influence on logical positivism and philosophy of language. |
| Stefan Banach       | 1892–1945 | Austria-Hungary (now Poland); Polish                                                                               | Functional analysis                      | Banach spaces; contraction mapping theorem; Lwów school and <i>The Scottish Book</i> .               |
| Norbert Wiener      | 1894–1964 | United States                                                                                                      | Analysis, probability                    | Wiener process and measure; harmonic analysis; founder of cybernetics.                               |
| Emil Artin          | 1898–1962 | Austria-Hungary (now Czechia) (birthplace or historical origin) → Germany and United States (later residence/work) | Algebra, number theory                   | Artin reciprocity; braid groups; modern algebraic number theory.                                     |

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| Mathematician     | Life      | Origin and career locations                                                                                     | Main fields                        | Quiz anchors                                                                             |
|-------------------|-----------|-----------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------|
| Oscar Zariski     | 1899–1986 | Russian Empire (now Belarus) (birthplace or historical origin) → Italy and United States (later residence/work) | Algebraic geometry                 | Zariski topology; foundations of modern algebraic geometry.                              |
| Juliusz Schauder  | 1899–1943 | Austria-Hungary (now Ukraine); Polish                                                                           | Functional analysis, PDE           | Schauder fixed-point theorem and Schauder estimates.                                     |
| Antoni Zygmund    | 1900–1992 | Russian Empire (now Poland) (birthplace or historical origin) → United States (later residence/work)            | Harmonic analysis                  | <i>Trigonometric Series</i> ; Chicago school of harmonic analysis.                       |
| Alfred Tarski     | 1901–1983 | Russian Empire (now Poland) (birthplace or historical origin) → United States (later residence/work)            | Logic, foundations                 | Semantic definition of truth; Tarski's theorem; Banach–Tarski paradox.                   |
| Andrey Kolmogorov | 1903–1987 | Russia / USSR                                                                                                   | Probability, dynamics              | Axioms of probability; Kolmogorov complexity; turbulence and KAM theory.                 |
| Alonzo Church     | 1903–1995 | United States                                                                                                   | Logic, computation                 | Lambda calculus; Church's theorem; Church–Turing thesis.                                 |
| John von Neumann  | 1903–1957 | Austria-Hungary (now Hungary) (birthplace or historical origin) → United States (later residence/work)          | Analysis, computation, game theory | Operator algebras; quantum mechanics; minimax theorem; stored-program architecture.      |
| Stanisław Mazur   | 1905–1981 | Austria-Hungary (now Ukraine); Polish                                                                           | Functional analysis                | Lwów school; Banach–Mazur theorem; Scottish Book basis problem and its live-goose prize. |
| Kurt Gödel        | 1906–1978 | Austria-Hungary (now Czechia) (birthplace or historical origin) → United States (later residence/work)          | Logic, foundations                 | Completeness and incompleteness theorems; consistency of choice and CH with ZF.          |
| André Weil        | 1906–1998 | France (birthplace or historical origin) → United States (later residence/work)                                 | Number theory, algebraic geometry  | Weil conjectures; adeles; founding member of Bourbaki.                                   |
| Hans Fitting      | 1906–1938 | Germany                                                                                                         | Group theory, algebra              | Fitting subgroup, lemma, decomposition, and ideals.                                      |
| Lars Ahlfors      | 1907–1996 | Finland (birthplace or historical origin) → United States (later residence/work)                                | Complex analysis                   | Riemann surfaces; Ahlfors theory; one of the first two Fields Medalists.                 |
| Claude Chevalley  | 1909–1984 | South Africa; French (birthplace or historical origin) → United States (later residence/work)                   | Algebra, number theory             | Chevalley groups; class field theory; founding member of Bourbaki.                       |
| Stanisław Ulam    | 1909–1984 | Austria-Hungary (now Ukraine); Polish (birthplace or historical origin) → United States (later residence/work)  | Set theory, probability            | Monte Carlo method; Ulam spiral; Scottish Book and Manhattan Project.                    |
| Saunders Mac Lane | 1909–2005 | United States                                                                                                   | Algebra, topology                  | Co-founder of category theory; Eilenberg–Mac Lane spaces.                                |

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| Mathematician          | Life      | Origin and career locations                                                                                    | Main fields                                 | Quiz anchors                                                                              |
|------------------------|-----------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------|
| Shiing-Shen Chern      | 1911–2004 | China (birthplace or historical origin) → United States (later residence/work)                                 | Differential geometry, topology             | Chern classes; Chern–Gauss–Bonnet theorem; Chern Medal namesake.                          |
| Alan Turing            | 1912–1954 | England                                                                                                        | Logic, computation                          | Turing machine; computability; codebreaking; Turing test.                                 |
| Paul Erdős             | 1913–1996 | Austria-Hungary (now Hungary); Hungarian                                                                       | Number theory, combinatorics                | Probabilistic method; Erdős number; prolific collaboration.                               |
| Israel Gelfand         | 1913–2009 | Russian Empire (now Ukraine) (birthplace or historical origin) → USSR and United States (later residence/work) | Functional analysis, representation theory  | Gelfand representation; representation theory; influential seminar.                       |
| Samuel Eilenberg       | 1913–1998 | Russian Empire (now Poland) (birthplace or historical origin) → United States (later residence/work)           | Algebraic topology                          | Co-founder of category theory and homological algebra; Eilenberg–Mac Lane spaces.         |
| Mark Kac               | 1914–1984 | Russian Empire (now Ukraine); Polish (birthplace or historical origin) → United States (later residence/work)  | Probability, mathematical physics           | Feynman–Kac formula; “Can one hear the shape of a drum?”                                  |
| George Dantzig         | 1914–2005 | United States                                                                                                  | Optimization, statistics                    | Linear programming; simplex method; the mistaken-homework anecdote.                       |
| Laurent Schwartz       | 1915–2002 | France                                                                                                         | Analysis                                    | Theory of distributions; 1950 Fields Medal.                                               |
| Claude Shannon         | 1916–2001 | United States                                                                                                  | Information theory                          | Entropy, channel capacity, and the mathematical theory of communication.                  |
| Atle Selberg           | 1917–2007 | Norway (birthplace or historical origin) → United States (later residence/work)                                | Analytic number theory                      | Selberg sieve and trace formula; elementary proof of the prime number theorem with Erdős. |
| Benoit Mandelbrot      | 1924–2010 | Poland (birthplace or historical origin) → France and United States (later residence/work)                     | Geometry, probability                       | Fractal geometry; Mandelbrot set; popularized the term “fractal.”                         |
| Louis Nirenberg        | 1925–2020 | Canada (birthplace or historical origin) → United States (later residence/work)                                | Analysis, PDE                               | Fundamental estimates for elliptic PDE; 2015 Abel Prize with John Nash.                   |
| Peter D. Lax           | 1926–2025 | Hungary (birthplace or historical origin) → United States (later residence/work)                               | Analysis, PDE                               | Lax pair and Lax equivalence theorem; numerical analysis; 2005 Abel Prize.                |
| Jean-Pierre Serre      | 1926–     | France                                                                                                         | Algebraic topology, geometry, number theory | Youngest Fields Medalist; first Abel laureate; Serre duality and spectral sequences.      |
| Alexander Grothendieck | 1928–2014 | Germany; stateless, later French                                                                               | Algebraic geometry                          | Schemes, topoi, étale cohomology; transformed modern algebraic geometry.                  |

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| Mathematician      | Life      | Origin and career locations                                                                | Main fields                          | Quiz anchors                                                                                                                                                                                    |
|--------------------|-----------|--------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| John Nash          | 1928–2015 | United States                                                                              | Geometry, game theory, PDE           | Nash equilibrium; Nash embedding theorem; 2015 Abel Prize.                                                                                                                                      |
| Michael Atiyah     | 1929–2019 | England; British                                                                           | Geometry, topology                   | Atiyah–Singer index theorem; K-theory; Fields and Abel prizes.                                                                                                                                  |
| Goro Shimura       | 1930–2019 | Japan (birthplace or historical origin) → United States (later residence/work)             | Number theory, automorphic forms     | Taniyama–Shimura modularity conjecture; Shimura varieties; work underlying the modular approach to Fermat’s Last Theorem.                                                                       |
| Stephen Smale      | 1930–     | United States                                                                              | Topology, dynamics                   | Higher-dimensional Poincaré conjecture; Smale horseshoe; Smale’s problems.                                                                                                                      |
| Lars Hörmander     | 1931–2012 | Sweden                                                                                     | Analysis, PDE                        | Linear partial differential operators; microlocal analysis; Fields Medal 1962.                                                                                                                  |
| John Milnor        | 1931–     | United States                                                                              | Topology, geometry, dynamics         | Exotic 7-spheres; Milnor fibration; Fields, Wolf, and Abel prizes.                                                                                                                              |
| Paul Cohen         | 1934–2007 | United States                                                                              | Logic, set theory                    | Forcing; independence of the continuum hypothesis and axiom of choice.                                                                                                                          |
| Yakov Sinai        | 1935–     | USSR (now Russia) (birthplace or historical origin) → United States (later residence/work) | Dynamical systems, probability       | Sinai billiards; Kolmogorov–Sinai entropy; 2014 Abel Prize.                                                                                                                                     |
| Robert Langlands   | 1936–     | Canada (birthplace or historical origin) → United States (later residence/work)            | Number theory, representation theory | Langlands program; reciprocity and functoriality; 2018 Abel Prize.                                                                                                                              |
| Vladimir Arnold    | 1937–2010 | USSR (now Ukraine); Russian                                                                | Dynamical systems, geometry          | KAM theory; Arnold conjecture; singularity theory.                                                                                                                                              |
| John Horton Conway | 1937–2020 | England (birthplace or historical origin) → United States (later residence/work)           | Algebra, combinatorics               | Surreal numbers; cellular automaton <i>Life</i> ; Conway groups and knot notation; look-and-say sequence<br>1, 11, 21, 1211, 111221, ... and Conway constant<br>$\lambda \approx 1.303577269$ . |
| Donald Knuth       | 1938–     | United States                                                                              | Algorithms, combinatorics            | <i>The Art of Computer Programming</i> ; TeX; analysis of algorithms.                                                                                                                           |
| Alan Baker         | 1939–2018 | England                                                                                    | Number theory                        | Linear forms in logarithms; transcendence theory; 1970 Fields Medal.                                                                                                                            |
| Dennis Sullivan    | 1941–     | United States                                                                              | Topology, dynamical systems          | Surgery theory, rational homotopy theory, and conformal dynamics; 2022 Abel Prize.                                                                                                              |
| Karen Uhlenbeck    | 1942–     | United States                                                                              | Geometric analysis                   | Gauge theory and harmonic maps; first woman Abel laureate.                                                                                                                                      |
| Mikhail Gromov     | 1943–     | USSR (now Russia) (birthplace or historical origin) → France (later residence/work)        | Geometry, topology                   | Gromov–Hausdorff distance; geometric group theory; 2009 Abel Prize.                                                                                                                             |

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| Mathematician     | Life          | Origin and career locations                                                                         | Main fields                                 | Quiz anchors                                                                                    |
|-------------------|---------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------|
| Per Enflo         | 1944–         | Sweden                                                                                              | Functional analysis                         | Solved Mazur’s Scottish Book basis problem and received the promised live goose.                |
| Pierre Deligne    | 1944–         | Belgium                                                                                             | Algebraic geometry                          | Proof of the Weil conjectures; Fields, Wolf, and Abel prizes.                                   |
| Grigory Margulis  | 1946–         | USSR (now Russia) (birthplace or historical origin) → United States (later residence/work)          | Lie groups, dynamics                        | Superrigidity, arithmeticity, and homogeneous dynamics; Fields and Abel prizes.                 |
| William Thurston  | 1946–<br>2012 | United States                                                                                       | Topology, geometry                          | Geometrization conjecture; eight geometries; low-dimensional topology.                          |
| Masaki Kashiwara  | 1947–         | Japan                                                                                               | Algebraic analysis, representation theory   | $D$ -modules, microlocal analysis, and crystal bases; 2025 Abel Prize.                          |
| Luis Caffarelli   | 1948–         | Argentina (birthplace or historical origin) → United States (later residence/work)                  | Nonlinear PDE                               | Regularity theory, free-boundary problems, and the Monge–Ampère equation; 2023 Abel Prize.      |
| László Lovász     | 1948–         | Hungary (birthplace or historical origin) → United States and Hungary (later residence/work)        | Combinatorics, theoretical computer science | Lovász local lemma, graph limits, and algorithms; 2021 Abel Prize.                              |
| Shing-Tung Yau    | 1949–         | China (birthplace or historical origin) → Hong Kong and United States (later residence/work)        | Geometric analysis                          | Calabi conjecture; Ricci-flat metrics; 1982 Fields Medal.                                       |
| Edward Witten     | 1951–         | United States                                                                                       | Mathematical physics                        | First physicist to receive the Fields Medal; quantum field theory, topology, and string theory. |
| Michel Talagrand  | 1952–         | France                                                                                              | Probability, functional analysis            | Concentration inequalities, stochastic processes, and spin glasses; 2024 Abel Prize.            |
| Adi Shamir        | 1952–         | Israel                                                                                              | Cryptography, theoretical computer science  | Co-inventor of RSA and Shamir secret sharing; Turing Award and Wolf Prize 2024.                 |
| Andrew Wiles      | 1953–         | England (birthplace or historical origin) → United States and United Kingdom (later residence/work) | Number theory                               | Fermat’s Last Theorem via modularity; special IMU plaque and Abel Prize.                        |
| Gerd Faltings     | 1954–         | West Germany (now Germany)                                                                          | Arithmetic geometry                         | Mordell conjecture; 1986 Fields Medal; 2026 Abel Prize.                                         |
| Ingrid Daubechies | 1954–         | Belgium (birthplace or historical origin) → United States (later residence/work)                    | Analysis, applied mathematics               | Orthogonal wavelets; Daubechies wavelets; 2023 Wolf Prize.                                      |
| Efim Zelmanov     | 1955–         | USSR (now Russia) (birthplace or historical origin) → United States (later residence/work)          | Algebra, group theory                       | Restricted Burnside problem; 1994 Fields Medal.                                                 |

Continued on the next page.

| Mathematician      | Life          | Origin and career locations                                                                     | Main fields                    | Quiz anchors                                                                                                       |
|--------------------|---------------|-------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Yitang Zhang       | 1955–         | China (birthplace or historical origin) → United States (later residence/work)                  | Analytic number theory         | Proved the first bounded-gap theorem for infinitely many pairs of primes, initiating rapid progress on prime gaps. |
| Avi Wigderson      | 1956–         | Israel (birthplace or historical origin) → United States (later residence/work)                 | Theoretical computer science   | Complexity theory, randomness, and pseudorandomness; 2021 Abel Prize and Turing Award.                             |
| Noga Alon          | 1956–         | Israel (birthplace or historical origin) → United States (later residence/work)                 | Combinatorics, graph theory    | Probabilistic method, algebraic combinatorics, and theoretical computer science; 2024 Wolf Prize.                  |
| Simon Donaldson    | 1957–         | England                                                                                         | Geometry, topology             | Gauge theory and smooth 4-manifolds; 1986 Fields Medal.                                                            |
| Maxim Kontsevich   | 1964–         | USSR (now Russia) (birthplace or historical origin) → France (later residence/work)             | Geometry, mathematical physics | Deformation quantization; mirror symmetry; 1998 Fields Medal.                                                      |
| Grigori Perelman   | 1966–         | USSR (now Russia)                                                                               | Geometry, topology             | Poincaré and geometrization conjectures via Ricci flow; declined the Fields Medal and Millennium Prize.            |
| Elon Lindenstrauss | 1970–         | Israel                                                                                          | Ergodic theory, number theory  | Measure rigidity and applications to number theory; 2010 Fields Medal.                                             |
| Cédric Villani     | 1973–         | France                                                                                          | PDE, optimal transport         | Boltzmann equation and Landau damping; 2010 Fields Medal; <i>Théorème vivant</i> .                                 |
| Manjul Bhargava    | 1974–         | Canada; Canadian–American                                                                       | Number theory                  | Higher composition laws; arithmetic statistics; 2014 Fields Medal.                                                 |
| Terence Tao        | 1975–         | Australia (birthplace or historical origin) → United States (later residence/work)              | Analysis, PDE, combinatorics   | Green–Tao theorem; harmonic analysis; 2006 Fields Medal.                                                           |
| Martin Hairer      | 1975–         | Switzerland; Austrian (birthplace or historical origin) → United Kingdom (later residence/work) | Probability, stochastic PDE    | Theory of regularity structures; 2014 Fields Medal.                                                                |
| Maryam Mirzakhani  | 1977–<br>2017 | Iran (birthplace or historical origin) → United States (later residence/work)                   | Geometry, dynamics             | Moduli spaces and Riemann surfaces; first woman Fields Medalist.                                                   |
| Caucher Birkar     | 1978–         | Iran; Kurdish (birthplace or historical origin) → United Kingdom (later residence/work)         | Algebraic geometry             | Birational geometry and boundedness of Fano varieties; 2018 Fields Medal.                                          |
| Akshay Venkatesh   | 1981–         | India (birthplace or historical origin) → Australia and United States (later residence/work)    | Number theory, dynamics        | Number theory and homogeneous dynamics; 2018 Fields Medal.                                                         |

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| Mathematician      | Life                           | Origin and career locations                                                                                  | Main fields                                 | Quiz anchors                                                                                                                                |
|--------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| June Huh           | 1983–                          | United States (birthplace or historical origin) → Republic of Korea and United States (later residence/work) | Combinatorics, algebraic geometry           | Hodge theory for matroids; log-concavity conjectures; 2022 Fields Medal.                                                                    |
| Alessio Figalli    | 1984–                          | Italy (birthplace or historical origin) → Switzerland (later residence/work)                                 | Analysis, optimal transport                 | Optimal transport and PDE; 2018 Fields Medal.                                                                                               |
| Maryna Viazovska   | 1984–                          | Ukraine (birthplace or historical origin) → Switzerland (later residence/work)                               | Number theory, discrete geometry            | Optimal sphere packing in dimensions 8 and 24; 2022 Fields Medal.                                                                           |
| Hugo Duminil-Copin | 1985–                          | France (birthplace or historical origin) → Switzerland (later residence/work)                                | Probability, mathematical physics           | Phase transitions and percolation; 2022 Fields Medal.                                                                                       |
| James Maynard      | 1987–                          | England                                                                                                      | Number theory                               | Small gaps between primes and Diophantine approximation; 2022 Fields Medal.                                                                 |
| Peter Scholze      | 1987–                          | East Germany (now Germany)                                                                                   | Arithmetic geometry                         | Perfectoid spaces and $p$ -adic geometry; 2018 Fields Medal.                                                                                |
| Jacob Tsimerman    | 1988–                          | USSR (now Russia) (birthplace or historical origin) → Israel and Canada (later residence/work)               | Arithmetic geometry                         | O-minimality in arithmetic and complex algebraic geometry; André–Oort and related conjectures; 2026 Fields Medal.                           |
| Yu Deng            | 1989–                          | China (birthplace or historical origin) → United States (later residence/work)                               | PDE, kinetic theory                         | Boltzmann equation from hard-sphere dynamics, wave kinetic equations, and probabilistic nonlinear Schrödinger equations; 2026 Fields Medal. |
| John Pardon        | 1989–                          | United States                                                                                                | Symplectic geometry, topology               | Virtual fundamental cycles, Fukaya categories, group actions on 3-manifolds, and knot theory; 2026 Fields Medal.                            |
| Hong Wang          | 1991–                          | China (birthplace or historical origin) → France and United States (later residence/work)                    | Harmonic analysis, geometric measure theory | Fourier restriction, local smoothing, and the three-dimensional Keakeya conjecture; 2026 Fields Medalist, third woman to receive the medal. |
| Botong Wang        | birth year not publicly listed | China (undergraduate education) → United States (doctoral study and later work)                              | Algebraic geometry, topology                | Topology of algebraic varieties and interactions with combinatorics and applied mathematics.                                                |

## Mathematical Institutions

The institutions below include learned societies, international unions, higher-education institutions, and research institutes. A research institute may host scholars and seminars without awarding degrees.

| Institution                                         | Founded                  | Location                       | Role and distinctive facts                                                                                                                             |
|-----------------------------------------------------|--------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Collège de France                                   | 1530                     | Paris, France                  | Public research chairs and lectures; admits listeners without enrollment and does not grant degrees.                                                   |
| École normale supérieure (ENS)                      | 1794                     | Paris, France                  | Selective higher-education and research institution; unlike IAS, IHES, and SLMath, it has degree-granting programs.                                    |
| London Mathematical Society (LMS)                   | 1865                     | London, United Kingdom         | Learned society that promotes mathematics through journals, meetings, grants, and prizes.                                                              |
| Société Mathématique de France (SMF)                | 1872                     | Paris, France                  | French learned society for mathematical research, publications, conferences, and professional exchange.                                                |
| American Mathematical Society (AMS)                 | 1888                     | Providence, United States      | Began as the New York Mathematical Society and adopted its present name in 1894; publishes <i>Mathematical Reviews</i> and MathSciNet.                 |
| Deutsche Mathematiker-Vereinigung (DMV)             | 1890                     | Germany                        | German mathematical society; supports research, communication, education, and public engagement.                                                       |
| Institut Mittag-Leffler                             | 1916                     | Djursholm, Sweden              | International research center of the Royal Swedish Academy of Sciences; associated with <i>Acta Mathematica</i> .                                      |
| International Mathematical Union (IMU)              | 1920; reconstituted 1951 | Secretariat in Berlin, Germany | International union that supports global cooperation, organizes the ICM, and awards the Fields Medal and other IMU prizes.                             |
| Institut Henri Poincaré (IHP)                       | 1928                     | Paris, France                  | International center for mathematics and theoretical physics; thematic programs, seminars, library, and outreach.                                      |
| Institute for Advanced Study (IAS)                  | 1930                     | Princeton, United States       | Independent institute for fundamental research; home to a School of Mathematics; has no degree programs.                                               |
| Mathematisches Forschungsinstitut Oberwolfach (MFO) | 1944                     | Oberwolfach, Germany           | International workshop and research center famous for small, intensive meetings and the Oberwolfach Research Institute for Mathematics.                |
| Tata Institute of Fundamental Research (TIFR)       | 1945                     | Mumbai, India                  | National center for fundamental research in mathematics and the sciences; central to the development of modern mathematics in India.                   |
| Institut des Hautes Études Scientifiques (IHES)     | 1958                     | Bures-sur-Yvette, France       | Private advanced research institute in mathematics and theoretical physics; hosts a small permanent faculty and many visitors; does not grant degrees. |
| Research Institute for Mathematical Sciences (RIMS) | 1963                     | Kyoto, Japan                   | <a href="#">Kyoto University</a> institute for mathematical research and collaboration; publishes <i>Publications of RIMS</i> .                        |

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| Institution                                            | Founded | Location                           | Role and distinctive facts                                                                                                                    |
|--------------------------------------------------------|---------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Max Planck Institute for Mathematics (MPIM)            | 1980    | Bonn, Germany                      | Max Planck Society institute for pure mathematics; long-term and visitor research programs.                                                   |
| Simons Laufer Mathematical Sciences Institute (SLMath) | 1982    | Berkeley, United States            | Founded as MSRI and renamed in 2022; collaborative research programs, workshops, and outreach; no degree programs.                            |
| Fields Institute for Research in Mathematical Sciences | 1992    | Toronto, Canada                    | Collaborative research institute named after John Charles Fields; distinct from the Fields Medal.                                             |
| Isaac Newton Institute for Mathematical Sciences       | 1992    | Cambridge, United Kingdom          | United Kingdom national and international visitor institute for long thematic programs in the mathematical sciences.                          |
| Korea Institute for Advanced Study (KIAS)              | 1996    | Seoul, Republic of Korea           | Government-founded institute for basic research in mathematics, physics, and computational sciences; no formal curriculum or degree programs. |
| Clay Mathematics Institute (CMI)                       | 1998    | Denver, United States              | Private foundation supporting research, fellowships, and publications; announced the seven Millennium Prize Problems in 2000.                 |
| Institute for Pure and Applied Mathematics (IPAM)      | 1999    | Los Angeles, United States         | UCLA-based institute connecting mathematics with science, engineering, technology, and society through long programs and workshops.           |
| Banff International Research Station (BIRS)            | 2003    | Banff, Canada                      | International mathematical research station for focused workshops, collaborative teams, and training programs.                                |
| National Institute for Mathematical Sciences (NIMS)    | 2005    | Daejeon, Republic of Korea         | Government research institute for mathematical research, applications, computation, and public engagement.                                    |
| IBS Center for Geometry and Physics (IBS-CGP)          | 2012    | POSTECH, Pohang, Republic of Korea | Institute for Basic Science center founded for geometry, mathematical physics, symplectic topology, and international collaboration.          |

## Chapter 14

# Mathematical History, Works, and Anecdotes

This chapter collects three complementary kinds of historical recall: chronological landmarks, mathematical works whose titles and authors are themselves worth recognizing, and well-known anecdotes. The three sections are kept distinct so that an event, a work, and a story do not become interchangeable kinds of facts.

### Mathematical History

The following chronology records changes that altered mathematical language, method, or subject matter. It is intentionally written as short recall statements rather than as a continuous historical essay.

- **c. 1800–1600 BCE.** Babylonian tablets already contained sophisticated arithmetic, quadratic procedures, and examples of Pythagorean triples.
- **c. 1650 BCE.** The Rhind Mathematical Papyrus, copied by the scribe Ahmes from an older source, recorded Egyptian arithmetic and geometric problems.
- **c. 300 BCE.** Euclid organized Greek geometry and number theory axiomatically in the thirteen books of *Elements*.
- **3rd century BCE.** Archimedes used the method of exhaustion to obtain rigorous area and volume results and proved
$$\frac{223}{71} < \pi < \frac{22}{7}.$$
- **263.** Liu Hui completed his influential commentary on *The Nine Chapters on the Mathematical Art* and refined polygon methods for approximating  $\pi$ .
- **5th century.** Zu Chongzhi obtained  $3.1415926 < \pi < 3.1415927$  and the approximation  $355/113$ .
- **628.** Brahmagupta's *Brahmasphutasiddhanta* gave systematic arithmetic rules involving zero and negative numbers.
- **c. ninth century.** The translation movement centered in Baghdad, including work associated with the House of Wisdom, transmitted and developed Greek, Persian, and Indian mathematics in Arabic and helped carry that knowledge into later Latin Europe.
- **c. 825.** Al-Khwarizmi's treatise on *al-jabr* gave algebra its name; the Latin form of his name produced the word *algorithm*.

- **1202.** Fibonacci's *Liber Abaci* promoted Hindu–Arabic numerals and place-value arithmetic in Latin Europe.
- **14th–16th centuries.** Madhava and the Kerala school developed infinite series for trigonometric functions and  $\pi$ , centuries before the European calculus.
- **1545.** Cardano published *Ars Magna*, containing methods for general cubic and quartic equations, based in part on work of del Ferro, Tartaglia, and Ferrari.
- **1557.** Robert Recorde introduced the equality sign  $=$  in *The Whetstone of Witte*.
- **1614.** John Napier published logarithms, transforming difficult multiplication and astronomical computation into addition.
- **1631.** Thomas Harriot's work, published posthumously, introduced the inequality signs  $<$  and  $>$ .
- **1637.** Descartes' *La Géométrie* joined algebra and geometry through coordinates and equations.
- **1654.** The Fermat–Pascal correspondence on games of chance helped establish mathematical probability.
- **1655.** John Wallis introduced the infinity symbol  $\infty$  in *De sectionibus conicis*.
- **1665–1666.** During the plague years, Newton developed major parts of his calculus of fluxions, series methods, mechanics, and gravitation.
- **1671.** James Gregory found a general power-series expansion method anticipating what is now called Taylor's theorem, more than four decades before Brook Taylor's publication.
- **1675.** Leibniz first used the elongated-*S* integral sign  $\int$  in a manuscript; his differential notation  $dx$  and  $dy$  became standard.
- **1684.** Leibniz published the first paper on differential calculus. Newton and Leibniz are now credited with independent developments of calculus.
- **1687.** Newton published the *Philosophiæ Naturalis Principia Mathematica*, placing mechanics and gravitation in a mathematical framework.
- **1706.** William Jones used the symbol  $\pi$  for the ratio of circumference to diameter; Euler later popularized it.
- **1713 and 1738.** Nicolaus Bernoulli proposed the St. Petersburg paradox, and Daniel Bernoulli analyzed it using expected utility.
- **1715.** Brook Taylor published the general expansion now called Taylor's theorem.
- **1730s.** Euler adopted  $e$  for the base of natural logarithms and  $f(x)$  for a function; these notations became standard through his work.
- **1734–1735.** Euler solved the Basel problem,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

- **1736.** Euler's Königsberg bridges paper founded graph theory and helped initiate topology.

- **1742.** Christian Goldbach wrote to Euler about the conjecture now usually stated as: every even integer greater than 2 is a sum of two primes.
- **1755.** Euler used capital sigma  $\Sigma$  for summation in *Institutiones calculi differentialis*.
- **1763.** Bayes' essay on inverse probability was published posthumously by Richard Price.
- **1777.** Euler used  $i$  for  $\sqrt{-1}$  in a manuscript published in 1794; Gauss later helped make the notation standard.
- **1795.** John Playfair published a widely used reformulation of Euclid's parallel postulate: through a point outside a line, exactly one parallel line can be drawn.
- **1796.** At age 19, Gauss discovered that the regular 17-gon is constructible with straight-edge and compass, linking Fermat primes with classical construction problems.
- **1799.** Gauss's doctoral dissertation supplied the first substantially complete proof of the fundamental theorem of algebra.
- **1801.** Gauss published *Disquisitiones Arithmeticae*, reorganizing number theory; in the same year his least-squares calculations helped recover the orbit of Ceres.
- **1821.** Cauchy's *Cours d'Analyse* made limits, convergence, and continuity central to rigorous analysis.
- **1824.** Abel proved that the general quintic equation cannot be solved by radicals.
- **1829–1832.** Lobachevsky and Bolyai published independent constructions of hyperbolic geometry. Gauss had developed related ideas without publishing a systematic account; Riemann's 1854 lecture later broadened geometry through manifolds and curvature.
- **1830s–1846.** Galois connected polynomial equations with permutation groups; Liouville published Galois' manuscripts posthumously in 1846.
- **1843.** Hamilton discovered the quaternions, with

$$i^2 = j^2 = k^2 = ijk = -1.$$

- **1844.** Liouville proved that transcendental numbers exist, providing the first explicit examples.
- **1847.** Boole published *The Mathematical Analysis of Logic*, treating logic by algebraic operations.
- **1854.** Riemann's habilitation lecture introduced a broad theory of manifolds and curvature, transforming geometry.
- **1858.** August Ferdinand Möbius and Johann Benedict Listing independently described the one-sided surface now called the Möbius strip.
- **1872.** Dedekind published Dedekind cuts and Cantor published a construction of the real numbers, helping complete the arithmetization of analysis.
- **1873.** Charles Hermite proved that  $e$  is transcendental.
- **1874.** Cantor proved that the real algebraic numbers are countable and that transcendental numbers are uncountable.

- **1882.** Ferdinand von Lindemann proved that  $\pi$  is transcendental. Consequently, squaring the circle with straightedge and compass is impossible.
- **1889.** Peano published axioms for the natural numbers in a symbolic logical language.
- **1895.** Poincaré published *Analysis Situs*, a foundational work of algebraic topology.
- **1896.** Hadamard and de la Vallée Poussin independently proved the prime number theorem,

$$\pi(x) \sim \frac{x}{\ln x}.$$

- **1899.** Hilbert's *Foundations of Geometry* gave a modern axiomatic treatment of Euclidean geometry.
- **1900.** At the second ICM in Paris, Hilbert presented ten of the twenty-three problems in his published list; the list guided much of twentieth-century mathematics.
- **1901.** Russell discovered the paradox of the set of all sets that are not members of themselves, exposing a crisis in naive set theory.
- **1904–1908.** Zermelo used the axiom of choice to prove the well-ordering theorem and then proposed an axiomatization of set theory, later refined into ZF and ZFC.
- **1910–1913.** Whitehead and Russell published the three volumes of *Principia Mathematica*.
- **1918.** Emmy Noether proved that continuous symmetries of an action correspond to conservation laws.
- **1922.** Banach's work established complete normed spaces as central objects of functional analysis; his contraction principle became a standard existence theorem.
- **1930.** The Institute for Advanced Study was founded in Princeton as an independent research institution without degree programs.
- **1931.** Gödel proved his incompleteness theorems: every sufficiently strong consistent effectively axiomatized formal system is incomplete and cannot prove its own consistency.
- **1933.** Kolmogorov axiomatized probability using measure theory.
- **1935.** The Bourbaki group began working under the fictional name Nicolas Bourbaki.
- **1935–1941.** Mathematicians of the Lwów school recorded problems in *The Scottish Book*.
- **1936.** Turing introduced the machine model now bearing his name and proved undecidability results for the Entscheidungsproblem; the modern halting problem is a closely related standard formulation. The first Fields Medals were awarded in Oslo.
- **1940 and 1963.** Gödel proved that CH is consistent with ZF if ZF is consistent; Cohen introduced forcing and proved that CH is independent of ZFC, subject to consistency.
- **1947.** George Dantzig introduced the simplex method for linear programming.
- **1948.** Shannon founded information theory by defining entropy and channel capacity mathematically.
- **1950.** Nash introduced the equilibrium concept now called Nash equilibrium.

- **1956.** Milnor discovered exotic smooth structures on the 7-sphere.
- **1958.** The Institut des Hautes Études Scientifiques was founded near Paris as a research institute without degree programs.
- **1963.** The Atiyah–Singer index theorem connected the analytic index of an elliptic operator with topological data.
- **1970.** Matiyasevich completed earlier work of Davis, Putnam, and Robinson, proving that Hilbert’s tenth problem has no algorithmic solution.
- **1976–1977.** Appel and Haken announced their computer-assisted proof of the Four Color Theorem in 1976 and published it in 1977, creating a landmark debate about computer-assisted proof.
- **1982.** The Mathematical Sciences Research Institute was founded in Berkeley; it was renamed the Simons Laufer Mathematical Sciences Institute in 2022.
- **1983.** Faltings proved the Mordell conjecture: a smooth projective curve of genus greater than 1 over a number field has only finitely many rational points.
- **1985.** Louis de Branges proved the Bieberbach conjecture on coefficients of univalent functions.
- **1994–1995.** Wiles, with the crucial repair developed with Richard Taylor, proved Fermat’s Last Theorem through the modularity of semistable elliptic curves.
- **1998 and 2014.** Hales announced a computer-assisted proof of the Kepler conjecture; the Flyspeck project later completed a formal verification.
- **2002–2003.** Perelman posted papers completing Hamilton’s Ricci-flow program and proving the Poincaré and geometrization conjectures.
- **2004.** Green and Tao proved that the prime numbers contain arbitrarily long arithmetic progressions.
- **2013.** Yitang Zhang proved the first finite bound on gaps between infinitely many pairs of primes; Maynard, Tao, and the Polymath project rapidly improved bounded-gap methods.
- **2014.** The 2014 ICM was held in Seoul, Republic of Korea, and Maryam Mirzakhani became the first woman to receive the Fields Medal.
- **2015 and 2017.** A fifteenth family of convex pentagons that tile the plane was found in 2015. In 2017 Michaël Rao announced an exhaustive computer-assisted search showing that there are no additional families beyond these 15.
- **2016.** Maryna Viazovska proved the optimality of the  $E_8$  lattice sphere packing in dimension 8; joint work soon settled dimension 24 using the Leech lattice.
- **2022.** June Huh received the Fields Medal for work joining combinatorics, algebraic geometry, and Hodge theory; Maryna Viazovska became the second woman Fields Medalist.
- **2025.** Hong Wang and Joshua Zahl proved the three-dimensionalakeya conjecture: a subset of  $\mathbb{R}^3$  that contains a unit line segment in every direction has full Hausdorff and Minkowski dimension 3.

**Fact 14.1. Extensions of the Number System**

The following historical associations organize several standard number systems without implying that each system was created by one person in a single step.

- **Zero and negative numbers:** Brahmagupta's *Brahmasphutasiddhanta* of 628 gave systematic arithmetic rules involving both.
- **Complex numbers:** Rafael Bombelli's *L'Algebra* of 1572 gave a systematic treatment of arithmetic with square roots of negative numbers.
- **Quaternions:** William Rowan Hamilton discovered them in 1843, with

$$i^2 = j^2 = k^2 = ijk = -1.$$

- **$p$ -adic numbers:** Kurt Hensel introduced the  $p$ -adic viewpoint at the end of the nineteenth century, producing completions of  $\mathbb{Q}$  different from  $\mathbb{R}$ .
- **Hyperreal numbers:** Edwin Hewitt introduced the term *hyper-real* in 1948; Abraham Robinson later developed nonstandard analysis using infinitesimal and infinitely large numbers.
- **Surreal numbers:** John Conway constructed the number system from combinatorial games around 1970; Donald Knuth introduced the name *surreal numbers*.

**Mathematical Works**

The works below are ordered chronologically. They are included because the title, author, mathematical content, or historical role is itself a useful Science Quiz association. Ancient dates and dates of long compilations are necessarily approximate; for a work first circulated in one form and published later, both dates are recorded when that distinction is useful.

| Date               | Work                              | Author / association                                | Main content                                                                                  | Historical significance                                                                                 |
|--------------------|-----------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| c. 1650 BCE        | <i>Rhind Mathematical Papyrus</i> | Ahmes, scribe; copied from an older Egyptian source | Arithmetic, Egyptian fractions, equations, progressions, areas, and volumes.                  | One of the principal surviving sources for ancient Egyptian mathematics.                                |
| c. 300 BCE         | <i>Elements</i>                   | Euclid                                              | Geometry, proportion, number theory, and the Euclidean algorithm in thirteen books.           | The archetype of the axiomatic method and one of the most influential mathematical works in history.    |
| c. 3rd century BCE | <i>On the Sphere and Cylinder</i> | Archimedes                                          | Surface area and volume of the sphere and cylinder by methods anticipating integral geometry. | Contains Archimedes' celebrated sphere-cylinder results and exemplifies the Greek method of exhaustion. |
| c. 200 BCE         | <i>Conics</i>                     | Apollonius of Perga                                 | Systematic theory of the ellipse, parabola, and hyperbola.                                    | Established much of the classical language and theory of conic sections.                                |

Continued on the next page.

| Date                             | Work                                                                   | Author / association                                       | Main content                                                                                                                | Historical significance                                                                                                               |
|----------------------------------|------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| c. 200 BCE–50 CE; commentary 263 | <i>The Nine Chapters on the Mathematical Art</i>                       | Anonymous Chinese compilation; major commentary by Liu Hui | Algorithms for arithmetic, fractions, areas, volumes, linear systems, and practical problems.                               | Foundational Chinese mathematical classic; Liu Hui's commentary supplied explanations and refined polygonal approximations of $\pi$ . |
| c. 250                           | <i>Arithmetica</i>                                                     | Diophantus of Alexandria                                   | Problems on polynomial and indeterminate equations with rational solutions.                                                 | A major source for Diophantine analysis; Fermat's famous Last Theorem note was written in a later edition of this work.               |
| c. 3rd–5th century               | <i>Sunzi Suanjing (Sunzi Mathematical Classic)</i>                     | Traditionally associated with Sunzi                        | Arithmetic algorithms and remainder problems.                                                                               | Contains the classical remainder problem that became a standard example of the Chinese Remainder Theorem.                             |
| 499                              | <i>Aryabhatiya</i>                                                     | Aryabhata                                                  | Arithmetic, algebra, trigonometry, astronomical computation, and approximations to $\pi$ .                                  | A landmark of classical Indian mathematics and astronomy; influential in the development of sine tables and computational methods.    |
| 628                              | <i>Brahmasphuta-siddhanta</i>                                          | Brahmagupta                                                | Arithmetic with zero and negative numbers, algebra, quadratic equations, geometry, and astronomy.                           | Gave systematic rules involving zero and signed numbers and includes Brahmagupta's formula for cyclic quadrilaterals.                 |
| c. 820                           | <i>Al-Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabala</i>            | Muhammad ibn Musa al-Khwarizmi                             | Systematic procedures for linear and quadratic equations.                                                                   | The word <i>algebra</i> derives from <i>al-jabr</i> in the title; al-Khwarizmi's name is also the source of <i>algorithm</i> .        |
| c. 1070                          | <i>Treatise on Demonstration of Problems of Algebra</i>                | Omar Khayyam                                               | Classification and geometric solution of cubic equations using conic sections.                                              | A major medieval synthesis of algebra and geometry before general symbolic solutions of cubics were known.                            |
| 1150                             | <i>Siddhanta Shiromani</i>                                             | Bhaskara II                                                | Arithmetic, algebra, planetary mathematics, and sphere mathematics; includes <i>Lilavati</i> and <i>Bijaganita</i> .        | One of the major works of medieval Indian mathematics, especially noted for algebraic and indeterminate-equation techniques.          |
| 1202                             | <i>Liber Abaci</i>                                                     | Leonardo of Pisa (Fibonacci)                               | Hindu–Arabic numerals, commercial arithmetic, algebraic problems, and the rabbit problem leading to the Fibonacci sequence. | Helped popularize Hindu–Arabic positional arithmetic in medieval Europe.                                                              |
| 1303                             | <i>The Precious Mirror of the Four Elements</i>                        | Zhu Shijie                                                 | Polynomial equations, elimination, finite sums, and a triangular array of binomial coefficients.                            | A high point of classical Chinese algebra, including methods for equations of high degree.                                            |
| 1494                             | <i>Summa de arithmetica, geometria, proportioni et proportionalita</i> | Luca Pacioli                                               | Comprehensive printed survey of arithmetic, algebra, geometry, proportion, and commercial calculation.                      | One of the earliest major printed mathematical encyclopedias of Renaissance Europe.                                                   |

Continued on the next page.

| Date                            | Work                                               | Author /<br>association   | Main content                                                                                                 | Historical significance                                                                                                             |
|---------------------------------|----------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1545                            | <i>Ars Magna</i>                                   | Girolamo Cardano          | General methods for cubic equations and Ferrari's solution of quartic equations.                             | A landmark in Renaissance algebra and an important step toward accepting complex quantities in algebraic calculation.               |
| 1572                            | <i>L'Algebra</i>                                   | Rafael Bombelli           | Algebraic rules for complex numbers and applications to cubic equations.                                     | Gave the first systematic practical treatment of arithmetic with square roots of negative numbers.                                  |
| 1585                            | <i>De Thiende</i>                                  | Simon Stevin              | Decimal fractions and their arithmetic.                                                                      | Helped establish decimal fractions as a practical computational language in Europe.                                                 |
| 1591                            | <i>In artem analyticem isagoge</i>                 | Francois Viète            | Symbolic algebra using letters for known and unknown quantities.                                             | A decisive step toward modern symbolic algebra and systematic manipulation of general equations.                                    |
| 1614                            | <i>Mirifici Logarithmorum Canonis Descriptio</i>   | John Napier               | Logarithms and numerical tables designed to simplify multiplication and division.                            | Introduced logarithms to a wide audience and transformed numerical computation in astronomy and science.                            |
| 1637                            | <i>La Géométrie</i>                                | René Descartes            | Algebraic treatment of geometric curves and coordinate methods.                                              | A founding work of analytic geometry, linking algebraic equations with geometric loci.                                              |
| 1654;<br>pub-<br>lished<br>1665 | <i>Traité du triangle arithmétique</i>             | Blaise Pascal             | Binomial coefficients and identities organized through the arithmetical triangle.                            | One of the classic early systematic treatments of Pascal's triangle and combinatorial identities.                                   |
| 1684                            | <i>Nova Methodus pro Maximis et Minimis</i>        | Gottfried Wilhelm Leibniz | Differential calculus, tangents, extrema, and rules of differentiation.                                      | The first published account of Leibniz's differential calculus and the source of notation that strongly influenced modern calculus. |
| 1687                            | <i>Philosophiæ Naturalis Principia Mathematica</i> | Isaac Newton              | Laws of motion, universal gravitation, celestial mechanics, and geometric forms of infinitesimal reasoning.  | One of the foundational works of mathematical physics and a defining achievement of the Scientific Revolution.                      |
| 1713                            | <i>Ars Conjectandi</i>                             | Jacob Bernoulli           | Combinatorics, probability, Bernoulli numbers, and the law of large numbers.                                 | A foundational treatise in probability theory and one of the earliest systematic works on the subject.                              |
| 1718                            | <i>The Doctrine of Chances</i>                     | Abraham de Moivre         | Probability calculations for games of chance, independence, and approximations to the binomial distribution. | A classic early probability text; later editions contain work closely related to the normal approximation.                          |
| 1748                            | <i>Introductio in analysin infinitorum</i>         | Leonhard Euler            | Functions, infinite series, exponential and logarithmic functions, trigonometry, and analytic methods.       | A foundational text in modern analysis that helped establish the function-centered language of mathematics.                         |

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| Date                         | Work                                                                         | Author /<br>association | Main content                                                                                                | Historical significance                                                                                                    |
|------------------------------|------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1788                         | <i>Mécanique analytique</i>                                                  | Joseph-Louis Lagrange   | Mechanics formulated through algebra, differential equations, and variational principles.                   | Transformed mechanics into a systematic branch of mathematical analysis and introduced the Lagrangian viewpoint.           |
| 1801                         | <i>Disquisitiones Arithmeticae</i>                                           | Carl Friedrich Gauss    | Congruences, quadratic residues, binary quadratic forms, cyclotomy, and construction of the regular 17-gon. | A foundational work of modern number theory; it systematized congruence notation and quadratic reciprocity.                |
| 1821                         | <i>Cours d'Analyse</i>                                                       | Augustin-Louis Cauchy   | Limits, continuity, derivatives, infinite series, and rigorous foundations of analysis.                     | A central nineteenth-century step toward the modern rigorous formulation of calculus and analysis.                         |
| 1822                         | <i>Théorie analytique de la chaleur</i>                                      | Joseph Fourier          | Heat equation and representation of functions by trigonometric series.                                      | Established Fourier series as a fundamental analytic tool and strongly influenced PDE, analysis, and mathematical physics. |
| 1832                         | <i>Appendix scientiam spatii absolute veram exhibens</i>                     | János Bolyai            | A systematic development of geometry independent of Euclid's parallel postulate.                            | One of the foundational publications of non-Euclidean geometry.                                                            |
| written 1831; published 1846 | <i>Mémoire sur les conditions de résolubilité des équations par radicaux</i> | Évariste Galois         | Permutations of roots and criteria for solvability of polynomial equations by radicals.                     | Laid the foundations of Galois theory and connected group structure with solvability of equations.                         |
| 1844                         | <i>Die lineale Ausdehnungslehre</i>                                          | Hermann Grassmann       | An algebra of directed and higher-dimensional geometric quantities.                                         | A precursor of modern vector spaces, multilinear algebra, and exterior algebra.                                            |
| 1847                         | <i>The Mathematical Analysis of Logic</i>                                    | George Boole            | Algebraic formalization of logical operations.                                                              | Helped create mathematical logic and the algebraic tradition that led to Boolean algebra and digital logic.                |
| 1854; published 1868         | <i>On the Hypotheses Which Lie at the Foundations of Geometry</i>            | Bernhard Riemann        | Manifolds, metrics, curvature, and the possibility of non-Euclidean geometries in higher dimensions.        | A foundational source for Riemannian geometry and the geometric language later used in general relativity.                 |
| 1859                         | <i>On the Number of Primes Less Than a Given Magnitude</i>                   | Bernhard Riemann        | The zeta function, complex analytic continuation, and the distribution of prime numbers.                    | Introduced the Riemann Hypothesis and permanently linked complex analysis with prime-number theory.                        |
| 1872                         | <i>Stetigkeit und irrationale Zahlen</i>                                     | Richard Dedekind        | A construction of the real numbers by Dedekind cuts and a precise treatment of continuity.                  | One of the classical arithmetizations of analysis and a rigorous foundation for the real number system.                    |
| 1872                         | <i>Erlangen Program</i>                                                      | Felix Klein             | Classification of geometries by transformation groups and invariants.                                       | Reframed geometry through group actions and unified Euclidean, projective, and non-Euclidean viewpoints.                   |

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| Date      | Work                                                                 | Author / association                     | Main content                                                                                                | Historical significance                                                                                                          |
|-----------|----------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1874      | <i>On a Property of the Collection of All Real Algebraic Numbers</i> | Georg Cantor                             | Countability of algebraic numbers and uncountability of the real numbers.                                   | Commonly regarded as the first paper of Cantor's set theory and the beginning of the theory of different infinite cardinalities. |
| 1879      | <i>Begriffsschrift</i>                                               | Gottlob Frege                            | A formal language for propositional and predicate logic with quantification.                                | A foundational work of modern mathematical logic and formal systems.                                                             |
| 1888      | <i>Was sind und was sollen die Zahlen?</i>                           | Richard Dedekind                         | Natural numbers, mappings, chains, and an abstract logical construction of arithmetic.                      | A foundational treatment of the natural numbers closely related to the later Peano axioms.                                       |
| 1889      | <i>Arithmetices principia, nova methodo exposita</i>                 | Giuseppe Peano                           | Symbolic logic and axioms for the natural numbers.                                                          | Gave the famous Peano-style axiomatization and helped standardize a symbolic language for foundations.                           |
| 1895      | <i>Analysis Situs</i>                                                | Henri Poincaré                           | Qualitative invariants of spaces, homology-type ideas, and the fundamental group.                           | A foundational paper of algebraic topology and the origin of much modern topological language.                                   |
| 1899      | <i>Grundlagen der Geometrie</i>                                      | David Hilbert                            | Axioms of geometry, independence, consistency, and incidence, order, congruence, parallels, and continuity. | A landmark in the axiomatic method and the modern foundations of geometry.                                                       |
| 1900      | <i>Mathematical Problems</i>                                         | David Hilbert                            | Twenty-three problems proposed as major directions for future mathematical research.                        | Hilbert's Paris ICM address became one of the most influential research agendas in twentieth-century mathematics.                |
| 1902      | <i>Intégrale, longueur, aire</i>                                     | Henri Lebesgue                           | Measure, measurable functions, and the Lebesgue integral.                                                   | Established the measure-theoretic integration framework that underlies modern real analysis and probability.                     |
| 1908      | <i>Untersuchungen über die Grundlagen der Mengenlehre I</i>          | Ernst Zermelo                            | An axiomatic system for set theory designed to avoid the paradoxes of naive set theory.                     | A direct ancestor of Zermelo–Fraenkel set theory, the standard foundational framework for modern mathematics.                    |
| 1910–1913 | <i>Principia Mathematica</i>                                         | Alfred North Whitehead; Bertrand Russell | Symbolic logic and a logicist derivation of substantial parts of mathematics.                               | A landmark of mathematical logic and foundations; its program formed part of the background to Gödel's incompleteness results.   |
| 1913      | <i>Die Idee der Riemannschen Fläche</i>                              | Hermann Weyl                             | Riemann surfaces treated through topology, analysis, and geometric structure.                               | A highly influential synthesis that helped shape the modern concept of a Riemann surface.                                        |
| 1914      | <i>Grundzüge der Mengenlehre</i>                                     | Felix Hausdorff                          | Set theory, topological spaces, neighborhoods, separation, and dimension-related ideas.                     | One of the foundational books of general topology; the Hausdorff separation condition bears the author's name.                   |

Continued on the next page.

| Date           | Work                                                                                                        | Author /<br>association                                                    | Main content                                                                                                | Historical significance                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1918           | <i>Invariante<br/>Variationsprobleme</i>                                                                    | Emmy Noether                                                               | Relations between continuous symmetries and conservation laws in variational systems.                       | Contains Noether's theorems, fundamental to modern mathematical physics and the calculus of variations.                                                             |
| 1930–<br>1931  | <i>Moderne Algebra</i>                                                                                      | B. L. van der Waerden                                                      | Groups, rings, fields, ideals, and the structural algebra developed by Noether, Artin, and others.          | A defining textbook of abstract algebra that helped standardize the structural viewpoint of twentieth-century algebra.                                              |
| 1931           | <i>Über formal<br/>unentscheidbare Sätze<br/>der Principia<br/>Mathematica und<br/>verwandter Systeme I</i> | Kurt Gödel                                                                 | Arithmetization of syntax and the first and second incompleteness phenomena for formal systems.             | Established fundamental limits on completeness and self-verification in sufficiently strong formal axiomatic systems.                                               |
| 1932           | <i>Théorie des opérations<br/>linéaires</i>                                                                 | Stefan Banach                                                              | Normed linear spaces, linear operators, functional analysis, and results now associated with Banach spaces. | A foundational monograph of functional analysis and the source of much of its modern structural language.                                                           |
| 1933           | <i>Grundbegriffe der<br/>Wahrscheinlichkeits-<br/>rechnung</i>                                              | Andrey Kolmogorov                                                          | Probability formulated axiomatically using measure theory.                                                  | Established the standard modern axioms of probability on a probability space.                                                                                       |
| 1935–<br>1941  | <i>The Scottish Book</i>                                                                                    | Mathematicians of the Lwów school; initiated around Stefan Banach's circle | Unsolved problems, conjectures, and prizes recorded at the Scottish Café.                                   | A unique mathematical notebook associated with Banach, Steinhaus, Ulam, Mazur, and others; Problem 153 famously carried a live-goose prize.                         |
| 1936           | <i>On Computable<br/>Numbers, with an<br/>Application to the<br/>Entscheidungsproblem</i>                   | Alan Turing                                                                | Turing machines, computable numbers, and undecidability of the Entscheidungsproblem.                        | A foundational paper of computability theory and theoretical computer science.                                                                                      |
| 1938           | <i>An Introduction to the<br/>Theory of Numbers</i>                                                         | G. H. Hardy; E. M. Wright                                                  | Classical elementary and analytic number theory.                                                            | One of the most famous enduring number-theory textbooks and a standard historical reference for classical results.                                                  |
| 1939<br>onward | <i>Éléments de<br/>mathématique</i>                                                                         | Nicolas Bourbaki                                                           | A systematic axiomatic treatment of major areas of modern mathematics.                                      | The collective project strongly influenced twentieth-century terminology, abstraction, and textbook style; Bourbaki is a pseudonymous group, not one mathematician. |
| 1940           | <i>A Mathematician's<br/>Apology</i>                                                                        | G. H. Hardy                                                                | A reflective essay on pure mathematics, creativity, beauty, and the mathematician's life.                   | One of the best-known literary works about mathematical culture and the values of pure mathematics.                                                                 |
| 1944           | <i>Theory of Games and<br/>Economic Behavior</i>                                                            | John von Neumann; Oskar Morgenstern                                        | Strategic games, utility, minimax ideas, and mathematical models of economic behavior.                      | Established game theory as a systematic mathematical discipline with major applications beyond mathematics.                                                         |

Continued on the next page.

| Date      | Work                                                                            | Author / association                     | Main content                                                                                            | Historical significance                                                                                                               |
|-----------|---------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1945      | <i>How to Solve It</i>                                                          | George Pólya                             | Heuristics for problem solving: understand the problem, devise a plan, carry it out, and look back.     | One of the most influential books on mathematical problem solving and mathematical education.                                         |
| 1948      | <i>A Mathematical Theory of Communication</i>                                   | Claude Shannon                           | Entropy, information, coding, channel capacity, and quantitative communication theory.                  | Founded information theory and supplied the mathematical framework underlying modern digital communication.                           |
| 1950–1951 | <i>Equilibrium Points in <math>N</math>-Person Games; Non-Cooperative Games</i> | John Nash                                | Existence and theory of equilibrium in finite non-cooperative games.                                    | Introduced the Nash equilibrium, a central solution concept in game theory and mathematical economics.                                |
| 1955      | <i>Faisceaux algébriques cohérents</i>                                          | Jean-Pierre Serre                        | Coherent sheaves and cohomological methods in algebraic geometry.                                       | The celebrated “FAC” paper helped establish sheaf cohomology as a fundamental tool of modern algebraic geometry.                      |
| 1960–1967 | <i>Éléments de géométrie algébrique</i>                                         | Alexander Grothendieck; Jean Dieudonné   | Schemes, morphisms, cohomological foundations, and a systematic language for algebraic geometry.        | The “EGA” series transformed the foundations and language of modern algebraic geometry.                                               |
| 1963      | <i>Solvability of Groups of Odd Order</i>                                       | Walter Feit; John G. Thompson            | Proof that every finite group of odd order is solvable.                                                 | A landmark 255-page group-theory paper and a major precursor to the classification of finite simple groups.                           |
| 1968      | <i>The Index of Elliptic Operators I</i>                                        | Michael Atiyah; Isadore Singer           | Index theory for elliptic differential operators connecting analysis, topology, and geometry.           | Part of the foundational development of the Atiyah–Singer Index Theorem, one of the central results of twentieth-century mathematics. |
| 1976      | <i>On Numbers and Games</i>                                                     | John Horton Conway                       | Combinatorial game theory and a vast ordered field containing the surreal numbers.                      | Introduced a unified mathematical treatment of games and Conway’s construction of the surreal numbers.                                |
| 1977      | <i>Every Planar Map Is Four Colorable</i>                                       | Kenneth Appel; Wolfgang Haken; John Koch | Reduction to finitely many configurations and computer checking in the proof of the Four Color Theorem. | One of the first celebrated computer-assisted proofs, changing discussion about computation and mathematical proof.                   |
| 1982      | <i>The Fractal Geometry of Nature</i>                                           | Benoit Mandelbrot                        | Fractals, self-similarity, scaling, and irregular geometric structures.                                 | Popularized fractal geometry and made the term <i>fractal</i> central in mathematics and science.                                     |
| 1995      | <i>Modular Elliptic Curves and Fermat’s Last Theorem</i>                        | Andrew Wiles                             | Modularity of semistable elliptic curves using Galois representations and modular forms.                | Together with the Taylor–Wiles companion paper, supplied the proof of Fermat’s Last Theorem.                                          |

Continued on the next page.

| Date | Work                                                                                         | Author / association             | Main content                                                                                    | Historical significance                                                                                                                |
|------|----------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1995 | <i>Ring-Theoretic Properties of Certain Hecke Algebras</i>                                   | Richard Taylor; Andrew Wiles     | The commutative-algebra and Hecke-algebra input needed to complete Wiles's modularity argument. | The companion paper that repaired the gap in the original argument and completed the published proof of Fermat's Last Theorem.         |
| 1998 | <i>Proofs from THE BOOK</i>                                                                  | Martin Aigner; Günter M. Ziegler | Elegant proofs in number theory, geometry, analysis, combinatorics, and graph theory.           | Inspired by Paul Erdős' metaphor of "The Book" containing the most beautiful proofs; many topics were suggested by Erdős.              |
| 2002 | <i>The Entropy Formula for the Ricci Flow and Its Geometric Applications</i>                 | Grigori Perelman                 | Entropy and monotonicity for Ricci flow, noncollapsing, and geometric applications.             | The first of Perelman's three preprints advancing Hamilton's Ricci-flow program toward geometrization.                                 |
| 2003 | <i>Ricci Flow with Surgery on Three-Manifolds</i>                                            | Grigori Perelman                 | Construction and analysis of Ricci flow with surgery on 3-manifolds.                            | The technical continuation that supplied the surgery mechanism central to the proof of geometrization.                                 |
| 2003 | <i>Finite Extinction Time for the Solutions to the Ricci Flow on Certain Three-Manifolds</i> | Grigori Perelman                 | Finite-time extinction for relevant Ricci flows with surgery.                                   | Completed the sequence of preprints from which the Poincaré conjecture and Thurston geometrization theorem were established.           |
| 2012 | <i>Théorème vivant</i>                                                                       | Cédric Villani                   | A narrative of research with Clément Mouhot on nonlinear Landau damping.                        | A rare modern research memoir whose title and story became part of Villani's public mathematical identity after his 2010 Fields Medal. |

## Mathematical Anecdotes

Some famous stories are documented, while others survive as traditional legends. A story whose details are uncertain is explicitly identified below; the uncertainty is part of the fact to remember.

### Fact 14.2. Frequently Named Mathematical Paradoxes

In quiz settings, a paradox is usually identified by its defining tension rather than by a full resolution.

- **Zeno's Achilles paradox.** Infinitely many remaining subintervals appear to prevent a faster runner from overtaking a slower one; convergent series resolve the apparent contradiction.
- **Aristotle's wheel paradox.** Concentric circles of different circumferences appear to roll through the same distance; the smaller circle does not undergo ordinary no-slip rolling against the line.
- **St. Petersburg paradox.** A game can have infinite expected monetary payoff while attracting only a modest entrance fee; expected utility explains the discrepancy.
- **Russell's or the barber paradox.** Unrestricted self-reference creates a contradiction, motivating axiomatic set theory.

- **Banach–Tarski paradox.** Assuming the axiom of choice, a solid ball in  $\mathbb{R}^3$  can be partitioned into finitely many nonmeasurable pieces and reassembled into two balls congruent to the original.
- **Birthday paradox.** Among 23 people, the probability that at least two share a birthday exceeds  $1/2$ , under the usual uniform independent model.
- **Coastline paradox.** The measured length of an irregular boundary increases as the measuring scale becomes finer; fractal dimension describes this scale dependence.
- **Newcomb’s paradox.** Dominance reasoning and expected utility reasoning recommend different choices when a highly accurate predictor has already filled two boxes.
- **Parrondo’s paradox.** Alternating or mixing two losing stochastic games can produce a winning process.

The following anecdotes are arranged chronologically. Legendary or uncertain details remain identified as such.

- **c. fifth century BCE. Pythagorean irrationality legend.** A famous but historically uncertain legend says that Hippasus was drowned after revealing the irrationality of  $\sqrt{2}$ . The discovery itself shattered the belief that every magnitude is a ratio of integers.
- **c. 300 BCE. No royal road.** Proclus reports that when Ptolemy asked for an easier route to geometry, Euclid replied that there is no royal road to geometry.
- **third century BCE. Archimedes and the crown.** Vitruvius tells the traditional story that Archimedes recognized how displacement could test a crown’s purity and ran out crying “Eureka!” The precise details are not independently verified.
- **212 BCE. Archimedes’ circles.** Ancient accounts say that Archimedes was killed during the Roman capture of Syracuse while absorbed in a geometric diagram. The familiar wording “Do not disturb my circles” is traditional.
- **75 BCE. Archimedes’ tomb.** Cicero reported rediscovering the tomb of Archimedes by recognizing the carved sphere and cylinder that commemorated Archimedes’ favorite result.
- **1539–1545. Cardano and Tartaglia.** Tartaglia disclosed his cubic method to Cardano under a promise of secrecy. Cardano later published it in *Ars Magna* after learning that del Ferro had found the method earlier, beginning a celebrated priority dispute.
- **c. 1637. Fermat’s margin.** Fermat wrote that he had a marvelous proof of his last theorem but that the margin was too small to contain it. No valid general proof by Fermat is known.
- **c. 1666. Newton’s apple.** Newton later told William Stukeley that seeing an apple fall prompted thoughts about gravity. The story does not require that an apple struck Newton’s head.
- **late seventeenth–early eighteenth centuries. The calculus priority dispute.** Newton and Leibniz developed calculus independently, but a bitter priority dispute divided British and continental mathematics for decades.

- **1736. Königsberg and Kaliningrad.** Euler's Seven Bridges problem used the city then called Königsberg, also Hilbert's birthplace. After World War II the city became Kaliningrad and is now part of Russia.
- **1766 onward. Euler's eyesight.** Euler continued producing mathematics at an extraordinary rate after losing almost all of his sight, dictating long calculations from memory.
- **c. 1780s. Gauss and the arithmetic sum.** A traditional schoolroom story says that young Gauss immediately summed  $1 + 2 + \cdots + 100$  by pairing terms to obtain 5050. The details vary among later accounts.
- **1790s. Sophie Germain's pseudonym.** Because women were excluded from the École Polytechnique, Germain studied its lecture notes and submitted work under the name "M. LeBlanc." Lagrange became a supporter after learning her identity.
- **1832. Galois' last night.** Galois wrote urgent mathematical notes before the duel that killed him at age 20. He did not create all of Galois theory in one night; the notes summarized and revised ideas developed earlier.
- **1843. Hamilton's bridge.** On October 16, 1843, Hamilton carved the quaternion relations

$$i^2 = j^2 = k^2 = ijk = -1$$

into Brougham Bridge in Dublin after the key idea came to him during a walk.

- **1900. Hilbert's problems.** Hilbert's Paris address did not orally present all twenty-three problems; ten were delivered in the lecture, while the complete list appeared in print.
- **1913. Ramanujan's letter to Hardy.** Working as a clerk in Madras, Ramanujan sent Hardy a letter containing many striking formulas and results. Hardy recognized his talent and helped bring him to Cambridge.
- **1915. Hilbert and Noether.** When objections were raised to Noether teaching at Göttingen because she was a woman, Hilbert was reported to have replied that the university was not a bathhouse; the exact wording is not preserved.
- **c. 1918–1919. The taxi-cab number.** Hardy told Ramanujan that taxi number 1729 seemed dull. Ramanujan immediately answered that it is the smallest number expressible as a sum of two positive cubes in two ways:

$$1729 = 1^3 + 12^3 = 9^3 + 10^3.$$

- **1935 onward. Nicolas Bourbaki.** Bourbaki is not one mathematician but the collective pseudonym of a group that sought a unified, axiomatic presentation of modern mathematics.
- **1935–1941. The Scottish Book.** Mathematicians of the Lwów school wrote problems in a notebook kept at the Scottish Café. Prizes ranged from coffee or beer to a live goose.
- **1936 and 1972. Mazur's goose.** Stanisław Mazur offered a live goose for Problem 153 in 1936. Per Enflo solved the basis problem in 1972, and Mazur presented the promised goose in a televised ceremony.
- **1939. Dantzig's homework.** George Dantzig arrived late to Jerzy Neyman's statistics class, copied two unsolved problems from the board believing they were homework, and later submitted solutions to both.

- **1947. Gödel’s citizenship hearing.** Before his 1947 US citizenship examination, Gödel found what he believed was a logical route to dictatorship in the Constitution. Einstein and Oskar Morgenstern accompanied him and helped prevent the hearing from becoming a constitutional debate.
- **1947. Hilbert’s hotel.** The infinite-hotel thought experiment, popularized by George Gamow, illustrates that a countably infinite full hotel can still accommodate more guests by moving each guest to another numbered room.
- **twentieth century. Erdős’ language.** Erdős called children “epsilons” and imagined an ideal book in which God keeps the most elegant proofs. The Erdős number measures collaboration distance from him.
- **1954. Turing’s apple.** Turing died from cyanide poisoning in 1954, and a bitten apple was found nearby. The popular claim that the apple was deliberately poisoned, or that it inspired the Apple logo, is not established.
- **1956. Milnor’s exotic spheres.** Milnor discovered that a topological 7-sphere can carry differentiable structures not equivalent to the standard one. The group of oriented homotopy 7-spheres has order 28.
- **1976. The computer proof controversy.** The 1976 proof of the Four Color Theorem required computer checking of many cases, forcing mathematicians to debate what it means for a proof to be humanly inspectable.
- **1993–1994. Wiles’ secret work.** Wiles worked privately for years on Fermat’s Last Theorem. After announcing a proof in 1993, a gap was found; Wiles and Richard Taylor repaired it in 1994.
- **2006 and 2010. Perelman’s refusals.** Perelman declined the 2006 Fields Medal and the Clay Millennium Prize for the Poincaré conjecture.
- **2010s. Villani’s public image.** Cédric Villani became recognizable for a cravat and spider brooch. His book *Théorème vivant* recounts work with Clément Mouhot on nonlinear Landau damping.
- **2018. Birkar’s stolen medal.** Caucher Birkar’s Fields Medal was stolen shortly after the 2018 ceremony in Rio de Janeiro; the IMU later presented him with a replacement.

## Chapter 15

# Famous Problems and Conjectures

Famous problems are usually not tested by asking for complete proofs. They are tested through the following information: the standard statement, the person or group attached to the problem, whether it is solved, and the main idea or keyword attached to the solution. The goal of this chapter is to make those associations immediate.

### Millennium Prize Problems

The Clay Mathematics Institute announced the seven Millennium Prize Problems in 2000. Each problem carries a prize of one million US dollars. The Poincaré conjecture is the only problem officially classified as solved by Clay. A Navier–Stokes solution was announced in September 2026, but Clay’s evaluation remained pending at this book’s cutoff.

| Problem                                | Standard mathematical form                                                                                                                          | Status                                         | Quiz anchors                                                   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------------|
| Poincaré Conjecture                    | Every closed simply connected 3-manifold is homeomorphic to $S^3$ .                                                                                 | Solved.                                        | Poincaré; Perelman; Ricci flow.                                |
| Riemann Hypothesis                     | Every nontrivial zero of $\zeta(s)$ has real part $1/2$ .                                                                                           | Open.                                          | Riemann; zeta function; prime numbers.                         |
| P versus NP                            | Is $P = NP$ ?                                                                                                                                       | Open.                                          | Cook, Karp; computational complexity.                          |
| Birch and Swinnerton-Dyer Conjecture   | For an elliptic curve $E$ , $\text{rk } E(\mathbb{Q})$ equals the order of vanishing of $L(E, s)$ at $s = 1$ .                                      | Open.                                          | Elliptic curves; $L$ -functions.                               |
| Hodge Conjecture                       | Rational Hodge classes on smooth projective complex varieties should come from algebraic cycles.                                                    | Open.                                          | Hodge theory; algebraic geometry.                              |
| Navier–Stokes Existence and Smoothness | Smooth initial data for the 3-dimensional incompressible Navier–Stokes equations should have global smooth solutions, or one must find a breakdown. | Solution announced; under official evaluation. | Fluid mechanics; PDE; finite-time singularity; September 2026. |

Continued on the next page.

| Problem                           | Standard mathematical form                                                                                           | Status | Quiz anchors                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------|
| Yang–Mills Existence and Mass Gap | Construct quantum Yang–Mills theory on $\mathbb{R}^4$ for compact simple gauge groups and prove a positive mass gap. | Open.  | Gauge theory; quantum field theory; mass gap. |

### Fact 15.1. Millennium Prize status check

As of September 15, 2026, Clay classifies the Poincaré conjecture as the only solved Millennium Prize Problem. Clay still lists Navier–Stokes as active while evaluating the announced solution; the other five remain unsolved.

*Remark.* Henri Poincaré posed the problem in the early twentieth century. Grigori Perelman proved it using Ricci flow and ideas from Hamilton’s program. The principal association to retain is

$$\text{Poincaré conjecture} \longleftrightarrow \text{Perelman} \longleftrightarrow \text{Ricci flow}.$$

### Hypothesis 15.2. Riemann Hypothesis

Let

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

be the Riemann zeta function, initially defined for  $\text{Re}(s) > 1$  and continued meromorphically. The Riemann Hypothesis states that every nontrivial zero of  $\zeta(s)$  has real part  $1/2$ .

*Idea.* The zeta function is connected to prime numbers by Euler’s product

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Thus the Riemann Hypothesis is a statement about the hidden regularity of the distribution of prime numbers.

### Open Problem 15.3. P versus NP

Determine whether

$$P = NP.$$

Here  $P$  is the class of decision problems solvable in polynomial time, and  $NP$  is the class of decision problems whose proposed solutions can be verified in polynomial time.

*Remark.* The usual slogan is: can every solution that can be checked quickly also be found quickly? The anchors are Cook, Karp, NP-completeness, and theoretical computer science.

### Conjecture 15.4. Birch and Swinnerton-Dyer Conjecture

For an elliptic curve  $E$  over  $\mathbb{Q}$ ,  $\text{rk } E(\mathbb{Q})$  equals the order of vanishing of its  $L$ -function  $L(E, s)$  at  $s = 1$ .

*Remark.* The problem connects rational points on elliptic curves with analytic behavior of  $L$ -functions. The phrase “rk equals order of vanishing” is the main quiz memory.

**Conjecture 15.5. Hodge Conjecture**

*On a smooth projective complex variety, every rational cohomology class of type  $(p, p)$  should be a rational linear combination of classes of algebraic cycles.*

*Remark.* This is one of the deepest bridges between topology and algebraic geometry. The principal association to retain is

$$\text{Hodge classes} \longleftrightarrow \text{algebraic cycles}.$$

**Open Problem 15.6. Navier–Stokes Existence and Smoothness: Status at the Cut-off**

For the 3-dimensional incompressible Navier–Stokes equations, determine whether smooth initial data always lead to global smooth solutions, or whether finite-time singularities can occur.

*Remark.* On September 8, 2026, OpenAI released a manuscript and Lean formalization claiming a smooth finite-energy forced solution that develops a finite-time singularity. On September 11, Clay said that the problem had “apparently been settled” while emphasizing that its deliberately unhurried evaluation and credit process remained pending. Thus the status current through September 15, 2026 is *solution announced; manuscript and formalization public; under official evaluation*, not an unconditional Clay-recognized solution.

**Open Problem 15.7. Yang–Mills Existence and Mass Gap**

Construct a mathematically rigorous quantum Yang–Mills theory on  $\mathbb{R}^4$  for every compact simple gauge group and prove that it has a positive mass gap.

*Remark.* The problem comes from quantum field theory. The relevant Quiz anchors are gauge theory, quantum field theory, and mass gap.

**Hilbert’s Problems**

David Hilbert published a list of 23 problems after his address at the 1900 ICM in Paris. He presented ten of them orally. The list became a program for twentieth-century mathematics, but its items are not all yes–no questions; several are broad research programs whose status cannot be reduced to a single word.

| No. | Subject                                 | Modern status and principal names                                                                                                                              |
|-----|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Continuum hypothesis and well-ordering  | CH is independent of ZFC, assuming consistency: Gödel proved relative consistency and Cohen proved independence using forcing.                                 |
| 2   | Consistency of arithmetic               | Gödel’s incompleteness theorems rule out the strongest form of Hilbert’s program; Gentzen later gave a relative consistency proof using transfinite induction. |
| 3   | Scissors congruence of polyhedra        | Solved negatively by Max Dehn through the Dehn invariant: equal volume does not imply equidecomposability.                                                     |
| 4   | Geometries with straight-line geodesics | A broad program rather than one fixed proposition; substantially developed through metric and convex geometry.                                                 |

Continued on the next page.

| No. | Subject                                           | Modern status and principal names                                                                                             |
|-----|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 5   | Continuous groups and Lie groups                  | Solved affirmatively by Gleason and by Montgomery–Zippin: locally Euclidean topological groups are Lie groups.                |
| 6   | Axiomatization of physics                         | An open-ended program involving probability, mechanics, and physical theories; no single final solution.                      |
| 7   | Transcendence of $a^b$                            | Solved by the Gelfond–Schneider theorem when $a$ and $b$ are algebraic, $a \neq 0, 1$ , and $b$ is irrational.                |
| 8   | Prime-number problems                             | Includes the Riemann Hypothesis and Goldbach-type questions; central parts remain open.                                       |
| 9   | General reciprocity laws                          | Largely answered by Artin reciprocity and class field theory.                                                                 |
| 10  | Algorithm for Diophantine equations               | Solved negatively by the Davis–Putnam–Robinson–Matiyasevich theorem: no such general algorithm exists over $\mathbb{Z}$ .     |
| 11  | Quadratic forms over algebraic number fields      | Essentially solved through Hasse’s local–global theory and later arithmetic developments.                                     |
| 12  | Explicit class field theory                       | Solved for important special cases, but the general <i>Kronecker Jugendtraum</i> remains open.                                |
| 13  | Seventh-degree equations and functions            | The continuous-function interpretation was solved by Kolmogorov and Arnold; stricter algebraic variants remain distinct.      |
| 14  | Finite generation of certain algebras             | Solved negatively in general by Nagata’s counterexample.                                                                      |
| 15  | Rigorous foundation of Schubert calculus          | Fulfilled through modern intersection theory and algebraic geometry.                                                          |
| 16  | Real algebraic curves and limit cycles            | Both parts contain major open questions; the limit-cycle problem is open even in low degrees.                                 |
| 17  | Nonnegative rational functions as sums of squares | Solved affirmatively by Emil Artin using sums of squares of rational functions.                                               |
| 18  | Space groups, tilings, and sphere packing         | Major parts were solved by Bieberbach and, for the three-dimensional Kepler conjecture, by Hales; it is a multi-part problem. |
| 19  | Regularity of variational solutions               | Solved through work of De Giorgi, Nash, Moser, and others.                                                                    |
| 20  | General boundary-value problems                   | A broad program developed through modern functional analysis and PDE; not one universally closed proposition.                 |
| 21  | Riemann–Hilbert problem                           | The unrestricted original existence claim is false by Bolibrukh’s counterexample; important variants are solved.              |
| 22  | Uniformization by automorphic functions           | Solved by Koebe and Poincaré through the uniformization theorem.                                                              |
| 23  | Further development of the calculus of variations | A research program that stimulated extensive modern theory rather than a single proposition with one completion date.         |

**Fact 15.8. Hilbert problem anchors**

Hilbert’s first problem concerns the continuum hypothesis; the second concerns consistency; the eighth includes the Riemann Hypothesis; the tenth concerns an algorithm for Diophantine equations; and the sixteenth includes the number and arrangement of limit cycles.

## Landau's Problems

At the 1912 ICM in Cambridge, Edmund Landau identified four elementary prime-number problems that he regarded as exceptionally difficult. All four remain open.

| Problem                      | Statement                                                                         | Status and anchor                   |
|------------------------------|-----------------------------------------------------------------------------------|-------------------------------------|
| Goldbach Conjecture          | Every even integer greater than 2 is a sum of two primes.                         | Open; additive number theory.       |
| Twin Prime Conjecture        | There are infinitely many primes $p$ for which $p + 2$ is prime.                  | Open; bounded prime gaps are known. |
| Legendre Conjecture          | For every positive integer $n$ , there is a prime between $n^2$ and $(n + 1)^2$ . | Open; primes in short intervals.    |
| Primes of the form $n^2 + 1$ | There are infinitely many positive integers $n$ for which $n^2 + 1$ is prime.     | Open; polynomial prime values.      |

## Other Famous Problems and Conjectures

The following problems are not all open problems. Some are solved theorems, but they remain famous because of their history, statements, or solution methods. They are collected together rather than divided into mathematical fields.

| Problem                            | Standard form                                                                                                                               | People and status                                        | Quiz anchors                              |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------|
| Basel Problem                      | Evaluate $\sum_{n=1}^{\infty} 1/n^2$ .                                                                                                      | Solved by Euler: $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$ . | Zeta value $\zeta(2)$ ; Euler.            |
| Fermat's Last Theorem              | $x^n + y^n = z^n$ has no positive integer solutions for $n \geq 3$ .                                                                        | Fermat; proved by Wiles with Taylor–Wiles.               | Elliptic curves and modular forms.        |
| Four Color Theorem                 | Every planar map can be colored with at most four colors.                                                                                   | Appel–Haken; later Robertson–Sanders–Seymour–Thomas.     | First famous computer-assisted proof.     |
| Kepler Conjecture                  | The maximum packing density of congruent spheres in $\mathbb{R}^3$ is $\pi/(3\sqrt{2})$ and is attained by the face-centered cubic packing. | Kepler; proved by Hales.                                 | Sphere packing; Flyspeck project.         |
| Thurston Geometrization Conjecture | Closed orientable 3-manifolds decompose into pieces modeled on eight geometries.                                                            | Thurston; completed through Perelman's work.             | Eight Thurston geometries.                |
| Continuum Hypothesis               | There is no cardinality strictly between $ \mathbb{N} $ and $ \mathbb{R} $ .                                                                | Cantor; independent of ZFC by Gödel and Cohen.           | Forcing; independence.                    |
| Collatz Conjecture                 | Iterating $n \mapsto n/2$ for even $n$ and $n \mapsto 3n + 1$ for odd $n$ eventually reaches 1.                                             | Collatz; open.                                           | Elementary iteration; nonlinear dynamics. |
| $abc$ Conjecture                   | If $a + b = c$ and $a, b, c$ are coprime, then $c$ is controlled by the radical of $abc$ , up to $\varepsilon$ .                            | Masser–Oesterlé; open in the standard sense.             | Radical, Diophantine equations.           |

**Theorem 15.9. Basel Problem**

*Euler evaluated the reciprocal-square series:*

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \zeta(2) = \frac{\pi^2}{6}.$$

*Remark.* The problem was first posed by Pietro Mengoli in 1644 and became associated with the Bernoulli family in Basel. Euler's solution was a landmark because an infinite sum over integers produced a value involving  $\pi$ .

*Remark.* Fermat wrote that he had a marvelous proof, but no proof appeared in his margin. Andrew Wiles proved the theorem through the modularity of semistable elliptic curves. The principal association to retain is

$$\text{Fermat's Last Theorem} \longleftrightarrow \text{Wiles} \longleftrightarrow \text{elliptic curves and modular forms.}$$

**Theorem 15.10. Four Color Theorem**

*Every planar map can be colored with at most four colors so that any two adjacent regions have different colors.*

*Remark.* The theorem was proved by Kenneth Appel and Wolfgang Haken in 1976 with extensive computer assistance. This made it one of the most famous early examples of a computer-assisted proof.

**Theorem 15.11. Kepler Conjecture**

*The maximum density of a packing of congruent spheres in  $\mathbb{R}^3$  is equal to*

$$\frac{\pi}{3\sqrt{2}}.$$

*This density is attained by the face-centered cubic packing and also by other optimal packings, including the hexagonal close packing.*

*Remark.* Thomas Hales proved the Kepler conjecture using a large computer-assisted argument. The Flyspeck project later produced a formal verification of the proof. In higher-dimensional sphere packing, Maryna Viazovska proved the optimality of the  $E_8$  lattice in dimension 8.

**Theorem 15.12. Bourgain's Slicing Problem**

*There exists a universal constant  $c > 0$  such that for every dimension  $n$  and every convex body  $K \subseteq \mathbb{R}^n$  of volume 1, there is an affine hyperplane  $H$  such that*

$$\text{vol}_{n-1}(K \cap H) \geq c.$$

*Remark.* Also called the hyperplane conjecture or slicing conjecture, this long-standing problem in high-dimensional convex geometry was resolved affirmatively by Boaz Klartag and Joseph Lehec in 2024. The main quiz association is

$$\begin{aligned} \text{Bourgain's slicing problem} &\longleftrightarrow \text{Klartag--Lehec,} \\ &\text{convex bodies} \longleftrightarrow \text{hyperplane sections.} \end{aligned}$$

*Remark.* The Poincaré conjecture is a special case of the geometrization program. The 2018 Science Quiz asked for the number of Thurston geometries; the answer is 8.

### **Hypothesis 15.13. Continuum Hypothesis**

*There is no set whose cardinality is strictly between the cardinality of the natural numbers and the cardinality of the real numbers. Equivalently,*

$$2^{\aleph_0} = \aleph_1.$$

*Remark.* Kurt Gödel proved that CH cannot be disproved from ZFC if ZFC is consistent. Paul Cohen proved that CH cannot be proved from ZFC if ZFC is consistent. Cohen introduced forcing and received the Fields Medal in 1966.

### **Open Problem 15.14. Goldbach Conjecture**

Every even integer greater than 2 is a sum of two prime numbers.

*Remark.* Goldbach is one of the easiest famous open problems to state. It belongs to additive number theory. The weak Goldbach conjecture, concerning sums of three odd primes, was proved by Harald Helfgott.

### **Open Problem 15.15. Twin Prime Conjecture**

There are infinitely many primes  $p$  such that

$$p + 2$$

is also prime.

*Remark.* The modern quiz anchors are bounded gaps between primes, Yitang Zhang, James Maynard, Terence Tao, and the Polymath project. These results do not prove the twin prime conjecture, but they prove that infinitely many prime pairs occur within some fixed bounded distance.

### **Open Problem 15.16. Collatz Conjecture**

Start with a positive integer  $n$ . If  $n$  is even, replace it by  $n/2$ . If  $n$  is odd, replace it by  $3n + 1$ . The Collatz conjecture states that repeated iteration always eventually reaches 1.

*Remark.* This problem is famous because the rule is elementary but the global behavior is still unknown. It is also called the  $3n + 1$  problem. In the reverse graph, every positive integer  $m$  has the predecessor  $2m$ . It also has the odd predecessor  $(m - 1)/3$  exactly when  $m \equiv 4 \pmod{6}$ .

### **Conjecture 15.17. abc Conjecture**

*For every  $\varepsilon > 0$ , there exists a constant  $K_\varepsilon$  such that whenever  $a, b, c$  are pairwise coprime positive integers satisfying  $a + b = c$ , we have*

$$c < K_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon},$$

*where  $\operatorname{rad}(n)$  is the product of the distinct prime divisors of  $n$ .*

*Remark.* The conjecture is powerful because it predicts that the additive relation  $a + b = c$  strongly restricts the repeated prime factors of  $abc$ . The keyword is “radical.” Shinichi Mochizuki announced a claimed proof in 2012, and the papers were later published, but no broadly accepted proof has been established; the conjecture is therefore recorded here as open.

**Fact 15.18. June Huh and combinatorial log-concavity**

June Huh's work connects combinatorics with algebraic geometry and Hodge theory. The central associations are:

- Read's conjecture  $\leftrightarrow$  chromatic polynomials.
- Brylawski's and Okounkov's conjectures  $\leftrightarrow$  matroid characteristic polynomials and log-concavity.
- Rota's conjecture  $\leftrightarrow$  log-concavity for matroids.
- Mason's conjecture  $\leftrightarrow$  independent sets of matroids.
- Hodge theory  $\leftrightarrow$  the geometric method behind these combinatorial log-concavity results.

In a quiz, if June Huh appears with a list of conjectures, the odd one out is often a famous conjecture from a different field, such as the Poincaré conjecture.

**Conjecture 15.19. Hadwiger Conjecture**

*Every finite graph  $G$  with chromatic number  $\chi(G) = k$  contains the complete graph  $K_k$  as a minor.*

*Remark.* The conjecture is known for  $k \leq 6$  and remains open in general. The case  $k = 5$  is equivalent to the Four Color Theorem.

**Open Problem 15.20. Hadwiger–Nelson Problem**

Determine the chromatic number of the Euclidean plane when points at distance exactly 1 are required to receive different colors.

*Remark.* The exact value is still unknown. The best general bounds are

$$5 \leq \chi(\mathbb{R}^2) \leq 7.$$

Aubrey de Grey's 2018 construction raised the lower bound from 4 to 5. This problem is distinct from Hadwiger's graph-minor conjecture.

**Conjecture 15.21. Hadamard Conjecture**

*For every positive integer  $n$ , there exists a  $4n \times 4n$  matrix  $\mathbf{H}$  with entries in  $\{-1, 1\}$  such that*

$$\mathbf{H}\mathbf{H}^T = 4n\mathbf{I}_{4n}.$$

*Remark.* A matrix satisfying this orthogonality condition is a Hadamard matrix. The necessary divisibility condition is known, but existence for every positive multiple of 4 remains open.

**Conjecture 15.22. Union-Closed Sets Conjecture**

*Let  $\mathcal{F}$  be a finite nonempty family of finite sets such that*

$$A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}.$$

*Then some element belongs to at least half of the sets in  $\mathcal{F}$ .*

*Remark.* The conjecture was proposed by Péter Frankl in 1979. It is also called Frankl's union-closed sets conjecture and remains open.

**Conjecture 15.23. Sunflower Conjecture**

For every fixed integer  $r \geq 3$ , there should exist a constant  $C_r$  such that every sufficiently large family of  $w$ -element sets with more than

$$C_r^w$$

members contains an  $r$ -sunflower:  $r$  sets whose pairwise intersections are all equal to their common intersection.

*Remark.* Erdős and Rado introduced sunflowers. A major 2019 breakthrough by Ryan Alweiss, Shachar Lovett, Kewen Wu, and Jiapeng Zhang improved the classical upper bound to roughly  $(\log w)^w$  for fixed  $r$ , but the conjectured exponential bound remains open.

**Conjecture 15.24. Erdős–Hajnal Conjecture**

For every fixed graph  $H$ , there exists a constant  $c(H) > 0$  such that every  $n$ -vertex graph containing no induced copy of  $H$  has either a clique or an independent set of size at least

$$n^{c(H)}.$$

*Remark.* The conjecture remains open in general. Tung Nguyen, Alex Scott, and Paul Seymour proved the five-vertex path case; this completed the Erdős–Hajnal property for every graph on five vertices. Work in 2026 pushed the method beyond the five-vertex path for additional six-vertex configurations.

**Conjecture 15.25. Lonely Runner Conjecture**

Suppose that  $n \geq 2$  runners move at distinct constant speeds around a unit circle. For each runner, there is a time at which that runner is at circular distance at least

$$\frac{1}{n}$$

from every other runner.

*Remark.* After subtracting the chosen runner's speed from every speed, the problem becomes one of simultaneous Diophantine approximation. The conjecture remains open in general; computer-assisted proofs announced in 2025 established the cases through ten runners.

**Open Problem 15.26. Odd Perfect Number Problem**

Determine whether there exists an odd positive integer  $n$  satisfying

$$\sigma(n) = 2n,$$

where  $\sigma(n)$  is the sum of the positive divisors of  $n$ .

*Remark.* Euclid and Euler characterized the even perfect numbers through Mersenne primes. No odd perfect number is known, and nonexistence has not been proved.

**Conjecture 15.27. Legendre Conjecture**

For every positive integer  $n$ , there is a prime  $p$  satisfying

$$n^2 < p < (n+1)^2.$$

*Remark.* This is stronger than what follows from the prime number theorem and remains open.

**Conjecture 15.28. Erdős–Straus Conjecture**

For every integer  $n \geq 2$ , there exist positive integers  $x, y, z$  such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

*Remark.* The conjecture concerns Egyptian-fraction decompositions and is verified computationally for enormous ranges, but no general proof is known.

**Conjecture 15.29. Beal Conjecture**

If positive integers satisfy

$$A^x + B^y = C^z, \quad x, y, z > 2,$$

then  $A$ ,  $B$ , and  $C$  have a common prime factor.

*Remark.* Fermat’s Last Theorem is the equal-exponent case with pairwise coprime bases. The Beal conjecture remains open.

**Conjecture 15.30. Schanuel Conjecture**

If  $z_1, \dots, z_n \in \mathbb{C}$  are linearly independent over  $\mathbb{Q}$ , then

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(z_1, \dots, z_n, e^{z_1}, \dots, e^{z_n}) \geq n.$$

*Remark.* Schanuel’s conjecture is a central organizing conjecture in transcendental number theory. It would imply many famous results and conjectures about algebraic independence.

**Open Problem 15.31. Infinitely Many Mersenne Primes**

Determine whether there are infinitely many primes of the form

$$2^p - 1,$$

where  $p$  is prime.

*Remark.* Every even perfect number has the Euclid–Euler form

$$2^{p-1}(2^p - 1)$$

with  $2^p - 1$  prime. Thus the infinitude of even perfect numbers is equivalent to the infinitude of Mersenne primes.

**Open Problem 15.32. Perfect Cuboid Problem**

Determine whether there exists a rectangular box with integer edge lengths  $a, b, c$  for which all three face diagonals and the space diagonal are integers:

$$a^2 + b^2, \quad b^2 + c^2, \quad c^2 + a^2, \quad a^2 + b^2 + c^2$$

are all perfect squares.

*Remark.* Boxes with integer edges and integer face diagonals are known as Euler bricks. Requiring the space diagonal to be integral gives the still-open perfect cuboid problem.

**Theorem 15.33. Catalan–Mihăilescu Theorem**

The only solution in integers  $x, y, a, b > 1$  of

$$x^a - y^b = 1$$

is

$$3^2 - 2^3 = 1.$$

*Remark.* Catalan posed the conjecture in 1844, and Preda Mihăilescu proved it in 2002. Thus 8 and 9 are the only consecutive positive perfect powers greater than 1.

**Open Problem 15.34. Smooth Four-Dimensional Poincaré Conjecture**

Determine whether every smooth closed 4-manifold that is homotopy equivalent to  $S^4$  is diffeomorphic to  $S^4$ .

*Remark.* The topological 4-dimensional Poincaré conjecture was proved by Freedman. The smooth 4-dimensional case remains open, while the other dimensions are settled.

**Theorem 15.35. Kakeya Conjecture in Three Dimensions**

If a set  $K \subseteq \mathbb{R}^3$  contains a unit line segment in every direction, then

$$\dim_{\mathbb{H}} K = 3,$$

and its Minkowski dimension is also 3.

*Remark.* Besicovitch showed that Kakeya sets may have Lebesgue measure 0. Hong Wang and Joshua Zahl proved the full dimension conjecture in  $\mathbb{R}^3$  in 2025. The analogous conjecture remains open in dimensions  $n \geq 4$ .

**Open Problem 15.36. Inscribed Square Problem**

Determine whether every Jordan curve in the plane contains four points that are the vertices of a square.

*Remark.* Also called the Toeplitz square-peg problem, it is known for many regular classes of curves but remains open for arbitrary Jordan curves.

**Open Problem 15.37. Moving Sofa Problem**

Determine the maximum area of a connected planar shape that can be moved around a right-angled corner in a corridor of unit width.

*Remark.* Gerver constructed a sofa of area approximately 2.21953166. The 2024 preprint *Optimality of Gerver's Sofa* (Jineon Baek) presents a proof that Gerver's construction is optimal. At the September 15, 2026 cutoff, the result remained a preprint, so its publication status should be distinguished from the mathematical claim made in the manuscript.

**Fact 15.38. Kissing Numbers**

The kissing number  $\tau_n$  is the maximum number of nonoverlapping unit spheres that can touch one unit sphere in  $\mathbb{R}^n$ . Famous exact values include

$$\tau_2 = 6, \quad \tau_3 = 12, \quad \tau_4 = 24, \quad \tau_8 = 240, \quad \tau_{24} = 196560.$$

Exact values are unknown in most dimensions.

**Open Problem 15.39. Hilbert–Smith Conjecture**

Every locally compact topological group that acts faithfully and continuously on a connected manifold should be a Lie group.

*Remark.* The problem asks whether manifolds admit genuinely non-Lie continuous symmetry groups. It remains open in full generality.

**Open Problem 15.40. Inverse Galois Problem**

Determine whether every finite group  $G$  occurs as

$$\mathrm{Gal}(K/\mathbb{Q})$$

for some finite Galois extension  $K$  of  $\mathbb{Q}$ .

*Remark.* The problem is solved for many classes of groups but remains open in general.

**Open Problem 15.41. Hilbert’s Tenth Problem over  $\mathbb{Q}$** 

Determine whether there is an algorithm that decides, for every polynomial

$$f \in \mathbb{Z}[x_1, \dots, x_n],$$

whether the equation  $f = 0$  has a solution in  $\mathbb{Q}^n$ .

*Remark.* Matiyasevich’s theorem gives a negative answer over  $\mathbb{Z}$ . The analogous decision problem over  $\mathbb{Q}$  remains open.

**Theorem 15.42. Classification of Finite Simple Groups**

*Every finite simple group is one of the following:*

- a cyclic group of prime order;
- an alternating group  $A_n$  with  $n \geq 5$ ;
- a finite simple group of Lie type;
- one of the 26 sporadic simple groups.

*Remark.* The proof is distributed across thousands of pages by many authors. The largest sporadic group is the Monster group.

**Theorem 15.43. Restricted Burnside Problem**

*For fixed positive integers  $d$  and  $n$ , there are only finitely many finite  $d$ -generated groups of exponent  $n$ , up to isomorphism.*

*Remark.* Efim Zelmanov solved the restricted Burnside problem using the theory of Lie algebras and received the 1994 Fields Medal. The unrestricted Burnside problem has a negative answer in general.

**Fact 15.44. Langlands Program**

The Langlands program predicts deep correspondences between arithmetic objects such as Galois representations and analytic objects such as automorphic representations. It is not one conjecture but a network of conjectures linking number theory, representation theory, and geometry.

**Open Problem 15.45. Three-Dimensional Euler Singularity Problem: Status at the Cutoff**

For the incompressible Euler equations in  $\mathbb{R}^3$ , determine whether smooth finite-energy initial data can develop a singularity in finite time.

*Remark.* The Euler equations model an ideal fluid without viscosity. Their global regularity problem is distinct from the Navier–Stokes Millennium problem. A September 2026 manuscript publicly claimed a finite-time singularity for the unforced problem. At the September 15 cutoff this remained an announced manuscript result rather than a settled published theorem.

**Conjecture 15.46. Falconer Distance Conjecture**

If a compact set  $E \subseteq \mathbb{R}^n$  satisfies

$$\dim_{\mathrm{H}} E > \frac{n}{2},$$

then its distance set

$$\Delta(E) = \{\|\mathbf{x} - \mathbf{y}\| \mid \mathbf{x}, \mathbf{y} \in E\}$$

has positive Lebesgue measure.

*Remark.* The conjecture connects harmonic analysis, geometric measure theory, and discrete distance problems. It remains open in general.

**Open Problem 15.47. Hilbert’s Sixteenth Problem**

The second part asks for a uniform upper bound, in terms of the degree  $n$ , on the number and arrangement of limit cycles of a planar polynomial vector field of degree  $n$ .

*Remark.* Even the quadratic case is not completely understood. The first part of Hilbert’s sixteenth problem concerns the topology of real algebraic curves.

**Theorem 15.48. Bieberbach Conjecture**

If

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n$$

is analytic and injective on the unit disk, then

$$|a_n| \leq n \quad (n \geq 2).$$

*Remark.* Ludwig Bieberbach conjectured the bound in 1916, and Louis de Branges proved it in 1985.

**Theorem 15.49. Kadison–Singer Problem**

Every pure state on the diagonal maximal abelian subalgebra of bounded operators on  $\ell^2$  has a unique extension to a state on all bounded operators.

*Remark.* Kadison and Singer posed the problem in 1959. Marcus, Spielman, and Srivastava solved it in 2013 through an equivalent finite-dimensional partition problem and the method of interlacing polynomials.

**Part III**

**Mathematics in 2026**

## Chapter 16

# Prizes and Congresses in 2026

This part records the mathematical events most relevant to the 2026 Science Quiz season. Its time window begins on January 1, 2025, and is current through September 15, 2026. Unlike Part II, which collects durable facts that should remain useful in future years, this part emphasizes why a person, prize, theorem, conjecture, institution, or competition became especially important during 2025 or 2026.

The guiding association is

$$\begin{array}{ccccc} \text{event} & \longleftrightarrow & \text{person or institution} & & \\ & \longleftrightarrow & \text{mathematical achievement} & \longleftrightarrow & \text{year and status.} \end{array}$$

### Fact 16.1. Status language in this part

Recent mathematics must be described with care.

- *Published* means that the work appeared in a scholarly publication.
- *Accepted* means that a journal announced acceptance, even if the final issue had not yet appeared.
- *Preprint* means that a manuscript was publicly posted but had not completed ordinary journal publication.
- *Announced* means that authors or a responsible institution publicly reported a result whose ordinary publication status must still be stated separately.
- *Formally verified* means that a proof assistant checked a formal proof term; this is different from ordinary peer review.
- *Scheduled* describes an officially announced future event or appointment.

Numerical evidence, an announced proof, a preprint, a formally verified argument, and a published theorem are therefore not interchangeable.

*Strategy.* Past Science Quiz questions frequently combine a recent event with an older mathematical fact. For each entry, remember the recent year and the shortest unmistakable mathematical anchor. Dates, locations, institutions, prize combinations, unusual records, and changes of affiliation are especially useful for multiple-choice elimination.

The chapter on mathematical prizes and congresses in Part II explains the Fields Medal, Abel Prize, and Wolf Prize. The present chapter records the season-specific results announced in 2025 and 2026.

| Year | Prize or congress                           | Laureate(s) or location                          | Principal Quiz anchors                                                                                              |
|------|---------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 2025 | Abel Prize                                  | Masaki Kashiwara                                 | Algebraic analysis, representation theory, $D$ -modules, crystal bases.                                             |
| 2025 | Breakthrough Prize in Mathematics           | Dennis Gaitsgory                                 | Geometric Langlands program in characteristic zero.                                                                 |
| 2025 | Shaw Prize in Mathematical Sciences         | Kenji Fukaya                                     | Fukaya category, symplectic topology, mirror symmetry, gauge theory.                                                |
| 2025 | ICTP–IMU Ramanujan Prize                    | Claudio Muñoz                                    | Nonlinear dispersive PDE, soliton stability, multi-soliton dynamics.                                                |
| 2025 | Basic Science Lifetime Awards (mathematics) | Shigefumi Mori; George Lusztig; Robert Tarjan    | Minimal model program; representation theory; algorithms and data structures.                                       |
| 2026 | Abel Prize                                  | Gerd Faltings                                    | Arithmetic geometry, Mordell conjecture, Lang conjectures, Diophantine geometry.                                    |
| 2026 | International Congress of Mathematicians    | Philadelphia, United States, July 23–30          | Fields Medals and the principal IMU prizes; IMU General Assembly in New York City.                                  |
| 2026 | Fields Medals                               | Yu Deng; John Pardon; Jacob Tsimerman; Hong Wang | PDE and kinetic theory; symplectic geometry; arithmetic geometry and o-minimality; harmonic analysis and $Kakeya$ . |
| 2026 | IMU Abacus Medal                            | Shayan Oveis Gharan                              | Algorithms, combinatorial optimization, approximation algorithms.                                                   |
| 2026 | Carl Friedrich Gauss Prize                  | Yurii Nesterov                                   | Convex optimization and accelerated first-order methods.                                                            |
| 2026 | Chern Medal Award                           | Graeme Segal                                     | Topology, geometry, and mathematical physics.                                                                       |
| 2026 | Leelavati Award                             | Hannah Fry                                       | Public communication and popular understanding of mathematics.                                                      |
| 2026 | Breakthrough Prize in Mathematics           | Frank Merle                                      | Nonlinear evolution equations, singularity formation, stability, solitons.                                          |
| 2026 | Shaw Prize in Mathematical Sciences         | Emmanuel Candès; Camillo De Lellis               | Compressed sensing, signal processing, statistics; geometric measure theory, singularities, and fluid dynamics.     |
| 2026 | Clay Research Awards                        | Three research groups                            | Furstenberg and $Kakeya$ sets; Telescope Conjecture counterexamples; Boltzmann equation from hard-sphere dynamics.  |
| 2026 | Rolf Schock Prize in Mathematics            | Tobias Holck Colding                             | Minimal surfaces and geometric flows.                                                                               |
| 2026 | Maryam Mirzakhani Prize in Mathematics      | Roman Bezrukavnikov                              | Geometric representation theory and modular representation theory.                                                  |

Continued on the next page.

| Year | Prize or congress                       | Laureate(s) or location                                       | Principal Quiz anchors                                                                              |
|------|-----------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 2026 | Michael and Sheila Held Prize           | Irit Dinur; Subhash Khot; Guy Kindler; Dor Minzer; Muli Safra | The 2-to-2 Games Theorem, approximation hardness, Unique Games.                                     |
| 2026 | International Congress of Basic Science | Beijing, China, August 9–21                                   | Inaugural ICBS Medals and Frontiers of Science Awards; mathematics, physics, and engineering.       |
| 2026 | ICBS Medals in Mathematics              | Claire Voisin; Horng-Tzer Yau; Shou-Wu Zhang                  | Hodge theory and algebraic geometry; random matrices and mathematical physics; arithmetic geometry. |

**Fact 16.2. 2026 ICM**

The 2026 International Congress of Mathematicians was held in Philadelphia from July 23 to July 30. The twentieth IMU General Assembly met in New York City immediately before the Congress, on July 20–21. The opening ceremony included the Fields Medals, IMU Abacus Medal, Gauss Prize, and Chern Medal Award; the Leelavati Award was associated with the closing of the Congress.

**Fact 16.3. 2026 ICBS and the inaugural mathematics medals**

The International Congress of Basic Science was held in Beijing from August 9 through August 21. The ICBS Medal was introduced in 2026 with three medals in mathematics:

|                |                       |                          |
|----------------|-----------------------|--------------------------|
| Claire Voisin  | $\longleftrightarrow$ | Emmy Noether Medal,      |
| Horng-Tzer Yau | $\longleftrightarrow$ | Shiing-Shen Chern Medal, |
| Shou-Wu Zhang  | $\longleftrightarrow$ | Andrew Wiles Medal.      |

The shortest mathematical anchors are Hodge theory and algebraic geometry for Voisin, random matrix theory and mathematical physics for Yau, and arithmetic geometry for Zhang. The Congress also presents the Frontiers of Science Awards, which recognize research papers rather than functioning as another single-laureate mathematics prize.

| 2026 Fields Medalist | Main area                                         | Season-specific Quiz anchors                                                                                                                                                                |
|----------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yu Deng              | Partial differential equations and kinetic theory | Rigorous derivation of kinetic equations from many-particle dynamics; hard spheres, the Boltzmann equation, wave kinetic equations, and a concrete part of Hilbert's sixth-problem program. |
| John Pardon          | Symplectic geometry and topology                  | Holomorphic curves, virtual fundamental cycles, Fukaya categories, symplectic topology, knot theory, and 3-manifolds.                                                                       |
| Jacob Tsimerman      | Arithmetic and complex algebraic geometry         | O-minimality, the André–Oort conjecture for Siegel modular varieties, period maps, and the Griffiths conjecture.                                                                            |
| Hong Wang            | Harmonic analysis and geometric measure theory    | Restriction theory, local smoothing, Furstenberg sets, and the three-dimensionalakeya conjecture. She became the third woman to receive a Fields Medal.                                     |

**Fact 16.4. 2025 and 2026 Abel Prizes**

The two Abel laureates in this preparation window are distinguished by sharply different anchors.

Masaki Kashiwara  $\longleftrightarrow$   $D$ -modules and crystal bases,

Gerd Faltings  $\longleftrightarrow$  Mordell conjecture and arithmetic geometry.

Faltings had already received the 1986 Fields Medal. Thus he joined the group of mathematicians who have received both the Fields Medal and the Abel Prize.

**Fact 16.5. 2026 Clay Research Awards**

The three recognized clusters are worth remembering as complete associations.

- Tuomas Orponen, Pablo Shmerkin, Hong Wang, and Joshua Zahl: Furstenberg sets in the plane and Kakeya sets in three-dimensional space.
- Robert Burklund, Jeremy Hahn, Ishan Levy, and Tomer Schlank: counterexamples to the Ravenel Telescope Conjecture in chromatic homotopy theory.
- Yu Deng and Zaher Hani: the long-time derivation of the Boltzmann equation from hard-sphere dynamics; Xiao Ma was a coauthor of the mathematical work.

*Remark.* Only officially confirmed prize results are recorded. In particular, no unannounced 2026 Wolf Prize in Mathematics laureate is inferred or supplied merely to complete the traditional prize triad.

## Chapter 17

# Mathematical Competitions in 2026

Competition questions differ from prize questions. The most useful chain is

competition  $\longleftrightarrow$  host and year  $\longleftrightarrow$  team result  $\longleftrightarrow$  individual record.

| Competition | Host and scale                                                                      | Republic of Korea                                                                                                       | Notable records                                                                                                             |
|-------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 2025 IMO    | Sunshine Coast, Australia; 110 countries and 630 contestants                        | Third place with 203 points; four gold and two silver medals                                                            | Gold cutoff 35; five perfect scores; six problems over two contest days.                                                    |
| 2026 IMO    | Shanghai, China; 117 teams and 666 contestants                                      | Sixth place with 167 points; three gold, two silver, and one honorable mention                                          | Seven perfect scores; Hyeonjun Lee of the Republic of Korea obtained 42/42.                                                 |
| 2025 APMO   | Thirty-seventh Asian Pacific Mathematical Olympiad                                  | Kyungjun Park scored 35/35 and received gold; the next nine Korean results ranged from silver through honorable mention | Five problems, each worth seven points; official results.                                                                   |
| 2026 APMO   | Thirty-eighth Asian Pacific Mathematical Olympiad; 43 countries and 397 contestants | Second place with 306 points; one gold, two silver, four bronze, and three honorable mentions                           | Seohyung Jo, Hyeonjun Lee, and Hyunjun Jang each scored 35/35; official awards were gold, silver, and silver, respectively. |

### Fact 17.1. Republic of Korea at the 2025 IMO

The six Korean contestants and their scores were

37, 36, 36, 35, 31, 28.

Kyungjun Park, Wooju Ham, Hyeonjun Lee, and Hyewon Yoon received gold; Hyunjun Jang and Hyungjune Cho received silver. The team total was

$$37 + 36 + 36 + 35 + 31 + 28 = 203.$$

### Fact 17.2. Republic of Korea at the 2026 IMO

The Republic of Korea placed sixth with 167 points. Its aggregate problem scores were

42, 20, 17, 40, 37, 11.

Hyeonjun Lee was one of seven contestants worldwide to obtain a perfect score of 42.

**Fact 17.3. Republic of Korea at the 2026 APMO**

The official 2026 APMO results place the Republic of Korea second among 43 participating countries, with a team total of 306. Seohyung Jo, Hyeonjun Lee, and Hyunjun Jang each obtained the perfect score 35/35. The official award labels were

Seohyung Jo: Gold,      Hyeonjun Lee: Silver,      Hyunjun Jang: Silver.

Thus equal perfect scores can receive different official award labels in the 2026 APMO; the numerical score and the award label should be remembered separately.

**Fact 17.4. 2026 IMO medal design**

The reverse of the 2026 IMO medal used Zhao Shuang's diagram for the Pythagorean theorem, a classical diagram associated with his commentary on the *Zhoubi Suanjing*. The medal had diameter 9 cm. This provides a direct connection between a current competition, the Pythagorean theorem, and the history of Chinese mathematics.

## Chapter 18

# Major Mathematical Developments in 2026

This chapter records developments that were prominent during 2025 and 2026. Complete proofs are generally far beyond the scope of Science Quiz. The intended memory unit is

standard problem or theorem + people + status + main method or field.

| Development                                    | Principal names                                                                                    | Status in this season                                                                                                  | Quiz anchors                                                                                         |
|------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Navier–Stokes existence and smoothness         | OpenAI; Clay Mathematics Institute evaluating the announcement                                     | September 2026 solution announced; manuscript and Lean formalization public; official evaluation pending at the cutoff | Forced three-dimensional finite-time singularity; not yet an unconditional Clay-recognized solution. |
| Three-dimensional Kakeya conjecture            | Hong Wang; Joshua Zahl                                                                             | 2025 proof preprint; recognized by a 2026 Clay Research Award                                                          | Full Hausdorff and Minkowski dimension 3; harmonic analysis and geometric measure theory.            |
| Hilbert’s tenth problem over rings of integers | Levent Alpöge; Manjul Bhargava; Wei Ho; Ari Shnidman; independently Peter Koymans and Carlo Pagano | One proof accepted and published; an independent preprint                                                              | Diophantine equations, undecidability, number fields, elliptic curves.                               |
| Boltzmann equation from hard spheres           | Yu Deng; Zaher Hani; Xiao Ma                                                                       | Long-time derivation; recognized by a 2026 Clay Research Award                                                         | Kinetic theory, many-particle dynamics, Hilbert’s sixth problem.                                     |

Continued on the next page.

| Development                                          | Principal names                                                                                                            | Status in this season                                                                                                                                | Quiz anchors                                                                                                                               |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Modularity of ordinary abelian surfaces              | George Boxer; Frank Calegari; Toby Gee; Vincent Pilloni                                                                    | 2025 preprint                                                                                                                                        | Abelian surfaces, Siegel modular forms, Galois representations, Langlands philosophy.                                                      |
| Mizohata–Takeuchi conjecture<br>counterexample       | Hannah Mira Cairo                                                                                                          | 2025 preprint                                                                                                                                        | Harmonic analysis, Fourier restriction, counterexample, logarithmic loss.                                                                  |
| Compact Bonnet pairs                                 | Alexander Bobenko; Tim Hoffmann; Andrew Sageman-Furnas                                                                     | Published in 2025                                                                                                                                    | Noncongruent immersed tori with the same metric and mean curvature.                                                                        |
| Optimal spectral gap for typical hyperbolic surfaces | Nalini Anantharaman; Laura Monk                                                                                            | 2025 preprint completing a multi-paper program                                                                                                       | Random hyperbolic surfaces, Weil–Petersson measure, Laplacian, asymptotically optimal spectral gap $1/4$ .                                 |
| A convex polyhedron without Rupert’s property        | Jakob Steiner; Sergey Yurkevich                                                                                            | 2025 preprint                                                                                                                                        | The Noperthedron; first proved convex polyhedron that cannot pass through a suitable straight hole in an identical copy of itself.         |
| Unknotting number is not additive                    | Mark Brittenham; Susan Hermiller                                                                                           | Accepted in 2026                                                                                                                                     | Knot theory, connected sum, explicit counterexamples.                                                                                      |
| Kissing-number lower bounds in dimensions 17–21      | Henry Cohn; Anqi Li                                                                                                        | Preprint widely publicized in 2025                                                                                                                   | Sphere packing, spherical codes, nonlattice configurations.                                                                                |
| Furstenberg set conjecture                           | Tuomas Orponen; Pablo Shmerkin and collaborators                                                                           | Major planar result; 2026 Clay recognition                                                                                                           | Geometric measure theory, sets containing line segments in many directions.                                                                |
| Ravenel Telescope Conjecture                         | Robert Burklund; Jeremy Hahn; Ishan Levy; Tomer Schlank                                                                    | Counterexamples recognized in 2026                                                                                                                   | Chromatic homotopy theory and stable homotopy theory.                                                                                      |
| Jacobian Conjecture                                  | Levent Alpöge; AI-assisted discovery involving Claude Fable                                                                | Explicit 3-dimensional counterexample announced and independently checked by direct calculation in 2026; counterexamples now exist for every $n > 2$ | Polynomial maps, constant Jacobian determinant, noninjectivity; the case $n = 2$ remains open.                                             |
| Riemann zeta zeros on the critical line              | Levent Alpöge; Ralph Furman; mathematical argument discovered and written by Claude; independent proof by Youness Lamzouri | August 2026 preprint with Lean formalization; independent preprint posted September 2                                                                | Unconditionally more than two thirds of the nontrivial zeros are simple and on $\text{Re}(s) = 1/2$ ; the Riemann Hypothesis remains open. |

Continued on the next page.

| Development                                              | Principal names                                                 | Status in this season                                                                                               | Quiz anchors                                                                                                                  |
|----------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Sendov's Conjecture                                      | Lech Mazur; Terence Tao                                         | AI-assisted proof in August 2026; human-digested exposition and Lean formalization                                  | Zeros of a polynomial in the unit disk and nearby critical points; Phelps–Rodriguez strengthening.                            |
| Banach's isometric conjecture                            | Xinbao Lu; Kaiwen Yang; Antonio Acuaviva; Tomasz Kania          | August 2026 preprints completing the real and complex cases and a quaternionic counterpart                          | Normed spaces, mutually isometric finite-dimensional subspaces, Hilbert and Hermitian inner-product norms.                    |
| Gromov's codimension-two volume-growth conjecture        | Jian Ge; independently Bochao Kong and Xingyu Zhu               | Independent August 2026 preprints                                                                                   | Nonnegative Ricci curvature, positive scalar curvature, and the sharp growth order $R^{n-2}$ .                                |
| Samuels' Conjecture                                      | Zhi Ling                                                        | August 2026 preprint                                                                                                | Sharp lower bound for sums of independent nonnegative random variables; Feige's conjecture follows from the equal-means case. |
| Erdős–Szemerédi sum–product conjecture over $\mathbb{R}$ | Thomas F. Bloom; Will Sawin; Carl Schildkraut; Dmitrii Zhelezov | Disproved by a May 2026 preprint                                                                                    | Additive combinatorics, sumsets and product sets, algebraic number fields; the integer version is not disproved.              |
| Cycle Double Cover Conjecture                            | OpenAI; expositions by Sang-il Oum and Jim Geelen               | Affirmative AI-generated proof announced in July 2026 and publicly expositied in preprints                          | Bridgeless graphs, cycle covers, every edge covered exactly twice.                                                            |
| Maxwell Conjecture on point charges                      | Philip Arathoon; Gavin Ball; Matthew D. Kvalheim                | Disproved by a July 2026 preprint                                                                                   | Five point charges with at least 24 nondegenerate critical points, exceeding Maxwell's proposed $(n-1)^2$ bound.              |
| Planar Pompeiu, Schiffer, and Berenstein conjectures     | Matthew J. Colbrook; George Stepaniants; Siavash Sadeghi        | Computer-assisted counterexample preprints in August 2026                                                           | Overdetermined eigenvalue problems, Fourier zero sets, noncircular real-analytic planar domains, interval arithmetic.         |
| Petersen Coloring Conjecture                             | Bryce Putman; Jorik Jooken                                      | Explicit 112-vertex counterexample posted in August 2026; SAT/DRAT verification followed by a human-checkable proof | Bridgeless cubic graphs, Petersen coloring, normal 5-edge-coloring.                                                           |

Continued on the next page.

| Development                         | Principal names                                                                                | Status in this season                                                                                          | Quiz anchors                                                                                               |
|-------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Heil–Ramanathan–Topiwala Conjecture | Markus Faulhuber; Philipp Petersen; Jordy Timo van Velthoven; Felix Voigtlaender; Vignon Oussa | Disproved in August 2026; a 12-shift counterexample was followed by a computer-assisted 4-shift counterexample | Time–frequency analysis, Gabor systems, linear dependence of time–frequency shifts of a Schwartz function. |

**Fact 18.1. September 2026 Navier–Stokes announcement**

On September 8, OpenAI released a manuscript and Lean formalization claiming a finite-time singularity for the forced three-dimensional incompressible Navier–Stokes equations. On September 11, the Clay Mathematics Institute described the problem as “apparently” settled while stating that its evaluation and credit process would be deliberately unhurried. Clay still listed Navier–Stokes as active at the September 15 cutoff. The precise quiz status is therefore *solution announced with a public Lean formalization; Clay evaluation pending*, not simply *solved*.

**Theorem 18.2. Three-Dimensional Kakeya Conjecture**

*A Kakeya set in  $\mathbb{R}^3$  contains a unit line segment in every direction. The 2025 result of Hong Wang and Joshua Zahl establishes that every such set has Hausdorff dimension and Minkowski dimension equal to 3.*

*Idea.* The theorem does not say that every Kakeya set has positive three-dimensional volume. It says that no Kakeya set can have dimension smaller than the ambient dimension. The main quiz association is

$$\text{Kakeya in } \mathbb{R}^3 \longleftrightarrow \text{Hong Wang and Joshua Zahl.}$$

Both the Hausdorff dimension and the Minkowski dimension are equal to 3.

**Fact 18.3. Hilbert’s tenth problem over number fields**

Hilbert’s tenth problem asks for an algorithm deciding whether an integer polynomial equation has an integer solution. Matiyasevich, building on Davis, Putnam, and Robinson, proved that no such general algorithm exists over  $\mathbb{Z}$ .

A paper by Levent Alpöge, Manjul Bhargava, Wei Ho, and Ari Shnidman was accepted and published during the 2025–2026 season and established the analogous negative answer for the ring of integers of every number field. Peter Koymans and Carlo Pagano posted an independent and more general undecidability result for infinite rings finitely generated over  $\mathbb{Z}$ .

*Remark.* The phrase “Hilbert’s tenth problem over every number field” refers to the corresponding ring of algebraic integers. It is not the same as the still distinct question of Hilbert’s tenth problem over the rational numbers  $\mathbb{Q}$ .

**Fact 18.4. Boltzmann equation and Hilbert’s sixth problem**

Yu Deng, Zaher Hani, and Xiao Ma developed a rigorous long-time derivation of the Boltzmann equation from deterministic hard-sphere dynamics. This is a major realization of the kinetic-theory part of Hilbert’s sixth-problem program.

*Remark.* Hilbert’s sixth problem is a broad request for an axiomatic treatment of physics. The hard-sphere-to-Boltzmann result is therefore a major solution of a concrete component, not a declaration that every aspect of Hilbert’s sixth problem has been completed.

**Fact 18.5. Modularity beyond elliptic curves**

George Boxer, Frank Calegari, Toby Gee, and Vincent Pilloni proved modularity for an important class of ordinary abelian surfaces over  $\mathbb{Q}$ , under explicit local and residual hypotheses. In particular, their work proves modularity for a positive proportion of abelian surfaces in a natural ordering.

*Idea.* An elliptic curve is one-dimensional as an abelian variety, whereas an abelian surface has dimension 2. The association to retain is

$$\text{elliptic curves and modular forms} \longrightarrow \text{abelian surfaces and Siegel modular forms.}$$

This is part of the broader Langlands philosophy connecting arithmetic objects with automorphic forms and representations.

**Fact 18.6. Hannah Cairo’s counterexample**

In 2025, Hannah Mira Cairo constructed a counterexample to the Mizohata–Takeuchi conjecture in Fourier restriction theory. Her result showed that the conjectured estimate fails in its original form and identified a logarithmic loss that cannot simply be removed.

**Fact 18.7. Compact Bonnet pairs**

A Bonnet pair consists of noncongruent surfaces with the same induced metric and the same mean-curvature function. In 2025, Alexander Bobenko, Tim Hoffmann, and Andrew Sageman-Furnas published the first compact examples: a pair of immersed tori.

**Fact 18.8. Unknotting number is not additive**

Let  $u(K)$  denote the unknotting number of a knot  $K$ . The long-standing additivity expectation was

$$u(K_1 \# K_2) = u(K_1) + u(K_2).$$

Brittenham and Hermiller constructed examples for which

$$u(K_1 \# K_2) < u(K_1) + u(K_2),$$

and their paper was accepted in 2026. Thus a connected sum may require fewer crossing changes than separately unknotting the two summands.

**Fact 18.9. Improved kissing configurations**

Henry Cohn and Anqi Li improved the best known lower bounds for kissing numbers in dimensions 17 through 21. Their constructions used spherical codes that need not come from lattices, reinforcing the lesson that the most symmetric construction need not be the best known one.

**Fact 18.10. More than two thirds of the zeta zeros are simple and on the critical line**

In an August 2026 preprint, Levent Alpöge and Ralph Furman proved unconditionally that at least two thirds of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, are simple and lie on the critical line

$$\operatorname{Re}(s) = \frac{1}{2}.$$

They also proved that at least five sixths are distinct. With the Montgomery–Taylor window, these lower bounds improve to approximately

$$0.6725 \quad \text{and} \quad 0.8362.$$

The proof replaces a positivity step that classically used the Riemann Hypothesis with a rank–trace inequality for a finite compression of Weil’s Hermitian form, together with Sylvester’s law of inertia. The paper includes a Lean 4 formalization.

*Remark.* This is a new unconditional record toward the Riemann Hypothesis, not a proof of the hypothesis. The paper states that the mathematical argument was discovered and written by Claude, with Alpöge and Furman verifying the proof and taking responsibility for its contents. On September 2, Youness Lamzouri posted an independent, shorter proof of the same numerical bounds. His argument replaces the finite-dimensional matrix framework with a Hilbert-space inequality and an unconditional form of Montgomery’s pair-correlation theorem.

**Fact 18.11. Sendov’s conjecture and the Phelps–Rodriguez strengthening**

Sendov’s conjecture concerns a polynomial whose zeros all lie in the closed unit disk: every zero must lie within distance 1 of some critical point. In August 2026, Lech Mazur obtained an AI-assisted proof for all degrees. Terence Tao then produced a human-digested exposition and a substantially shorter Lean formalization, and observed that the same argument proves the stronger Phelps–Rodriguez statement.

**Fact 18.12. Banach’s isometric conjecture**

Banach asked in 1932 whether a real normed space must be a Hilbert space if, for some fixed dimension  $n > 1$ , all of its  $n$ -dimensional subspaces are mutually isometric. Gromov had proved the even-dimensional case. In an August 2026 preprint, Xinbao Lu and Kaiwen Yang proved every remaining odd-dimensional case, completing the conjecture over the real field. A follow-up preprint by Antonio Acuaviva and Tomasz Kania completed the complex version, where the norm is forced by a Hermitian inner product, and also established the quaternionic counterpart.

**Fact 18.13. Gromov’s codimension-two volume-growth conjecture**

In 1986, Gromov asked whether a complete noncompact  $n$ -dimensional Riemannian manifold with nonnegative Ricci curvature and scalar curvature bounded below by 1 must satisfy

$$\text{vol}(B(p, R)) \leq C_n R^{n-2}$$

for all centers  $p$  and radii  $R > 0$ . Independent August 2026 preprints by Jian Ge and by Bochao Kong with Xingyu Zhu gave affirmative resolutions. The exponent  $n - 2$  is the reason for the name “codimension-two volume growth.”

**Fact 18.14. Samuels’ conjecture and Feige’s conjecture**

Samuels’ conjecture asks for the sharp smallest possible probability that a sum of independent nonnegative random variables stays below a given threshold when their means are prescribed. In an August 2026 preprint, Zhi Ling proved the conjectured sharp bound and showed that it is attained. Feige’s conjecture follows immediately from the equal-means case, giving the most useful short association

$$\text{Samuels} \longrightarrow \text{Feige as the equal-means case.}$$

**Fact 18.15. Sum–product conjecture over the reals**

For a finite set  $A$ , write

$$A + A = \{a + b \mid a, b \in A\}, \quad AA = \{ab \mid a, b \in A\}.$$

In May 2026, Thomas F. Bloom, Will Sawin, Carl Schildkraut, and Dmitrii Zhelezov constructed arbitrarily large finite  $A \subseteq \mathbb{R}$  satisfying

$$\max\{|A + A|, |AA|\} \leq |A|^{2-c}$$

for an absolute constant  $c > 0$ . This disproves the Erdős–Szemerédi sum–product conjecture over  $\mathbb{R}$ . Because their sets live in algebraic number fields of degree growing like  $\log |A|$ , the result does not by itself settle the original integer case.

**Fact 18.16. Maxwell’s proposed bound for electrostatic equilibria is false**

Maxwell proposed that the field of  $n$  point charges should have at most  $(n - 1)^2$  nondegenerate critical points. Philip Arathoon, Gavin Ball, and Matthew D. Kvalheim gave a configuration of five point charges whose electrostatic potential has at least 24 nondegenerate critical points. Since  $(5 - 1)^2 = 16$ , the proposed bound is false.

**Fact 18.17. Planar Pompeiu–Schiffer and Berenstein counterexamples**

In August 2026, Matthew J. Colbrook and George Stepaniants constructed a bounded simply connected noncircular planar domain with real-analytic boundary that is simultaneously a counterexample to the planar Schiffer conjecture and the corresponding Pompeiu conjecture. Colbrook, Siavash Sadeghi, and Stepaniants then adapted the method to disprove the unrestricted planar Berenstein conjecture. Both results are computer-assisted preprints with rigorous interval-arithmetic certification.

**Fact 18.18. Petersen coloring counterexample**

Bryce Putman’s August 2026 preprint gives an explicit simple bridgeless cubic graph on 112 vertices with no Petersen coloring and no normal 5-edge-coloring. The computational proof supplies SAT instances and checked DRAT certificates. Jorik Jooker subsequently posted a human-checkable proof for the 112-vertex counterexample.

**Fact 18.19. The HRT conjecture is false**

The Heil–Ramanathan–Topiwala conjecture asserted that every finite set of distinct time–frequency shifts of a nonzero function in  $L^2(\mathbb{R})$  is linearly independent. In August 2026, Markus Faulhuber, Philipp Petersen, Jordy Timo van Velthoven, and Felix Voigtlaender disproved the conjecture using 12 time–frequency shifts of a Schwartz function. Vignon Oussa then produced a computer-assisted counterexample using only 4 shifts, again with a nonzero Schwartz function.

*Strategy.* For a newly announced result, do not memorize only the headline “problem solved.” Memorize whether the event was a proof, a counterexample, an improved bound, a construction, or a new framework. These distinctions are common sources of plausible but incorrect answer choices.

## Chapter 19

# Artificial Intelligence and Mathematics in 2026

The years 2025 and 2026 marked a transition from isolated demonstrations to sustained use of AI in olympiad reasoning, algorithm discovery, formal proof, and mathematical research. The correct comparison is not merely “human versus machine.” One must identify the type of mathematical output and the manner in which it was checked.

| Type of claim          | What was produced                                                              | What the label does and does not mean                                                                                         |
|------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Gold-medal-level score | A score at or above the human gold-medal cutoff on a specified problem set     | It is a benchmark statement, not automatically a formal proof or a new research theorem.                                      |
| Natural-language proof | A human-readable written argument                                              | It may be judged by expert graders, but it is not mechanically checked merely because a model produced it.                    |
| Formal proof           | A proof term accepted by a proof assistant such as Lean                        | The kernel checks the formal derivation; the formal statement and its relation to the intended theorem must still be correct. |
| AI-assisted discovery  | A construction, algorithm, conjecture, counterexample candidate, or proof idea | Human mathematical validation, literature search, attribution, and publication status remain essential.                       |
| Numerical candidate    | Computational evidence suggesting a phenomenon or singularity                  | It is not a theorem until an appropriate rigorous argument is supplied.                                                       |

### Fact 19.1. Gold-medal-level performance on the 2025 IMO

Google DeepMind reported that an advanced version of Gemini with Deep Think solved five of the six 2025 IMO problems and received 35 out of 42 points under an official evaluation process. Since the human gold cutoff was also 35, the result met the gold-medal standard. The system worked from the natural-language problem statements and produced natural-language proofs.

*Remark.* The phrase “gold-medal-level” is exact but limited: an AI system was not an official contestant and did not receive an IMO medal. The statement compares its evaluated score with the human cutoff.

**Fact 19.2. Two AI systems on the 2025 IMO**

OpenAI separately reported that a general-purpose reasoning model also scored 35 out of 42 on the 2025 IMO problem set. Thus both OpenAI and Google DeepMind announced gold-medal-level natural-language performance in 2025, using different systems. Neither system was an official contestant, and neither received an IMO medal.

**Fact 19.3. AlphaProof and formal mathematics**

A paper on AlphaProof appeared in *Nature* in 2025. AlphaProof combines reinforcement learning with the Lean proof assistant and produces machine-checkable formal proofs. Its well-known benchmark used the 2024 IMO problem set and reached silver-medal-level performance when combined with a geometry system.

**Fact 19.4. AlphaEvolve**

Google DeepMind introduced AlphaEvolve in 2025 as an evolutionary coding agent for discovering and improving algorithms. Its mathematical significance lies in searching large spaces of executable constructions and evaluating candidates automatically, rather than merely generating prose solutions.

**Fact 19.5. ICPC-level algorithmic performance**

In 2025, Gemini Deep Think was also evaluated on the ICPC World Finals problem set and achieved a gold-medal-level result. This is primarily an achievement in algorithm design and competitive programming, whereas the IMO emphasizes proof-based olympiad mathematics.

**Fact 19.6. GPT-5.5 and off-diagonal Ramsey numbers**

In April 2026, OpenAI reported that an internal GPT-5.5 system with a custom research harness found a new proof of a longstanding asymptotic fact about off-diagonal Ramsey numbers. The argument was subsequently verified in Lean. For quiz purposes, the important association is

$$\text{GPT-5.5} \longleftrightarrow \text{off-diagonal Ramsey numbers} \longleftrightarrow \text{Lean verification.}$$

**Fact 19.7. GPT-5.6 Sol and the Cycle Double Cover Conjecture**

The Cycle Double Cover Conjecture asks whether every finite bridgeless loopless multigraph admits a multiset of cycles in which every edge appears exactly twice. In July 2026, OpenAI released the prompt and materials for an affirmative proof generated by GPT-5.6 Sol Ultra. Sang-il Oum and Jim Geelen posted independent expository write-ups of the argument. The season-specific association is therefore

$$\begin{aligned} \text{Cycle Double Cover Conjecture} &\longleftrightarrow \text{GPT-5.6 Sol Ultra} \\ &\longleftrightarrow \text{Oum and Geelen expositions.} \end{aligned}$$

The public proof and expositions should still be distinguished from the separate question of ordinary journal publication.

**Fact 19.8. First Proof**

In February 2026, OpenAI released proof attempts by an internal model for all ten problems in the research-level First Proof challenge. After expert feedback, OpenAI assessed at least five attempts—Problems 4, 5, 6, 9, and 10—as having a high chance of being correct, while other attempts remained under review. Its initially favored attempt for Problem 2 was later acknowledged to be incorrect. The episode is a useful reminder that a persuasive research proof attempt still requires independent mathematical checking.

**Fact 19.9. AI disproof of the Erdős unit-distance conjecture**

Let  $u(n)$  be the largest possible number of pairs at unit distance among  $n$  points in the plane. Erdős had conjectured that

$$u(n) = n^{1+o(1)}.$$

In May 2026, OpenAI released a proof generated by an internal general-purpose model that disproved this conjecture. The construction gives, for infinitely many  $n$  and some fixed  $\delta > 0$ ,

$$u(n) \geq n^{1+\delta}.$$

The argument uses class-field towers and other ideas from algebraic number theory, and external mathematicians checked the released proof. The result disproves the conjectured near-linear upper bound; it does not determine the exact asymptotic order of  $u(n)$ .

**Fact 19.10. Ten AI-generated research advances**

On August 1, 2026, OpenAI released manuscripts for ten results generated by an internal version of its Astra model. Humans prepared the manuscripts with the same model, after which the model formalized each argument as a Lean certificate. The announced results were:

- (1) new upper bounds for high-dimensional sphere packing down to the Cohn–Elkies threshold;
- (2) exponentially improved bounds for binary and spherical codes;
- (3) a construction establishing the existence of non-sofic groups;
- (4) a disproof of Connes’s rigidity conjecture;
- (5) new arithmetic-circuit lower bounds for the permanent, including an arithmetic-formula lower bound of order  $n^4/\log n$ ;
- (6) an exponential parallel-repetition theorem for general two-player quantum games;
- (7) polynomial-factor hardness of approximation for the closest vector problem;
- (8) a resolution of Ehrhart’s volume conjecture;
- (9) a superexponential lower bound for multicolor triangle Ramsey numbers, resolving Erdős Problem 183;
- (10) results on compactness and degeneracy in extremal graph theory, resolving Erdős Problems 146 and 180.

At the September 15, 2026 cutoff, these remained recently released manuscripts with formal certificates. Formal verification of the stated Lean theorems and long-term independent assessment by the mathematical community are distinct stages.

**Fact 19.11. Palomar registry for Lean-verified mathematics**

On August 18, 2026, a group including Jeremy Avigad, Terence Tao, Ravi Vakil, and Akshay Venkatesh announced Palomar, a public registry of Lean-verified mathematics incubated by ICARM and the Lean Focused Research Organization. Palomar records fixed source versions, exact formal statements, dependencies, and review comments. Its central methodological point is equally important: successful kernel checking verifies a formal derivation from stated assumptions, but by itself does not certify novelty, mathematical importance, ordinary peer review, or that an informal claim has been formalized exactly as intended.

**Fact 19.12. Fermat’s Last Theorem in Lean**

On September 4, 2026, Anthropic released a computer-checked formalization of Fermat’s Last Theorem produced with Claude and Prove2Me. It follows the Darmon–Diamond–Taylor exposition of Wiles’s argument and uses Lean’s standard axioms. This is a milestone in formal verification, not the first proof of Fermat’s Last Theorem.

**Fact 19.13. Prime Gaps at Most 186: Conditional Formalization**

The September 2026 PrimeGaps186 project targets

$$\liminf_{n \rightarrow \infty} (p_{n+1} - p_n) \leq 186,$$

where  $p_n$  is the  $n$ th prime. Its Lean derivation is conditional on three explicit input axioms: two cited exponential-sum estimates and numerical integral bounds. The accompanying numerical certificate does not itself discharge those axioms in Lean. The precise label is therefore *conditional formalization with a numerical certificate*; the twin prime conjecture remains open.

**Fact 19.14. AI, Navier–Stokes, and the limits of status labels**

The September 2026 Navier–Stokes release included both a public manuscript and a Lean formalization. Kernel checking certifies the formal derivation from its encoded assumptions; it does not by itself complete Clay’s evaluation, determine credit, or constitute journal publication. The full cutoff status is recorded in the Major Mathematical Developments chapter.

**Fact 19.15. Claude and the Riemann zeta critical-line record**

An August 2026 paper by Levent Alpöge and Ralph Furman states that its mathematical argument was discovered and written by Claude, an AI system developed by Anthropic; the listed authors verified the proof and take responsibility for it. The mathematical result itself is recorded in the Major Mathematical Developments chapter: an unconditional lower bound of more than two thirds for nontrivial zeta zeros that are both simple and on the critical line, accompanied by Lean 4 formalization. On September 2, Youness Lamzouri posted an independent proof of the same bounds using a shorter Hilbert-space argument. The Riemann Hypothesis itself remains open.

**Fact 19.16. Leiden Declaration on AI and Mathematics**

The Leiden Declaration, released in 2026 and endorsed by the IMU, articulated principles for responsible AI-assisted mathematics. The central requirements are transparent disclosure of significant AI use, accurate attribution, openness about methods and evidence, and continued human responsibility for mathematical correctness and scholarship.

*Strategy.* A Science Quiz question may deliberately mix the following terms:

IMO score, formal verification, peer review, AI-assisted discovery, new theorem.

Treat each as a separate claim. A result may satisfy more than one, but none implies all the others.

## Chapter 20

# Mathematicians and Institutions in 2026

Recent-person questions often use a move, retirement, death, or unusual career event as the first clue, followed by an older theorem or prize. The following tables record the principal institutional changes and memorial events of the season.

| Mathematician                             | 2025–2026 event                                                 | Institution                                                                | Quiz anchors                                                             |
|-------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Hong Wang                                 | Became a permanent professor in 2025                            | Institut des Hautes Études Scientifiques, jointly with New York University | Harmonic analysis, Kakeya, 2026 Fields Medal.                            |
| Jacob Tsimerman                           | Announced in July 2026 that he would begin a position at OpenAI | OpenAI; on leave from the University of Toronto                            | Arithmetic geometry, o-minimality, 2026 Fields Medal, AI safety.         |
| Tim Roughgarden                           | Joined the faculty on July 1, 2026                              | Institute for Advanced Study                                               | Algorithms, algorithmic game theory, mechanism design.                   |
| James Maynard                             | Appointed Regius Professor; scheduled to begin October 1, 2026  | University of Oxford                                                       | Prime gaps, sieve methods, 2022 Fields Medal; successor to Andrew Wiles. |
| Ryan Chen; Alex Cohen; Anna Skorobogatova | Began Clay Research Fellowships in 2025                         | Clay Mathematics Institute appointments hosted at research institutions    | Clay Research Fellowships and early-career research appointments.        |
| Oliver Edtmair; Qiuyu Ren                 | Began Clay Research Fellowships in 2026                         | Clay Mathematics Institute appointments hosted at research institutions    | Clay Research Fellowships and early-career research appointments.        |

### Fact 20.1. Jacob Tsimerman and OpenAI

After receiving a 2026 Fields Medal, Jacob Tsimerman announced at the 2026 ICM that he would begin a position at OpenAI. He went on leave from the University of Toronto to work on AI safety, while remaining a professor and faculty member there. The accurate season-specific association is therefore a Fields Medalist moving temporarily from ordinary university duties into AI-safety research, not an unqualified claim that he quit mathematics or resigned from the university.

| Mathematician    | Death                      | Principal associations                                                                                                                         |
|------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Peter D. Lax     | May 16, 2025,<br>aged 99   | Partial differential equations, conservation laws, shock waves, numerical analysis, Lax equivalence theorem, 1987 Wolf Prize, 2005 Abel Prize. |
| Heisuke Hironaka | March 18, 2026,<br>aged 94 | Resolution of singularities in characteristic zero, algebraic geometry, 1970 Fields Medal.                                                     |
| Michael O. Rabin | April 14, 2026             | Finite automata, probabilistic algorithms, Rabin–Karp string search, 1976 Turing Award shared with Dana Scott.                                 |
| Cleve Moler      | May 20, 2026,<br>aged 86   | Numerical linear algebra and numerical computing; creator of MATLAB; cofounder of MathWorks.                                                   |

**Fact 20.2. Peter D. Lax**

Peter D. Lax spent his academic career at New York University’s Courant Institute. His work connected partial differential equations, numerical methods, fluid dynamics, shock waves, scattering theory, and integrable systems. The Lax equivalence theorem states, in its standard numerical analysis setting, that a consistent finite-difference approximation of a well-posed linear initial-value problem is convergent if and only if it is stable.

**Fact 20.3. Heisuke Hironaka**

Hironaka received the 1970 Fields Medal for proving resolution of singularities for algebraic varieties over fields of characteristic zero. The season-specific clue is his death in 2026; the durable mathematical clue is

Hironaka  $\longleftrightarrow$  resolution of singularities.

**Fact 20.4. Cleve Moler and MATLAB**

Cleve Moler was a pioneer of numerical computing and the creator of MATLAB, originally developed to give students convenient interactive access to matrix software. He later cofounded MathWorks. His death in May 2026 creates a particularly short quiz association:

Cleve Moler  $\longleftrightarrow$  MATLAB and numerical computing.

**Fact 20.5. From Wiles to Maynard at Oxford**

Andrew Wiles was the inaugural holder of Oxford’s modern Regius Professorship of Mathematics. James Maynard was appointed as his successor, with the new appointment scheduled to begin on October 1, 2026. This creates a memorable association between Fermat’s Last Theorem and modern work on prime gaps.

## Chapter 21

# Mathematics in the Republic of Korea in 2026

This chapter records events whose specifically Korean institutional or biographical connection makes them useful for the **POSTECH–KAIST** Science War. International results already discussed above are not duplicated merely because a Korean mathematician or contestant was involved.

| Year | Person or event                                                          | Korean connection                                                                                         | Principal Quiz anchors                                                                                                                |
|------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Sug Woo Shin received the Samsung Ho-Am Prize in Physics and Mathematics | Distinguished professor at KIAS; professor at the University of California, Berkeley                      | Number theory, automorphic forms, the Langlands program.                                                                              |
| 2025 | Jae Kyoung Kim delivered a plenary lecture at the SIAM Annual Meeting    | Chief investigator at the IBS Biomedical Mathematics Group                                                | First Korean plenary speaker at a SIAM Annual Meeting; mathematical and computational biology.                                        |
| 2025 | Nam-Gyu Kang received the Korean Mathematical Society DI Prize           | Professor at KIAS                                                                                         | Probability, complex analysis, conformal field theory.                                                                                |
| 2025 | Inhyeok Choi received the Sangsan Prize and joined KIAS                  | KIAS fellow and professor                                                                                 | Geometric group theory, random walks, hyperbolic groups.                                                                              |
| 2025 | Jae Choon Cha received the Korea Science Award                           | <b>POSTECH</b> professor; Ph.D. from <b>KAIST</b>                                                         | Geometric topology; Disk Embedding Theory in dimension four; transfinite invariants resolving a long-standing problem of John Milnor. |
| 2025 | Cheolwoo Park received the Choi Seok-jeong Award                         | Professor at <b>KAIST</b>                                                                                 | Statistical machine learning and statistical methodology.                                                                             |
| 2026 | Sung-Jin Oh received the Samsung Ho-Am Prize in Physics and Mathematics  | <b>KAIST</b> alumnus; professor at KIAS and associate professor at the University of California, Berkeley | Nonlinear hyperbolic PDE, general relativity, instability inside black holes.                                                         |

Continued on the next page.

| Year      | Person or event                                                                                    | Korean connection                                                                                                                                      | Principal Quiz anchors                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026      | Bo-Hae Im and JungHwan Park received Korean Mathematical Society Paper Awards                      | Professors at <a href="#">KAIST</a>                                                                                                                    | Positive-characteristic multiple zeta values and the Zagier–Hoffman conjectures; knot concordance and the $(2, 1)$ -cable of the figure-eight knot. |
| 2026      | Bo-Hae Im, Hojin Kim, and Tuan Ngo Dac posted a preprint on Thakur’s conjecture                    | Bo-Hae Im is a professor at <a href="#">KAIST</a>                                                                                                      | Positive-characteristic multiple zeta values; independence through weight $2q$ and a unique relation at weight $2q + 1$ .                           |
| 2026      | <a href="#">POSTECH</a> Arithmetic Geometry Lab selected for the Basic Research Laboratory Program | First collaborative arithmetic-geometry research group organized in the Republic of Korea; Sungmun Cho is principal investigator                       | Langlands program, Diophantine geometry, arithmetic geometry.                                                                                       |
| 2026      | Kyeongsu Choi received the Science Prize at the POSCO TJ Park Prize                                | Professor at KIAS                                                                                                                                      | Mean-curvature flow, Gauss-curvature flow, geometric PDE.                                                                                           |
| 2026      | IMU Executive Committee met in Seoul                                                               | Hosted by KIAS on March 13–14                                                                                                                          | International Mathematical Union and Korean mathematical institutions.                                                                              |
| 2026      | Inaugural ASIACOMB conference and ASIACOMB Prize in Daejeon, August 24–28                          | Held at the Daejeon Convention Center; organized as a new Asian combinatorics forum with IBS participation; Huy Tuan Pham received the inaugural prize | Biennial combinatorics conference; the prize recognizes outstanding young researchers of Asian origin or based in Asia.                             |
| 2025–2026 | Jineon Baek and the moving sofa problem                                                            | <a href="#">POSTECH</a> alumnus; June E Huh Fellow at KIAS                                                                                             | Gerver’s sofa, optimal area approximately 2.21953166, preprint status.                                                                              |

### Fact 21.1. Jae Choon Cha and the 2025 Korea Science Award

Jae Choon Cha of [POSTECH](#) was recognized for work in geometric topology. Two especially memorable anchors are his Disk Embedding Theory for four-dimensional topological manifolds and transfinite invariants of three-manifolds, the latter leading to a resolution of a problem posed by John Milnor that had remained open for more than sixty years. Cha received his Ph.D. from [KAIST](#), giving this entry a direct connection to both universities in the Science War.

### Fact 21.2. Two [KAIST](#) faculty members and the 2026 KMS Paper Awards

The two [KAIST](#) recipients have sharply different mathematical anchors.

- Bo-Hae Im was recognized for work determining algebraic relations among multiple and alternating multiple zeta values in positive characteristic and completely establishing the positive-characteristic Zagier–Hoffman conjectures.
- JungHwan Park was recognized for work showing that the  $(2, 1)$ -cable of the figure-eight knot is not smoothly slice, resolving a problem that had remained open for about forty years and connecting directly to four-dimensional topology and the slice–ribbon conjecture.

**Fact 21.3. A 2026 preprint on Thakur's conjecture**

Bo-Hae Im, Hojin Kim, and Tuan Ngo Dac posted the preprint *The threshold for linear independence of multiple zeta values in positive characteristic*. For a finite field  $\mathbb{F}_q$ , they prove linear independence through weight  $2q$ , while at weight  $2q + 1$  there is a unique explicit  $\mathbb{F}_q$ -linear relation. Thus the preprint gives the first counterexample to Thakur's 2009 conjecture and identifies the precise threshold at which the conjectured independence fails. Because this work remained a preprint at the September 15, 2026 cutoff, its status should be remembered together with the result.

**Fact 21.4. Two consecutive Samsung Ho-Am mathematics laureates**

The Physics and Mathematics laureates in the two-year window are

2025 : Sug Woo Shin  $\longleftrightarrow$  Langlands program,  
 2026 : Sung-Jin Oh  $\longleftrightarrow$  nonlinear hyperbolic PDE and black holes.

The prize category combines physics and mathematics, so the laureate's own discipline must be identified from the citation rather than from the category name alone.

**Fact 21.5. 2026 ASIACOMB in Daejeon**

The inaugural ASIACOMB conference was held at the Daejeon Convention Center from August 24 through August 28. ASIACOMB is a biennial conference in Asia devoted to combinatorics and international collaboration. Huy Tuan Pham of Caltech and the University of Chicago received the inaugural ASIACOMB Prize, which recognizes outstanding research contributions in combinatorics by researchers of Asian origin or based in Asia who are not older than 35.

*Strategy.* Questions connected to the Republic of Korea are often easiest when three levels are kept separate:

nationality or origin,      current institution,      institution connected to the event.

A Korean or Korean-American mathematician may work abroad, hold a joint appointment, or receive a Korean prize for research performed in an international setting.

Part IV

Past Problems

**Problem 2010.1.** Determine whether there exist positive integers  $x$  and  $y$  satisfying

$$x^2 + y^2 = 963.$$

*Solution.* No such positive integers exist. Every integer is even or odd, so its square satisfies

$$(a^2 \equiv 0 \pmod{4}) \vee (a^2 \equiv 1 \pmod{4}).$$

Hence the sum of two integer squares can be congruent only to

$$0, \quad 1, \quad \text{or} \quad 2 \pmod{4}.$$

However,

$$963 \equiv 3 \pmod{4}.$$

Thus 963 cannot be written as the sum of two integer squares.

Therefore, the answer is no.

□

**Problem 2010.2.** Identify the mathematician who became famous for refusing the Fields Medal after solving the Poincaré conjecture.

*Solution.* Grigori Perelman proved the Poincaré conjecture using Ricci flow and declined the 2006 Fields Medal. These are the two identifying clues in the Fields Medal and Poincaré conjecture entries in Part II. Therefore, the answer is Grigori Perelman.

□

**Problem 2010.3.** Determine the country of origin of John Charles Fields, after whom the Fields Medal is named.

*Solution.* John Charles Fields was a Canadian mathematician. The medal's name honors him, as recorded in the Fields Medal section. Therefore, the answer is Canada.

□

**Problem 2010.4.** Determine the maximum possible number of distinct eigenvalues of a  $5 \times 5$  matrix.

*Solution.* The eigenvalues of a  $5 \times 5$  matrix  $\mathbf{A}$  are the roots of its characteristic polynomial

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_5 - \mathbf{A}).$$

This polynomial has degree 5. Therefore it can have at most 5 distinct roots, and hence  $\mathbf{A}$  can have at most 5 distinct eigenvalues. This bound is attained by  $\text{diag}(1, 2, 3, 4, 5)$ , whose five diagonal entries are its eigenvalues.

Therefore, the answer is 5.

□

**Problem 2010.5.** In 1742, a Prussian mathematician wrote to Euler and proposed a conjecture. Euler later stated it in the following form:

Every even integer greater than 2 can be expressed as the sum of two primes.

This is an old unsolved problem in number theory. It has been verified for very large numbers, but it has not been proved in general. Identify this conjecture.

*Solution.* The statement that every even integer greater than 2 can be expressed as the sum of two primes is known as Goldbach's conjecture.

Therefore, the answer is Goldbach's conjecture.

□

**Problem 2010.6.** Brownian motion is modeled by a stochastic process named after the American mathematician who developed its mathematical theory and later founded cybernetics. For this process, increments over disjoint time intervals are independent, and an increment over an interval of length  $t$  is Gaussian with mean 0 and variance  $t$ . Identify the mathematician.

*Solution.* The process described is the Wiener process, the standard mathematical model of Brownian motion. It is named after Norbert Wiener.

Hence the mathematician is Norbert Wiener.

□

**Problem 2010.7.** This set is a closed uncountable set and is also perfect. Although it is uncountable, it has Lebesgue measure 0. It is self-similar and is often used in the construction of fractals. It is obtained from the closed interval  $[0, 1]$  by repeatedly removing the open middle third, then doing the same to the remaining intervals. It is named after the mathematician widely known as the founder of set theory. Identify this set.

*Solution.* The set obtained by repeatedly removing the open middle third from  $[0, 1]$  is the Cantor set. It is closed, perfect, uncountable, self-similar, and has Lebesgue measure 0.

Therefore, the answer is Cantor set.

□

**Problem 2010.8.** Determine the year in which Terence Tao received the Fields Medal.

*Solution.* The Fields Medal table in Facts to Memorize places Terence Tao among the 2006 medalists at the International Congress of Mathematicians.

Therefore, the answer is 2006.

□

**Problem 2011.1.** Identify the international mathematical congress, scheduled to be held in the Republic of Korea in 2014, at which the Fields Medal is awarded.

*Solution.* The Fields Medal is awarded at the International Congress of Mathematicians (ICM). The congress scheduled for 2014 was in Seoul, Republic of Korea. Therefore, the answer is

International Congress of Mathematicians (ICM).

□

**Problem 2011.2.** Let  $\mathbf{A}$  and  $\mathbf{B}$  be  $n \times n$  matrices satisfying

$$\mathbf{AB} - \mathbf{I}_n = \mathbf{0},$$

where  $\mathbf{I}_n$  is the  $n \times n$  identity matrix. Determine

$$\mathbf{BA} - \mathbf{I}_n.$$

*Solution.* The condition is  $\mathbf{AB} = \mathbf{I}_n$ . Taking determinants gives  $\det(\mathbf{A})\det(\mathbf{B}) = 1$ , so  $\mathbf{A}$  is invertible. Multiplying on the left by  $\mathbf{A}^{-1}$  yields  $\mathbf{B} = \mathbf{A}^{-1}$ . Hence

$$\mathbf{BA} - \mathbf{I}_n = \mathbf{A}^{-1}\mathbf{A} - \mathbf{I}_n = \mathbf{0}.$$

□

**Problem 2011.3.** Differentiate  $(1+x)^{10}$  three times, substitute  $x = 0$ , and determine the resulting value.

*Solution.* We compute the third derivative:

$$\frac{d^3}{dx^3}(1+x)^{10} = 10 \cdot 9 \cdot 8(1+x)^7.$$

Substituting  $x = 0$ , we obtain

$$10 \cdot 9 \cdot 8(1+0)^7 = 720.$$

Therefore, the answer is  $\boxed{720}$ .

□

**Problem 2011.4.** Give at least two real-valued functions defined on  $\mathbb{R}$  that remain unchanged after being differentiated twice. In other words, give at least two functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfying

$$f'' = f.$$

*Solution.* Two simple examples are

$$f(x) = e^x \quad \text{and} \quad g(x) = e^{-x}.$$

Indeed,

$$\frac{d^2}{dx^2}e^x = e^x$$

and

$$\frac{d^2}{dx^2}e^{-x} = e^{-x}.$$

Thus both functions remain unchanged after being differentiated twice.

Therefore, two valid examples are  $\boxed{e^x \text{ and } e^{-x}}$ .

□

**Problem 2011.5.** Let  $\mathbf{A}$  be a  $5 \times 5$  matrix s.t. every entry in the  $i$ th row is equal to  $i$ . Find the sum of the five eigenvalues of  $\mathbf{A}$ , counted with multiplicity.

*Solution.* By Proposition 3.62, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to  $\text{tr}(\mathbf{A})$ .

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad 4, \quad 5.$$

Therefore,

$$\text{tr}(\mathbf{A}) = 1 + 2 + 3 + 4 + 5 = 15.$$

Hence the sum of the five eigenvalues of  $\mathbf{A}$ , counted with multiplicity, is  $\boxed{15}$ .

□

**Problem 2011.6.** A train takes 22 seconds to completely cross an 82 m bridge. When its speed is doubled, it takes 50 seconds to completely cross a 706 m bridge. Determine the length of the train.

*Solution.* Let  $L$  be the length of the train, and let  $v$  be its original speed in meters per second.

To completely cross a bridge, the train must travel the length of the bridge plus its own length. Hence

$$\frac{L + 82}{v} = 22.$$

When the speed is doubled, the train takes 50 seconds to completely cross a 706 m bridge, so

$$\frac{L + 706}{2v} = 50.$$

Equivalently,

$$L + 82 = 22v, \quad L + 706 = 100v.$$

Subtracting the first equation from the second gives

$$624 = 78v,$$

so

$$v = 8.$$

Therefore,

$$L + 82 = 22 \cdot 8 = 176,$$

and hence

$$L = 94.$$

Thus the length of the train is 94 m.

□

**Problem 2011.7.** Form an integer using the digits

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Each of these digits must be used at least once, and digits may be repeated. Determine the smallest such integer divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

*Solution.* The number must be divisible by

$$\text{lcm}(1, 2, 3, 4, 6, 7, 8, 9) = 504.$$

Since

$$\text{lcm}(7, 8, 9) = 504,$$

it is enough to enforce divisibility by 7, 8, and 9 separately.

If each digit is used exactly once, then the sum of the digits is

$$1 + 2 + 3 + 4 + 6 + 7 + 8 + 9 = 40,$$

which is not divisible by 9. Hence no 8-digit number works.

If we use exactly one extra digit  $d$ , then the digit sum becomes  $40 + d$ . For divisibility by 9, we need

$$40 + d \equiv 0 \pmod{9},$$

so

$$d \equiv 5 \pmod{9}.$$

But the digit 5 is not allowed. Hence no 9-digit number works.

Therefore, the answer must have at least 10 digits. For a 10-digit number, we need to add two extra digits. Their sum must be congruent to 5 modulo 9. Among the allowed digits, the possible extra pairs are

$$(1, 4), \quad (2, 3), \quad (6, 8), \quad (7, 7).$$

Only the extra pair (1, 4) allows the number to start with 11; the other pairs allow only one occurrence of 1. Thus any valid arrangement starting with 11 is smaller than every arrangement from the other pairs. We therefore first consider the digits

$$1, 1, 2, 3, 4, 4, 6, 7, 8, 9.$$

With these digits, the smallest possible beginning is

$$112344.$$

It remains to arrange the last four digits 6, 7, 8, 9 so that the whole number is divisible by 504.

Since the digit sum is already divisible by 9, it remains to check divisibility by 7 and 8. The last three digits must be divisible by 8. Among the permutations of three of the digits 6, 7, 8, 9, the possible last three digits divisible by 8 are

$$768, \quad 896, \quad 968, \quad 976.$$

Hence the only candidates with prefix 112344 are

$$1123449768, \quad 1123447896, \quad 1123447968, \quad 1123448976.$$

Checking divisibility by 7, only

$$1123449768$$

is divisible by 7. Indeed,

$$1123449768 = 504 \cdot 2229067.$$

Therefore it is divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Thus the smallest possible integer is 1123449768.

□

**Problem 2011.8.** Determine the greatest integer less than or equal to

$$-\frac{1}{\ln\left(\frac{365}{366}\right)}.$$

*Hint.* Use the Taylor approximation of  $\ln(1-x)$ .

*Solution.* We write

$$\frac{365}{366} = 1 - \frac{1}{366}.$$

Using

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots,$$

with  $x = \frac{1}{366}$ , we obtain

$$-\ln\left(\frac{365}{366}\right) = \frac{1}{366} + \frac{1}{2 \cdot 366^2} + \frac{1}{3 \cdot 366^3} + \cdots.$$

For  $0 < x < 1$ , the logarithm series is strictly larger than  $x$  and strictly smaller than the geometric series  $\sum_{k=1}^{\infty} x^k = x/(1-x)$ , because  $1/k < 1$  for  $k \geq 2$ . With  $x = 1/366$ , this gives

$$\frac{1}{366} < -\ln\left(\frac{365}{366}\right) < \frac{1}{365}.$$

Taking reciprocals gives

$$365 < -\frac{1}{\ln\left(\frac{365}{366}\right)} < 366.$$

Therefore, the greatest integer less than or equal to the given number is  $\boxed{365}$ .

□

**Problem 2011.9.** Evaluate the integral

$$\int_0^{\pi/2} \frac{1}{1 + \tan^{\sqrt{2}} t} dt.$$

*Hint.* Use the substitution  $x = \frac{\pi}{2} - t$ .

*Solution.* Let

$$I = \int_0^{\pi/2} \frac{1}{1 + \tan^{\sqrt{2}} t} dt.$$

Using the reflection substitution for definite integrals, set

$$x = \frac{\pi}{2} - t.$$

Then

$$\tan t = \tan\left(\frac{\pi}{2} - x\right) = \cot x = \frac{1}{\tan x}.$$

Hence

$$I = \int_0^{\pi/2} \frac{1}{1 + \cot^{\sqrt{2}} x} dx = \int_0^{\pi/2} \frac{\tan^{\sqrt{2}} x}{1 + \tan^{\sqrt{2}} x} dx.$$

Adding this expression to the original expression for  $I$ , we obtain

$$2I = \int_0^{\pi/2} \left( \frac{1}{1 + \tan^{\sqrt{2}} x} + \frac{\tan^{\sqrt{2}} x}{1 + \tan^{\sqrt{2}} x} \right) dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

Therefore, the answer is  $\boxed{\frac{\pi}{4}}$ .

□

**Problem 2011.10.** This French mathematician was born on December 8, 1865, and died on October 17, 1963, living to the age of 97. He is well known for his proof of the prime number theorem, which states that the number of primes less than or equal to  $x$  is asymptotically approximated by  $\frac{x}{\ln x}$  for large  $x$ . He was also personally connected to the Dreyfus affair and became a strong supporter of Zionism.

Identify this mathematician.

*Hint.* The following clues are useful:

- (1) The Cauchy–\_\_\_\_\_ theorem gives the radius of convergence of a power series.
- (2) A \_\_\_\_\_ matrix is a square matrix whose entries are all  $+1$  or  $-1$  and whose rows are mutually orthogonal; such matrices are also used in error-correcting codes.

*Solution.* The mathematician described in the problem is Jacques Hadamard.

The first hint refers to the Cauchy–Hadamard theorem, which gives the radius of convergence of a power series. The second hint refers to a Hadamard matrix, a square matrix whose entries are all  $+1$  or  $-1$  and whose rows are mutually orthogonal.

Hadamard is also famous for proving the prime number theorem in 1896, independently of Charles Jean de la Vallée Poussin. Therefore, the answer is Jacques Hadamard.

□

**Problem 2012.1.** This British mathematical genius died of cyanide poisoning shortly before his 42nd birthday. A famous account says that he died after eating an apple laced with cyanide. Identify this mathematician.

*Solution.* The mathematician described in the problem is Alan Turing.

Turing was a British mathematician, logician, and computer scientist. He made fundamental contributions to computability theory through the concept of the Turing machine, and he also played a crucial role in codebreaking during World War II.

Turing died in 1954 from cyanide poisoning, shortly before his 42nd birthday. The story involving an apple laced with cyanide is widely known, although some details of the event remain historically uncertain.

Therefore, the answer is Alan Turing.

□

**Problem 2012.2.** Evaluate the infinite series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots.$$

*Solution.* The given series is

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

This is the famous Basel problem. Euler proved that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the value of the series is  $\frac{\pi^2}{6}$ .

□

**Problem 2012.3.** Identify the mathematician who first proved the law of quadratic reciprocity and discovered more than seven proofs of it.

*Solution.* The Law of Quadratic Reciprocity in Mathematical Concepts is one of the central theorems in elementary number theory. It describes a deep symmetry between the solvability of

$$x^2 \equiv p \pmod{q} \quad \text{and} \quad x^2 \equiv q \pmod{p}$$

for odd primes  $p$  and  $q$ .

Carl Friedrich Gauss gave the first complete proof of the law of quadratic reciprocity. He regarded it as one of the jewels of number theory and found several different proofs of the theorem.

Therefore, the answer is Carl Friedrich Gauss.

□

**Problem 2012.4.** Let  $\mathbf{A}$  be a  $10 \times 10$  matrix s.t. every entry in the  $k$ th row is equal to  $k$ . Find the sum of the eigenvalues of  $\mathbf{A}$ , counted with multiplicity.

*Solution.* By Proposition 3.62, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to  $\text{tr}(\mathbf{A})$ .

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad \dots, \quad 10.$$

Therefore,

$$\text{tr}(\mathbf{A}) = 1 + 2 + 3 + \dots + 10 = \frac{10 \cdot 11}{2} = 55.$$

Hence the sum of the eigenvalues of  $\mathbf{A}$ , counted with multiplicity, is 55.

□

**Problem 2012.5.** This philosopher was born as the youngest son of a wealthy Austrian steel magnate who supported many artists. He showed talent in several fields, especially mathematics and philosophy. His work had a major influence on logical positivism and on the philosophy of language, and it also influenced the humanities, social sciences, and the arts. His later philosophy is especially known for emphasizing ordinary language use.

Identify this person.

*Solution.* The person described in the problem is Ludwig Wittgenstein.

Wittgenstein was born into a wealthy Austrian family and studied logic, mathematics, and philosophy. His early work, especially the *Tractatus Logico-Philosophicus*, strongly influenced logical positivism and the Vienna Circle.

Later, in works such as *Philosophical Investigations*, Wittgenstein emphasized the importance of ordinary language and argued that the meaning of a word is closely connected to its use in language.

Therefore, the answer is Ludwig Wittgenstein.

□

**Problem 2013.1.** A complex number is called *algebraic* if it is a root of a nonzero polynomial with integer coefficients; otherwise, it is called *transcendental*. Joseph Liouville first proved the existence of transcendental numbers in 1844. Hilbert's seventh problem, one of the 23 problems presented at the 1900 International Congress of Mathematicians in Paris, also concerns transcendental numbers. It was solved independently by Aleksandr Gelfond and Theodor Schneider in the 1930s, and Alan Baker received the 1970 Fields Medal in part for major advances in this area.

Which mathematician proved in 1873 that  $e$ , the base of the natural logarithm, is transcendental? Ferdinand von Lindemann later generalized this mathematician's method and proved in 1882 that  $\pi$  is transcendental.

*Solution.* Charles Hermite proved the transcendence of  $e$  in 1873. Lindemann proved the transcendence of  $\pi$  in 1882. This is the Hermite–Lindemann chronological pair in the mathematical history section. Therefore, the answer is Charles Hermite.

□

**Problem 2013.2.** Beginning around 1935, mathematicians associated with the Lwów school gathered in cafés near the university and recorded problems in a notebook. The names appearing in it include Banach, Mazur, Ulam, Kac, Zygmund, Schauder, Erdős, Dugundji, Steinhaus, and Ahlfors. Some problems carried prizes. In particular, on November 6, 1936, Stanisław Mazur

offered a live goose for Problem 153, later associated with the basis problem. Per Enflo solved the problem in 1972, and Mazur presented him with the promised goose in a televised ceremony.

Determine the title by which this famous notebook is known.

*Solution.* The notebook is known as *The Scottish Book*, named after the Scottish Café in Lwów, where members of the Lwów school of mathematics frequently met and discussed problems.

Therefore, the answer is *The Scottish Book*.

□

**Problem 2013.3.** Using exactly three occurrences of the digit 9, together with standard arithmetic operations, parentheses, powers, and square roots, construct each of the numbers

$$1, \quad 6, \quad 8.$$

*Solution.* One set of constructions is

$$\left(\frac{9}{9}\right)^9 = 1, \quad \frac{9+9}{\sqrt{9}} = 6, \quad 9 - \frac{9}{9} = 8.$$

Each expression uses exactly three occurrences of the digit 9. Alternative constructions include

$$9^{9-9} = 1, \quad \sqrt{9 \cdot 9} - \sqrt{9} = 6, \quad 9 - \frac{\sqrt{9}}{\sqrt{9}} = 8.$$

□

**Problem 2014.1.** This twentieth-century German mathematician is well known for a theorem stating that symmetries of Hamiltonian dynamical systems correspond to conservation laws. Identify this mathematician.

*Solution.* The theorem described in the problem is Noether's theorem. It states, roughly speaking, that continuous symmetries of a physical system correspond to conserved quantities. For example, time-translation symmetry corresponds to conservation of energy, and spatial-translation symmetry corresponds to conservation of momentum.

The mathematician associated with this theorem is Emmy Noether, one of the most important algebraists of the twentieth century.

Therefore, the answer is Emmy Noether.

□

**Problem 2014.2.** In the famous Chinese mathematical classic *The Nine Chapters on the Mathematical Art*, often compared with Euclid's *Elements* as a foundational mathematical classic, the value of  $\pi$  appears as 3. Later, Liu Hui, who wrote a commentary on *The Nine Chapters*, obtained the approximation 3.14 by refining the method of inscribed polygons. Identify the ancient Chinese mathematician who further refined this method and obtained the approximations

$$\frac{355}{113} \quad \text{and} \quad \frac{22}{7}$$

for  $\pi$ .

*Solution.* The mathematician described in the problem is Zu Chongzhi.

Liu Hui improved the approximation of  $\pi$  by using polygons with many sides. Zu Chongzhi later obtained the very accurate bounds

$$3.1415926 < \pi < 3.1415927,$$

and gave two famous rational approximations:

$$\frac{22}{7} \quad \text{and} \quad \frac{355}{113}.$$

The approximation  $\frac{355}{113}$  is especially accurate and is traditionally known as the Milü, or “accurate ratio.”

Therefore, the answer is Zu Chongzhi.

□

**Problem 2014.3.** Consider the  $2014 \times 2014$  matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

*Solution.* Use the all-ones matrix formula (Formula 3.65). Let  $\mathbf{A}$  be the given matrix, and let  $\mathbf{J}_{2014}$  be the  $2014 \times 2014$  matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J}_{2014} - \mathbf{I}_{2014}.$$

The matrix  $\mathbf{J}_{2014}$  has eigenvalue 2014 for the vector

$$\mathbf{1} = (1, 1, \dots, 1)^\top,$$

since

$$\mathbf{J}_{2014}\mathbf{1} = 2014\mathbf{1}.$$

Therefore,

$$\mathbf{A}\mathbf{1} = (\mathbf{J}_{2014} - \mathbf{I}_{2014})\mathbf{1} = 2014\mathbf{1} - \mathbf{1} = 2013\mathbf{1}.$$

Hence 2013 is an eigenvalue of  $\mathbf{A}$ .

Now take any vector  $\mathbf{v} = (v_1, \dots, v_{2014})^\top$  s.t.

$$v_1 + \dots + v_{2014} = 0.$$

Then  $\mathbf{J}_{2014}\mathbf{v} = \mathbf{0}$ , so

$$\mathbf{A}\mathbf{v} = (\mathbf{J}_{2014} - \mathbf{I}_{2014})\mathbf{v} = -\mathbf{v}.$$

Thus  $-1$  is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2013, so the multiplicity of  $-1$  is 2013.

Therefore, the eigenvalues are

$$\boxed{2013 \text{ with multiplicity } 1, \quad -1 \text{ with multiplicity } 2013}.$$

□

**Problem 2014.4.** Let  $m \geq 3$  and  $n \geq 3$ . Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq \mathbb{R}^m, \quad \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} \subseteq \mathbb{R}^n.$$

Assume that

$$V = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$$

is a 2-dimensional subspace of  $\mathbb{R}^m$ , and that

$$W = \text{span}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$$

is a 3-dimensional subspace of  $\mathbb{R}^n$ . Determine  $\text{rk}(\mathbf{A})$  for the  $m \times n$  matrix

$$\mathbf{A} = \mathbf{v}_1\mathbf{w}_1^\top + \mathbf{v}_2\mathbf{w}_2^\top + \mathbf{v}_3\mathbf{w}_3^\top.$$

*Solution.* Let

$$\mathbf{P} = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3), \quad \mathbf{Q} = (\mathbf{w}_1 \quad \mathbf{w}_2 \quad \mathbf{w}_3).$$

Then

$$\mathbf{A} = \mathbf{P}\mathbf{Q}^\top.$$

Since  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$  span a 2-dimensional subspace, we have

$$\text{rk}(\mathbf{P}) = 2.$$

Since  $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$  span a 3-dimensional subspace, they are linearly independent, and hence

$$\text{rk}(\mathbf{Q}) = 3.$$

Transposition preserves rank, so  $\mathbf{Q}^\top$  has rank 3. Its output space is  $\mathbb{R}^3$ , and thus it can produce every vector in  $\mathbb{R}^3$ : the map is surjective.

It follows that the image of  $\mathbf{A} = \mathbf{P}\mathbf{Q}^\top$  is

$$\text{im}(\mathbf{A}) = \mathbf{P}(\text{im}(\mathbf{Q}^\top)) = \mathbf{P}(\mathbb{R}^3) = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3).$$

This space has dimension 2. Hence

$$\text{rk}(\mathbf{A}) = 2.$$

Therefore, the answer is 2.

□

**Problem 2015.1.** Read the following passage and fill in the blank with the title of the mathematical classic, either in English or in Korean.

For 2000 years, \_\_\_\_\_ was not only the core of mathematical education but also at the heart of Western culture. The most glowing tributes to \_\_\_\_\_ do not, in fact, come from mathematicians, but from philosophers, politicians, and others.

Here are some examples:

- (1) He studied and nearly mastered the six books of \_\_\_\_\_ since he was a member of Congress. He regrets his want of education, and does what he can to supply the want.

Abraham Lincoln, *Short Autobiography*

- (2) He studied \_\_\_\_\_ until he could demonstrate with ease all the propositions in the six books.

Herndon's *Life of Lincoln*

- (3) At the age of eleven, I began \_\_\_\_\_. This was one of the great events of my life, as dazzling as first love. I had not imagined there was anything so delicious in the world.

Bertrand Russell, *Autobiography*, Vol. 1

*Solution.* The quotations refer to Euclid's *Elements*, whose first six books treat plane geometry. Its organization around definitions, postulates, and deductive reasoning made it a model for mathematical exposition for centuries. Therefore, the answer is Euclid's *Elements*.

□

**Problem 2015.2.** Let  $\mathbf{A}$  be an invertible square matrix. Suppose that, for a suitable permutation matrix  $\mathbf{P}$ , Gaussian elimination gives an LU-decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

It may also happen that another permutation matrix  $\tilde{\mathbf{P}}$  gives another LU-decomposition

$$\tilde{\mathbf{P}}\mathbf{A} = \tilde{\mathbf{L}}\tilde{\mathbf{U}}.$$

Show that the sets of absolute values of the pivots can be different. In other words, show by example that

$$\{|u_{kk}| \mid 1 \leq k \leq n\} \neq \{|\tilde{u}_{kk}| \mid 1 \leq k \leq n\}.$$

*Hint.* Look for a counterexample in the  $2 \times 2$  case.

*Solution.* Consider the invertible matrix

$$\mathbf{A} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{3} & 1 \end{pmatrix}.$$

With no row exchange, we have the LU-decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{5}{6} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{1, \frac{5}{6}\right\}.$$

Now let  $\tilde{\mathbf{P}}$  be the permutation matrix that swaps the two rows. Then

$$\tilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} \frac{1}{3} & 1 \\ 1 & \frac{1}{2} \end{pmatrix}.$$

This matrix has the LU-decomposition

$$\tilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 1 \\ 0 & -\frac{5}{2} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

Therefore,

$$\left\{1, \frac{5}{6}\right\} \neq \left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

This shows that the set of absolute values of the pivots can depend on the chosen row permutation.

□

**Problem 2015.3.** In  $\mathbb{R}^2$ , the so-called Mercedes-Benz system is defined by

$$\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2\}, \quad \mathbf{v}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \end{pmatrix} \quad (k = 0, 1, 2).$$

Find vectors  $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^3$  s.t. they have the same length, are mutually orthogonal, have positive  $z$ -coordinates, and satisfy

$$\mathbf{v}_k = P_{xy}(\mathbf{w}_k) \quad (k = 0, 1, 2),$$

where  $P_{xy}$  denotes the projection onto the  $xy$ -plane. Here  $\mathbb{R}^2$  is regarded as the subspace of  $\mathbb{R}^3$  with  $z$ -coordinate equal to 0.

*Solution.* Each projection  $\mathbf{v}_k$  has length 1. If the positive height of  $\mathbf{w}_k$  is  $c_k$ , then  $\|\mathbf{w}_k\|^2 = 1 + c_k^2$ . Equal lengths and positive heights therefore force  $c_0 = c_1 = c_2 = c > 0$ . Thus each  $\mathbf{w}_k$  has the form

$$\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ c \end{pmatrix}$$

for some positive constant  $c$ .

For  $i \neq j$ , the angle between  $\mathbf{v}_i$  and  $\mathbf{v}_j$  is  $120^\circ$ , so

$$\mathbf{v}_i \cdot \mathbf{v}_j = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}.$$

Therefore,

$$\mathbf{w}_i \cdot \mathbf{w}_j = \mathbf{v}_i \cdot \mathbf{v}_j + c^2 = -\frac{1}{2} + c^2.$$

To make the vectors mutually orthogonal, we need

$$-\frac{1}{2} + c^2 = 0,$$

so

$$c = \frac{1}{\sqrt{2}}.$$

Hence one possible choice is

$$\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ \frac{1}{\sqrt{2}} \end{pmatrix} \quad (k = 0, 1, 2).$$

These vectors all have squared length

$$1 + \frac{1}{2} = \frac{3}{2},$$

have positive  $z$ -coordinates, project to the given vectors  $\mathbf{v}_k$ , and are mutually orthogonal.  $\square$

**Problem 2015.4.** This French mathematician was born in 1973 and received the Fields Medal in 2010 at the age of 36. On his website, he described himself as “Mathématicien, Professeur de l’Université de Lyon, Directeur de l’Institut Henri Poincaré.” He is also famous for his stylish clothing.

Together with Clément Mouhot, he completed a theorem that played a central role in his Fields Medal-winning work. He later wrote a book about the process of developing this theorem, and the title of the book is also one of his nicknames.

Write the name of this mathematician, using the correct French spelling, and the title of the book.

*Solution.* The mathematician described in the problem is Cédric Villani.

Villani received the Fields Medal in 2010 for his work on nonlinear Landau damping and the Boltzmann equation. He is also widely recognized for his distinctive style, often including a cravat and a spider brooch.

The book mentioned in the problem is *Théorème vivant*, which may be translated as *Living Theorem*. It describes the process through which Villani and Clément Mouhot developed the theorem that contributed to Villani's Fields Medal-winning work.

Therefore, the answer is Cédric Villani, *Théorème vivant*.

□

**Problem 2015.5.** On the front side of the Fields Medal, there is a profile of Archimedes together with the phrase by Marcus Manilius:

“TRANSIRE SUUM PECTUS MUNDOQUE POTIRI.”

On the back side, there is another inscription together with the two solid figures that were engraved on Archimedes' tomb.

Identify the two solid figures and determine their volume ratio.

*Solution.* The two solid figures are a sphere and its circumscribed cylinder.

Suppose the sphere has radius  $R$ . Then the volume of the sphere is

$$V_S = \frac{4}{3}\pi R^3.$$

The circumscribed cylinder has radius  $R$  and height  $2R$ , so its volume is

$$V_C = \pi R^2 \cdot 2R = 2\pi R^3.$$

Therefore,

$$V_S : V_C = \frac{4}{3}\pi R^3 : 2\pi R^3 = 2 : 3.$$

Thus the two figures are a sphere and its circumscribed cylinder, and the volume ratio is

2 : 3.

□

**Problem 2015.6.** Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a periodic function with period  $2\pi$ . For  $x \in (0, 2\pi]$ , define

$$f(x) = \lceil \max(3, |\pi - x|) - 3 \rceil,$$

and extend  $f$  periodically, so that

$$f(x + 2\pi) = f(x)$$

for all  $x \in \mathbb{R}$ . Determine whether the following limit converges:

$$\lim_{n \rightarrow \infty} f(n).$$

*Hint.* Use  $3.14 < \pi < 3.145$ .

*Solution 1.* For  $x \in (0, 2\pi]$ , the function  $f$  takes only the values 0 and 1. More precisely,

$$f(x) = 1$$

when

$$0 < x < \pi - 3 \quad \text{or} \quad \pi + 3 < x \leq 2\pi,$$

and

$$f(x) = 0$$

when

$$\pi - 3 \leq x \leq \pi + 3.$$

Thus  $f$  is a  $2\pi$ -periodic function which is 1 near multiples of  $2\pi$ , and 0 on the large middle interval.

By the density of irrational rotations (Theorem 2.141), since  $1/(2\pi)$  is irrational, the fractional parts

$$\left\{ \frac{n}{2\pi} \right\}, \quad n = 1, 2, 3, \dots,$$

are dense in  $[0, 1]$ . Equivalently, the representatives of the integers modulo  $2\pi$  are dense in  $(0, 2\pi]$ . Hence there are infinitely many positive integers  $n$  whose representatives modulo  $2\pi$  lie in

$$(0, \pi - 3) \cup (\pi + 3, 2\pi],$$

and for these integers we have

$$f(n) = 1.$$

There are also infinitely many positive integers  $n$  whose residues modulo  $2\pi$  lie in

$$(\pi - 3, \pi + 3),$$

and for these integers we have

$$f(n) = 0.$$

The sequence  $f(n)$  therefore has a subsequence equal to 1 and another subsequence equal to 0, so it does not converge as  $n \rightarrow \infty$ .

Therefore, the answer is the limit does not exist.

□

*Solution 2.* We follow the hint and use an elementary step-size argument. Put

$$\alpha = \pi - 3, \quad \delta = 19 - 6\pi.$$

From  $3.14 < \pi < 3.145$ ,

$$0.14 < \alpha < 0.145, \quad 0.13 < \delta < 0.16,$$

and in particular

$$0 < \delta < 2\alpha.$$

On one period,  $f = 1$  on

$$(0, \alpha) \cup (2\pi - \alpha, 2\pi]$$

and  $f = 0$  on the complementary middle interval

$$[\alpha, 2\pi - \alpha].$$

Fix any  $N \in \mathbb{N}$  and consider the arithmetic progression

$$n_k = N + 19k, \quad k = 0, 1, 2, \dots$$

Since

$$19 = 3(2\pi) + \delta,$$

passing from  $n_k$  to  $n_{k+1}$  advances its representative modulo  $2\pi$  by exactly  $\delta$ .

If  $r_0$  is one representative of  $N$  modulo  $2\pi$ , the unwrapped quantities  $r_0 + k\delta$  cross multiples of  $2\pi$  infinitely often. During any such passage across the endpoint  $2\pi \equiv 0$ , let  $r_-$  be the last representative before the passage and  $r_+$  the first one after it. Then

$$(2\pi - r_-) + r_+ = \delta < 2\alpha.$$

Hence at least one of  $2\pi - r_-$  and  $r_+$  is smaller than  $\alpha$ , so at least one of the corresponding terms satisfies  $f(n_k) = 1$ .

Starting from the first representative after such a passage, if it is already at least  $\alpha$ , then it lies in the interval where  $f = 0$ . Otherwise, add steps of length  $\delta$  until the representative first reaches  $\alpha$ . That first representative is smaller than

$$\alpha + \delta < 0.305 < 2\pi - \alpha,$$

so it also lies in the interval where  $f = 0$ .

Therefore, beyond every  $N$  there are integers  $n_1, n_2 > N$  with

$$f(n_1) = 0, \quad f(n_2) = 1.$$

Thus  $f(n)$  cannot converge.

Therefore, the answer is the limit does not exist.

□

**Problem 2015.7.** This person is believed to have lived approximately from 624 BCE to 546 BCE. He is often regarded as one of the earliest figures worthy of being called a mathematician. He introduced the method of proof to geometric facts that had been known empirically through land surveying, helping to lay the foundations of geometry that were later systematized in Euclid's *Elements*.

The theorems attributed to him include:

- (1) Vertical angles are equal.
- (2) The base angles of an isosceles triangle are equal.
- (3) Triangle congruence criteria such as side-angle-side and angle-side-angle.
- (4) An angle inscribed in a semicircle is a right angle.

Identify this person.

*Solution.* The person described in the problem is Thales of Miletus.

Thales is traditionally regarded as one of the earliest Greek mathematicians. He is especially important because he is associated not merely with observing geometric facts, but with proving them. Several elementary geometric results are attributed to him, including the theorem that an angle inscribed in a semicircle is a right angle.

Therefore, the answer is Thales of Miletus.

□

**Problem 2015.8.** Determine which of the following mathematicians was born earliest.

- (1) Henri Poincaré
- (2) Karl Weierstrass
- (3) Julius Wilhelm Richard Dedekind
- (4) Andrei Andreyevich Markov
- (5) Georg Cantor

*Solution.* We compare their years of birth:

|                                 |       |
|---------------------------------|-------|
| Karl Weierstrass                | 1815, |
| Julius Wilhelm Richard Dedekind | 1831, |
| Georg Cantor                    | 1845, |
| Henri Poincaré                  | 1854, |
| Andrei Andreyevich Markov       | 1856. |

Among these mathematicians, the one born earliest is Karl Weierstrass.

Therefore, the answer is Karl Weierstrass.

□

**Problem 2015.9.** The first International Congress of Mathematicians at which the Fields Medal was awarded was held in 1936. Determine the country in which it was held.

*Solution.* The Fields Medal was first awarded at the 1936 International Congress of Mathematicians. The 1936 ICM was held in Oslo, Norway.

Therefore, the answer is Norway.

□

**Problem 2015.10.** Find the sum of the distinct prime numbers appearing in the prime factorization of  $15!$ .

*Solution.* The prime numbers appearing in the prime factorization of  $15!$  are exactly the primes less than or equal to 15:

$$2, \quad 3, \quad 5, \quad 7, \quad 11, \quad 13.$$

Therefore, their sum is

$$2 + 3 + 5 + 7 + 11 + 13 = 41.$$

Therefore, the answer is 41.

□

**Problem 2015.11.** In three-dimensional space, suppose that the lengths of the orthogonal projections of a line segment  $\ell$  onto the  $yz$ -plane, the  $zx$ -plane, and the  $xy$ -plane are 5, 5, and 6, respectively. Determine the length of  $\ell$ .

*Solution.* Let the displacement vector of the line segment be

$$\mathbf{v} = (a, b, c)^T.$$

Then the length of  $\ell$  is

$$\|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2}.$$

The projection of  $\mathbf{v}$  onto the  $yz$ -plane has length

$$\sqrt{b^2 + c^2} = 5.$$

Similarly, the projections onto the  $zx$ -plane and the  $xy$ -plane give

$$\sqrt{c^2 + a^2} = 5, \quad \sqrt{a^2 + b^2} = 6.$$

Squaring and adding these three equations, we obtain

$$(b^2 + c^2) + (c^2 + a^2) + (a^2 + b^2) = 5^2 + 5^2 + 6^2.$$

Hence

$$2(a^2 + b^2 + c^2) = 25 + 25 + 36 = 86,$$

so

$$a^2 + b^2 + c^2 = 43.$$

Therefore, the length of  $\ell$  is

$$\|\mathbf{v}\| = \sqrt{43}.$$

Therefore, the answer is  $\boxed{\sqrt{43}}$ .

□

**Problem 2015.12.** In three-dimensional space, consider the matrix obtained by rotating space by  $30^\circ$  around the line passing through the origin and the point  $(1, 1, 2)$ . Determine the number of real eigenvalues of this matrix.

*Solution.* As in Formula 3.61, a rotation in  $\mathbb{R}^3$  around an axis fixes every vector lying on the rotation axis. Therefore, the direction vector

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

is an eigenvector with eigenvalue 1.

On the plane perpendicular to the rotation axis, the matrix acts as a  $30^\circ$  rotation. A nontrivial rotation in a two-dimensional plane has no real eigenvalues unless the rotation angle is  $0^\circ$  or  $180^\circ$ . Since the angle here is  $30^\circ$ , the two remaining eigenvalues are non-real complex conjugates.

Hence the only real eigenvalue is 1. Therefore, the number of real eigenvalues is  $\boxed{1}$ .

□

**Problem 2015.13.** A right circular cone has base radius 3 and height 6. Determine the maximum possible volume of a cylinder inscribed in this cone.

*Solution.* Let  $x$  be the radius of the inscribed cylinder, and let  $h$  be its height. By similarity of triangles in the cone, the radius of the horizontal cross-section decreases linearly from 3 to 0 as the height increases from 0 to 6. Hence

$$x = 3 - \frac{h}{2}.$$

Equivalently,

$$h = 6 - 2x.$$

Therefore, the volume of the cylinder is

$$V(x) = \pi x^2 h = \pi x^2 (6 - 2x) = 6\pi x^2 - 2\pi x^3,$$

where  $0 \leq x \leq 3$ .

Differentiating, we obtain

$$V'(x) = 12\pi x - 6\pi x^2 = 6\pi x(2 - x).$$

Thus the only critical point in the open interval  $(0, 3)$  is

$$x = 2.$$

The endpoint volumes are  $V(0) = V(3) = 0$ , so the positive volume at  $x = 2$  is the global maximum. At this point,

$$h = 6 - 2 \cdot 2 = 2,$$

so the maximum volume is

$$V(2) = \pi \cdot 2^2 \cdot 2 = 8\pi.$$

Therefore, the maximum possible volume is  $\boxed{8\pi}$ .

□

**Problem 2015.14.** Suppose that a polynomial equation of degree 7 with real coefficients has three non-real complex roots whose absolute values are all different. Determine the number of real roots of the equation.

*Solution.* By the conjugate-root theorem (Theorem 9.53), every non-real root of a real-coefficient polynomial appears together with its complex conjugate, with the same multiplicity.

Suppose the three given non-real roots are

$$\alpha_1, \quad \alpha_2, \quad \alpha_3,$$

and their absolute values are all different. Then their complex conjugates

$$\overline{\alpha_1}, \quad \overline{\alpha_2}, \quad \overline{\alpha_3}$$

are also roots. Since

$$|\alpha_i| = |\overline{\alpha_i}|,$$

and the three given roots have distinct absolute values, these conjugate roots give three additional non-real roots.

Thus the polynomial has at least 6 non-real roots. Since the degree is 7, there is exactly one remaining root, counted with multiplicity. It must be real: an additional non-real root, or a repeated one among the six, would require its conjugate with the same multiplicity and exceed degree 7.

Therefore, the equation has  $\boxed{1}$  real root.

□

**Problem 2015.15.** This mathematician was born in 1882 and died in 1935. When she was being considered for a professorship at the University of Göttingen, there was strong opposition. In response, Hilbert famously argued, “I do not see that the sex of the candidate is an argument against her admission as a Privatdozent. After all, we are a university, not a bathhouse.”

She made remarkable contributions to several areas of mathematics, especially modern algebra, and much of twentieth-century abstract algebra was deeply influenced by her work. Identify this mathematician.

*Solution.* The dates 1882–1935, work on rings and ideals, and Hilbert’s defense at Göttingen identify Emmy Noether. The mathematical history section records the same anecdote. Therefore, the answer is  $\boxed{\text{Emmy Noether}}$ .

□

**Problem 2015.16.** Determine the number of distinct terms in the expansion of

$$(x_1 + x_2 + \cdots + x_7)^4.$$

*Solution.* Each term in the expansion has the form

$$x_1^{k_1} x_2^{k_2} \cdots x_7^{k_7},$$

where

$$k_1 + k_2 + \cdots + k_7 = 4$$

and each  $k_i$  is a nonnegative integer.

Therefore, the number of distinct terms is the number of nonnegative integer solutions to

$$k_1 + k_2 + \cdots + k_7 = 4.$$

By stars and bars (Formula 5.6), this number is

$$\binom{4+7-1}{7-1} = \binom{10}{6} = 210.$$

Therefore, the answer is  $\boxed{210}$ .

□

**Problem 2015.17.** Let

$$\mathbf{v} = \mathbf{w} = (1 \ 2 \ 3 \ 4)$$

be row vectors. Consider the  $4 \times 4$  matrix obtained as the product

$$\mathbf{v}^T \mathbf{w}.$$

Find the sum and the product of the eigenvalues of this matrix.

*Solution.* The matrix is

$$\mathbf{v}^T \mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} (1 \ 2 \ 3 \ 4).$$

By Proposition 3.62, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to its trace. Therefore,

$$\sum_{i=1}^4 \lambda_i = \text{tr}(\mathbf{v}^T \mathbf{w}) = 1^2 + 2^2 + 3^2 + 4^2 = 30.$$

The product of the eigenvalues is equal to the determinant. Since  $\mathbf{v}^T \mathbf{w}$  is an outer product of two vectors, we have  $\text{rk}(\mathbf{v}^T \mathbf{w}) = 1$ . In particular, as a  $4 \times 4$  matrix, it is not invertible, so

$$\det(\mathbf{v}^T \mathbf{w}) = 0.$$

Hence the product of the eigenvalues is

$$\prod_{i=1}^4 \lambda_i = 0.$$

Therefore, the sum and product of the eigenvalues are  $\boxed{30 \text{ and } 0}$ , respectively.

□

**Problem 2015.18.** Evaluate the infinite series

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}.$$

*Hint.* Use

$$\frac{e^x - 1}{x} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!},$$

then differentiate and substitute  $x = 1$ .

*Solution 1.* We use the telescoping identity

$$\frac{n}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!}.$$

Therefore,

$$\sum_{n=1}^N \frac{n}{(n+1)!} = \sum_{n=1}^N \left( \frac{1}{n!} - \frac{1}{(n+1)!} \right) = 1 - \frac{1}{(N+1)!}.$$

Taking  $N \rightarrow \infty$ , we obtain

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1.$$

Therefore, the answer is  $\boxed{1}$ .

□

*Solution 2.* We use the given power series

$$g(x) = \frac{e^x - 1}{x} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!}.$$

Because this power series has infinite radius of convergence, the term-by-term differentiation theorem from Mathematical Concepts gives

$$g'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{(n+1)!}.$$

Differentiating the closed form yields

$$g'(x) = \frac{xe^x - (e^x - 1)}{x^2} = \frac{(x-1)e^x + 1}{x^2}.$$

Substituting  $x = 1$  gives

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = g'(1) = 1.$$

Therefore, the answer is  $\boxed{1}$ .

□

**Problem 2017.1.** Determine all possible positive integers that can occur as the number of faces of a regular polyhedron, that is, a Platonic solid.

*Solution.* There are exactly five Platonic solids:

| Solid        | Number of faces |
|--------------|-----------------|
| Tetrahedron  | 4               |
| Cube         | 6               |
| Octahedron   | 8               |
| Dodecahedron | 12              |
| Icosahedron  | 20              |

Therefore, the possible numbers of faces of a regular polyhedron are

$$\boxed{4, 6, 8, 12, 20}.$$

□

**Problem 2017.2.** Let  $(F_n)$  be the Fibonacci sequence defined by

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+1} = F_n + F_{n-1} \quad (n \geq 1).$$

Determine

$$\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n}.$$

*Solution.* Applying the constant-coefficient recurrence formula (Formula 2.5) to the roots of  $t^2 - t - 1 = 0$  gives Binet's formula:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Since  $|\psi/\varphi| < 1$ ,

$$\frac{F_{n+1}}{F_n} = \varphi \frac{1 - (\psi/\varphi)^{n+1}}{1 - (\psi/\varphi)^n} \rightarrow \varphi.$$

Therefore,

$$\boxed{\frac{1 + \sqrt{5}}{2}}.$$

□

**Problem 2017.3.** For a positive integer  $n$ , define the  $n \times n$  matrix  $\mathbf{C}$  by

$$c_{ij} = \binom{i+j-2}{i-1} \quad (1 \leq i, j \leq n).$$

Now define another  $n \times n$  matrix  $\mathbf{B}$  by

$$b_{ij} = \begin{cases} c_{ij}, & (i, j) \neq (n, n), \\ c_{ij} + 2016, & (i, j) = (n, n). \end{cases}$$

Determine  $\det(\mathbf{B})$ .

*Solution.* For  $n = 1$ ,  $\mathbf{B} = (2017)$  and the answer is immediate. Assume  $n \geq 2$ . We first compute the determinant of  $\mathbf{C}$ . By Vandermonde's identity (Formula 5.4),

$$\binom{i+j-2}{i-1} = \sum_{k=1}^{\min(i,j)} \binom{i-1}{k-1} \binom{j-1}{k-1}.$$

Hence

$$\mathbf{C} = \mathbf{L}\mathbf{L}^\top,$$

where

$$l_{ik} = \begin{cases} \binom{i-1}{k-1}, & k \leq i, \\ 0, & k > i. \end{cases}$$

The matrix  $\mathbf{L}$  is lower triangular and all of its diagonal entries are 1. Therefore,

$$\det(\mathbf{L}) = 1,$$

and so

$$\det(\mathbf{C}) = \det(\mathbf{L}\mathbf{L}^\top) = (\det(\mathbf{L}))^2 = 1.$$

The matrix  $\mathbf{B}$  differs from  $\mathbf{C}$  only in the  $(n, n)$ -entry. By cofactor expansion, the determinant is affine in a single entry when all other entries are fixed, we have

$$\det(\mathbf{B}) = \det(\mathbf{C}) + 2016 \cdot M_{nn},$$

where  $M_{nn}$  is the  $(n, n)$ -minor of  $\mathbf{C}$ .

This minor is the determinant of the upper-left  $(n-1) \times (n-1)$  submatrix of  $\mathbf{C}$ , which has the same form as  $\mathbf{C}$  with size  $n-1$ . Therefore,

$$M_{nn} = 1.$$

Hence

$$\det(\mathbf{B}) = 1 + 2016 \cdot 1 = 2017.$$

Therefore, the answer is 2017.

□

**Problem 2017.4.** In 2017, this prize was awarded to the French mathematician Yves Meyer. The Norwegian Academy of Science and Letters announced on March 21 that Meyer would receive the prize, worth six million Norwegian kroner, in recognition of his decisive contributions to the development of wavelet theory, which is used in areas ranging from gravitational wave detection to movie file compression.

Identify this prize, which is often called the Nobel Prize of mathematics.

*Solution.* The prize described in the problem is the Abel Prize.

The Abel Prize is awarded by the Norwegian Academy of Science and Letters and is often described as one of the most prestigious prizes in mathematics. In 2017, it was awarded to Yves Meyer for his fundamental contributions to the mathematical theory of wavelets.

Therefore, the answer is Abel Prize.

□

**Problem 2017.5.** Express the integral

$$\int_0^1 x^m (1-x)^n dx$$

in terms of a binomial coefficient closely related to the positive integers  $m$  and  $n$ .

*Solution.* This is a beta integral (Formula 2.53). For positive integers  $m$  and  $n$ , we have

$$\int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!}.$$

This is the factorial form of the answer.

To express it using a binomial coefficient, recall that

$$\binom{m+n}{m} = \frac{(m+n)!}{m!n!}.$$

Hence

$$\frac{m!n!}{(m+n+1)!} = \frac{1}{m+n+1} \cdot \frac{m!n!}{(m+n)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore,

$$\int_0^1 x^m (1-x)^n dx = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore, the answer is  $\boxed{\frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}}.$

□

**Problem 2018.1.** The following people are this year's Fields Medalists. Excluding choice (5), identify the second-oldest medalist among them.

- (1) Caucher Birkar
- (2) Alessio Figalli
- (3) Peter Scholze
- (4) Akshay Venkatesh
- (5) The person who stole Birkar's medal

*Solution.* The 2018 Fields Medalists were

- |                      |               |
|----------------------|---------------|
| (1) Caucher Birkar   | born in 1978, |
| (2) Alessio Figalli  | born in 1984, |
| (3) Peter Scholze    | born in 1987, |
| (4) Akshay Venkatesh | born in 1981. |

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

$$\boxed{(4)}.$$

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal.

□

**Problem 2018.2.** According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of  $n$  possible geometric structures. Determine  $n$ .

*Solution.* By the Thurston Geometrization Theorem and the accompanying list in Mathematical Concepts, the 3-dimensional geometric structures are the eight Thurston geometries:

$$S^3, \quad \mathbb{R}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}}_2(\mathbb{R}), \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence  $n = 8$ , so the answer is

$$\boxed{8}.$$

□

**Problem 2018.3.** Find the sum of the following infinite series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots.$$

*Solution.* The given series is one half of the alternating harmonic series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2} \left( 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right).$$

Recall the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

At  $x = 1$ , the alternating series converges. Abel's theorem identifies its sum with the limit of  $\ln(1+x)$  as  $x \rightarrow 1^-$ , so

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

Therefore,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{\ln 2}{2}.$$

Therefore, the answer is  $\boxed{\frac{\ln 2}{2}}$ .

□

**Problem 2018.4.** Which of the following five Fields Medalists does not have a Korean doctoral student?

- (1) William Thurston
- (2) Maxim Kontsevich
- (3) Grigory Margulis
- (4) Terence Tao
- (5) Efim Zelmanov

*Solution.* In 2018, the intended answer was Maxim Kontsevich. Therefore, the answer is

$$\boxed{(2) \text{ Maxim Kontsevich}}.$$

□

**Problem 2019.1.** Consider the  $2019 \times 2019$  matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

*Solution.* Use the all-ones matrix formula (Formula 3.65). Let  $\mathbf{A}$  be the given matrix, and let  $\mathbf{J}_{2019}$  be the  $2019 \times 2019$  matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J}_{2019} - \mathbf{I}_{2019}.$$

First, consider the vector

$$\mathbf{1} = (1, 1, \dots, 1)^\top.$$

Since

$$\mathbf{J}_{2019}\mathbf{1} = 2019\mathbf{1},$$

we obtain

$$\mathbf{A}\mathbf{1} = (\mathbf{J}_{2019} - \mathbf{I}_{2019})\mathbf{1} = 2019\mathbf{1} - \mathbf{1} = 2018\mathbf{1}.$$

Hence 2018 is an eigenvalue.

Now take any vector  $\mathbf{v} = (v_1, \dots, v_{2019})^\top$  satisfying

$$v_1 + \dots + v_{2019} = 0.$$

Then

$$\mathbf{J}_{2019}\mathbf{v} = \mathbf{0},$$

and therefore

$$\mathbf{A}\mathbf{v} = (\mathbf{J}_{2019} - \mathbf{I}_{2019})\mathbf{v} = -\mathbf{v}.$$

Thus  $-1$  is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2018, so the multiplicity of  $-1$  is 2018.

Therefore, the eigenvalues are

|                           |                              |
|---------------------------|------------------------------|
| 2018 with multiplicity 1, | $-1$ with multiplicity 2018. |
|---------------------------|------------------------------|

□

**Problem 2019.2.** Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2} dx.$$

*Solution.* Let

$$u = 2x.$$

Then

$$dx = \frac{1}{2} du.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2} dx = \int_{-\infty}^{\infty} \frac{8}{1+u^2} \cdot \frac{1}{2} du = 4 \int_{-\infty}^{\infty} \frac{1}{1+u^2} du.$$

Since

$$\int_{-\infty}^{\infty} \frac{1}{1+u^2} du = \pi,$$

we obtain

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2} dx = 4\pi.$$

Therefore, the answer is  $\boxed{4\pi}$ .

□

**Problem 2019.3.** Compute the value of  $\pi + e$  up to five decimal places.

*Solution.* Using the approximations

$$\pi \approx 3.14159265, \quad e \approx 2.71828183,$$

we obtain

$$\pi + e \approx 3.14159265 + 2.71828183 = 5.85987448.$$

Therefore, up to five decimal places,

$$\pi + e \approx 5.85987.$$

Therefore, the answer is  $\boxed{5.85987}$ .

□

**Problem 2020.1.** The International Congress of Mathematicians, abbreviated as ICM, is a congress for mathematicians from around the world. It is organized every four years by the International Mathematical Union and may be thought of as a kind of Olympic Games for mathematicians. The 2014 ICM was held in the Republic of Korea, and the 2018 ICM was held in Brazil. As of 2019, determine the country in which the 2022 ICM was scheduled to be held.

*Solution.* ICM stands for the International Congress of Mathematicians. It is the main international congress in mathematics and is organized every four years by the International Mathematical Union.

The 2014 ICM was held in Seoul, Republic of Korea, and the 2018 ICM was held in Rio de Janeiro, Brazil. In 2019, the 2022 ICM was scheduled to be held in Saint Petersburg, Russia. Therefore, the answer is Russia.

The congress was later changed to a virtual event, with the 2022 awards ceremony held in Helsinki, Finland. Thus the answer records the host country that was scheduled at the time. □

**Problem 2020.2.** A fair six-sided die has the numbers  $-2, 0, 2, 3, 4, 5$  written on its faces. The die is rolled 10 times, and  $X$  is defined to be the product of the 10 numbers obtained. Determine  $\mathbb{E}[X]$ .

*Solution.* Let  $Y$  be the number obtained from one roll of the die. Since all six faces are equally likely,

$$\mathbb{E}[Y] = \frac{-2 + 0 + 2 + 3 + 4 + 5}{6} = 2.$$

Now let  $Y_1, Y_2, \dots, Y_{10}$  be the numbers obtained from the 10 rolls. Then

$$X = Y_1 Y_2 \cdots Y_{10}.$$

Since the rolls are independent, the random variables  $Y_1, Y_2, \dots, Y_{10}$  are independent. Therefore,

$$\mathbb{E}[X] = \mathbb{E}[Y_1 Y_2 \cdots Y_{10}] = \mathbb{E}[Y_1] \mathbb{E}[Y_2] \cdots \mathbb{E}[Y_{10}].$$

All the  $Y_i$  have the same distribution as  $Y$ , so

$$\mathbb{E}[X] = (\mathbb{E}[Y])^{10} = 2^{10} = 1024.$$

Therefore, the answer is 1024. □

**Problem 2020.3.** Which of the following Fields Medalists is not a professor?

- (1) Cédric Villani
- (2) Elon Lindenstrauss
- (3) Peter Scholze
- (4) Martin Hairer
- (5) Terence Tao

*Solution.* The intended answer is (1) Cédric Villani. In 2020, Villani was serving as a member of the French National Assembly; his parliamentary term ran from 2017 to 2022. The question refers to his political role at the time. Its wording should not be read as saying that he had never held, or had permanently lost, a professorship. □

**Problem 2020.4.** In a three-dimensional gravitational field, when an object moves from a point  $\mathbf{a}$  to another point  $\mathbf{b}$ , the change in its potential energy is independent of the path taken from  $\mathbf{a}$  to  $\mathbf{b}$ . Determine which of the following mathematical facts is least directly related to this physical phenomenon.

- (1) On a simply connected domain, a continuously differentiable curl-free vector field is the gradient of a potential function.
- (2) Stokes' theorem
- (3) The Laplacian of  $\frac{1}{\|\mathbf{x}\|}$  is a delta distribution, up to a constant factor.
- (4) The curl of a gradient is identically zero.

*Solution.* Path independence means that the gravitational force field is conservative. If  $\mathbf{F} = -\nabla U$ , then the finite change

$$\Delta U := U(\mathbf{b}) - U(\mathbf{a}) = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{r}$$

depends only on the endpoints. Here  $\Delta U$  denotes a finite change; in the distributional identity below,  $\Delta$  acts on a function and denotes the Laplacian. The conservative-field criterion (Proposition 2.103) connects this statement with choice (1), while

$$\nabla \times \nabla U = \mathbf{0}$$

gives choice (4). Stokes' theorem relates zero curl to vanishing circulation around closed curves, so choice (2) is also directly tied to path independence.

The distributional identity

$$\Delta \left( \frac{1}{\|\mathbf{x}\|} \right) = -4\pi\delta_0$$

is fundamental in Newtonian gravitation and Poisson's equation, but it is not the direct reason that line integrals of a conservative field are path-independent. Therefore, the intended answer is

$$\boxed{(3)}.$$

□

**Problem 2020.5.** Determine which of the following mathematics-related academic institutions is a degree-granting school.

- (1) Institut des Hautes Études Scientifiques, IHES
- (2) École normale supérieure
- (3) Mathematical Sciences Research Institute, MSRI
- (4) Institute for Advanced Study, IAS
- (5) Collège de France

*Solution.* École normale supérieure offers degree programs and its own ENS diploma. IHES, MSRI, and IAS are research institutes rather than degree-granting schools; Collège de France offers research and public lectures without awarding degrees. Therefore, the answer is

$$\boxed{(2) \text{ École normale supérieure}}.$$

□

**Problem 2021.1.** Determine which of  $A$  and  $B$  is larger, where

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{1+4}}}, \quad B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1+3}}}.$$

*Solution.* First, we compute  $A$ :

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{5}}} = 1 + \frac{1}{1 + \frac{1}{\frac{4}{5}}} = 1 + \frac{1}{1 + \frac{5}{4}} = 1 + \frac{1}{\frac{9}{4}} = 1 + \frac{4}{9} = \frac{13}{9}.$$

Next, we compute  $B$ :

$$B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{4}}} = 1 + \frac{1}{1 + \frac{1}{\frac{5}{4}}} = 1 + \frac{1}{1 + \frac{4}{5}} = 1 + \frac{1}{\frac{9}{5}} = 1 + \frac{5}{9} = \frac{14}{9}.$$

Since

$$\frac{13}{9} < \frac{14}{9},$$

we have  $A < B$ . Therefore, the answer is  $\boxed{B}$ .

□

**Problem 2021.2.** Let

$$A(n) = \frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^{n+1}.$$

Determine the integer closest to  $A(7)$ .

*Solution.* By Binet's formula, obtained from Formula 2.5,

$$F_8 = \frac{\varphi^8 - \psi^8}{\sqrt{5}}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

Hence

$$A(7) = \frac{\varphi^8}{\sqrt{5}} = F_8 + \frac{\psi^8}{\sqrt{5}}.$$

Since  $|\psi| < 1$ ,

$$0 < \frac{|\psi|^8}{\sqrt{5}} < \frac{1}{2}.$$

Thus the nearest integer is  $F_8 = 21$ , and therefore

$$\boxed{21}.$$

□

**Problem 2021.3.** Choose all groups among the following that are not finitely generated:

- (1) The fundamental group of an open unit disk with finitely many punctures.
- (2)  $\mathrm{SL}_2(\mathbb{Z})$ , the group of  $2 \times 2$  integer matrices with determinant 1.
- (3)  $\mathbb{C}^\times$ , the group of nonzero complex numbers under multiplication.
- (4) The Monster group, the largest sporadic finite simple group.

*Solution.* By Definition 9.3, it is enough to exhibit a finite generating set or to prove that none can exist. First, consider an open unit disk with finitely many punctures. If the removed points are  $p_1, \dots, p_n$ , then the fundamental group is isomorphic to the free group on  $n$  generators:

$$\pi_1(D \setminus \{p_1, \dots, p_n\}, x_0) \cong F_n.$$

Here  $x_0$  is a base point in the punctured disk, and a loop around each puncture gives one generator. Hence the group in choice (1) is finitely generated.

The group  $\mathrm{SL}_2(\mathbb{Z})$  is also finitely generated. For example, it is generated by the two matrices

$$\mathbf{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus choice (2) is finitely generated.

On the other hand,  $\mathbb{C}^\times$  is an uncountable abelian group. By Proposition 9.4, every finitely generated group is at most countable. Hence  $\mathbb{C}^\times$  cannot be finitely generated: finite words in a finite list of generators and their inverses form an at most countable set. Therefore, choice (3) is not finitely generated.

Finally, the Monster group is a finite group. Every finite group is finitely generated, since one may simply take all of its elements as generators. Hence choice (4) is finitely generated.

Therefore, the only group in the list that is not finitely generated is (3)  $\mathbb{C}^\times$ .

□

**Problem 2021.4.** Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find all eigenvalues of  $\mathbf{A}$  together with their multiplicities. Also determine whether  $\mathbf{A}$  is diagonalizable.

*Solution.* We use rank–nullity (Theorem 3.13) and the trace identity (Proposition 3.62). First, write  $\mathbf{A}$  as

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} (1 \ 2 \ 3).$$

Thus  $\mathrm{rk}(\mathbf{A}) = 1$ .

Let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 6\mathbf{v}.$$

Hence 6 is an eigenvalue of  $\mathbf{A}$ .

Since  $\mathrm{rk}(\mathbf{A}) = 1$ , its nullity is 2. Therefore, the eigenspace corresponding to the eigenvalue 0 has dimension 2. Indeed,

$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ x + 2y + 3z \\ x + 2y + 3z \end{pmatrix},$$

so

$$E_0(\mathbf{A}) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid x + 2y + 3z = 0 \right\}.$$

Thus the geometric multiplicity of 0 is 2.

Now

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 = 6.$$

Since the sum of the eigenvalues, counted with algebraic multiplicity, is equal to  $\operatorname{tr}(\mathbf{A})$ , the eigenvalues are

$$6, \quad 0, \quad 0.$$

Hence the algebraic multiplicities are

$$6 : 1, \quad 0 : 2.$$

Finally, the eigenspace for 6 has dimension 1, and the eigenspace for 0 has dimension 2. The total dimension of the eigenspaces is therefore

$$1 + 2 = 3.$$

Hence there exists a basis of  $\mathbb{R}^3$  consisting of eigenvectors of  $\mathbf{A}$ . Therefore,  $\mathbf{A}$  is diagonalizable.

Thus the distinct eigenvalues are 6 and 0, with algebraic multiplicities 1 and 2, respectively, and  $\mathbf{A}$  is diagonalizable.

□

**Problem 2022.1.** Determine the largest of the following numbers.

- (1)  $e^\pi$
- (2)  $\pi^e$
- (3)  $7\pi$
- (4)  $\binom{7}{2}$
- (5)  $4!$

*Solution.* We compare the values directly:

$$e^\pi \approx 23.1407, \quad \pi^e \approx 22.4592, \quad 7\pi \approx 21.9911.$$

Also,

$$\binom{7}{2} = \frac{7 \cdot 6}{2} = 21, \quad 4! = 24.$$

Therefore, the largest number among the choices is (5)  $4!$ .

□

**Problem 2022.2.** Three people store a jointly owned treasure in a safe and want to lock it. They want no single person to be able to open the safe alone, but any two or more people should always be able to open it together. This can be done by putting  $n$  locks on the safe, making  $m$  copies of the key for each lock, and distributing the keys appropriately among the three people. Find  $n$  and  $m$  so that the number of locks  $n$  is minimized.

*Solution.* Let the three people be  $A$ ,  $B$ , and  $C$ .

Since any two people should be able to open the safe, the key to each lock must be given to at least two people. If a certain lock had its key given to only one person, then the other two people would not be able to open that lock together.

A lock whose key is given to all three people restricts no one and is unnecessary. Each useful lock can therefore exclude exactly one person. On the other hand, no single person should be able to open the safe alone. Therefore, each person must be missing the key to at least one lock.

Hence we need at least three locks: one lock whose key is not given to  $A$ , one lock whose key is not given to  $B$ , and one lock whose key is not given to  $C$ . This lower bound is achievable by distributing the keys as follows:

|       | $A$        | $B$        | $C$        |
|-------|------------|------------|------------|
| $L_A$ | $\times$   | $\bigcirc$ | $\bigcirc$ |
| $L_B$ | $\bigcirc$ | $\times$   | $\bigcirc$ |
| $L_C$ | $\bigcirc$ | $\bigcirc$ | $\times$   |

That is, the key to  $L_A$  is given to  $B$  and  $C$ , the key to  $L_B$  is given to  $A$  and  $C$ , and the key to  $L_C$  is given to  $A$  and  $B$ .

Then each person is unable to open one of the locks, so no one can open the safe alone. However, any two people together can open all three locks. Therefore, the minimum is achieved by  $\boxed{n = 3, m = 2}$ .

□

**Problem 2022.3.** Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2 + 7} dx.$$

*Solution.* Let

$$u = x - 2.$$

Then the limits of integration remain  $-\infty$  and  $\infty$ , so

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2 + 7} dx = 3 \int_{-\infty}^{\infty} \frac{1}{u^2 + 7} du.$$

For  $a > 0$ , we use the arctangent integral (Formula 2.49)

$$\int_{-\infty}^{\infty} \frac{1}{u^2 + a^2} du = \frac{\pi}{a}.$$

Here  $a = \sqrt{7}$ , and therefore

$$3 \int_{-\infty}^{\infty} \frac{1}{u^2 + 7} du = 3 \cdot \frac{\pi}{\sqrt{7}} = \frac{3\pi}{\sqrt{7}}.$$

Therefore, the answer is  $\boxed{\frac{3\pi}{\sqrt{7}}}$ .

□

**Problem 2022.4.** A couple has two children.

- Given that at least one child is a daughter, determine the probability that both children are daughters.
- Given that a specified child is a daughter named Youngmi, determine the probability that the other child is also a daughter.

*Solution.* Use the usual model in which the two children's sexes are independent, each with probability  $1/2$  of being a girl. Distinguishing the children by birth order gives four equally likely outcomes:

$$GG, \quad GB, \quad BG, \quad BB.$$

In part (a), conditioning on at least one daughter removes  $BB$ . The remaining three outcomes are equally likely, and only  $GG$  has two daughters. Thus

$$\mathbb{P}(GG \mid \text{at least one daughter}) = \frac{1}{3}.$$

In part (b), the specified child is already identified. Independence leaves the other child's probability of being a girl equal to  $1/2$ ; the name does not change this calculation under the stated interpretation. Therefore, the answers are

$$\boxed{\text{(a) } \frac{1}{3}, \quad \text{(b) } \frac{1}{2}}.$$

□

**Problem 2022.5.** Determine which of the following conjectures was not solved by Professor June Huh.

- (1) Read's conjecture
- (2) Brylawski's conjecture
- (3) Poincaré conjecture
- (4) Rota's conjecture
- (5) Okounkov's conjecture

*Solution.* Professor June Huh is known for connecting combinatorics with algebraic geometry and using these ideas to solve several major conjectures in combinatorics. His work is related to conjectures such as Read's conjecture, Brylawski's conjecture, Rota's conjecture, and Okounkov's conjecture.

On the other hand, the Poincaré conjecture is a famous problem in 3-dimensional topology. It was proved by Grigori Perelman using Ricci flow, not by June Huh.

Therefore, the conjecture in the list that was not solved by June Huh is

$$\boxed{\text{(3) Poincaré conjecture}}.$$

□

**Problem 2022.6.** This mathematician, often called the father of linear programming, is famous for solving difficult problems while he was a doctoral student. One day, while studying at UC Berkeley, he arrived late to a class. The statistics professor Jerzy Neyman had written two famous unsolved problems on the blackboard, but the student thought they were homework problems. He took them home, solved them, and submitted complete solutions a few days later, thinking only that he had turned in the homework late. Identify this mathematician.

*Solution.* George Dantzig developed the simplex method for linear programming. The two statistics problems he mistook for homework were posed in Jerzy Neyman's class at UC Berkeley. This is the Dantzig anecdote recorded in Part II. Therefore, the answer is George Dantzig.

□

**Problem 2022.7.** Two fair coins are each tossed three times, with all six tosses independent. Let  $X$  and  $Y$  be the numbers of heads obtained from the two coins, respectively. Determine  $\mathbb{P}(X = Y)$ .

*Solution.* Each count has the binomial distribution introduced in Probability and Statistics, so

$$X, Y \sim \text{Bin}\left(3, \frac{1}{2}\right),$$

and  $X$  and  $Y$  are independent. Therefore,

$$\begin{aligned} \mathbb{P}(X = Y) &= \sum_{k=0}^3 \mathbb{P}(X = k)\mathbb{P}(Y = k) \\ &= \frac{1}{64} \sum_{k=0}^3 \binom{3}{k}^2 \\ &= \frac{1 + 9 + 9 + 1}{64} = \boxed{\frac{5}{16}}. \end{aligned}$$

□

**Problem 2023.1.** Determine the year in which the Fields Medal, received by Professor June Huh, was first awarded.

*Solution.* The Fields Medal is one of the most prestigious awards in mathematics, and it is awarded by the International Mathematical Union. Professor June Huh received the Fields Medal in 2022.

The Fields Medal was first awarded in 1936. The first two recipients were Lars Ahlfors and Jesse Douglas.

Therefore, the answer is  $\boxed{1936}$ .

□

**Problem 2023.2.** Consider the  $6 \times 6$  square grid with vertices  $(0, 0)$ ,  $(6, 0)$ ,  $(0, 6)$ , and  $(6, 6)$ . A path starts at  $(0, 0)$  and ends at  $(6, 6)$ , moving only one unit to the right or one unit upward at each step. Determine the number of such paths that never go above the line  $x = y$ . Equivalently, determine the number of such paths that stay in the region

$$\{(x, y) \in \mathbb{R}^2 \mid y \leq x\}.$$

*Solution.* A path contains six right steps and six upward steps, so there are  $\binom{12}{6}$  unrestricted paths. To count the paths that cross above  $y = x$ , reflect the part up to the first point with  $y = x + 1$ , exchanging right and upward steps. This gives a bijection with paths having seven right steps and five upward steps, of which there are  $\binom{12}{5}$ . This is the reflection principle in Theorem 5.8. Subtracting the forbidden paths gives

$$\binom{12}{6} - \binom{12}{5} = 924 - 792 = \boxed{132}.$$

Equivalently, this is the Catalan number  $C_6 = \binom{12}{6}/7$ .

□

**Problem 2023.3.** A point on the surface of the Earth has the following property: starting from that point, one moves 5 km south, then 5 km east, and then 5 km north, and returns to the original point. Determine the number of such points on the surface of the Earth.

Assume that the surface of the Earth is a perfect two-dimensional sphere, and that the North Pole and the South Pole are antipodal points. At the North Pole, the directions north, east, and west are not defined, and at the South Pole, the directions south, east, and west are not defined. Starting points for which one of these undefined directions occurs during the movement are excluded.

*Solution.* The North Pole is one such point. If we start at the North Pole, move 5 km south, then move 5 km east along a latitude circle, and finally move 5 km north, we return to the North Pole.

There are also many such points near the South Pole. Let  $Q$  be the point reached after moving 5 km south from the starting point. Suppose that the latitude circle passing through  $Q$  has circumference

$$\frac{5}{n} \text{ km}$$

for some positive integer  $n$ . Then moving 5 km east from  $Q$  goes exactly  $n$  full times around that latitude circle and returns to  $Q$ . After that, moving 5 km north returns to the original starting point.

Near the South Pole, the circumferences of latitude circles approach 0. Hence, for every positive integer  $n$ , there is a latitude circle near the South Pole whose circumference is exactly  $\frac{5}{n}$  km. For each such latitude circle, every point lying 5 km north of it gives a valid starting point.

Therefore, there are not just one but infinitely many such points on the Earth's surface. In fact, this construction gives entire latitude circles of valid starting points, so there are uncountably many such points. In particular, the number of such points is

$$\boxed{\text{infinitely many}}.$$

□

**Problem 2023.4.** Let  $f: (-1, 1) \rightarrow \mathbb{R}$  be a continuous function. For each  $\lambda > 0$ , define

$$f_\lambda(x) := \lambda f\left(\frac{x}{\lambda}\right),$$

where  $f_\lambda$  is defined for  $x \in (-\lambda, \lambda)$ .

Suppose that for every sequence  $\lambda_i \rightarrow \infty$ , the functions  $f_{\lambda_i}$  converge locally uniformly to 0 on  $\mathbb{R}$ . Determine whether the following statement is true or false:

$$(f \text{ is differentiable at } 0) \wedge (f'(0) = 0).$$

*Solution.* Locally uniform convergence implies convergence at each fixed point once that point belongs to the domains. We only need the points 0, 1, and  $-1$ .

At  $x = 0$ , the assumption gives  $\lambda f(0) \rightarrow 0$  as  $\lambda \rightarrow \infty$ , so  $f(0) = 0$ . At  $x = 1$  and  $x = -1$ , it gives

$$\lambda f(1/\lambda) \rightarrow 0, \quad \lambda f(-1/\lambda) \rightarrow 0.$$

These limits hold as  $\lambda \rightarrow \infty$  because the assumption holds for every sequence tending to infinity.

To compute the right-hand derivative, put  $h = 1/\lambda > 0$ :

$$\frac{f(h) - f(0)}{h} = \lambda f(1/\lambda) \rightarrow 0.$$

For the left-hand derivative, put  $h = -1/\lambda < 0$ :

$$\frac{f(h) - f(0)}{h} = -\lambda f(-1/\lambda) \rightarrow 0.$$

Both one-sided derivatives exist and agree. Therefore  $f$  is differentiable at 0 with  $f'(0) = 0$ , and the answer is True.

□

**Problem 2024.1.** Determine the number of occurrences of the digit 0 when all natural numbers from 100 to 999 are written consecutively.

*Solution.* Every natural number from 100 to 999 is a three-digit number.

In a three-digit number, the hundreds digit cannot be 0. Therefore, the digit 0 can appear only in the tens digit or the units digit.

First, count the numbers whose tens digit is 0. The hundreds digit can be one of  $1, 2, \dots, 9$ , and the units digit can be one of  $0, 1, \dots, 9$ . Hence there are

$$9 \cdot 10 = 90$$

such numbers.

Next, count the numbers whose units digit is 0. The hundreds digit can be one of  $1, 2, \dots, 9$ , and the tens digit can be one of  $0, 1, \dots, 9$ . Hence there are again

$$9 \cdot 10 = 90$$

such numbers.

A number such as 100 contributes twice, correctly counting its two zeros. Therefore, the total number of appearances of the digit 0 is

$$90 + 90 = 180.$$

Therefore, the answer is 180.

□

**Problem 2024.2.** Determine which of the following people was born latest.

- (1) John von Neumann
- (2) Alan Turing
- (3) John Nash
- (4) Pierre de Fermat

*Solution.* We compare their years of birth:

- (1) John von Neumann: born in 1903,
- (2) Alan Turing: born in 1912,
- (3) John Nash: born in 1928,
- (4) Pierre de Fermat: early seventeenth century.

Among these four, the person born the latest is John Nash.

Therefore, the answer is (3) John Nash.

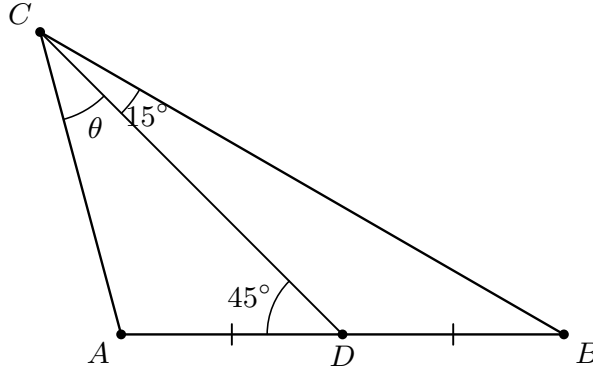
□

**Problem 2024.3.** In triangle  $ABC$ , the point  $D$  is the midpoint of  $\overline{AB}$ . Suppose that

$$\angle BCD = 15^\circ \quad \text{and} \quad \angle ADC = 45^\circ.$$

Determine

$$\theta = \angle ACD.$$



*Solution 1.* Let  $E$  be the foot of the perpendicular from  $C$  to the line  $AB$ . Since  $A, D, B$  are collinear and  $\angle ADC = 45^\circ$ , we have

$$\angle CDE = 45^\circ.$$

Thus  $\triangle CED$  is a  $45^\circ$ – $45^\circ$ – $90^\circ$  triangle. Set

$$CE = \ell.$$

Then

$$ED = \ell.$$

Also,

$$\angle BDC = 180^\circ - \angle ADC = 135^\circ,$$

so in  $\triangle BCD$ ,

$$\angle CBD = 180^\circ - 135^\circ - 15^\circ = 30^\circ.$$

Hence  $\triangle CEB$  is a  $30^\circ$ – $60^\circ$ – $90^\circ$  triangle, and therefore

$$EB = \sqrt{3}\ell.$$

The equality  $\angle CDE = 45^\circ$  places  $E$  on the  $A$ -side of  $D$ , so  $D$  lies between  $E$  and  $B$ . Hence

$$DB = EB - ED = (\sqrt{3} - 1)\ell.$$

Because  $D$  is the midpoint of  $\overline{AB}$ ,

$$AD = DB = (\sqrt{3} - 1)\ell.$$

In particular  $AD < ED$ , so the points occur in the order  $E, A, D, B$ , and

$$EA = ED - AD = (2 - \sqrt{3})\ell.$$

Since  $\angle ECD = 45^\circ$  and  $\angle ACD = \theta$ ,

$$\angle ECA = 45^\circ - \theta.$$

Applying the tangent ratio in the right triangle  $CEA$  gives

$$\tan(45^\circ - \theta) = \frac{EA}{CE} = 2 - \sqrt{3}.$$

On the other hand, the tangent subtraction formula gives

$$\tan 15^\circ = \tan(45^\circ - 30^\circ) = \frac{1 - 1/\sqrt{3}}{1 + 1/\sqrt{3}} = 2 - \sqrt{3}.$$

Since  $0^\circ < 45^\circ - \theta < 45^\circ$ , we conclude that

$$\boxed{\theta = 30^\circ}.$$

□

*Solution 2.* Since  $A, D, B$  are collinear,

$$\angle BDC = 180^\circ - \angle ADC = 135^\circ.$$

Therefore, in  $\triangle BCD$ ,

$$\angle CBD = 180^\circ - 135^\circ - 15^\circ = 30^\circ.$$

By the Sine Rule,

$$\frac{DB}{CD} = \frac{\sin 15^\circ}{\sin 30^\circ} = 2 \sin 15^\circ.$$

Let  $\theta = \angle ACD$ . In  $\triangle ACD$ ,

$$\angle CAD = 180^\circ - 45^\circ - \theta = 135^\circ - \theta.$$

Again by the Sine Rule,

$$\frac{AD}{CD} = \frac{\sin \theta}{\sin(135^\circ - \theta)}.$$

Since  $AD = DB$ , we obtain

$$\frac{\sin \theta}{\sin(135^\circ - \theta)} = 2 \sin 15^\circ.$$

Expanding the denominator gives

$$\sin \theta = \sqrt{2} \sin 15^\circ (\sin \theta + \cos \theta).$$

Since

$$\sqrt{2} \sin 15^\circ = \frac{\sqrt{3} - 1}{2},$$

it follows that

$$\tan \theta = \frac{1}{\sqrt{3}}.$$

As  $\theta$  is an interior angle of the triangle, we conclude that

$$\boxed{\theta = 30^\circ}.$$

□

**Problem 2024.4.** Let  $C$  be a convex subset of  $\mathbb{R}^3$ . Here convex means that for any two points  $\mathbf{p}, \mathbf{q} \in C$ , the line segment joining  $\mathbf{p}$  and  $\mathbf{q}$  is also contained in  $C$ .

Suppose that the following two facts are known:

- (1) The intersection of  $C$  with the  $xy$ -plane is

$$C \cap \{z = 0\} = \{(x, y, 0) \mid x^2 + 4y^2 \leq 1\}.$$

- (2) The set  $C$  contains the  $z$ -axis.

Determine the volume of  $C$  in the region  $-1 \leq z \leq 1$ , or determine that the given information is insufficient.

- (1) This cannot be determined.

- (2)  $\pi$

- (3)  $2\pi$

- (4)  $4\pi$

*Solution.* Let

$$E = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 \leq 1\}.$$

By condition (1),

$$C \cap \{z = 0\} = E \times \{0\}.$$

The ellipse  $E$  has semiaxes 1 and  $\frac{1}{2}$ , so its area is

$$\text{area}(E) = \pi \cdot 1 \cdot \frac{1}{2} = \frac{\pi}{2}.$$

For each height  $z$ , define the horizontal cross-section

$$C_z = \{(x, y) \in \mathbb{R}^2 \mid (x, y, z) \in C\}.$$

By Cavalieri's principle (Theorem 7.66), the volume is determined by these cross-sectional areas. We will show that every  $C_z$  contains the interior of  $E$  and is contained in  $E$ . Possible differences on the ellipse boundary do not affect area.

First, we show that  $C_z \subseteq E$ . Suppose that  $(x, y, z) \in C$  and

$$x^2 + 4y^2 > 1.$$

Since  $C$  contains the  $z$ -axis, the point  $(0, 0, t)$  belongs to  $C$  for every  $t \in \mathbb{R}$ . By convexity, the line segment joining  $(x, y, z)$  and  $(0, 0, t)$  is contained in  $C$ .

For any  $\alpha \in (0, 1)$ , we may choose

$$t = -\frac{\alpha z}{1 - \alpha}$$

so that this segment intersects the  $xy$ -plane at the point

$$(\alpha x, \alpha y, 0).$$

Since  $\alpha$  can be chosen sufficiently close to 1, we may arrange that

$$(\alpha x)^2 + 4(\alpha y)^2 > 1.$$

This would give a point of  $C \cap \{z = 0\}$  lying outside  $E \times \{0\}$ , contradicting condition (1). Therefore,

$$C_z \subseteq E$$

for every  $z$ .

Conversely, we show that the interior of  $E$  is contained in every  $C_z$ . Take  $(x, y)$  with

$$x^2 + 4y^2 < 1.$$

Choose  $\alpha \in (0, 1)$  sufficiently close to 1 so that

$$\left(\frac{x}{\alpha}\right)^2 + 4\left(\frac{y}{\alpha}\right)^2 \leq 1.$$

Then

$$\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) \in C.$$

Also, since  $C$  contains the  $z$ -axis,

$$\left(0, 0, \frac{z}{1-\alpha}\right) \in C.$$

By convexity, their convex combination also belongs to  $C$ , and

$$\alpha \left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) + (1-\alpha) \left(0, 0, \frac{z}{1-\alpha}\right) = (x, y, z).$$

Hence  $(x, y) \in C_z$ .

Thus, for every  $z$ ,

$$E^\circ \subseteq C_z \subseteq E.$$

Since the boundary of the ellipse has area 0, we have

$$\text{area}(C_z) = \text{area}(E) = \frac{\pi}{2}$$

for every  $z$ . Therefore, the volume of  $C$  in the region  $-1 \leq z \leq 1$  is

$$\int_{-1}^1 \text{area}(C_z) \, dz = \int_{-1}^1 \frac{\pi}{2} \, dz = \pi.$$

Therefore, the answer is  $\boxed{(2) \pi}$ .

□

**Problem 2025.1.** A fair coin is tossed repeatedly until either two heads in a row or two tails in a row appear. Determine the expected number of tosses before the process stops.

*Solution.* The first toss cannot stop the process. After any non-stopping history, the next toss matches the previous outcome with probability  $1/2$ . Let  $G$  count the additional tosses, including the matching toss. For  $k \in \mathbb{N}$ , surviving  $k-1$  additional tosses and then stopping gives

$$\mathbb{P}(G = k) = \left(\frac{1}{2}\right)^{k-1} \frac{1}{2}.$$

Thus  $G \sim \text{Geom}(1/2)$  under the convention in the distribution table: the successful trial is counted. Since  $\mathbb{E}[G] = 1/(1/2) = 2$ , the expected total number of tosses is

$$1 + \mathbb{E}[G] = \boxed{3}.$$

□

**Problem 2025.2.** The nineteenth-century mathematicians Cauchy and Weierstrass established rigorous definitions of limits. Identify the first mathematician to prove that the sum of the reciprocals of the squares of all positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

- (1) David Hilbert
- (2) Georg Cantor
- (3) Bernhard Riemann
- (4) Leonhard Euler

*Solution.* The problem of finding the exact value of the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is known as the Basel problem.

This problem had been known since the seventeenth century, and it was Leonhard Euler who first found and proved the famous identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the answer is (4) Leonhard Euler.

□

**Problem 2025.3.** Find the equation of the smallest sphere tangent to all four planes

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y + z = 3.$$

*Solution.* If a sphere of radius  $r > 0$  is tangent to the three coordinate planes, its center has the form

$$\mathbf{c} = (\sigma_1 r, \sigma_2 r, \sigma_3 r)^T, \quad \sigma_1, \sigma_2, \sigma_3 \in \{-1, 1\}.$$

Put  $k = \sigma_1 + \sigma_2 + \sigma_3$ , so

$$k \in \{-3, -1, 1, 3\}.$$

By the point-to-plane distance formula (Formula 7.74), tangency to  $x + y + z = 3$  requires

$$\frac{|kr - 3|}{\sqrt{3}} = r.$$

For  $k = 3$ , the positive solutions are

$$r = \frac{3}{3 + \sqrt{3}} \quad \text{and} \quad r = \frac{3}{3 - \sqrt{3}}.$$

For  $k = 1$ , the only positive solution is

$$r = \frac{3}{1 + \sqrt{3}},$$

and for  $k = -1$ , the only positive solution is

$$r = \frac{3}{\sqrt{3} - 1}.$$

The case  $k = -3$  has no positive solution. Therefore the smallest radius is

$$r = \frac{3}{3 + \sqrt{3}} = \frac{3 - \sqrt{3}}{2},$$

attained when  $\sigma_1 = \sigma_2 = \sigma_3 = 1$ . Hence the center is

$$\mathbf{c} = \left( \frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2} \right)^T,$$

and the required sphere is

$$\left[ \left( x - \frac{3 - \sqrt{3}}{2} \right)^2 + \left( y - \frac{3 - \sqrt{3}}{2} \right)^2 + \left( z - \frac{3 - \sqrt{3}}{2} \right)^2 = \left( \frac{3 - \sqrt{3}}{2} \right)^2 \right].$$

□

**Problem 2025.4.** There are 200 cards numbered from 1 to 200, all initially facing up. For each integer from 1 to 200, we call out that integer and flip every card whose number is a multiple of it. Determine the number of cards facing up after all 200 calls.

- (1) 142
- (2) 153
- (3) 164
- (4) 175
- (5) 186

*Solution.* A card numbered  $n$  is flipped once for each positive divisor of  $n$ . Hence it is flipped  $\tau(n)$  times, where  $\tau(n)$  is the number of positive divisors of  $n$ .

Divisors pair as  $d$  and  $n/d$ ; the only unpaired divisor occurs when  $d = \sqrt{n}$ . Thus  $\tau(n)$  is odd if and only if  $n$  is a perfect square. Therefore, exactly the cards numbered

$$1^2, \quad 2^2, \quad 3^2, \quad \dots, \quad 14^2$$

are flipped an odd number of times, since

$$14^2 = 196 \leq 200 < 225 = 15^2.$$

These 14 cards end up facing down. The remaining

$$200 - 14 = 186$$

cards are facing up.

Therefore, the answer is (5) 186.

□

**Problem 2025.5.** Determine which of  $11^9$  and  $9^{11}$  is larger.

*Solution.* Taking logarithms, it is enough to compare

$$9 \ln 11 \quad \text{and} \quad 11 \ln 9.$$

Equivalently, compare

$$\frac{\ln 11}{11} \quad \text{and} \quad \frac{\ln 9}{9}.$$

The comparison-of-powers method uses  $f(x) = \ln x/x$ . Since  $f'(x) = (1 - \ln x)/x^2 < 0$  for  $x > e$ , this function is decreasing there. Since  $9 < 11$  and both are greater than  $e$ , we have

$$\frac{\ln 9}{9} > \frac{\ln 11}{11}.$$

Hence

$$11 \ln 9 > 9 \ln 11,$$

so

$$9^{11} > 11^9.$$

Therefore, the larger number is  $\boxed{9^{11}}$ .

□

**Problem 2025.6.** Roll a fair six-sided die repeatedly until a 6 appears. Determine the expected sum of all rolls, including the final 6.

*Solution.* If  $N$  is the stopping time, then  $N \sim \text{Geom}(1/6)$  and  $\mathbb{E}[N] = 6$ . The total sum is at most  $6N$ , so its expectation is finite. Let  $E$  be the desired expected sum. On the first roll, if the result is 6, the process stops and contributes 6. If the result is one of 1, 2, 3, 4, 5, then the process returns to the same situation after contributing that roll.

Therefore,

$$E = \frac{1}{6} \cdot 6 + \frac{1}{6} \sum_{k=1}^5 (k + E).$$

Hence

$$E = 1 + \frac{15 + 5E}{6} = \frac{7}{2} + \frac{5}{6}E.$$

Thus

$$\frac{1}{6}E = \frac{7}{2}, \quad E = 21.$$

Therefore, the answer is  $\boxed{21}$ .

□

**Problem 2025.7.** Determine the probability that at least two people in a group of 10 share the same birthday. Assume a 365-day year and that each day is equally likely.

*Solution.* Under the standard independent-birthday model, use the complement as in Formula 6.30. It is easier to compute the probability that all 10 birthdays are distinct. The first person may have any birthday, the second must avoid that birthday, the third must avoid the first two birthdays, and so on. Hence

$$\mathbb{P}(\text{all birthdays are distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}.$$

Therefore,

$$\mathbb{P}(\text{at least two people share a birthday}) = 1 - \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}.$$

Therefore, the answer is

$$\boxed{1 - \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}}.$$

□

**Problem 2025.8.** Find the largest of the following numbers:

- (1)  $e^\pi$
- (2)  $\pi^e$
- (3)  $\sqrt{500}$
- (4)  $7\pi$

*Solution.* Using the mathematical constants listed at the beginning of the book gives

$$e^\pi \approx 23.14, \quad \pi^e \approx 22.46, \quad \sqrt{500} \approx 22.36, \quad 7\pi \approx 21.99.$$

Therefore, the answer is (1)  $e^\pi$ .

□

**Problem 2025.9.** There are three doors. One hides a car and the other two hide goats. After the contestant chooses a door, a host who knows the car's location must open an unchosen door hiding a goat; when two such doors are available, the host follows a rule independent of the car's location. The contestant may then keep the original choice or switch to the only other unopened door. Determine which choice gives the higher probability of winning the car.

- (1) The probabilities are the same.
- (2) It is better not to switch.
- (3) It cannot be determined.
- (4) It is better to switch.

*Solution.* The original choice contains the car with probability  $1/3$  and is wrong with probability  $2/3$ . Under the stated host protocol, switching wins exactly when the original choice was wrong. Hence staying wins with probability  $1/3$ , whereas switching wins with probability  $2/3$ .

Therefore, the answer is

(4) It is better to switch.

□

**Problem 2025.10.** Determine which of the following Fields Medalists majored in history rather than mathematics as an undergraduate.

- (1) June Huh
- (2) Caucher Birkar
- (3) Edward Witten
- (4) Simon Donaldson
- (5) William Thurston

*Solution.* Edward Witten is famous as a mathematical physicist and was the first physicist to receive the Fields Medal. As an undergraduate, he majored in history rather than mathematics.

Therefore, the answer is (3) Edward Witten.

□

**Problem 2025.11.** Determine which of the following mathematicians was not alive during the twentieth century.

- (1) Sophus Lie
- (2) Wilhelm Killing
- (3) Alfred Young

- (4) Hans Fitting
- (5) Botong Wang

*Solution.* Sophus Lie lived from 1842 to 1899, so he died before the twentieth century began. The other listed mathematicians were alive at some point during the twentieth century.

Therefore, the answer is (1) Sophus Lie.

□

**Problem 2025.12.** Königsberg was the capital of East Prussia and a major center of European scholarship for many years. It was also the hometown of Immanuel Kant, David Hilbert, and Jürgen Moser. The famous Seven Bridges of Königsberg problem, in which Leonhard Euler proved that no solution exists, marked the beginning of both topology and graph theory. Determine the present-day country containing the former city of Königsberg.

- (1) Germany
- (2) Poland
- (3) Lithuania
- (4) Russia
- (5) Republic of Korea

*Solution.* The historical city Königsberg is now Kaliningrad. Kaliningrad is a city in Russia.

Therefore, the answer is (4) Russia.

□

**Problem 2025.13.** A fair coin is flipped 50 times, and each outcome is recorded as either heads ( $H$ ) or tails ( $T$ ). A run is a maximal consecutive sequence of the same outcome. For example, if the result is  $HHTTTH$ , there are 3 runs:  $HH$ ,  $TTT$ , and  $H$ . Determine the expected number of runs.

- (1) 24.5
- (2) 25
- (3) 25.5
- (4) 26

*Solution.* The first toss always starts one run. For each position from the second toss onward, a new run begins exactly when the current toss differs from the previous toss.

For a fair coin,

$$\mathbb{P}(\text{current toss differs from previous toss}) = \frac{1}{2}.$$

For  $2 \leq i \leq 50$ , let

$$I_i = \mathbf{1}_{\{\text{a new run starts at position } i\}}.$$

The indicator  $I_i$  is 1 when a new run starts and 0 otherwise. If  $R$  denotes the total number of runs, then

$$R = 1 + \sum_{i=2}^{50} I_i.$$

Since  $\mathbb{E}[I_i] = \mathbb{P}(I_i = 1) = 1/2$ , linearity of expectation (Proposition 6.12) gives

$$\mathbb{E}[R] = 1 + \sum_{i=2}^{50} \mathbb{E}[I_i] = 1 + 49 \cdot \frac{1}{2} = 25.5.$$

Therefore, the answer is (3) 25.5.

□

**Problem 2025.14.** Determine which of the following results appeared first historically.

- (1) Bayes' Theorem
- (2) Intermediate Value Theorem
- (3) Cauchy–Schwarz Inequality
- (4) Fundamental Theorem of Calculus

*Solution.* Interpreting the question by the earliest recognizable form of each result, the inverse relationship between differentiation and integration was established in seventeenth-century work associated with Barrow, Newton, and Leibniz. Bayes' theorem was published in the eighteenth century, while rigorous forms of the Intermediate Value Theorem and the Cauchy–Schwarz inequality appeared in the nineteenth century.

Therefore, the intended answer is

(4) Fundamental Theorem of Calculus

□

**Problem 2025.15.** Which of the following amounts is closest to the Abel Prize money?

- (1) None
- (2) About \$1,000,000
- (3) About \$1,500,000
- (4) About \$2,000,000

*Solution.* In 2025, the Abel Prize amounted to NOK 7.5 million. Using the exchange rate relevant at the time, this was closest to one million US dollars among the listed choices.

Therefore, the answer is (2) About \$1,000,000.

□

**Problem 2025.16.** Determine which of the following mathematicians has received both the Fields Medal and the Abel Prize.

- (1) John Milnor
- (2) Terence Tao
- (3) June Huh
- (4) No one

*Solution.* John Milnor received the Fields Medal in 1962 and the Abel Prize in 2011. Therefore he has received both prizes.

Therefore, the answer is (1) John Milnor.

□

**Problem 2025.17.** A fair six-sided die is rolled until all 6 faces have appeared. Determine the probability that no odd-numbered face appears before all even-numbered faces have appeared.

- (1) 20%
- (2) 15%
- (3) 10%
- (4) 5%

*Solution.* Ignore repeated rolls and look only at the order in which new faces appear. The condition is that the first three distinct faces to appear are exactly the even faces 2, 4, 6 in some order.

The probability that the first new face is even is  $3/6$ . Given that one even face has appeared, the probability that the next new face is also even is  $2/5$ . Given that two even faces have appeared, the probability that the next new face is the remaining even face is  $1/4$ . Therefore the desired probability is

$$\frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{20} = 5\%.$$

Therefore, the answer is (4) 5%.

□

**Problem 2025.18.** In the Tower of Hanoi puzzle, 5 disks are stacked on peg  $A$  from largest to smallest. The objective is to move all disks to peg  $C$  using auxiliary peg  $B$ , subject to the following rules:

- (1) Only one disk may be moved at a time.
- (2) A larger disk may not be placed on a smaller disk.
- (3) Peg  $B$  may be used freely.

Determine the minimum number of moves required.

- (1) 25
- (2) 31
- (3) 32
- (4) 63
- (5) 120

*Solution.* Let  $T(n)$  be the minimum number of moves needed for  $n$  disks. To move  $n$  disks, first move the top  $n - 1$  disks to the auxiliary peg, then move the largest disk once, and finally move the  $n - 1$  disks onto the largest disk. These two transfers are also necessary before and after the largest disk reaches its destination, so the construction attains the lower bound. Hence

$$T(n) = 2T(n - 1) + 1, \quad T(1) = 1.$$

This is the Tower of Hanoi recurrence in Formula 5.14; solving it gives

$$T(n) = 2^n - 1.$$

Therefore,

$$T(5) = 2^5 - 1 = 31.$$

Therefore, the answer is (2) 31.

□

**Problem 2025.19.** Determine the last two digits of  $7^{123}$ , expressed as a two-digit number.

- (1) 37
- (2) 41
- (3) 43
- (4) 49

*Solution.* Work modulo 100. Since

$$7^4 = 2401 \equiv 1 \pmod{100},$$

and

$$123 = 4 \cdot 30 + 3,$$

we have

$$7^{123} = (7^4)^{30} 7^3 \equiv 1^{30} \cdot 343 \equiv 43 \pmod{100}.$$

Therefore, the answer is  $\boxed{(3) \ 43}$ .

□